Mathematical Methods for Physics 3a

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Contents

Goal of these notes is to make miserable the life of the reader(s) for the next 24 hours, which is, alas, equivalent to 3 credits at the University of Pavia. The detailed evil plan is based on the idea of outlining the following key concepts in differential geometry and algebraic topology which play a key role in several constructions in theoretical physics:

1. De Rham theory of differential forms on manifolds,
2. Integration on manifolds,
3. fiber and principal bundles together with their applications.

The plan is ambitious (no it is not, but, at this stage, you would not believe me), but it is based on a rich and exhaustive literature. Personally, I think that the most effective text is [14] except for the theory of integration which is not discussed there. Yet, it is fair to say, that the most complete text in algebraic topology (as far as differential forms are concerned) is [3] while, in differential geometry is [11]. For gauge theories and principal bundles alone, I recommend strongly [8] and [1] (more modern, but not as exhaustive). Finally, for a broader perspective on the theory of fiber bundles, refer to [7].

DISCLAIMER

It is assumed that the reader, also known as student or, for short, beloved reader, is familiar ... deeply familiar with the basic concepts of differential geometry, namely topological and smooth manifolds, tangent and cotangent spaces, vector fields and integral curves... no exception!

1 The Algebra of Differential Forms on $\mathbb{R}^n$

In this section we start discussing the theory of differential forms as discussed by de Rham. As it should be clear now (now meaning, after having learned enough differential geometry), we start our construction from $\mathbb{R}^n$, $n \geq 1$ and we generalize it subsequently to an arbitrary smooth manifold. In order to be self contained, at the beginning there will be a significant overlap with concepts introduced in the first part of the course. Yet, we feel that, ultimately, the pros are more than the cons. In the following we follow alternatively [3] and [14].

Definition 1. *Let $\{e_1, ..., e_n\}$, $n > 1$, be a collection of arbitrary abstract elements. We call* **alternating (or exterior) algebra** $\mathcal{A}_n$*, the algebra over the reals generated by the abstract elements via the relation*

$$e_i \cdot e_j = -e_j \cdot e_i, \quad \forall i, j = 1, ..., n$$

where $\cdot$ stands for the algebra product.

Henceforth we will omit writing explicitly the product symbol $\cdot$. It is immediate to observe the following consequences:

$$e_i^2 = 0, \quad \forall i = 1, ..., n.$$

In addition we have required that $n > 1$ since, with only one generator, the algebra structure becomes trivial. In the following we will refer to $\mathcal{A}_1 = span_{\mathbb{R}}(e_1, ..., e_n)$ which is not an algebra, but simply a vector space isomorphic to $\mathbb{R}^n$. A related concept is the following:

Definition 2. *Let $\mathcal{A}$ be an algebra (over a commutative ring C) and let G be a group. We say that $\mathcal{A}$ is G-graded if, for every $g \in G$ there exists a C-module $\mathcal{A}_g \subseteq \mathcal{A}$ such that*

1. *$\mathcal{A}_g \mathcal{A}_h \subseteq \mathcal{A}_{gh}$, for all $g, h \in G$,*
2. *$\mathcal{A} = \bigoplus_{g \in G} \mathcal{A}_g$ as module.*

In particular we call every element $a \in \mathcal{A}_g$, homogeneous of degree g. *If $G = \mathbb{Z}$ and $A_n = 0$ for all $n < 0$, we call it* **positively graded**.

Proposition 3. *For every $n > 1$, the alternating algebra $\mathcal{A}_n$ is positively graded*

Proof. By using the algebra relations, one realizes that $\mathcal{A}_n$ is a vector space of dimension 2^n with a canonical basis $\{1, e_1, ..., e_n, \bigcup\limits_{i_1 < i_2} (e_{i_1} e_{i_2}), \bigcup\limits_{i_1 < i_2 < i_3} (e_{i_1} e_{i_2} e_{i_3}), ..., e_1 e_2 ... e_n\}$. Hence, for any $k \in \mathbb{Z}$ we can set $(\mathcal{A}_n)_k = \{0\}$ if $k < 0$ or $k > n$, while $(\mathcal{A}_n)_0 = \mathbb{R}$ and $(\mathcal{A}_n)_k = span_{\mathbb{R}}(\bigcup\limits_{i_1 < i_2 < ... < i_k} (e_{i_1} e_{i_2} ... e_{i_k}))$ if $0 < k \leq n$. Using Definition 1, both condition 1. and 2. in Definition 2 hold true either by direct inspection or using the defining relation $e_i e_j = -e_j e_i$ for all $i, j = 1, ..., n$. $\square$

In order to simplify the notation we introduce

$$(\mathcal{A}_n)_k \equiv \mathcal{A}_n^k \Longrightarrow \mathcal{A}_n = \bigoplus_{k=0}^{n} \mathcal{A}_n^k.$$

We are now ready to define the main structure of this section:

Definition 4. *We call k-***differential forms on** $\mathbb{R}^n$ *for $n \geq 1$ the elements of the vector space $\mathcal{A}^k(\mathbb{R}^n) \doteq C^\infty(\mathbb{R}^n; \mathcal{A}_n^k)$.*

Notice that the notion of smoothness is meaningful since $\mathcal{A}_n^k$ is per construction a vector space of dimension $\binom{k}{n}$, hence isomorphic to $\mathbb{R}^{\binom{k}{n}}$ which can be endowed with the standard differential structure. The space of k-forms (differential will be henceforth implicit) admits several equivalent characterizations. The following exercise will be particularly instructive (if you solve it):

Exercise 1. There exists an isomorphism of topological vector spaces between $C^\infty(\mathbb{R}^n; \mathcal{A}_n^k)$ and $C^\infty(\mathbb{R}^n) \otimes \mathcal{A}_n^k$.

Warning: In the literature the following symbols often appear to indicate the same space:

$$\Omega^k(\mathbb{R}^n) \equiv \Lambda^k(\mathbb{R}^n) = \mathcal{A}^k(\mathbb{R}^n).$$

We might hop from one notation to the other depending on which one is more commonly used in the case in hand.

It is immediate to observe that the space of all possible differential forms inherits the alternating algebra structure.

Proposition 5. *Let $\mathcal{A}^\bullet(\mathbb{R}^n) = \bigoplus_{k=0}^n \mathcal{A}^k(\mathbb{R}^n)$. This is a positively graded alternating algebra if endowed with the* **wedge product** *structure $\wedge : \mathcal{A}^l(\mathbb{R}^n) \times \mathcal{A}^{l'}(\mathbb{R}^n) \to \mathcal{A}^{l+l'}(\mathbb{R}^n)$, $l, l' \in \mathbb{Z}$ completely and uniquely specified by the relation*

$$(f \otimes a) \wedge (f' \otimes b) = ff' \otimes (a \cdot b),$$

where $f, f' \in C^\infty(\mathbb{R}^n)$, while $a \in \mathcal{A}_n^l$, $b \in \mathcal{A}_n^{l'}$ and $\cdot$ is the algebra product in $\mathcal{A}_n$

The proof is fairly straightforward since it requires only a direct verification that the defining properties of a positively graded alternating algebra are met. In addition, we remark that, when working with k-forms it is often customary to introduce explicitly the *degree* map:

$$\deg : \mathcal{A}^l(\mathbb{R}^n) \to \mathbb{Z}, \quad \omega \mapsto \deg(\omega) = l, \tag{1}$$

from which it descends the famous relation between differential forms

$$\omega \wedge \omega' = (-1)^{\deg(\omega)\deg(\omega')} \omega' \wedge \omega. \quad \forall \omega \in \mathcal{A}^l(\mathbb{R}^n) \text{ and } \forall \omega' \in \mathcal{A}^{l'}(\mathbb{R}^n), \tag{2}$$

where both l and l' are arbitrary. Having introduced differential forms on $\mathbb{R}^n$, there are three natural questions to be asked:

1. what is the alternating algebra encoding as information?
2. does there exist a geometric characterization of k-forms?
3. what are k-forms good for?

In the next section we shall give an answer first of all to the second question starting from the simplest example of differentiable manifold, namely $\mathbb{R}^n$. Subsequently we shall generalize the whole construction to arbitrary differentiable manifolds, giving an answer to the third and most important question.

Before proceeding in our endeavor (which is also the name given to one of the space shuttles), we focus instead on the first question. As a matter of fact, Definition 1 in particular is nothing but the abstract algebraic realization of the canonical anticommutation relations. It goes without saying that, despite the temptation, the reason for introducing these structures does not stem from Fermions or from the Pauli exclusion principle. On the contrary, in many different instances, in mathematics one introduces operations which are anti-symmetric with respect to their arguments, the canonical example being the cross product between vectors. It is therefore natural to try to encode this property in the most abstract possible way within an algebraic structure. This is encoded by Definition 1 and 2.

It is worth emphasizing that, from the viewpoint of an algebraist, the alternating algebra is very informative and useful since it allows for an abstract characterization of one of the key operations in linear algebra: *the determinant.*

Definition 6. *Let $\mathcal{A}_n^1 = span_{\mathbb{R}}\{e_1, e_2, ..., e_n\}$ be the vector space, composed of homogeneous elements of $\mathcal{A}_n$ of degree 1. We call* **determinant** *the map*

$$\det : \mathcal{L}(\mathcal{A}_n^1) \to \mathbb{R} \quad K \mapsto \det(K),$$

which is completely characterized by the defining relation

$$K(e_1) \cdot K(e_2) \cdot ... \cdot K(e_n) = \det(K) e_1 \cdot e_2 \cdot ... \cdot e_n, \tag{3}$$

where $\cdot$ is the algebra product as in Definition 1.

In order to convince ourselves that this is a meaningful definition, we must compare this with the standard notion of determinant of a matrix proving that they are equivalent.

Proposition 7. *Let $\mathcal{K} \in M(n, \mathbb{R})$ and let $K \in \mathcal{L}(\mathcal{A}_n^1)$ be the linear map which is defined as the unique linear extension of the application*

$$K(e_i) = \sum_{j=1}^{n} \mathcal{K}_{ij} e_j, \quad \forall i = 1, ..., n$$

Then $\det(K) = \det_{M(n,\mathbb{R})}(\mathcal{K})$ where the right hand side stands for the determinant of a matrix.

Proof. We can proceed by direct inspection, since, using Definition 1, it holds

$$K(e_1) \cdot K(e_2) \cdot ... \cdot K(e_n) = \sum_{j_1=1}^{n} \mathcal{K}_{1j_1} e_{j_1} \cdot ... \cdot \sum_{j_n=1}^{n} \mathcal{K}_{nj_n} e_{j_n} = \sum_{\sigma \in S_n} \prod_{i=1}^{n} \mathrm{sign}(\sigma) \mathcal{K}_{i\sigma(i)} e_1 \cdot ... \cdot e_n,$$

where S_n is the permutation group of n-elements $(1, 2, ..., n)$, while $\mathrm{sign}(\sigma)$ is the sign of the permutation and $\sigma(i)$ is the i-th entry of the permutation $\sigma(1, 2, ..., n)$. From standard linear algebra and from Equation (3), we obtain

$$\det(K) = \sum_{\sigma \in S_n} \prod_{i=1}^{n} \mathrm{sign}(\sigma) \mathcal{K}_{i\sigma(i)} = \det_{M(n,\mathbb{R})}(\mathcal{K}),$$

which concludes the proof. $\square$

We leave to the reader as an exercise of linear algebra to prove a converse of Proposition 7, namely that every $n \times n$ real matrix induces an element of $\mathcal{L}(\mathcal{A}_n^1)$ and the determinants coincide. To conclude this first part we stress, that, while the notion of determinant might appear just a slightly more abstract generalization of the standard one, this will become especially useful in the theory of integration on manifolds.

1.1 Geometric realization of k-forms on $\mathbb{R}^n$

In the previous part, we have introduced k-differential forms $\mathcal{A}^k(\mathbb{R}^n)$ in Definition 4, endowing $\mathcal{A}^\bullet(\mathbb{R}^n)$ with the structure of an alternating, positively graded, algebra via the wedge product in Proposition 5. In this construction there is a purely algebraic component coming from $\mathcal{A}_n$ and a geometric one represented by $C^\infty(\mathbb{R}^n)$. In the following we show that one can give a concrete realization to the generating elements $\{e_1, e_2, ..., e_n\}$ in terms of suitable geometric structures. This procedure will make more manifest the correlation between our approach to k-forms and the standard one. At the beginning the whole construct might appear baroque and ostentatious, but it will be clear at a later stage that it allows for a rather straightforward generalization to arbitrary differentiable manifolds (well... if you have come to my lectures on the theory of Lie groups, you should better know by now...).

The starting point of our analysis is the observation of the following non trivial interplay between vectors of $\mathbb{R}^n$ and functions lying in $C^\infty(\mathbb{R}^n)$, where smoothness is required only for simplicity, since it does not lead to a loss of information in the procedure. More precisely, observe that given the standard Euclidean coordinates $\{x_1, ..., x_n\}$ of $\mathbb{R}^n$ and given a vector $v \in \mathbb{R}^n$ whose components with respect to the standard basis are $v_1, ..., v_n$, then v identifies a *directional derivative* at the base point of the vector ($x = 0$ in this case) as follows:

$$\mathbb{R}^n \ni v \mapsto D_v : C^\infty(\mathbb{R}^n) \to \mathbb{R} \text{ where } f \mapsto D_v(f) = \sum_{i=1}^n v_i \, \partial_i f|_{x=0}\,, \tag{4}$$

This motivates the following definition, which aims at allowing to vary smoothly the base point of the vector and at keeping the interpretation of a vector as a directional derivative:

Definition 8. *We call* **smooth vector field** *on* $\mathbb{R}^n$ *any linear map* $v : C^\infty(\mathbb{R}^n) \to C^\infty(\mathbb{R}^n)$ *which satisfies the Leibnitz rule:*

$$v(ff')(x) = f(x)v(f')(x) + f'(x)v(f)(x), \quad \forall x \in \mathbb{R}^n$$

We call the collection of all smooth vector fields $\aleph(\mathbb{R}^n)$*.*

A standard result in differential geometry, see e.g. [11, Chap.3], entails that, once chosen a coordinate system on $\mathbb{R}^n$, say $\{x_1, ..., x_n\}$, not necessarily the Cartesian one, then, there must exist $v_i \in C^\infty(\mathbb{R}^n)$, $i = 1, ..., n$ such that

$$v = \sum_{i=1}^n v_i \frac{\partial}{\partial x_i}. \tag{5}$$

This alternative viewpoint on the role of vectors has the main advantage of being easily generalized to arbitrary differentiable manifolds. Here we shall not focus on this aspect. On the contrary we emphasize that, while in Definition 8 the rationale is that vector fields act on an arbitrary smooth function to give another function, suitably interpreted as a derivative, there exists a dual, albeit equivalent, interpretation. This calls for defining the action of a smooth function on a vector field via the following structure.

Definition 9. *Let $f \in C^\infty(\mathbb{R}^n)$. We call* **differential** *of f, the linear map*

$$df : \aleph(\mathbb{R}^n) \to C^\infty(\mathbb{R}^n) \quad v \mapsto df(v) \doteq v(f).$$

If a reader is already familiar with the geometric notion of differential of a function (push-forward), we stress that it is equivalent to Definition 9. We will focus on this aspect when we will generalize this construction to arbitrary differentiable manifolds.

It is important to give a characterization of the differential in terms of an arbitrary but fixed coordinate system $\{x_1, ..., x_n\}$. Per linearity it holds that, using Equation (5) and Definition 9, for any $f \in C^\infty(\mathbb{R}^n)$,

$$df(v) = \sum_{i=1}^n v_i df\left(\frac{\partial}{\partial x_i}\right) = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i},$$

At the same time, if we take into account that each coordinate over $\mathbb{R}^n$ identifies a smooth map $x_i : \mathbb{R}^n \to \mathbb{R}$ which reads for each $x \in \mathbb{R}^n$ its i-th component with respect to the chosen coordinates, it holds, that, for all $v \in \aleph(\mathbb{R}^n)$,

$$dx_j(v) = \sum_{i=1}^n \frac{dx_j}{dx_i} = v_j.$$

Putting together these infos, we have shown that, for any coordinate system $\{x_1, ..., x_n\}$ over $\mathbb{R}^n$ we have written

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i \tag{6}$$

We have finally all the ingredients necessary to give a geometric realization of k-forms. As a matter of fact, the differentials $\{dx_i\}_{i=1,...,n}$ are the natural candidate to play the role of the abstract basis $\{e_1, ..., e_n\}$ in Definition 1.

Remark 1. *In order to disentangle the abstract realization of k-differential forms from the concrete one, outlined in the last paragraph, we use a different notation. We call Ω_n the* exterior algebra *generated by $\{dx_1, ..., dx_n\}$ and the vector space built out of elements of grade* 1 *is*

$$\Omega^1 \equiv \Omega^1_n = span_{\mathbb{R}}(dx_1, ..., dx_n) \tag{7}$$

and consequently, in view of Definition 4, we have given a concrete and geometric realization of k-differential forms as

$$\Omega^k(\mathbb{R}^n) \doteq C^\infty(\mathbb{R}^n; \Omega^k_n) \equiv C^\infty(\mathbb{R}^n) \otimes \Omega^k_n,$$

where we used Exercise 1. With such notation, it is immediate to realize that Equation (6) identifies an element

$$df \in C^\infty(\mathbb{R}^n) \otimes \Omega^1 \equiv \Omega^1(\mathbb{R}^n).$$

Finally we recall that, in view of Proposition 5, if we introduce

$$\Omega^\bullet(\mathbb{R}^n) = \bigoplus_{k=0}^{n} \Omega^k(\mathbb{R}^n) \equiv C^\infty(\mathbb{R}^n) \otimes \Omega^\bullet, \quad \Omega^\bullet \doteq \bigoplus_{k=0}^{n} \Omega^k_n, \tag{8}$$

this is an exterior algebra with respect to the wedge product, which is still indicated as $\wedge$.

Exercise 2. Prove that the construction in Remark 1 does not depend on the choice of the coordinates $\{dx_1, \dots, dx_n\}$ in the sense, that different coordinates yield isomorphic algebras.

At this stage the question is the following: which is the advantage in having given a geometric realization of the alternating/exterior algebra?

1.2 The de Rham complex

In this section we give an answer to the question which concluded the previous one. The starting point is the observation that, in the process of constructing $\Omega^k(\mathbb{R}^n)$, $k = 0, ..., n$, we have defined a new relevant operation, namely in Definition 9, we have implicitly introduced a map

$$d_0 : C^\infty(\mathbb{R}^n) \equiv \Omega^0(\mathbb{R}^n) \to \Omega^1(\mathbb{R}^n) \quad f \mapsto d_0(f) = df.$$

It is straightforward to observe that this operation can be generalized to arbitrary k-forms as follows:

Definition 10. *We call* **exterior differentiation** *the linear map* $d_k : \Omega^k(\mathbb{R}^n) \to \Omega^{k+1}(\mathbb{R}^n)$, $k \in \mathbb{N} \cup \{0\}$ *such that, for any coordinate system* $\{x_1, ..., x_n\}$ *on* $\mathbb{R}^n$, *it holds*

1. *if* $k = 0$ $d_0(f) = df$ *as in Equation (6),*
2. *if* $k > 1$ *then, for every* $\omega \in \Omega^p(\mathbb{R}^n)$, *decompsed as* $\omega = \sum\limits_{i_1 < i_2 < ... < i_k} c_{i_1 ... i_k}(x) dx_{i_1} \wedge ... \wedge dx_{i_k}$,

$$d_k\omega = \sum_{j=1}^{n} \sum_{i_1 < i_2 < ... < i_k} \frac{\partial \omega_{i_1 ... i_k}(x)}{\partial x_j} dx_j \wedge dx_{i_1} \wedge ... \wedge dx_{i_k}.$$

The $(k+1)$*-form* $d_k\omega$, $\omega \in \Omega^k(\mathbb{R}^n)$ *is the called the* **exterior derivative** *of* ω.

It is immediate to show that the exterior differentiation satisfies the following notable properties:

Corollary 11. *Let* d_k *be the exterior differentiation. It holds that*

1. *for all* $k \geq 0$, $d_{k+1} \circ d_k = 0$,
2. *for all* $k, k' \in \mathbb{N} \cup \{0\}$, *it holds*

$$d_{k+k'}(\omega \wedge \omega') = d_k\omega \wedge \omega' + (-1)^{\deg \omega} \omega \wedge d_{k'}\omega', \quad \omega \in \Omega^k(\mathbb{R}^n) \text{ and } \omega' \in \Omega^{k'}(\mathbb{R}^n).$$

Proof. Let us prove the two items separately.

1. Consider any $\omega \in \Omega^k(\mathbb{R}^n)$. From Definition 10, it descends that, chosen any basis of $\mathbb{R}^n$, $\{x_1, ..., x_n\}$, we can write $\omega = \sum\limits_{i_1<i_2<...<i_k} c_{i_1...i_k}(x) dx_{i_1} \wedge ... \wedge dx_{i_k}$. It descends

$$(d_{k+1} \circ d_k)(\omega) = \sum_{j=1}^{n} \sum_{i_1<i_2<...<i_k} \frac{\partial^2 \omega_{i_1...i_k}(x)}{\partial x_i \partial x_j} dx_i \wedge dx_j \wedge dx_{i_1} \wedge ... \wedge dx_{i_k} = 0,$$

since $dx_i \wedge dx_j = -dx_j \wedge dx_i$, while the second derivative is symmetric under exchange of x_i and x_j by Schwartz theorem.

2. Since the exterior differentiation is linear, it suffices to prove the statement for $\omega = a(x)dx_{i_1} \wedge ... \wedge dx_{i_k}$ and $\omega' = b(x)dx_{j_1} \wedge ... \wedge dx_{j_{k'}}$, where $a, b \in C^\infty(\mathbb{R}^n)$ while $(i_1, ..., i_k)$ (*resp.* $(j_1, ..., j_{k'})$) are two arbitrary sequences of k (*resp.* k') increasing integers between 0 and n. Using Definition 10 it holds that

$$d_k\omega = \sum_{i=1}^{n} \frac{\partial a(x)}{\partial x_i} dx_i \wedge dx_{i_1} \wedge ... \wedge dx_{i_k}, \quad d_{k'}\omega' = \sum_{i=1}^{n} \frac{\partial b(x)}{\partial x_i} dx_i \wedge dx_{j_1} \wedge ... \wedge dx_{j_{k'}}.$$

Hence

$$\begin{aligned} d_k\omega \wedge \omega' + (-1)^{\deg \omega} \omega \wedge d_{k'}\omega' &= \sum_{i=1}^{n} \frac{\partial a(x)}{\partial x_i} b(x) dx_i \wedge dx_{i_1} \wedge ... \wedge dx_{i_k} \wedge dx_{j_1} \wedge ... \wedge dx_{j_{k'}} + \\ &+ (-1)^{\deg \omega} \sum_{i=1}^{n} \frac{\partial b(x)}{\partial x_i} a(x) dx_{i_1} \wedge ... \wedge dx_{i_k} \wedge dx_i \wedge dx_{j_1} \wedge ... \wedge dx_{j_{k'}} = \\ &= \sum_{i=1}^{n} \left(\frac{\partial a(x)}{\partial x_i} b(x) + \frac{\partial b(x)}{\partial x_i} a(x) \right) dx_i \wedge dx_{i_1} \wedge ... \wedge dx_{i_k} \wedge dx_{j_1} \wedge ... \wedge dx_{j_{k'}}, \end{aligned}$$

where in the last equality we used that each wedge product is antisymmetric and that $\deg \omega = k$. A direct inspection unveils that the last expression is nothing but $d_{k+k'}(\omega \wedge \omega')$. □

On account of the second property proven in this last corollary, it is said that the exterior differentiation is an *antiderivation*.

Remark 2. *Observe that it many textbooks, the subscript k in the exterior differential is omitted. This is not a slight abuse of notation. On the contrary, since each $\Omega^k(\mathbb{R}^n) \hookrightarrow \Omega^\bullet(\mathbb{R}^n)$, we can introduce*

$$\underline{d} : \Omega^\bullet(\mathbb{R}^n) \to \Omega^\bullet(\mathbb{R}^n),$$

which is defined as follows: For all $\underline{\omega} \in \Omega^\bullet(\mathbb{R}^n)$, there exists $\omega_k \in \Omega^k(\mathbb{R}^n)$, $k = 0, ..., n$ such that $\underline{\omega} = (\omega_0, ..., \omega_n)$ and

$$\underline{d\omega} \doteq (0, d_0\omega_0, ..., d_{n-1}\omega_{n-1}).$$

A direct application of Corollary 11, entails that, for all $\omega \in \Omega^k(\mathbb{R}^n)$ and for all $\omega' \in \Omega^{k'}(\mathbb{R}^n)$ it holds

$$\underline{d}^2 = \underline{d} \circ \underline{d} = 0, \text{ and } \underline{d}(\omega \wedge \omega') = \underline{d}\omega \wedge \omega' + (-1)^{\deg \omega} \omega \wedge \underline{d}\omega'. \tag{9}$$

The real (slight) abuse of notation originates from the habit of writing d *in place of* $\underline{d}$, *using thus a symbol which appears in several different contexts with slightly different meanings.*

In reading the above construction, one might have the feeling that the exterior differentiation is an operation which is not intrinsic on the space of differentiable forms of any order and it depends on the coordinate chosen on $\mathbb{R}^n$. In the following theorem we will prove that a uniqueness result which should dispel any doubt. Here we adapt to the case in hand [11, Th. 14.24]:

Theorem 12. *Let $\Omega^\bullet(\mathbb{R}^n) = \bigoplus_{k=0}^n \Omega^k(\mathbb{R}^n)$. There exists a unique operator $\underline{d} = (d_0, ..., d_n)$ acting component by component on any $\underline{\omega} \in \Omega^\bullet(\mathbb{R}^n)$ in such a way that the following four properties are satisfied:*

1. *$\underline{d}$ is $\mathbb{R}$-linear,*
2. *$\underline{d}^2 = 0$,*
3. *for all $\omega \in \Omega^k(M)$ and for all $\omega' \in \Omega^{k'}(\mathbb{R}^n)$*

$$\underline{d}(\omega \wedge \omega') = \underline{d}\omega \wedge \omega' + (-1)^{\deg \omega} \omega \wedge \underline{d}\omega'$$

4. *for every $\underline{\omega} = (f, 0, ..., 0)$ where $f \in C^\infty(\mathbb{R}^n) \equiv \Omega^0(\mathbb{R}^n)$, it holds*

$$\underline{d\omega} = (0, d_0 f, 0, ..., 0),$$

where $d_0 f = df$ coincides with the differential of f as per Definition 9.

Proof. Suppose that there exists a second operator $\underline{d}'$ which satisfies condition 1.-4. besides the one built in Remark 2. On account of item 4., $\underline{d}$ and $\underline{d}'$ coincide when acting on $\Omega^0(\mathbb{R}^n)$. Focusing on $\Omega^1(\mathbb{R}^n)$ and choosing any coordinate system $\{x_1, ..., x_n\}$ on $\mathbb{R}^n$, linearity (item 2.) entails that it suffices to compare the two operators on $f(x)dx_i$, where $f(x) \in C^\infty(\mathbb{R}^n)$, while the index i is arbitrary but fixed. Yet item 3. guarantees that

$$\underline{d}(f(x)dx_i) = df \wedge dx_i = \underline{d}'(f(x)dx_i),$$

where we used implicitly that $\underline{d}^2 = (\underline{d}')^2 = 0$. We can proceed by induction. Suppose that the two operators coincide on k-forms. In order to check what happens on $(k+1)$-forms, it suffices per linearity to check their action on $\omega_{k+1} = h(x)dx_{i_1} \wedge ... \wedge dx_{i_{k+1}}$ where $h \in C^\infty(\mathbb{R}^n)$, while $(i_1, ...i_{k+1})$ is a collection of arbitrary $(k+1)$ integer indices in increasing order. If we define $\omega_k = h(x)dx_{i_1} \wedge ... \wedge dx_{i_k}$, it holds $\omega_{k+1} = \omega_k \wedge dx_{i_{k+1}}$ from which it descends from item 3. that

$$\underline{d}\omega_{k+1} = \underline{d}\omega_k \wedge dx_{i_{k+1}} = \underline{d}'\omega_k \wedge dx_{i_{k+1}} = \underline{d}'\omega_{k+1},$$

where we used the induction hypothesis. As a consequence the operators $\underline{d}$ and $\underline{d}'$ coincide on all $\Omega^k(\mathbb{R}^n)$, $k \in \mathbb{N} \cup \{0\}$ and they are the same operator. □

We can reinterpret the information extracted from the exterior differentiation in terms of suitable algebraic structures. We recall here

Definition 13. *Let $C^\bullet$ be a graded vector space, i.e., there exists a countable family of vector spaces $\{C_k\}_{k\in\mathbb{N}}$, such that $C^\bullet \simeq \oplus_{k\in\mathbb{N}} C_k$. We call $C^\bullet$ a* **differential (or cochain) complex** *if there exists a map $\underline{d} : C^\bullet \to C^\bullet$, called the* **differential** *(of the complex), such that $\underline{d} = \{d_k\}_{k\in\mathbb{N}}$ is a collection of linear maps $d_k : C_{k-1} \to C_k$*

$$... \longrightarrow C_{k-1} \xrightarrow{d_{k-1}} C_k \xrightarrow{d_k} C_{k+1} \longrightarrow ... \quad , \tag{10}$$

subject to the constraint $d_{k+1} \circ d_k = 0$ for all $k \in \mathbb{N}$.

Corollary 14. *The pair $(\Omega^\bullet(\mathbb{R}^n), \underline{d})$ built out of differential forms and the exterior differentiation is a differential complex, called the* **de Rham complex**

$$... \longrightarrow \Omega^{k-1}(\mathbb{R}^n) \xrightarrow{d_{k-1}} \Omega^k(\mathbb{R}^n) \xrightarrow{d_k} \Omega^{k+1}(\mathbb{R}^n) \longrightarrow ... \quad , \tag{11}$$

Proof. This is a direct consequence of Definition 4, Remark 1 together with Corollary 11. □

We recall now the standard nomenclature used for differential forms, adapting it to a generic differential complex $(C^\bullet, \underline{d})$. We say that $c \in C_k$ is

- **closed (or a cocycle)** if $d_k c = 0$. We call $\mathbf{C^k_d}$ the vector subspace of C_k made of closed k-elements.
- **exact (or a coboundary)** if there exists $a \in C^{k-1}$ such that $c = d_{k-1}a$. The collection of all exact k-elements is a vector subspace of C_k indicated as $\mathbf{d[C_{k-1}]}$ where the subscript $k-1$ can be dropped from the differential since no confusion can arise.

In view of these distinguished subspaces, we can introduce the following notable quotient of vector spaces:

Definition 15. *Let $(C^\bullet, \underline{d})$ be any differential complex. For every $k \in \mathbb{N}$, we call* **k-th cohomology** *of the complex the vector space defined as the quotient*

$$H^k_{\underline{d}}(C^\bullet) \doteq \frac{C^k_d}{d[C_{k-1}]}.$$

If we consider in particular the de Rham complex, then, for every $k \in \mathbb{N} \cup \{0\}$, we call **k-th de Rham cohomology** *of $\mathbb{R}^n$ the vector space defined as the quotient*

$$H^k_{dR}(\mathbb{R}^n) \doteq \frac{\Omega^k_d(\mathbb{R}^n)}{d[\Omega^{k-1}(\mathbb{R}^n)]}.$$

Exercise 3. Prove with a direct computation that, for all $n \in \mathbb{N}$, it holds

$$H^k_{dR}(\mathbb{R}^n) \simeq \begin{cases} \mathbb{R} & \text{if k=0} \\ 0 & \text{otherwise} \end{cases},$$

where 0 stands for the trivial vector space.

Observe that the subscript dR (de Rham) is omitted unless there is a possible source of confusion. We will stick henceforth to this policy

Remark 3. *Exercise 3 seems to indicate that de Rham cohomology is rather uninformative since it is trivial unless $k = 0$, namely the only functions (0-forms) with vanishing differential are the constant ones. Yet, observe that the notion of de Rham complex can be straightforwardly extended to any open subset $U \subseteq \mathbb{R}^n$ by restriction since, from Equation 8, it is natural to set*

$$\Omega^\bullet(U) = \bigoplus_{k=0}^{n} \Omega^k(U) \quad \textit{where} \quad \Omega^k(U) = \Omega^k(\mathbb{R}^n)\Big|_U \equiv C^\infty(U) \otimes \Omega^\bullet,$$

while exterior differentiation is left untouched. While the de Rham complex has a rather obvious behaviour under restriction, this is not the case for de Rham cohomology. As a matter of fact, let

- *$\Omega^k_d(U)$ be the space of closed k-forms on U,*
- *$d[\Omega^{k-1}(U)]$ be the space of exact $k-1$ forms on U.*

We define

$$H^k_{dR}(U) \doteq \frac{\Omega^k_d(U)}{d[\Omega^{k-1}(U)]}.$$

The following exercise is very instructive

Exercise 4. Let $U \subset \mathbb{R}$. Prove that

$$H^k_{dR}(U) \simeq \begin{cases} \mathbb{R}^m & \text{if k=0} \\ 0 & \text{otherwise} \end{cases},$$

where m stands for the number of different connected components of U.

In other words Exercise 4 seems to indicate that de Rham cohomology is a more sophisticated tool than one could infer at first glance. As a matter of fact the 0-th cohomology yields a topological information on the underlying space, counting the number of its disconnected components.

To conclude this subsection, we remark that we have learned that differential k-forms and the de Rham complex carry information on the topology of the underlying space U, here limited to being a subset of $\mathbb{R}^n$, though only through $k = 0$. On the one hand this is a consequence of the renown Poincaré lemma which we will prove later, while, on the other hand, it reminds us why, on Minkowski spacetime, the formulation of Maxwell's equation via the Faraday tensor $F \in \Omega^2(\mathbb{R}^4)$ admits an equivalent description in terms of the vector potential. More precisely

$$d_2 F = 0 \Longleftrightarrow [F] \in H^2_{dR}(\mathbb{R}^4) = \{0\} \Longrightarrow F = d_1 A,$$

where $A \in \Omega^1(\mathbb{R}^n)$. Yet the de Rham complex from Corollary 14 tells us that A is not uniquely defined since we can make the identification $A \simeq A + d_0\chi$, $\chi \in C^\infty(\mathbb{R}^4)$. This is nothing but the gauge freedom of electromagnetism stemming from the first of the two Maxwell's equations, *i.e.* $d_2 F = 0$. In addition, we have just got a glimpse of another important fact: This last equation is not dynamical, since it encodes only an information on the topology of the underlying space via an object $\underline{d}$, the exterior differentiation, built out of the differentiable structure of $\mathbb{R}^4$. This the main reason why $dF = 0$ does not carry dynamical information on electromagnetism.

This leaves us with two additional natural questions:

1. does there exist a generalization of the de Rham complex for other gauge theories?
2. does there exist a generalization of the de Rham complex to an arbitrary differentiable manifold and which information is encoded in the associated de Rham cohomology?

We will start by giving an answer to the second question.

1.3 Short and Long exact sequences

Before addressing the extension of the previous construction to an arbitrary differentiable manifold, we need to take a short detour defining a few relevant, additional algebraic structures, which will be useful in the following sections.

Definition 16. *Let $G_0, ..., G_n$ be a collection of n groups and let $f_i \in Hom(G_i, G_{i+1})$, $i = 0, ..., n-1$, be e collection of n group homomorphisms. We call the sequence*

$$G_0 \xrightarrow{f_0} G_1 \xrightarrow{f_1} ... \xrightarrow{f_{n-2}} G_{n-1} \xrightarrow{f_{n-1}} G_n$$

exact *if, for every i, $\mathrm{Im}(f_i) = \ker(f_{i+1})$. If an exact sequence is of the form*

$$0 \longrightarrow G_1 \xrightarrow{f_1} G_2 \xrightarrow{f_2} G_3 \longrightarrow 0$$

this is called a **short exact sequence**. *Here 0 refers to the trivial group composed only by the identity element.*

Notice that, in the definition of a short exact sequence, the arrow $0 \to$ entails that $\ker(f_1) = \{e_1\}$ where e_1 is the identity element of G_1, while the arrow $\to 0$ entails that $\mathrm{Im}(f_2) = G_3$. A particular instance where Definition 16 applies is when we choose a collection of vector spaces $V_0, ..., V_n$ seen as Abelian groups under the sum. In this case the maps f_i $i = 0, ..., n-1$ are simply linear maps. It is rather obvious that Definition 16 can be easily adapted to other algebraic structures, *e.g.* modules and it is not limited to groups. The following exercise shows that short exact sequences between vector spaces carry a lot of relevant information:

Exercise 5. Prove the split lemma, namely that, given a short exact sequence of finite dimensional vector spaces:

$$0 \longrightarrow V_1 \xrightarrow{f_1} V_2 \xrightarrow{f_2} V_3 \longrightarrow 0$$

then $V_2 \simeq V_1 \oplus V_3$.

An interesting case to work with is that in which each vector space V_i is in turn a differential complex. Observe that, although, at first glance one might think that a differential complex is by itself an example of an exact sequence, this is not the case in general. As a matter of fact, given a differential complex $(C^\bullet, \underline{d})$ as per Definition 13, the constraint $d_{k+1} \circ d_k = 0$ entails only that $\mathrm{Im}(d_k) \subseteq \ker(d_{k+1})$. It is not hard to see that requiring the equality to hold true for all k, is tantamount to saying that the cohomologies $H^k_{\underline{d}}(C^\bullet)$ – see Definition 15 – ought to be all trivial. Already the simple example of de Rham cohomology on $\mathbb{R}^n$ fails to satisfy this property at order 0.

A more intriguing possibility is to consider instead maps between differential complexes. Yet a compatibility condition between the associated differentials is necessary.

Definition 17. *Let $(A^\bullet, \underline{d}_A)$ and $(B^\bullet, \underline{d}_B)$ be two differential complexes. A linear map $\underline{f} : A^\bullet \to B^\bullet$ of the form $\underline{f} = \{f_k\}_{k\in\mathbb{N}}$ where $f_k : A_k \to B_k$ is called a* **chain map** *if*

$$\underline{f} \circ \underline{d}_A = \underline{d}_B \circ \underline{f}; \quad \text{i.e.} \quad f_k \circ (d_A)_k = (d_B)_k \circ f_k,\ \forall k \in \mathbb{N} \cup \{0\}. \tag{12}$$

A rather interesting scenario arises when we consider three differential complexes $(A^\bullet, \underline{d}_A)$, $(B^\bullet, \underline{d}_B)$, $(C^\bullet, \underline{d}_C)$ and a short exact sequence

$$0 \longrightarrow A^\bullet \xrightarrow{\underline{f}} B^\bullet \xrightarrow{\underline{g}} C^\bullet \longrightarrow 0 \tag{13}$$

where both $\underline{f}$ and $\underline{g}$ are chain maps. Equation (13) gives rise to the following diagram:

$$\begin{tikzcd}
 & \cdots & \cdots & \cdots & \\
0 \arrow[r] & A_{k+1} \arrow[u] \arrow[r, "f_{k+1}"] & B_{k+1} \arrow[u] \arrow[r, "g_{k+1}"] & C_{k+1} \arrow[u] \arrow[r] & 0 \\
0 \arrow[r] & A_k \arrow[u, "(d_A)_k"'] \arrow[r, "f_k"] & B_k \arrow[u, "(d_B)_k"'] \arrow[r, "g_k"] & C_k \arrow[u, "(d_C)_k"'] \arrow[r] & 0 \\
 & .. \arrow[u] & \cdots \arrow[u] & \cdots \arrow[u] &
\end{tikzcd}$$

where each of the square sub-diagrams is commuting. As next step we observe that each horizontal map in the above diagram gives rise to a suitable counterpart at a cohomology level:

Proposition 18. *Consider a short exact sequence of differential complexes*

$$0 \longrightarrow A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \longrightarrow 0$$

For every $k \in \mathbb{N} \cup \{0\}$ it induces an exact sequence

$$H^k_{\underline{d}_A}(A^\bullet) \xrightarrow{\tilde{f}_k} H^k_{\underline{d}_B}(B^\bullet) \xrightarrow{\tilde{g}_k} H^k_{\underline{d}_C}(C^\bullet)$$

where

$$\tilde{f}_k : H^k_{\underline{d}_A}(A^\bullet) \to H^k_{\underline{d}_B}(B^\bullet), \quad [a_k] \mapsto \tilde{f}_k[a_k] \doteq [f_k(a_k)],$$

while $\tilde{g}_k : H^k_{\underline{d}_B}(B^\bullet) \to H^k_{\underline{d}_C}(C^\bullet)$ is similarly defined.

Proof. First of all we observe that, for every $k \in \mathbb{N} \cup \{0\}$, the map $\tilde{f}_k$ is well-defined. As a matter of fact, for every $a_k \in A^k_d$, $f_k(a_k) \in B^k_d$ since

$$d_B \circ f_k(a_k) = f_k(d_A(a_k)) = f_k(0) = 0,$$

while, for every a'_{k-1},

$$f_k(d_A(a'_{k-1})) = d_B(f_{k-1}(a'_{k-1})) \in d[B_{k-1}].$$

Here we used repeatedly Equation 12 for chain maps. To conclude we need to show that each sequence is exact. In view of the defining properties, it holds that, for every $[a_k] \in H^k_{\underline{d}_A}(A^\bullet)$,

$$(\tilde{g}_k \circ \tilde{f}_k)[a_k] = [(g_k \circ f_k)(a_k)] = [0],$$

since per hypothesis $g_k \circ f_k = 0$. In other words we have shown that $\mathrm{Im}(\tilde{f}_k) \subseteq \ker(\tilde{g}_k)$.
To prove the converse observe that

$$\ker(\tilde{g}_k) = \{[b_k] \in H^k_{\underline{d}_B}(B^\bullet) \mid \exists c_{k-1} \in C_{k-1} \text{ such that } g_k(b_k) = (d_C)_{k-1}(c_{k-1})\},$$

while

$$\mathrm{Im}(\tilde{f}_k) = \{[b_k] \in H^k_{\underline{d}_B}(B^\bullet) \mid \exists a_k \in A_k \text{ and } b_{k-1} \in B_{k-1} \text{ such that } b_k = f_k(a_k) + (d_B)_{k-1}(b_{k-1})\}.$$

Yet, on account of the exactness of the short sequences for each $k \in \mathbb{N} \cup \{0\}$, we know that each map g_k is surjective. In particular this entails, that, for every $c_{k-1} \in C_{k-1}$, there exists $b_{k-1} \in B_{k-1}$ such that $c_{k-1} = g_{k-1}(b_{k-1})$. As a consequence we have rewritten the kernel as

$$\ker(\tilde{g}_k) = \{[b_k] \in H^k_{\underline{d}_B}(B^\bullet) \mid \exists b_{k-1} \in b_{k-1} \text{ such that } g_k(b_k) = g_k((d_B)_{k-1}(b_{k-1}))\},$$

where we used Equation (12). In other words $b_k = (d_B)_{k-1}(b_{k-1})$ up to an element in the kernel of g_k, which coincides with the image of f_k since, per hypothesis the sequence of differential complexes is exact. A direct inspection shows that we have shown the sought statement. $\square$

Lemma 19 (Zig-Zag)**.** *Given a short exact sequence of differential complexes as in Equation* (13)*, for every $k \in \mathbb{N} \cup \{0\}$ there exists a linear map $\tilde{d}_k : H^k_{\underline{d}_C}(C^\bullet) \to H^{k+1}_{\underline{d}_A}(A^\bullet)$ so that the following sequence is exact:*

$$
\begin{aligned}
\cdots &\longrightarrow H^k_{\underline{d}_A}(A^\bullet) \xrightarrow{\tilde{f}_k} H^k_{\underline{d}_B}(B^\bullet) \xrightarrow{\tilde{g}_k} H^k_{\underline{d}_C}(C^\bullet) \longrightarrow \\
&\xrightarrow{\tilde{d}_k} H^{k+1}_{\underline{d}_A}(A^\bullet) \xrightarrow{\tilde{f}_{k+1}} H^{k+1}_{\underline{d}_B}(B^\bullet) \xrightarrow{\tilde{g}_{k+1}} H^{k+1}_{\underline{d}_C}(C^\bullet) \longrightarrow \\
&\xrightarrow{\tilde{d}_{k+1}} H^{k+2}_{\underline{d}_A}(A^\bullet) \longrightarrow \cdots
\end{aligned}
\tag{14}
$$

This is called a **long exact sequence***.*

Proof. To start with, we need to construct the map $\tilde{d}_k$ for all $k \in \mathbb{N} \cup \{0\}$. Using Equation 13, we can infer that, being g_k surjective, for every $c_k \in C_k$, there exists $b_k \in B_k$ such that $c_k = g_k(b_k)$. If in addition we assume that $(d_C)_k(c_k) = 0$, then Equation (12) entails that $(d_C)_k(g_k(b_k)) = g_k((d_B)_k(b_k)) = 0$. The exactness of the sequence entails that $\ker(g_k) = \mathrm{Im}(f_k)$. Hence there must exist $a_{k+1} \in A_{k+1}$ such that $(d_B)_k(b_k) = f_{k+1}(a_k + 1)$. Still using Equation (12) we observe that $(d_A)_{k+1}a_{k+1} = 0$ since

$$
f_{k+1}((d_A)_{k+1}a_{k+1}) = (d_B)_{k+1}(f_{k+1}a_{k+1}) = ((d_B)_{k+1} \circ (d_B)_k)(b_k) = 0.
$$

Yet, since the sequence is exact, f_{k+1} is injective, which entails that a_{k+1} is closed. Mutatis mutandis, it turns out that, if we start from $c_k = (d_C)_{k-1}c_{k-1}$, then the ensuing element a_{k+1} is also exact. Hence we have constructed the map

$$
\tilde{d}_k : H^k_{\underline{d}_C}(C^\bullet) \to H^{k+1}_{\underline{d}_A}(A^\bullet) \quad [c_k] \mapsto [a_{k+1}].
$$

It remains to be shown that the sequence (14) is exact at each $H^k_{\underline{d}_C}(C^\bullet)$, that is $\ker(\tilde{d}_k) = \mathrm{Im}(\tilde{g}_k)$. Suppose $[c_k] \in \ker(\tilde{d}_k)$. Using Equation 13, we can infer that, being g_k surjective, for any representative $c_k \in C_k$, there exists $b_k \in B_k$ such that $c_k = g_k(b_k)$. Hence $[c_k] = [g_k(b_k)] = \tilde{g}_k([b_k])$ where we used the definition of $\tilde{g}_k$ in Proposition 18. This implies that $\ker(\tilde{d}_k) \subseteq \mathrm{Im}(\tilde{g}_k)$. Suppose now that $[c_k] = \tilde{g}_k([b_k])$. The above construction shows that, since b_k is closed by assumption, $f_{k+1}(a_{k+1}) = 0$. Since f_{k+1} is injective per hypothesis, $a_{k+1} = 0$. This implies $\tilde{d}_k([c_k]) = 0$, namely $\mathrm{Im}(\tilde{g}_k) \subseteq \ker(\tilde{d}_k)$. This concludes the proof. □

2 Vector Bundles

Our goal is to generalize differential forms to arbitrary differentiable manifolds. We assume here that the reader, which is probably a student who has to take the exam at some point (if not...

why are you wasting your time? Go, have some fun!), is familiar with the standard notion of topological and differentiable manifold. Here we will adopt notation and conventions from [11].

To meet our goal, there are two standard approaches. The first and more pragmatic calls for a definition of differential forms on a generic differentiable manifold M via a partition of unity argument. In other words, one considers a cover of M in open subsets diffeomorphic to counterparts in $\mathbb{R}^n$, $n = \dim M$. In each of these subsets differential forms can be defined locally as per Definition 4 and Remark 3. Subsequently these local forms are glued together via a partition of unity defining a global differential form. This is the viewpoint adopted, *e.g.* in [3] and partly in [14]. Yet, in the opinion of the author (yeah, I know, who cares of my opinion?!), there are two drawbacks in this approach:

- The gluing argument misses any geometric interpretation of a global differential k-forms. For computational purposes this might not be relevant, but, it is in the global aspects that lies the real difference between a k-form on $\mathbb{R}^n$ and on a generic differentiable manifold,

- The partition of unity construction leads in many cases to subtle convergence issues, as soon as we consider a cover of the underlying manifold which does not include only a finite collection of open subsets. In addition, there is also the problem of addressing whether the definition of a k-form depends on the chosen cover.

It goes without saying that both questions can be addressed in a satisfactory way. Yet, in order to bypass this hurdle, we go for the second option, namely giving an intrinsically global definition of a differentiable form, checking subsequently that, on $\mathbb{R}^n$, $n \geq 1$, it matches the standard definition. Yet, to this end, it is convenient to introduce a global geometric structure which has uncountable applications, ranging from classical mechanics to field theory: *vector bundles*. In the following we shall work with the field of real numbers. Yet, mutatis mutandis, all constructions can be adapted to other choices, $\mathbb{C}$ in particular. We will follow as a guideline [11, Chap. 10], although a reader who wants to have a more advanced knowledge on bundles is encouraged to refer to [7].

Definition 20. *Let M be a topological space. We call (real)* **vector bundle** *of rank m over M (*base space*), a topological space E (*total space*) together with a surjective continuous map $\pi : E \to M$ (*projection*) such that*

- *for each $p \in M$, the* fiber *$E_p \doteq \pi^{-1}(p)$ is a real vector space of (finite) dimension m*

- *for each $p \in M$, there exists an open subset $U \subseteq M$ such that $p \in U$ and a* local trivialization *of E over U, namely an homeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^m$ such that*

  – *$\pi_U \circ \Phi = \pi$, where $\pi_U : U \times \mathbb{R}^m \to U$ is the projection on the first factor,*

  – *for each $q \in U$, the restriction $\Phi|_q : E_q \to \{q\} \times \mathbb{R}^m$ is a vector space isomorphism.*

If both M and E are smooth differentiable manifolds, $\pi \in C^\infty(E; M)$ and the local trivializations can be chosen to be smooth diffeomorphisms, then we call E a real, smooth vector bundle. A vector bundle is denoted as $E \equiv E[M; \pi, \mathbb{R}^m]$.

Remark 4. *Observe that the requirement of considering finite dimensional vector spaces as typical fiber could be relaxed, though with some effort. As a matter of fact, if we consider an infinite dimensional (real or complex) vector space, one also needs to assign a topology in order to make sense of the requirement that the local trivializations are continuous maps. On the contrary, if the rank m is finite, we always use implicitly the standard topology on $\mathbb{R}^m$.*

In the following, we will be mainly interested in smooth vector bundles, but it is convenient to discuss first two standard examples. The first highlights that one can easily construct an infinite class of vector bundles, which play the role of the trivial example.

Example 21. Consider any topological manifold M and let $E_m \doteq M \times \mathbb{R}^m$, $m \in \mathbb{N}$. This is a vector bundle of rank m if endowed with the map $\pi : E \to M$ which is the continuous map projecting on the first factor. It is immediate to observe that, for all $p \in M$, $\pi^{-1}(p) = \{p\} \times \mathbb{R}^m$ and that, regardless of the point $p \in M$, the role of U can be played by M itself, with Φ being the identity map. Any vector bundle E_m is called a **product bundle**.

This example prompts the following definition

Definition 22. *Let E be a vector bundle of rank m. We call it* **trivial (*resp.* smoothly trivial)** *if there exists an homeomorphism (*resp. *a diffeomorphism) $\Phi : E \to E_m = M \times \mathbb{R}^m$.*

Example 21 proves that the definition of a real vector bundle is non empty, but it raises the question whether there exists vector bundles which are not trivial. The following example dispels any doubt

Example 23. Consider $\mathbb{R}^2$ and the equivalence relation $(x, y) \sim (x', y')$ if there exists $n \in \mathbb{Z}$ such that

$$x' = x + n, \quad y' = (-1)^n y.$$

We let $E \doteq \mathbb{R}^2 / \sim$ and we observe that, up to the identification of the closed interval $[0, 1]$ with $\mathbb{S}^1$, there is a natural projection $\pi_E : E \to \mathbb{S}^1$ over the first factor in the equivalence classes. The total space E is called **Möbius band**, which can be shown to be a compact, smooth manifold. In addition E has the structure of a smooth fiber bundle called the **Möbius bundle**. We leave as an exercise to the reader to prove that, for all $p \in \mathbb{S}^1$.

$$\pi_E^{-1}(p) \simeq \mathbb{R},$$

and that, choosing any open subset $U \subset \mathbb{S}^1$ such that $p \in U$, then there exists a local trivialization $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}$. If you find it too difficult check [11, Example 10.3]

Remark 5. *One might wonder whether vector bundles are a simply a useful geometrical, additional structure or whether they carry a deeper physical significance. Indeed they do for the following reason. In many physical models, ranging from classical mechanics, to fluid dynamics, electromagnetism or, more generally field theory, the main object of investigation needs to carry often at least two bits of information: one concerning the behaviour of the system in different* *space (or spacetime) points, the second concerning internal degrees of freedom, e.g. the spin of a particle. A vector bundle is a natural geometric structure encoding these data. In other words, given a vector bundle E as per Definition 20, one can think of M as the physical space (or spacetime), while the fibers $\pi^{-1}(p)$, encode the internal degrees of freedom. Here we are assuming that these are best described by a vector space, which is a reasonable assumption to start with, if we want to account for a superposition principle on each fiber. Yet, as we will see in the next sections, there are several instances, e.g. Yang-Mills gauge theories, where one might wish to allow for a more general and rich structure at each $p \in M$. Furthermore, observe that, in all instances where internal degrees of freedom appear, the key kinematic and dynamical object is a field seen as a map from the underlying space (or spacetime) to the counterpart carrying the information on the additional degrees of freedom (think for example of a wave function as a map $\psi : \mathbb{R}^3 \to \mathbb{C}^{2j+1}$ or of the velocity field in fluid dynamics as $u : \mathbb{R}^3 \to \mathbb{R}^3$ or of a Dirac field in Minkowski $\Psi : \mathbb{R}^4 \to \mathbb{C}^4$). We will show that these maps appear naturally in the theory of vector bundles and of their generalizations.*

In the following we prove (well... by now... you should already know it, else... there is indeed a big problem somewhere) that there is a canonical vector bundle associated to each smooth manifold M.

Proposition 24. *Let M be a smooth, differentiable manifold of dimension $\dim M = n \geq 1$. Let $TM \doteq \bigsqcup_{p\in M} T_pM$ be the tangent space of M. This is a smooth vector bundle of rank n, called the* **tangent bundle** *of M.*

Proof. Standard results in differential geometry, *cf.* [11, Prop. 3.18], show that TM has a natural topology and smooth structure inherited from those of M, that make it into a differentiable manifold of dimension $\dim TM = 2n$. Furthermore the same result entails that there exists a natural, $\pi \in C^\infty(TM; M)$ such that

$$\pi(p, v) = p, \quad \forall (p, v) \in TM.$$

Per construction it holds that, for all $p \in M$, $\pi^{-1}(p) = T_pM$ which is a real vector space of dimension n, *cf.* [11, Prop. 3.15]. To conclude we need only to check the existence of a local trivialization. For any $p \in M$, consider (U, φ) a local smooth chart and let $\{x_i\}_{i=1\dots,n}$ be local coordinates thereon. For every $q \in U$, we know that $\{\frac{\partial}{\partial x_i}\}_{i=1,\dots,n}$ is basis of T_qM, i.e.

$$v = \sum_{i=1}^{n} v_i \frac{\partial}{\partial x_i}, \quad \forall v \in \pi^{-1}(U),$$

where each $v_i \in \mathbb{R}$. Hence we can set

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^n, \quad v \mapsto \Phi(v) = (p, (v_1, \dots, v_n)).$$

A direct calculations shows that $\pi_U \circ \Phi = \pi$ and, for each $q \in U$, $\Phi|_q$ is an isomorphism of vectors spaces. To prove that Φ is a diffeomorphism, we observe that, if we introduce the map

$$(\varphi, \mathrm{id}) : U \times \mathbb{R}^n \to \mathbb{R}^{2n}, \quad (p, (v_1, ..., v_n)) \mapsto (\varphi(p), (v_1, ..., v_n)),$$

then $\widetilde{\varphi} \doteq (\varphi, \mathrm{id}) \circ \Phi : \pi^{-1}(U) \to \mathbb{R}^{2n}$ is the natural coordinate map on $TM|_U$ induced from (U, φ). This is a diffeomorphism, *cf.* [11, Prop. 3.18]. Since also (φ, id) is a diffeomorphism, so is Φ which can be written as $\Phi = (\varphi, \mathrm{id})^{-1} \circ \widetilde{\varphi}$. This concludes the verification that TM meets all defining criteria of a smooth vector bundle. □

Exercise 6. Let M be a smooth, differentiable manifold of dimension $\dim M = n$. Let $T^*M = \bigsqcup\limits_{p \in M} T_p^*M$ where T_p^*M is the cotangent space at $p \in M$. Prove that T^*M is a real vector bundle of rank n. Prove that, more generally, if $E = E[M; \pi, \mathbb{R}^n]$ is a smooth, real vector bundle, then $E^* \doteq \bigsqcup\limits_{p \in M} E_p^*$ is a smooth, real vector bundle. Here E_p^* is the dual vector space of $E_p \doteq \pi^{-1}(p)$. E^* is called the **dual vector bundle** to E.

As next step we focus on a notable aspect of smooth vector bundles, concerning the form of two trivializations when they overlap. Notice that, when defining a manifold, a regularity condition is required on the transition functions between two smooth local charts. In Definition 20 no counterpart appears for the following reason:

Proposition 25. *Let $E = E[M, \pi, \mathbb{R}^m]$ be a smooth vector bundle of rank m and let $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^m$ and $\Psi : \pi^{-1}(V) \to V \times \mathbb{R}^m$ be two local trivializations such that $U \cap V \neq \emptyset$. Then, there exists a map $A : U \cap V \to \mathrm{GL}(m, \mathbb{R})$ such that, for all $(p, v) \in (U \cap V) \times \mathbb{R}^m$,*

$$(\Phi \circ \Psi^{-1})(p, v) = (p, A(p)v).$$

The map A is called the **transition function** *between the local trivializations Φ and Ψ.*

Proof. Consider the restriction of both trivializations Φ and Ψ to $U \cap V$, using the same symbol with a slight abuse of notation. On account of Definition 20 it holds

$$\pi_{U \cap V} \circ (\Phi \circ \Psi^{-1}) = \pi_{U \cap V}.$$

This entails that there must exist $\sigma \in C^\infty((U \cap V) \times \mathbb{R}^m; \mathbb{R}^m)$ such that, for all $(p, v) \in (U \cap V) \times \mathbb{R}^m$,

$$(\Phi \circ \Psi^{-1})(p, v) = (p, \sigma(p, v)).$$

Since, for fixed $p \in M$, transition functions are vector space isomorphisms, it descends that there must exist a map $A : U \cap V \to \mathrm{GL}(m, \mathbb{R})$ such that $\sigma(p, v) = A(p)v$ for every $v \in \mathbb{R}^m$. The smoothness of A is inherited from that of σ, observing that, given any basis $\{v_i\}_{i=1,\ldots,m}$ of $\mathbb{R}^m$, the generic entry of A can be written as $A_{ij}(p) = (v_i, \sigma(p, v_j))$ where $(,)$ stands for the standard Euclidean scalar product on $\mathbb{R}^m$. □

We conclude this short outline of the basic properties of vector bundles by stating a result which guarantees that, akin to the definition of tangent space, one can construct vector bundles by assigning a manifold M and a vector space E_p for each $p \in M$. For the proof, refer to [11, Lemma 10.6]

Proposition 26. *Let M be a smooth manifold and let $M \ni p \to E_p$ be an assignment of a real vector space of dimension m for each $p \in M$. Let $E \doteq \bigsqcup_{p\in M} E_p$ and let $\pi : E \to M$ be the natural projection map. Suppose that the following additional data are assigned:*

1. *an open cover $\{U_\alpha\}_{\alpha\in I}$ of M where I is a countable index set,*
2. *for every $\alpha \in I$, a bijective map $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^m$ whose restriction to each E_p, $p \in U_\alpha$ is a vector space isomorphism from E_p to $\{p\} \times \mathbb{R}^m$,*
3. *for each $\alpha, \beta \in I$ such that $U_\alpha \cap U_\beta \neq \emptyset$, $A_{\alpha\beta} \in C^\infty(U_\alpha \cap U_\beta; \mathrm{GL}(m, \mathbb{R}))$ fulfilling*

$$(\Phi_\alpha \circ \Phi_\beta^{-1})(p, v) = (p, A_{\alpha\beta}(v)), \quad \forall (p, v) \in (U_\alpha \cap U_\beta) \times \mathbb{R}^m.$$

Then E admits a unique topology and smooth structure which realizes it as a smooth vector bundle of rank m with π as projection and $\{(U_\alpha, \Phi_\alpha)\}_{\alpha\in I}$ as local trivializations.

To conclude the section, we outline a procedure to construct, starting from two vector bundles, a third one.

Definition 27. *Let $E \equiv E[M; \pi_E, \mathbb{R}^m]$ and $F \equiv F[M; \pi_F, \mathbb{R}^n]$ be two (smooth) vector bundles of rank m and n respectively. We call $E \oplus F$* **Whitney sum** *of E and F the real vector bundle of rank $m + n$ whose base space is M and $E \oplus F = \bigsqcup_{p\in M} (E_p \oplus F_p)$.*

It is immediate to notice that Definition 27 is based on Proposition 26. As a matter of fact one can consider an open cover $\{U_\alpha\}_{\alpha\in I}$ of M where I is a countable index set, such that, for each $\alpha \in I$ there exists Φ_α and Ψ_α, local trivializations respectively of E and F. Therefore the map

$$\Upsilon_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^{m+n} \; (p, v, v') \mapsto (p, \Phi_\alpha(v), \Psi_\alpha(v')),$$

is a bijective map which restricts to a vector space isomorphism at each $p \in U_\alpha$. Finally, for each $\alpha, \beta \in I$ such that $U_\alpha \cap U_\beta \neq \emptyset$ a direct computation shows that, given two trivializations Υ_α and Υ_β on U_α and U_β respectively, their restriction to $U_\alpha \cap U_\beta$ satisfies

$$(\Upsilon_\alpha \circ \Upsilon_\beta^{-1})(p, v, v') = (p, A(p)(v, v')),$$

where $A(p) = \mathrm{diag}(A_m(p), A_n(p))$ is the block-diagonal matrix whose entries are the transition functions between Φ_α (*resp.* Ψ_α) and Φ_β (*resp.* Ψ_β).

Definition 27 teaches us that it is possible to construct different vector bundles exploiting pointwisely the natural operation between vector spaces. This will play a key role in defining differential forms on manifolds, but, to start with, we strongly encourage the reader/student to make some practice solving the following exercise:

Exercise 7. Let $E \equiv E[M; \pi_E, \mathbb{R}^m]$ and $F \equiv F[M, \pi_F, \mathbb{R}^n]$ be two (smooth) vector bundles of rank m and n respectively. Prove that $E \otimes F \doteq \bigsqcup_{p\in M} (E_p \otimes F_p)$ can be endowed with the structure of real vector bundle of rank mn. This is called the **tensor product** between E and F.

Remark 6. *Exercise 7 acquires special significance when we consider a smooth differentiable manifold M and the associated tangent and cotangent bundles TM and T^*M. In this case it becomes natural introducing the following associated vector bundles:*

- *the* **bundle of contravariant k-tensors on M**, $\mathcal{T}^k TM \doteq \bigsqcup_{p\in M} T_pM^{\otimes k}$, $k \geq 1$,
- *the* **bundle of covariant k-tensors on M**, $\mathcal{T}^k T^*M \doteq \bigsqcup_{p\in M} (T_p^*M)^{\otimes k}$, $k \geq 1$,
- *the* **bundle of mixed tensors of type (k, l)** $\mathcal{T}^{(k,l)}TM \doteq \bigsqcup_{p\in M} \left(T_pM^{\otimes k} \otimes (T_p^*M)^{\otimes l}\right)$, *where* $k, l \geq 1$.

A less obvious construction which appears in several applications is the following:

Exercise 8. Let $E \equiv E[M; \pi_E, \mathbb{R}^m]$ and $F \equiv F[M', \pi_F, \mathbb{R}^n]$ be two (smooth) vector bundles of rank m and n respectively. Prove that $E \boxtimes F \doteq \bigsqcup_{(p,q)\in M\times M'} (E_p \otimes F_q)$ can be endowed with the structure of real vector bundle of rank nm with base space $M \times M'$. This is called the **external tensor product** between E and F.

2.1 Sections of a vector bundle

As outlined in Remark 5, vector bundles are the natural geometric structure encoding the information of internal degrees of freedom in a physical system besides those associated to the dependence from space (or spacetime) points. Yet, we still miss a key tool which allows the assignment to each space (or spacetime) point of a specific internal degree of freedom. In the language of classical mechanics and, more specifically of field theory, we are asking ourselves what plays the role of kinematic configurations in this language. An answer to this question is given in the next definition.

Definition 28. *Let $E \equiv E[M; \pi, \mathbb{R}^m]$ be a vector bundle of rank m. We call* **(cross) section** *of E a map $\sigma \in C^0(M; E)$ such that $\pi \circ \sigma = \mathrm{id}_M$ where id_M stands for the identity map on M. If the bundle is smooth and $\sigma \in C^\infty(M; E)$, we call it a smooth (cross) section. The collection of all sections if indicated as $\Gamma(E)$ (*resp. *$\Gamma^\infty(E)$).*

Remark 7. *A more general concept is that of* **local section** *of E, that is $\sigma \in C^0(U; E)$ where $U \subseteq M$ and $\pi \circ \sigma = \mathrm{id}_U$. To distinguish between the two cases, sometimes sections as in Definition 28 are called* global. *Notice that every local section of E is a global one of the bundle built as the* **restriction** *of E to U. This is a vector bundle of rank m, $E|_U \equiv E|_U[U; \pi|_U, \mathbb{R}^m]$, where the total space $E|_U = \pi^{-1}(U)$, while the projection map is the restriction of π to U.*

The following exercise has the goal to make the reader acquainted with the structural properties of sections.

Exercise 9. Let $E \equiv E[M;\pi,\mathbb{R}^m]$ be a vector bundle of rank m. Prove that $\Gamma(E)$ (*resp.* $\Gamma^\infty(E)$) is a vector space, as well as a module over the ring $C^\infty(M)$ with respect to pointwise multiplication:

$$\cdot : C^\infty(M) \times \Gamma(E) \to \Gamma(E), \quad (f,\sigma) \mapsto (f\cdot\sigma)(p) \doteq f(p)\sigma(p).$$

Two observations are in due course before focusing on a notable class of sections:

- Given a (smooth) vector bundle E of rank m, the space of (smooth) sections is never empty. One can always construct the so-called *0-section*:

  $$\sigma_0 : M \to E, \quad p \mapsto \sigma_0(p) = 0 \in E_p.$$

  The map σ_0 is continuous (or smooth) since, for any local trivialization of E, say Φ, it holds that $\Phi \circ \sigma_0(p) = (p, 0)$ for every $p \in U$. This map is manifestly continuous (or smooth) and, therefore, since Φ is an homeomorphism (or a diffeomorphims) on U, hence invertible, σ_0 has the desired property on each U. By considering a cover of M made of open subsets admitting a local trivialization, continuity (or smoothness) of σ_0 holds globally.

- Given any smooth manifold M and the associated tangent bundle TM, $\Gamma^\infty(TM)$, the space of smooth sections, coincides with the **smooth vector fields** over M.

- Given any smooth manifold M and using the notation of Remark 6, we call

  - **(covariant) k-tensor field** any element of $\Gamma^\infty(\mathcal{T}^k T^*M)$,
  - **(contravariant) k-tensor field** any element of $\Gamma^\infty(\mathcal{T}^k TM)$,
  - **(mixed) $(\mathbf{k},\mathbf{l})$-tensor field** any element of $\Gamma^\infty(\mathcal{T}^{(k,l)} TM)$.

  We shall not discuss how to represent locally a section of these bundles in terms of an arbitrary coordinate system. We shall discuss this problem in Section 3 for differential forms, leaving as an exercise for a reader the adaptation to the cases here discussed.

In the following proposition we prove existence of non trivial global sections for a rather special class of vector bundles.

Proposition 29. *Given any smooth manifold M, there exists an isomorphism of vector spaces between $C^\infty(M;\mathbb{R}^m)$ and $\Gamma^\infty(E)$ where $E = M \times \mathbb{R}^m$ is the trivial smooth vector bundle of rank m.*

Proof. Consider the trivial bundle $E = M \times \mathbb{R}^m$. On account of Definition 28, a smooth section is a map $\sigma \in C^\infty(M;E)$, which is a right inverse of the projection map on the first factor. In other words, for every $p \in M$, $\sigma(p) = (p, v)$, which identifies canonically $f \in C^\infty(M;\mathbb{R}^m)$ such that $f(p) = v$. Conversely, for every smooth map $\tilde{f} \in C^\infty(M;\mathbb{R}^m)$, we can define a smooth section of E, as $\sigma(p) = (p, \tilde{f}(p))$. The above assignments are per construction giving the sought bijection. □

In view of this result, it might be tempting to claim that a smooth vector bundle of rank m is trivial if and only if the space of its smooth sections is isomorphic to $C^\infty(M;\mathbb{R}^m)$. This is indeed the case as we shall discuss now. Observe that, henceforth, we will only analyze the smooth scenario, but, this requirement can always be dropped and all results hold true in the generic case.

Disclaimer: Henceforth we shall omit to state explicitly that the vector bundles are smooth and that their rank is $m < \infty$. This is always implicitly assumed. The same applies to sections.

To this end, we start by observing that, since a section of a vector bundle E over M, maps each $p \in M$ to a vector space E_p, we can generalize a standard definition, by claiming that, given a collection of local sections $\{\sigma_i : U \to E\}_{i=1,\ldots,k}$, we call them *linearly independent* if, for each $p \in U$, the vectors $\{\sigma_i(p)\}_{i=1,\ldots,k}$ are linearly independent elements of E_p. This motivates the following definition

Definition 30. *Let $E \equiv E[M;\pi,\mathbb{R}^m]$ be a vector bundle. We call* **local frame** *for E over U, an ordered m-tuple of linearly independent local sections $\{\sigma_i : U \to E\}_{i=1,\ldots,m}$. If $U = M$ we call it a* **global frame** *for E over M.*

The wise reader (what is real wisdom?!) should compare this definition with the standard notion of frame which is used in physics and in mechanics in particular. One can notice that Definition 30 is a generalization and that the notion, we are familiar with, can be recovered replacing E with TM. A curious idiosyncrasy is that, in this case, one always speaks of a global frame for M rather than for TM.

Remark 8. *Given a vector bundle $E \equiv E[M;\pi,\mathbb{R}^m]$ and a local trivialization $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^m$ over an open subset $U \subseteq M$, there exists always a local frame for E over U. This can be constructed as follows:*

1. *consider $U \times \mathbb{R}^m$ as a trivial vector bundle over U. Fix any ordered basis of $\mathbb{R}^m$, say $\{v_1, ..., v_m\}$ and consider the maps $f_i \in C^\infty(U;\mathbb{R}^m)$ such that $f_i(p) = v_i$. In view of Proposition 29, these functions induce m smooth sections $\sigma_i \in C^\infty(U;U \times \mathbb{R}^m)$ such that $\sigma_i(p) = (p, f_i(p))$. Per construction we have individuated a global frame of the trivial vector bundle $U \times \mathbb{R}^m$.*

2. *Since Φ is a diffeomorphism, define*

$$\widetilde{\sigma}_i = \Phi^{-1} \circ \sigma_i \in C^\infty(U;E)$$

These are sections since, per Definition 20

$$\pi \circ \widetilde{\sigma}_i = \pi \circ \Phi^{-1} \circ \sigma_i = \pi_U \circ \sigma_i = \mathrm{id}|_U.$$

Still on account of Φ being a diffeomorphism, for all $p \in U$, the vectors $\{\widetilde{\sigma}_i(p)\}_{i=1,\ldots,m}$ are linearly independent. Hence we have constructed a **local frame associated with Φ**.

In other words we have shown that, to a local trivialization, it corresponds always a local frame. One of the key initial results in the theory of vector bundles is that a converse of this statement holds true.

Theorem 31. *Given a vector bundle $E \equiv E[M;\pi,\mathbb{R}^m]$ and a local frame, there exists an associated local trivialization.*

Proof. Suppose that $\{\sigma_i : U \to E\}_{i=1,\ldots,m}$ is a local frame where $U \subseteq M$. We introduce the auxiliary maps $\widetilde{e}_i \in C^\infty(U;\mathbb{R}^m)$ such that, for all $p \in U$, $\widetilde{e}_i(p) = e_i$, where e_i is the i-th element of the standard basis of $\mathbb{R}^m$. Combining these data, we can construct

$$\Psi : U \times \mathbb{R}^m \to \pi^{-1}(U) \quad (p, v_1, \ldots, v_m) \mapsto \Psi(p, v_1, \ldots, v_m) = \sum_{i=1}^{m} v_i \sigma_i(p).$$

The map Ψ is a bijection since for all $p \in U$, $\{\sigma_i(p)\}_{i=1,\ldots,m}$ is a basis of E_p and, in addition,

$$\sigma_i = \Psi \circ \widetilde{e}_i. \tag{15}$$

We have found the candidate to play the role of the local trivialization. Let us show that Ψ^{-1} verifies all defining criteria, *cf.* Definition 20. Observe that

- Since each σ_i and $\widetilde{e}_i$, $i = 1, \ldots, m$ is a local section of E and of $U \times \mathbb{R}^m$ respectively, per Definition 28 it holds

$$\pi^{-1} \circ \pi \circ \sigma_i = \Psi \circ \pi_U^{-1} \circ \pi_U \circ \widetilde{e}_i \Longrightarrow \pi^{-1} = \Psi \circ \pi_U^{-1},$$

  which entails $\pi \circ \Psi^{-1} = \pi_U$,

- since Ψ is a bijection, for each $p \in U$, $\Psi^{-1}\big|_p : E_p \to \{p\} \times \mathbb{R}^m$ is a vector space isomorphism.

It remains to be proven that Ψ^{-1} is a diffeomorphism. We focus on Ψ and for any but fixed $p \in U$, we choose a local trivialization (V, Φ) centered at p. In addition it holds that $V \subseteq U$. Since Φ is a diffeomorphism, it suffices to prove that $\Phi \circ \Psi$ is a diffeomorphism on $V \times \mathbb{R}^m$ to conclude the proof. Here we omit writing explicitly $\Psi|_{V \times \{\mathbb{R}^m\}}$. Observe that, for every $(p, v_1, \ldots, v_m) \in V \times \mathbb{R}^m$,

$$(\Phi \circ \Psi)(p, v_1, \ldots, v_m) = \Phi\left(\sum_{i=1}^{m} v_i \sigma_i(p)\right) = \left(p, \sum_{i=1}^{m} v_i \Phi|_p(\sigma_i(p))\right).$$

Since each $\sigma_i \in \Gamma(E|_U)$, it descends that $\Phi \circ \sigma_i \in C^\infty(V; V \times \mathbb{R}^m)$ and $\pi_1 \circ (\Phi \circ \sigma_i) = \mathrm{id}|_V$ for all $i = 1, \ldots, m$, where π_1 is the projection on the first factor on $V \times \mathbb{R}^m$. Hence, for each $i = 1, \ldots, m$ there must exist $\widetilde{\sigma}_{ij} \in C^\infty(V)$, $j = 1, \ldots, m$ such that, for all $p \in V$,

$$(\Phi \circ \sigma_i)(p) = (p, \widetilde{\sigma}_{i1}(p), \ldots, \widetilde{\sigma}_{im}(p)).$$

As a consequence, for all $(p, v_1, \ldots, v_m) \in V \times \mathbb{R}^m$

$$(\Phi \circ \Psi)(p, v_1, \ldots, v_m) = \left(p, \sum_{i,j=1}^{m} v_i \widetilde{\sigma}_{ij}(p) \right),$$

which is smooth since such is each $\widetilde{\sigma}_{ij}$. To conclude we need to show that also $(\Phi \circ \Psi)^{-1}$ is smooth. Yet, the matrix $\widetilde{\sigma}_{ij}$ is invertible at each $p \in V$ since $\{\widetilde{\sigma}_i(p)\}_{i=1,\ldots,m}$ is a basis for E_p while Φ is a vector space isomorphism at each $p \in V$. If we interpret $p \mapsto \widetilde{\sigma}_{ij}(p)$ as a smooth map from V to the Lie group $\mathrm{GL}(m, \mathbb{R})$, it suffices to recall that, given any Lie group G, the map $G \ni g \mapsto g^{-1}$ is smooth. Hence the assignment $p \mapsto \widetilde{\tau}_{ij}(p) \doteq \left(\widetilde{\sigma}^{-1}\right)_{ij}(p)$ is smooth. A direct computation shows that, for all $(p, w_1, \ldots, w_m) \in V \times \mathbb{R}^m$

$$(\Phi \circ \Psi)^{-1}(p, w_1, \ldots w_m) = \left(p, \sum_{i=1}^{m} w_i \tau_{i1}(p), \ldots, \sum_{i=1}^{m} w_i \tau_{im}(p) \right),$$

which is manifestly a smooth map. □

A direct consequence of this theorem is the answer to the question which has prompted this analysis.

Corollary 32. *A real vector bundle is trivial if and only if it admits a global frame.*

Proof. One direction has been proven in Remark 8, while the other is a direct consequence of Theorem 31 replacing in the proof U with M. □

Corollary 33. *Given a vector bundle $E \equiv E[M; \pi, \mathbb{R}^m]$, (V, φ) a local chart of M endowed with coordinates $x_i : V \to \mathbb{R}$, $i = 1, \ldots n = \dim M$, and $\{\sigma_j\}_{j=1,\ldots,m}$ a local frame for E over V, then there exists an associated coordinate chart $(\pi^{-1}(V), \psi)$ such that, for all $q \in \pi^{-1}(V)$,*

$$\psi(q) = (x_1(p), \ldots, x_n(p), v_1, \ldots, v_m), \quad \pi(q) = p \text{ and } q = \sum_{j=1}^{m} v_j \sigma_j(p),$$

where each $v_j \in \mathbb{R}$.

Proof. It suffices to notice that $\psi : \pi^{-1}(V) \to \mathbb{R}^{n+n}$ and that, following the construction of Theorem 31, it is invertible. □

Exercise 10. Prove that, given a smooth manifold M, the standard smooth structure assigned to TM coincides with that induced by Corollary 33. Prove in addition that this is the unique smooth structure which realizes TM as a smooth vector bundle.

Remark 9. *It is worth trying to recapitulate this subsection from a physical viewpoint. Given a system on a space (spacetime) M with internal degrees of freedom modeled as a vector space V, the rationale is that the geometric structure encoding this information is a vector bundle $E =$*

$E[M;\pi;\mathbb{R}^m]$ where $m = \dim V$ and, for each $p \in M$, $\pi^{-1}(p) \simeq V$. Accepting this interpretation, the main object encoding this datum is a field, *seen as map which assigns to each $p \in M$ an internal degree of freedom. This is what sections are good for, that is*

$$\text{kinematical configurations} \equiv \Gamma^\infty(E).$$

This identification is not as harmless as it might seem at first glance. The idea of using global sections, combines the necessity of using natural geometric structures associated with the bundle E with that of interpreting a field as a global object. In a sense, if we were using only local sections, it might occur that a field exists only in a neighbourhood of a point $p \in M$, just to disappear in thin air as soon as we consider a second point outside of the domain of definition of the local sections.

Yet, now two different scenarios occur:

1. *E is trivial, hence, per Proposition 29, $\Gamma^\infty(E) \simeq C^\infty(E;V)$,*

2. *E is not trivial.*

In the first case we recover the heuristic idea that a field accounting for internal degrees of freedom is nothing but a smooth map from the space (spacetime) M to the vector space V. This is the rationale behind quantum mechanics when a wave functions is read as $\psi \in L^2(\mathbb{R}^3;\mathbb{C}^{2j+1})$. In the second case, instead, the space of smooth sections is made of elements which only locally look like ordinary smooth maps. Yet their global structure is quite different.

Notice, that since, given a smooth manifold M encoding information on space (or spacetime) and a finite dimensional vector space V encoding the internal degrees of freedom, one can always construct a trivial bundle $M \times V$, the non trivial counterparts must differ at the level of global structure, an aspect which we can expect to be encoded at a topological level. One might wonder whether there is any physical reason for looking into this possibility. Actually there are several scenarios where topological effects occur and the underlying model is built via a non trivial vector bundle, e.g. in the theory of Berry phases. Yet the main example where non trivial bundles do occur is electromagnetism, seen as a theory of differential forms on a manifold. In this case the key geometric object is the cotangent bundle T^*M which, in general, is far from being trivial. It is also known that using this rationale leads to consequences which are experimentally proven, namely the Aharonov-Bohm effect*.*

This reasoning opens now an important question with several ramifications in building physical models: given a smooth manifold M, how many nonequivalent ways exist of constructing a rank m vector bundle on top of it? The answer lies in the theory of classification of vector bundles and in the associated notion of characteristic classes, see e.g. [14, Ch. 5]

Remark 10. *One very special scenario is that of a real or complex scalar field over a manifold M. In this case the underlying geometric structure is a* **line bundle** *E over M, that is a real (or complex) vector bundle of rank 1. Since we are interested in the existence of at least a nowhere vanishing global section (else the model would be trivial), Corollary 32 tells us that we are forced to consider the trivial bundle $M \times \mathbb{R}$ (or $\mathbb{C}$). As a consequence the space of kinematic configurations coincides with $C^\infty(M)$.*

2.2 Morphisms of vector bundles and induced bundles

To conclude our introduction to theory of vector bundles, we need to address the question of what is a good notion of maps between vector bundles. Implicitly we have already introduced and used them in the preceding subsections, but now it has become imperative to define them carefully, at least to make precise the notion of when we can call two vector bundles equivalent. Here we refer still to [3, Chap. 10], but also to [7, Chap. 3]. We define the key objects of this subsection.

Definition 34. *Let $E \equiv E[M;\pi,\mathbb{R}^m]$ and $E' \equiv E'[M';\pi',\mathbb{R}^{m'}]$ be two real (smooth) vector bundles. A* **(vector) bundle homomorphism** *is a map $F \in C^0(E;E')$ (*resp. *$F \in C^\infty(E;E')$) such that there exists $f : M \to M'$ satisfying the following two conditions:*

1. *for each $p \in M$, $F|_{E_p} \in \mathcal{L}(E_p; E'_{f(p)})$,*
2. *the following is a commutative diagram:*

$$\begin{tikzcd} E \arrow[r, "F"] \arrow[d, "\pi"'] & E' \arrow[d, "\pi'"] \\ M \arrow[r, "f"'] & M' \end{tikzcd} \tag{16}$$

We say that **F covers f**. *In addition, if F in invertible and F^{-1} is a bundle homomorphism, we say that F is a* **bundle isomorphism**. *When F is also a diffeomorphism, we call it a* **smooth bundle isomorphism** *and we say that E and E' are smoothly isomorphic.*

Disclaimer: Henceforth we will only look into smooth bundle homomorphisms and therefore the word smooth will not be written explicitly, although it is implicitly assumed.

A special instance of this definition occurs when the base space of the two bundles are the same:

Definition 35. *Let $E \equiv E[M;\pi,\mathbb{R}^m]$ and $E' \equiv E'[M;\pi',\mathbb{R}^{m'}]$ be two real (smooth) vector bundles over the same base space M. A* **bundle homomorphism over M** *is a bundle homomorphism $F \in C^\infty(E;E')$ which covers the identity $\mathrm{id}_M : M \to M$, that is the following diagram is commutative:*

$$\begin{tikzcd} E \arrow[rr, "F"] \arrow[dr, "\pi"'] & & E' \arrow[dl, "\pi'"] \\ & M & \end{tikzcd} \quad \equiv \quad \begin{tikzcd} E \arrow[r, "F"] \arrow[d, "\pi"'] & E' \arrow[d, "\pi'"] \\ M \arrow[r, "\mathrm{id}_M"'] & M \end{tikzcd}$$

If F is a diffeomorphism, we say that E and E' are **isomorphic over M**

In order to make the reader (which I am still prone to believe that it might also be a student) acquainted with the above concepts, we recommend the following instructive exercise:

Exercise 11. Let M and N be smooth manifolds and let $F \in C^\infty(M;N)$. Prove that the push-forward $F_* \equiv dF : TM \to TN$ is a smooth bundle homomorphism covering F.

The following theorem gives a clear characterization of bundle homomorphisms over a base space M and it is the main result of this subsection.

Theorem 36. *Let $E \equiv E[M;\pi,\mathbb{R}^m]$ and $E' \equiv E'[M;\pi',\mathbb{R}^{m'}]$ be two real (smooth) vector bundles over the same base space M. A functional $\mathcal{F} : \Gamma(E) \to \Gamma(E')$ is linear over the ring $C^\infty(M)$ if and only if there exists a smooth bundle homomorphism $F \in C^\infty(E;E')$ such that, for all $\sigma \in \Gamma(E)$*

$$\mathcal{F}(\sigma) = F \circ \sigma. \tag{17}$$

Proof. $\Longleftarrow$ Suppose $F \in C^\infty(E;E')$ is a bundle homomorphism. The map $F \circ \sigma \in \Gamma(E')$ since it is smooth, being the composition of smooth maps and, moreover, the commutativity of the diagram in Definition 35 guarantees that

$$\pi' \circ (F \circ \sigma) = F \circ (\pi \circ \sigma) = F.$$

This entails that $F \circ \sigma$ is a right inverse of π' at each $F(p)$, $p \in M$. Linearity is a direct consequence of this observation: for all $a \in C^\infty(M)$ and for all $\sigma, \tau \in \Gamma(E)$, it holds that, for each $p \in M$

$$\mathcal{F}(a\sigma + \tau)(p) = (F \circ (a\sigma + \tau))(p) = F(a(p)\sigma(p) + \tau(p)) = a(p)F(\sigma(p)) + F(\tau(p)),$$

where, in the last equality we used that a bundle homomorphism is a fiberwise linear map, *cf.* Definition 34.

$\Longrightarrow$ Suppose that $\mathcal{F}$ is linear as per hypothesis. We split the proof in different steps:

Step 1: $\mathcal{F}$ acts locally, that is, if $\sigma, \sigma' \in \Gamma(E)$ are such that, there exists $U \subseteq M$, $\sigma|_U = \sigma'|_U$ then, $\mathcal{F}(\sigma)|_U = \mathcal{F}(\sigma')|_U$. To this end it suffices to consider $\tau \in \Gamma(E)$ such that $\tau|_U = 0$. Let $p \in U$ and let $\psi \in C_0^\infty(M)$ be a bump function[1], *i.e.* such that $\psi(p) = 1$ and $\psi \neq 0$ on U while $\psi = 0$ on $M \setminus U$. Then, since $\psi\tau = 0$, per linearity of $\mathcal{F}$, it holds that $\mathcal{F}(\psi\tau) = 0$ and, consequently

$$\mathcal{F}(\psi\tau)(p) = \psi(p)F(\tau(p)) = F(\tau(p)) = 0.$$

The arbitrariness of $p \in U$ entails that $\mathcal{F}(\tau)|_U = 0$.

Step 2: $\mathcal{F}$ acts pointwisely, that is, if $\sigma, \sigma' \in \Gamma(E)$ are such that, there exists $p \in M$ for which $\sigma(p) = \sigma'(p)$, then $\mathcal{F}(\sigma)(p) = \mathcal{F}(\sigma')(p)$. As in the previous step, this amounts to showing that, if $\tau \in \Gamma(E)$ is such that $\tau(p) = 0$, $p \in M$, then $\mathcal{F}(\tau)(p) = 0$. Consider $U \subseteq M$ such that $p \in U$ and there exists a local trivialization of E over U. Per Remark (7), there must exist a local frame for E over U, say $\{\sigma_i\}_{i=1,\dots,m}$ such that, for all $q \in U$, $\tau(q) = \sum_{i=1}^{m} v_i(q)\sigma_i(q)$, where $v_i \in C^\infty(U)$ for all $i = 1, \dots, m$. Setting $p = q$, since $\tau(p) = 0$, it must hold by linear independence of all σ_i, that

[1] Existence of bump functions is a consequence of that of a partition of unity, *cf.* [3, Prop. 2.25]

$v_i(p) = 0$ for all i. By a simple extension argument, we can find a collection of global sections $\widetilde{\sigma}_i$ and of functions $\widetilde{v}_i$, $i = 1, \ldots, m$, coinciding respectively with σ_i and with v_i, $i = 1, \ldots, m$ on U. Since on U τ coincides with $\sum_{i=1}^{m} \widetilde{v}_i \widetilde{\sigma}_i$, it holds that at $p \in U$,

$$\mathcal{F}(\tau)(p) = \sum_{i=1}^{m} \widetilde{v}_i(p) \mathcal{F}(\widetilde{\tau}_i)(p) = 0,$$

since $\widetilde{v}_i(p) = 0$ for all $i = 1, \ldots m$.

Step 3: We are ready to construct a bundle homomorphism starting from $\mathcal{F}$. For any $p \in M$ and for any $w \in E_p$, we set $F|_p : E_p \to E'_p$ as

$$F(p, w) = F|_{E_p}(w) \doteq \mathcal{F}(\widetilde{w})(p),$$

where $\widetilde{w} \in \Gamma(E)$ is any global section such that $\widetilde{w}(p) = w$. By step 2 we know that this definition does not depend on the choice of $\widetilde{w}$. In addition F is linear at E_p inheriting this property from $\mathcal{F}$. To see that F covers the identity on M, for all $(p, w) \in E$,

$$(\pi' \circ F)(p, w) = \pi'(\mathcal{F}(\widetilde{w})(p)) = p,$$

since $\mathcal{F}(\widetilde{w}) \in \Gamma(E')$.

Step 4: We have shown that F verifies all algebraic properties of a bundle morphism and Equation (17) per construction. It remains to be seen that it is smooth. It suffices to prove that this holds true in a neighbourhood of an arbitrary point of the base space. Let $p \in M$ and let $U \subseteq M$ be such that there exists $\{\sigma_i\}_{i=1,\ldots,m}$ and $\{\sigma'_j\}_{j=1,\ldots,m'}$ two local frames respectively for E and E' on U. Let $\{\widetilde{\sigma}_i\}_{i=1,\ldots,m}$ be global sections of E such that there exists $V \subseteq U$ where $\widetilde{\sigma}_i|_V = \sigma_i|_V$, for all $i = 1, \ldots, m$. We can now apply $\mathcal{F}$ to the global sections and using the linear independence of the elements of a local section, we conclude that, for each $i = 1, \ldots, m$ and $j = 1, \ldots, m'$, there must exist $A_{ij} \in C^\infty(V)$ such that

$$\mathcal{F}(\widetilde{\sigma}_i)|_V = \sum_{j=1}^{m'} A_{ij} \sigma'_j.$$

Hence, for each $v \in E_q$, $q \in V$, there must exist m real numbers v_i, $i = 1, \ldots, m$ such that, writing $v = \sum_{i=1}^{m} v_i \sigma_i(q)$, it holds

$$F(\sum_{i=1}^{m} v_i \sigma_i(q)) = \mathcal{F}(\sum_{i=1}^{m} v_i \widetilde{\sigma}_i)(q) = \sum_{i=1}^{m} v_i \mathcal{F}(\widetilde{\sigma}_i)(q) = \sum_{i=1}^{m} \sum_{j=1}^{m'} A_{ij}(q) v_i \sigma'_j(q).$$

As a consequence, a direct computation shows that, calling Φ and Φ' the local trivializations associated to the local frames $\{\sigma_i\}_{i=1,\ldots,m}$ and $\{\sigma'_j\}_{j=1,\ldots,m'}$ respectively, it holds that, for all

$(p, v_1, \ldots v_m) \in V \times \mathbb{R}^m$,

$$(\Phi' \circ F \circ \Phi)(p, v_1, \ldots, v_m) = (p, \sum_{i=1}^{m} A_{i1} v_i, \cdots \sum_{i=1}^{m} A_{im'} v_i).$$

By direct inspection this is a smooth map since each $A_{ij} \in C^\infty(V)$. Since Φ, Φ' are diffeomorphisms, hence possessing a smooth inverse, $F \in C^\infty(E; E')$. □

Having in mind concrete scenarios, the notion of bundle homomorphism as discussed above, has the limitation that one starts with a map between two vector bundles. In many practical instances, the starting point is very similar to that of Exercise 11, namely we have a smooth map between two manifolds, *i.e.* $f : M \to N$. The following definition shows that, if we have a vector bundle over N, there is a canonical assignment of a counterpart over M together with a bundle homomorphism which covers f.

Definition 37. *Let $E \equiv E[M; \pi, \mathbb{R}^m]$ be a real vector bundle and let $f \in C^\infty(N; M)$ where N is an arbitrary smooth manifold. We call* **pull-back (or induced) bundle** $f^*E \equiv f^*E[N; \pi', \mathbb{R}^m]$, *the real vector bundle whose base space is N, while the total space is*

$$f^*E \doteq \{(p, x) \in N \times E \mid \quad f(p) = \pi(x)\},$$

and the projection is

$$\pi' : f^*E \to N, \quad (p, x) \mapsto \pi'(p, x) = p.$$

In order to see that the definition is well-posed, first of all observe that, for all, $p \in N$,

$$(f^*E)_p = (\pi')^{-1}_p \doteq \{x \in E \mid \pi(x) = f(p)\} \simeq E|_{f(p)},$$

that is each fiber has the structure of a real vector space of the same dimension of the rank of E. Finally, to check the existence of a local trivialization for f^*E, let $p \in N$ and let $U \subseteq M$ be an open subset such that $f(p) \in U$ and such that there exists a diffeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^m$. Since f is smooth, hence continuous in particular, $f^{-1}(U)$ is an open subset of N containing p. Let

$$\Psi : f^{-1}(U) \times \mathbb{R}^m \to (\pi')^{-1}(U), \quad (q, v) \mapsto \Psi(q, v) = (q, \Phi^{-1}(f(q), v)).$$

At each $q \in U$ this is a vector bundle isomorphism inheriting the property from that of Φ^{-1}, while Ψ is a diffeomorphism since it is built as a composition of Φ and of f.

The next exercise proves that induced bundles are quite useful when studying the tangent space of manifolds which are built as crossed-products. For those of you (hopefully many) who have some background knowledge in general relativity, the prototypical example, that you should have in mind, is a globally hyperbolic spacetime M which is known to be isometric to a cross-product of the form $\mathbb{R} \times \Sigma$.

Exercise 12. Let M be a smooth manifold diffeomorphic to $N \times K$ where N, K are in turn smooth manifolds. Let $\pi_1 : N \times K \to N$ and $\pi_2 : N \times K \to K$ be the two natural projections. Prove that the tangent bundle TM is smoothly isomorphic as vector bundle to the Whitney sum $\pi_1^* TN \oplus \pi_2^* TK$.

The following proposition shows that the construction of induced bundles behaves well also when looking at sections.

Proposition 38. *Let $E \equiv E[M; \pi, \mathbb{R}^m]$ be a real vector bundle and let $f \in C^\infty(N; M)$ where N is an arbitrary smooth manifold. For every $\sigma \in \Gamma(E)$, there exists an induced section $f^*\sigma \in \Gamma(f^*E)$ such that, for all $p \in N$,*

$$f^*\sigma(p) = (\sigma \circ f)(p).$$

Proof. We observe that $f^*\sigma : N \to f^*E$ since $\pi \circ (\sigma \circ f)(p) = f(p)$ for all $p \in N$. In addition $f^*\sigma \in C^\infty(N; f^*E)$ since it is the composition of two smooth maps. Finally, calling π' the projection of f^*E as per Definition 37, it holds per construction that $(\pi' \circ f^*\sigma)(p) = p$. $\square$

Remark 11. *If we keep thinking from the viewpoint of building physical models encompassing information on both space (or spacetime) and internal degrees of freedom, we have built the following dictionary:*

- *Geometric background $\longrightarrow$ vector bundle E on a manifold M.*
- *Kinematic configurations $\longrightarrow$ smooth section $\Gamma(E)$.*
- *Equations of motion $\longrightarrow$ Differential operators on E which is by assumption a manifold.*

Yet, in order to set up a full-fledged dynamical problem, besides the equation of motion, we need also to assign initial or boundary data depending on the specific scenario, we are interested in. It is not difficult to realize that this amounts to making sense of a section on the restriction of E to a codimension 1 submanifold of M. Yet, Definition 37 offers the solution to this quandary, since initial data hypersurfaces or boundaries can be read as submanifold of M together with the standard smooth inclusion map.

We conclude this section, observing that the wise reader, being wise, should not make confusion between the notion of induced bundle and that of subbundle which we specify now.

Definition 39. *Let $E \equiv E[M; \pi, \mathbb{R}^m]$ be a vector bundle over M and let D be an embedded smooth submanifold of E. This is called a* **subbundle** *of E with projection map $\pi_D = \pi|_D : D \to M$ if the subsets $D_p = D \cap E_p$ are linear subspaces of E_p.*

We limit ourselves to giving the definition so to make clear the difference between induced bundles and subbundles. The interested reader can refer to [3, Chap. 10] to study further properties.

2.3 A generalization: fiber and principal bundles

In the analysis of vector bundles, *cf.* Definition 20, it is immediate to notice that many concepts and results do not rely on the linear structure of each fiber. This prompts the possibility of generalizing the notion of vector bundle, which becomes a necessity in many concrete models, *e.g.* gauge theories.

Definition 40. *Let M and F be two topological (*resp. *smooth) spaces. We call a topological (*resp. *smooth) space E a* **fiber bundle (over M with model space F)** *if there exists a surjective continuous (*resp. *smooth) map $\pi : E \to M$ (*projection*) such that*

- *for each $p \in M$, there exists an open subset $U \subseteq M$ such that $p \in U$ and a* local trivialization *of E over U, namely an homeomorphism (*resp. *diffeomorphism) $\Phi : \pi^{-1}(U) \to U \times F$, such that the following is a commutative diagram*

$$\begin{tikzcd} \pi^{-1}(U) \arrow[rr, "\Phi"] \arrow[dr, "\pi"'] & & U \times F \arrow[dl, "\pi_U"] \\ & U & \end{tikzcd} ,$$

  where $\pi|_U : U \times F \to U$ is the projection on the first component.

Disclaimer: When indicating a generic fiber bundle, in the literature one writes usually $E[M; \pi, F]$. Only if the typical fiber is obvious from the context, one adopts the symbol E or $\pi : E \to M$.

In full analogy to Definition 20 we shall refer to E as the *total space*, to M as the *base space* and to F as the *(typical) fiber*. The net advantage of this generalization is that we are no longer forced to linear spaces on top of each point of M. Yet, this calls for the question what of Section 2.1 and 2.2 can be adapted to this more general scenario. Luckily the answer is "almost everything", although we postpone the discussion to later section. Here we limit to (re-)introduce two key concepts.

Definition 41. *Let $E \equiv E[M; \pi, F]$ be a fiber bundle. We call it* **trivial (*resp.* smoothly trivial)** *if there exists an homeomorphism (*resp. *a diffeomorphism) $\Phi : E \to E = M \times F$.*

Much in the spirit of Definition 34 we need a criterion to compare two fiber bundles. The following appears to be a natural concept:

Definition 42. *Let $E = E[M; \pi, F]$ and $E' = E'[M'; \pi', F']$ be two fiber bundles. We call* **bundle morphism** *a map $\Phi \in C^\infty(E; E')$ such that there exists $\phi \in C^\infty(M; M')$ making the following diagram commutative*

$$\begin{tikzcd} E \arrow[r, "\Phi"] \arrow[d, "\pi"'] & E' \arrow[d, "\pi'"] \\ M \arrow[r, "\phi"'] & M' \end{tikzcd}$$

If in addition Φ is a diffeomorphism, we call it an isomorphism of fiber bundles and we write $P \simeq P'$.

Observe that Φ is automatically fiber-preserving, namely, for every $p \in M$, it holds that $\Phi(E|_p) = E|_{\phi(p)}$. Yet, in comparison to Definition 34, we have to impose one requirement less since there is no obvious counterpart to asking for the bundle morphism to be linear unless one is dealing with vector bundles. The following exercise aims at better characterizing bundle isomorphisms:

Exercise 13. Let $E = E[M;\pi,F]$ and $E' = E'[M;\pi',F']$ be two fiber bundles. Prove that, if $\Phi \in C^\infty(E;E')$ is a bundle morphism and if, for every $p \in M$, the restriction $\Phi|_p : E_p \to E_{\phi(p)}$ is a diffeomorphism, then Φ is a bundle isomorphism.

Finally also the concept of section admits a straightforward generalization:

Definition 43. *Let $E \equiv E[M;\pi,F]$ be a fiber bundle. We call* **(cross) section** *of E a map $\sigma \in C^0(M;E)$ such that $\pi \circ \sigma = \mathrm{id}_M$ where id_M stands for the identity map on M. If the bundle is smooth and $\sigma \in C^\infty(M;E)$, we call it a smooth (cross) section. The collection of all sections if indicated with $\Gamma(E)$ (*resp. *$\Gamma^\infty(E)$)*

Remark 12. *One of the key results in the basic theory of fiber bundle consists of Corollary 32 which connects the notion of global sections and of trivial vector bundle. One might wonder whether a similar statement exists in the case in hand. The answer is negative in general since the structure of $\Gamma(E)$ is completely different and more complicated for a generic bundle, e.g. the 0-section is no longer well-defined. We shall see that the situation improves greatly if F is chosen to be a (topological) Lie group. This leads to the distinguished notion of a principal G-bundle.*

We are now ready to start a journey with the ultimate goal of dispelling the myth of what is the gauge group in any physical theory. As first step let us consider an arbitrary smooth fiber bundle $E \equiv E[M;\pi,F]$ and let us consider an open cover of the base space M, namely $\{U_\alpha\}_{\alpha\in I}$ where I is a countable index set, while $U_\alpha \subseteq M$ for all $\alpha \in I$ and $\bigcup\limits_{\alpha\in I} U_\alpha = M$. Without loss of generality we can also take these open sets so that there exists a local trivialization $\varphi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times F$.

In view of Definition 40, we can construct for any $\alpha, \beta \in I$, the diffeomorphism

$$\varphi_\alpha \circ \varphi_\beta^{-1} : (U_\alpha \cap U_\beta) \times F \to (U_\alpha \cap U_\beta) \times F, \quad (p,a) \mapsto (\varphi_\alpha \circ \varphi_\beta^{-1})(p,a) = (p,b).$$

As a consequence since on the first entry these maps act as the identity, we can extract the collection of **transition functions**

$$h_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{Diff}(F), \quad p \mapsto h_{\alpha\beta}(p) = (\varphi_\alpha \circ \varphi_\beta^{-1})(p,\cdot),$$

where $\mathrm{Diff}(F)$ is the group of diffeomorphisms of F and where the transition functions are subject to the *cocycle condition*

$$h_{\alpha\beta} h_{\beta\gamma} = h_{\alpha\gamma} \text{ on } U_\alpha \cap U_\beta \cap U_\gamma.$$

Remark 13. *Observe that in the case of a real or complex vector bundle, where the role of F is played either by $\mathbb{R}^n$ or by $\mathbb{C}^n$, $n \geq 1$, then $\mathrm{Diff}(F) = \mathrm{GL}(n, \mathbb{R})$ or $\mathrm{GL}(n, \mathbb{C})$, cf. Proposition 25.*

In the next proposition we see that transition functions do encode all data necessary to fully identify an underlying fiber bundle. As a matter of fact, some authors (not me!) take the following as the definition of a fiber bundle. Observe that we are still working with smooth structures. There exists a rather straightforward counterpart when we only require the underlying objects and maps to be continuous, but we feel that presenting it would be redundant. The reader is wise and we are confident in his ability of figuring out the correct generalization.

Proposition 44. *Let M and F be two smooth manifolds and let $\{U_\alpha\}_{\alpha\in I}$, where I is a countable index set, be an open cover of M, Suppose that, for all $U_\alpha \cap U_\beta \neq \emptyset$, it is assigned a smooth map*

$$h_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{Diff}(F),$$

subject to the cocycle condition

$$h_{\alpha\beta} h_{\beta\gamma} = h_{\alpha\gamma},$$

whenever $U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset$. Then there exists a fiber bundle $E \equiv E[M; \pi, F]$ whose transition functions are the maps $\{h_{\alpha\beta}\}_{\alpha,\beta\in I}$.

Proof. Consider the given open cover of M and define $\widetilde{E} = \bigsqcup\limits_{\alpha\in I} (U_\alpha \times F)$. In order to account for the overlap between the open subsets covering M, we introduce the equivalence relation $(\alpha, p, a) \simeq (\beta, q, b)$, $\alpha, \beta \in I$, $p \in U_\alpha$, $q \in U_\beta$, $a, b \in F$ if

$$U_\alpha \cap U_\beta \neq \emptyset, \quad q = p \text{ and } b = (h_{\alpha\beta}(p))(a).$$

The sought bundle is defined as $E \doteq \widetilde{E}/\sim$. Endowing E with the quotient topology, since $U_\alpha \times F$ is an open subset of E, it acquires a differentiable structure induced from $\widetilde{E}$. Recalling that a point in E is an equivalence class $[\alpha, p, a]$ we define the projection map

$$\pi : E \to M, \quad [\alpha, p, a] \mapsto \pi[\alpha, p, a] = p,$$

which can be read as the composition between the restriction from E to $U_\alpha \times F$ and the projection on the first factor. These are both smooth maps and so is π. Following Definition 40 we have verified that E is indeed a fiber bundle, which concludes the proof. ☐

While in the case of a real vector bundle the diffeomorphism group of the fiber can be reduced to be a finite dimensional Lie group on account of Proposition 25, this option seems to disappear as soon as we allow for more general manifolds as typical fiber. Our beloved reader should think for example to the unit n-sphere $\mathbb{S}^n$. In this case, considering a fiber bundle with $\mathbb{S}^n$ as typical fiber, we should work with transition functions whose image lies in $\mathrm{Diff}(\mathbb{S}^n)$. Yet, in full analogy to the case where the fiber is $\mathbb{R}^n$, we would like to be able to restrict our attention to $\mathrm{SO}(n+1)$ seen as a subgroup of $\mathrm{Diff}(\mathbb{S}^n)$. Unfortunately this is not always possible, but when this is the case, the underlying structures are distinguished and they deserve to be characterized independently.

Definition 45. *Let $E = E[M, \pi, F]$ be e fiber bundle and let $\{U_\alpha\}_{\alpha \in I}$ where I is a countable index set, be an open cover of M such that there exists a local trivialization $\varphi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times F$. Assume in addition that the transition functions are*

$$h_{\alpha\beta} : U_\alpha \cap U_\beta \to G,$$

where G is a Lie group realized as a subgroup of $\mathrm{Diff}(F)$. Then we say that E is endowed with a G-structure and we call G the **structure group**. *This scenario is indicated via $E = E[M; \pi, F, G]$.*

The above definition allows to select a distinguished class of fiber bundles and actually they enjoy a lot of rather natural properties. Since the proofs are identical to others already given, this is the perfect chance to propose a few exercises which allow the reader(s) to revise what explained up to now.

Exercise 14. Let M and F be two smooth manifolds and let $\{U_\alpha\}_{\alpha \in I}$, where I is a countable index set, be an open cover of M, Suppose that, for all $U_\alpha \cap U_\beta \neq \emptyset$, it is assigned a smooth map

$$h_{\alpha\beta} : U_\alpha \cap U_\beta \to G,$$

where G is a Lie subgroup of $\mathrm{Diff}(F)$. Suppose in addition that it holds the cocycle condition

$$h_{\alpha\beta} h_{\beta\gamma} = h_{\alpha\gamma},$$

whenever $U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset$. Then there exists a fiber bundle $E \equiv E[M, \pi, F, G]$ whose transition functions are the maps $\{h_{\alpha\beta}\}$ and whose structure group is G.

It occurs in many instances that the transition functions take value in H, a Lie subgroup of G. In this case it means that we have considered a structure group too big ans we say that it is **reducible**. In general this feature is rather irrelevant except for one property which is at the hearth of the following exercise

Exercise 15. Let $E = E[M, \pi, F, G]$ be a fiber bundle with structure group G. Prove that E is trivial if and only if the structure group is reducible to the trivial one composed only by the identity element.

Remark 14. *A reader motivated in understanding the applications of differential geometry to physical theories should prick up his hears because it starts becoming manifest that, when in the physics literature, it is considered a G-gauge theory, say $G = U(1)$ or $SU(2)$, this is a reference to a structure group. Hence the natural question which arises is what is in reality the gauge group. We shall give an answer in the next subsection.*

In the previous discussion we have highlighted a deep connection between fiber bundles and Lie groups via transition functions. The main drawback comes from the fact that endowing a bundle with a G-structure is not a generic feature needing a case by case analysis. Yet, if we give a closer look at Definition 40 and at Proposition 44, it is rather easy to realize that one can identify a distinguished class of fiber bundles which automatically admits a G-structure, namely those whose typical fiber is itself a Lie group. This leads to introducing a basic structure with countless applications in physical models, especially gauge theories. While the topics that we shall discuss till the end of the section appear in many textbooks, in my opinion, the best analysis is present in [8]. This is a classic from 1963... oh... I am sorry... were you thinking that we are discussing modern mathematics and its application to physics? Cute.... nope, this is a theory which was already well-established in the sixties to make it in textbooks and it was mainly developed in the forties. To put thing in the correct perspective, the theory of distributions was developed mainly in the fifties....

Definition 46. *Let G be a Lie group endowed and let M be a smooth manifold. We call* **principal (G-)bundle** *a fiber bundle $P \equiv P[M;\pi,G]$ with model space G together with a, fiber preserving, free and transitive right action of G on P, namely a map $\varphi : G \times P \to P$, $(g,x) \mapsto \varphi(g,x) \equiv xg^{-1}$ such that, calling e the identity element of G, then*

1. *for all $x \in P$ and for all $g,h \in G$*

$$\varphi(e,x) = x, \quad \varphi(g,\varphi(h,x)) = \varphi(gh,x),$$
$$\varphi(g,\cdot) \in \mathrm{Diff(P)},$$

2. *for every $x,y \in P$, there exists $g \in G$ such that $y = \varphi(g,x)$ [Transitive Action],*
3. *for any $x \in P$, $\varphi(g,x) = x$ if and only if $g = e$ [Free Action],*
4. *for every $g \in G$, $\pi \circ \varphi(g,\cdot) = \pi$. [Fiber Preserving Action]*

If the reader is familiar with the theory of Lie groups, it will not come as a surprise that the choice of the right action is purely conventional and one could equivalently use the left action. We stick with the convention more commonly used nowadays. Observe in addition that, per definition, the action of the Lie group on P is free, namely, if there exists $x \in P$ and $g \in G$, such that $xg^{-1} = x$, then $g = e$. Before studying the properties of principal bundles, it is convenient to look at the notion of morphism between principal bundles. It is worth comparing Definition 34 and Definition 42 to understand that, whenever the typical fiber has additional structures, one seeks for maps compatible with them. This translates to the following definition.

Definition 47. *Let $P \equiv P[M;\pi,G]$ and $P' \equiv P'[M';\pi',G]$ be two principal bundles. We call $\Phi \in C^\infty(P;P')$ a* **morphism of principal bundles** *if it is a morphism of fiber bundles as per Definition 42 and, in addition, it is an equivariant map, namely for every $x \in P$ and for every g in G, it holds*

$$\Phi(xg^{-1}) = \Phi(x)g^{-1}.$$

If in addition Φ admits a smooth inverse than we call it an ***isomorphism of principal bundles*** *or an* ***automorphism*** *in the special case $P = P'$.*

In this definition we have assumed a priori that the structure groups of both principal bundles coincide. In principle one might consider $P' \equiv P'[M';\pi',G']$ and the notion of bundle morphism can still be meaningful provided that G is a Lie subgroup of G'. In between the collection of all automorphisms of a principal bundle, we can single out a distinguished subgroup which plays a pivotal role in all physical models. The reader is warned that the following concept might contradict their expectations, but, in later sections we shall apply it and it will become manifest that it is indeed what it is supposed to be.

Definition 48. *Let $P \equiv P[M;\pi,G]$ be a principal bundle and let $\mathrm{Aut}(P)$ denote the group of all its automorphisms as per Definition 47. We call* **gauge group** *of P, $\mathrm{Gau}(P) \subset \mathrm{Aut}(P)$ the subgroup of automorphisms of P covering the identity on M, i.e. $\Phi \in \mathrm{Gau}(P)$ if*

$$\begin{tikzcd} P \arrow[rr, "\Phi"] \arrow[dr, "\pi"'] & & P \arrow[dl, "\pi"] \\ & M & \end{tikzcd} .$$

Equivalently $\mathrm{Gau}(P)$ is characterized by the left exact sequence (not necessarily right exact)

$$0 \longrightarrow \mathrm{Gau}(P) \xrightarrow{\iota} \mathrm{Aut}(P) \xrightarrow{\widetilde{\pi}} \mathrm{Diff}(M)$$

where $\iota : \mathrm{Gau}(P) \to \mathrm{Aut}(P)$ is the inclusion map while $\widetilde{\pi} : \mathrm{Aut}(P) \to \mathrm{Diff}(M)$ is the map assigning to a bundle automorphism the associated diffeomorphism over the base manifold as per Definition 42. Here 0 has to be interpreted as the trivial group consisting only of the identity element.

Principal bundles are rather distinguished geometric objects since they enjoy some notable properties which we discuss now.

Proposition 49. *Let $P \equiv P[M;\pi,G]$ be a principal bundle. Then it holds that*

1. *P is endowed with a G-structure,*
2. *M is diffeomorphic to the quotient manifold P/G.*

Proof. 1. Consider an open cover $\{U_\alpha\}_{\alpha\in I}$, I being an index set, of M such that there exists a local trivialization $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times G$. For any $\alpha, \beta \in I$ such that $U_\alpha \cap U_\beta \neq \emptyset$, consider $(x,h) \in (U_\alpha \cap U_\beta) \times G$. Let $u = \Phi_\beta^{-1}(x,h)$. Then there exists $h' \in G$ such that $\Phi_\alpha(u) = (x,h')$. Calling φ the fiber-preserving, free and transitive action, there must exist $g \in G$ such that $\varphi(g,(x,h)) = (x,h')$. A direct inspection shows that the transition functions coincide with the maps $h_{\alpha\beta} : U_\alpha \cap U_\beta \to G$ such that $x \mapsto h_{\alpha\beta}(x) = g$. The cocycle condition is automatically satisfied and we have constructed a G-structure.

2. Since for any $g \in G$, $\varphi(g,\cdot)$, the action of the Lie group on P, is a diffeomorphism, then it is a proper map. Hence we can apply the quotient manifold theorem [11, Th. 21.10] to conclude that P/G is endowed with a unique smooth structure such that the canonical projection map $\pi_G : P \to P/G$ is smooth. Since the action of the group is per definition fiber preserving, it is immediate to observe that the action of π_G coincide with that of π. More precisely p, p' are two representatives of $[p] \in P/G$ if and only if there exists $g \in G$ such that $p' = \varphi(g, p)$. Yet, since the action is free and transitive $\pi_P(p) = [p] = [(z, e)]$, e being the identity of the group. At the same time $\pi(p) = z$ which is in 1:1 correspondence with $[(z, e)]$. Hence, up to the identification map, π coincides with π_G and thus P/G is diffeomorphic to M. □

The following exercise reminds to our readers that a bundle can be constructed unambiguously starting only from the transition functions. This paradigm has been shown to hold true for vector bundles, *cf.* Proposition 26, for fiber bundles, *cf.* Proposition 44 and we now extend it to principal bundles.

Exercise 16. Let M be a smooth manifold and let G be a Lie group. Suppose that an open cover of M, $\{U_\alpha\}_{\alpha \in I}$, I being an index set, is assigned together with a family of smooth functions $h_{\alpha\beta} : U_\alpha \cap U_\beta \to G$. Prove that, if the cocycle condition $h_{\alpha\beta} h_{\beta\gamma} = h_{\alpha\gamma}$ is satisfied whenever $U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset$, then there exists a principal G-bundle over M, admitting $\{h_{\alpha\beta}\}_{\alpha,\beta \in I}$ as transition functions.

We can prove now the main result concerning the structural properties of a principal bundle and it concerns the existence of a special interplay between global smooth sections and triviality of the bundle.

Theorem 50. *Let $P \equiv P[M; \pi, G]$ be a principal bundle. Then P is trivial, i.e. $P \simeq M \times G$, if and only if it admits a global section, namely $\sigma \in C^\infty(M; P)$ such that $\pi \circ \sigma = id|_M$, $id|_M$ being the identity map on M.*

Proof. $\Longrightarrow$ If P is trivial, then for any $f \in C^\infty(M; G)$, we can construct an induced global section as

$$\sigma_f : M \to P \quad x \mapsto \sigma_f(x) = (x, f(x)).$$

$\Longleftarrow$ Suppose that P is a principal G-bundle admitting a global section σ. Recalling that, per Definition 46 there exists a transitive action of G which acts fiberwise, we can define

$$\Phi : P \to M \times G \quad p \mapsto \Phi(p) = (\pi(p), g),$$

where g is the unique group element connecting on $P|_{\pi(p)}$ the points p and $\sigma(\pi(p))$. Reading $M \times G$ as the trivial principal bundle over M, since both π and σ are smooth maps, $\Phi \in C^\infty(P, P')$. In addition Φ is an equivariant map, since for every $h \in G$

$$\Phi(ph^{-1}) = (\pi(p), gh^{-1}) = \Phi(p)h^{-1},$$

where we used that the group action on P does not change the base point. Hence Φ is a morphism of principal bundles as per Definition 47 and it is invertible since, for every $(z, g) \in M \times G$, $\Phi^{-1}(z, g) = \sigma(z)g$. □

To conclude the section we introduce a structure which is of paramount relevance in the physical applications of principal bundle. As a matter of fact, an attentive reader might have already observed that, having in mind applications to field theory, principal bundles cannot be the unique geometric structure in the game. The reason lies in the fact that, in many instances, we would like to deal with a linear space in order to make sense of the superposition principle via the linear structure. This is not always the case, *e.g.* in the so-called non-linear sigma models, but it is certainly true for all free and interacting field theories at the heart of the standard model of elementary particles (well... yes... there exists another standard model... the cosmological one). To solve this quandary we introduce the following additional structure.

Proposition 51. *Let $P = P[M;\pi,G]$ be a principal bundle and let F be a smooth manifold together with a smooth (left) action $\rho : G \times F \to F$. Then we call*

$$E = P \times_\rho F$$

the set of equivalence classes $[(p,x)]$ where $P \times F \ni (p',x') \simeq (p,x)$ if and only if there exists $g \in G$ such that $p' = \varphi(g,p) \equiv pg^{-1}$ and $x' = \rho(g,x) \equiv gx$. Then E is a fiber bundle over M, called **the associated bundle (to P via ρ)** *and it is indicated with $E \equiv E[M;\widehat{\pi},F,\rho]$ where*

$$\widehat{\pi} : E \to M, \quad [(p,x)] \mapsto \widehat{\pi}([p,x]) = \pi(p).$$

Proof. Following Definition 40 we need to find a local trivialization of E. Since P is a principal bundle per hypothesis, there exists a cover $\{U_\alpha\}_{\alpha\in I}$, I being an index set, of M together with diffeomorphism $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times G$ such that

$$\pi^{-1}(U_\alpha) \ni p \mapsto \Phi_\alpha(p) = (\pi(p), \varphi_\alpha(p)),$$

where $\varphi_\alpha : U_\alpha \to G$. This induces in turn a map

$$\Psi_\alpha : \widehat{\pi}^{-1}(U_\alpha) \to U_\alpha \times F, \quad [(p,x)] \mapsto \Psi_\alpha([p,x]) = (\pi(p), \varphi_\alpha(p)x).$$

It is immediate to observe that the definition is independent from the choice of representative in $[(p,x)]$ on account of the equivariance of the map Φ_α. Since both π and φ_α are smooth per construction, we can conclude that $\Psi_\alpha \in C^\infty(\widehat{\pi}^{-1}(U_\alpha); U_\alpha \times F)$. As last step observe that

$$\Psi_\alpha^{-1} : U_\alpha \times F \to \widehat{\pi}(U_\alpha), \quad (z,x) \mapsto \Psi_\alpha^{-1}(z,x) = [\pi^{-1}(z), (\varphi_\alpha(\pi^{-1}z))^{-1}x].$$

This map is smooth since it inherits this property from π^{-1} and φ_α. □

Remark 15. *Proposition 51 has a rather wide range of applicability, but it does not come as a surprise that the most interesting scenario is that where F is a vector space, so that E becomes a vector bundle. One might prone to believe in this assertion by thinking immediately to a fancy example such as Yang-Mills field theory. This is certainly a possibility (although ultimately not the best one) but, by far, it is not the first one. When the goal is to construct a relativistic field theory, it is true that vector bundles are the natural geometric structure since their sections play* *the role of kinematic configurations of the underlying theory, cf. Remark 9. Yet, in this way, we could hardly distinguish a theory describing an elementary relativistic particle from another one encoding for example the behaviour of the velocity vector of a Newtonian fluid. The reason lies in the fact that a relativistic field theory knows about the existence of Lorentz invariance and, if the ambient space is Minkwoski, also of the Poincaré group. This is formalized by means of a principal and of an associated bundle. If you are skeptical, please answer to this question: what is the spin, a.k.a. spin structure, on a generic Lorentzian manifold? What would you call a Dirac field? Why would you still say that a photon propagating through the Universe has helicity* 1 *if this is a concept tied to exact Minkowski spacetime?*

2.4 Notable Examples

In this section we shall discuss a few examples of principal and of associated bundles which play a distinguished role in many applications both in differential geometry and in physical models.

The Frame Bundle

The first example of principal bundle is noteworthy not only for its relevance in many constructions and structures in differential geometry, but also for its physical interpretation. As a matter of fact, in classical mechanics, the first step in all models is the identification of the configuration space. This is typically an affine space $\mathbb{A}_n$, where $n \geq 1$ is the dimension of the associated vector space V_n. The second step consists of the realization that restricting the attention to V_n is tantamount to choosing a point of $\mathbb{A}_n$ to play the role of the origin. In the language of classical mechanics this amounts to picking an observer for the system. At this stage, one needs to assign to V_n a frame, that is an orthonormal basis of V_n, which, from a physical viewpoint, amounts to picking distinguished orthogonal directions with which events or point objects will be labeled.

The key and somehow implicit point in this idea is that, despite the fact that there are infinite possible choices for an origin/observer, nonetheless it suffices to pick a frame for one specific choice of the origin and by translation invariance of V_n, an associated counterpart can be found for any other choice of the origin of $\mathbb{A}_n$. Observe that this is not related to Galilean invariance which, in turn, establishes when two frames are equivalent. In addition the Galileo group contains dynamical information since it knows in its most general form about the relative velocity between two frames.

At this stage, if you think of repeating this reasoning when the configuration space is no longer an affine space but rather a smooth, connected and oriented manifold, the situation appears to be more complicated. It is clear that the choice of an observer becomes that of a generic point $x \in M$, but it is by no means obvious what is a frame.

To bypass this hurdle, consider a generic smooth manifold M and for any $x \in M$, consider $F_x(M) = \{(X_1, \ldots, X_n)\}$ that is the collection of all ordered basis of T_xM where $n = \dim M \geq 1$. Define

$$F[M] \doteq \bigsqcup_{x \in M} F_x(M).$$

We want to investigate whether $F[M]$ can be endowed with the structure of a fiber bundle. First of all, we observe that we can define an action of $\mathrm{GL}(n,\mathbb{R})$ over $F[M]$ as follows:

$$\varphi : \mathrm{GL}(n,\mathbb{R}) \times F(M) \to F(M)$$
$$(A,(x,(X_1,\ldots,X_n))) \mapsto \varphi((A,(x,(X_1,\ldots,X_n)))) = (x, A(X_1,\ldots,X_n)),$$

where $A(X_1,\ldots,X_n)$ is defined as the action of the matrix A over the vector $(X_1,\ldots X_n) \in \mathbb{R}^n \simeq T_xM$. It is a standard result in linear algebra that this action is both free and transitive. As next step we assign a differentiable structure to $F(M)$. To this end, for any but fixed x, consider a local chart (U,ϕ) so that $x \in U$ and consider any local coordinate system $\{x_1,\ldots,x_n\}$. Since $X_i \in T_xM$, we can decompose each of these elements of T_xM as

$$X_i = \sum_{j=1}^{n} X_{ij}\frac{\partial}{\partial x_j}, \quad X_{ij} \in \mathbb{R},\ \forall i,j = 1,\ldots,n$$

where $\frac{\partial}{\partial x_j}$ is the standard basis of T_xM induced by the chosen coordinate system. In addition since both $(X_1,\ldots,X_n)$ and $\left\{\frac{\partial}{\partial x_i}\right\}_{i=1,\ldots,n}$ are a basis of T_xM, the determinant of the matrix X, whose entries are X_{ij}, is non vanishing. In other words $X \in \mathrm{GL}(n,\mathbb{R})$. This proves that, if we call $\pi : F[M] \to M$, the natural projection map, for all $x \in M$,

$$\pi^{-1}(x) = F_x(M) \simeq \mathrm{GL}(n,\mathbb{R}).$$

In addition since TM is a vector bundle, *cf.* Proposition 24, we can choose U so that $TM|_U \simeq U \times \mathbb{R}^n$. In this open set, we can repeat the above construction obtaining

$$\pi^{-1}(U) \simeq U \times \mathrm{GL}(n,\mathbb{R}).$$

Hereon we can introduce the coordinate system (x_i, X_{jk}) where $i,j,k = 1,\ldots,n$ and repeating the procedure for all $x \in M$, we have assigned a differentiable structure to $F[M]$. At the end of the day we have constructed more than a simple fiber bundle as we summarize in the following proposition, whose proof is nothing but the construction juts outlined

Proposition 52. *Let M be a smooth differentiable manifold and let $F[M] \doteq \bigsqcup_{x\in M} F_x(M)$ where $F_x(M)$ is the collection of all ordered bases of T_xM. This is a principal bundle $F[M] \equiv F[M][M;\pi,\mathrm{GL}(n,\mathbb{R})]$ over M called the* **frame bundle (of M)** *(or also the bundle of linear frames of M).*

There are two notable exercises which will help our beloved readers to get really acquainted with the notion of a frame bundle

Exercise 17. Let $E = E[M;\pi,\mathbb{R}^n]$ be a real smooth vector bundle over a smooth manifold M. Prove that $L[M] \doteq \bigsqcup_{x\in M} L_x(M)$, where $L_x(M)$ is the collection of all ordered basis of $\pi^{-1}(x)$, is a principal $\mathrm{GL}(n,\mathbb{R})$-bundle where $n = \dim M \geq 1$. This is called the bundle of linear frames of E.

The following exercise unveil instead a deeper connection between the frame bundle and the tangent bundle.

Exercise 18. Let M be a smooth manifold and let $F[M]$ be the frame bundle. Prove that calling $\rho : \mathrm{GL}(n,\mathbb{R}) \times \mathbb{R}^n \to \mathbb{R}^n$ the natural action of matrices on vectors, then there exists an isomorphism of vector bundles

$$F[M] \times_\rho \mathbb{R}^n \simeq TM,$$

where $F[M] \times_\rho \mathbb{R}^n$ is the associated bundle to $F[M]$ via ρ, *cf.* Proposition 51.

To conclude this example, we consider a more specialized scenario, namely we assume that the underlying manifold M is endowed with a Riemannian or with a Lorentzian smooth metric g. In this case the construction above is left unchanged except for the fact that, at each $x \in M$, we are allowed to consider only orthonormal bases of the tangent space. If we recall that, furthermore, any Riemannian or Lorentzian smooth metric can be reduced at each point $x \in M$ either to the standard Euclidean or Minkowski metric, we can restrict the group $\mathrm{GL}(n,\mathbb{R})$ either to $\mathrm{O}(n)$ or to $\mathrm{O}(1, n-1)$, namely respectively the orthogonal and the Lorentz group in n-dimensions. This can be summarized in the following definition:

Definition 53. *Let (M, g) be a Lorentzian or a Riemannian smooth manifold of dimension $n = \dim M \geq 1$. We call* **bundle of orthonormal frames** $\mathfrak{O}[M] = \mathfrak{O}[M][M; \widetilde{\pi}, G]$*, where $G = \mathrm{O}(n)$ if M is Riemannian while $G = \mathrm{O}(1, n-1)$ if M is Lorentzian. This is principal bundle whose fiber at each $x \in M$ is the collection of all orthonormal basis of T_xM with respect to the metric g. In addition, if M is oriented, we can further restrict the structure group to the component connected to the identity, that is $G = \mathrm{SO}(n)$ and $G = \mathrm{SO}_0(1, n-1)$. In this case we call $\mathfrak{O}[M]$,* **the bundle of oriented orthonormal frames***.*

Remark 16. *The notion of frame bundle seems rather redundant and not particularly informative, but this would be an understatement. As a matter of fact, from a geometric viewpoint, a section of a frame bundle is tantamount to an assignment of a reference system to any observer/point in M. The question is whether we can do it globally in a smooth way and Theorem 50 tells us that this is possible only if the frame bundle or, more appropriately, the bundle of orthonormal frames is trivial. The answer in general is absolutely not. Yet, there are two classes of manifolds where this actually becomes possible and a reader with some experience in general relativity can appreciate their relevance: three dimensional metric complete Riemannian manifolds and four dimensional globally hyperbolic spacetimes.*

The Dirac Bundle and the Spin Structure

A rather surprising consequence of the frame bundle is its connection in the definition or rather in the identification of what is the correct translation of the notion of spin on a generic Riemannian or Lorentzian manifold. This leads to the possibility of correctly identifying the vector bundles encoding the kinematic configurations of all relativistic free fields. We shall succinctly outline the procedure focusing on the notable example of spinors. The first step is the identification of a group encoding the information about spin in arbitrary dimensions

Definition 54. *Let $p, q \in \mathbb{N}\cup\{0\}$. We call* **spin group** $\mathbf{Spin(p,q)}$ *the double cover of $SO_0(p,q)$ such that there exists a short exact sequence of groups, cf. Definition 16*

$$0 \longrightarrow \mathbb{Z}_2 \xrightarrow{\iota} \mathrm{Spin}(p,q) \xrightarrow{\Lambda} SO_0(p,q) \longrightarrow 0\ ,$$

where 0 has to be interpreted as the trivial group composed by the lone identity element, while Λ is the group covering homomorphism. The map ι is an embedding of $\mathbb{Z}_2$ in $\mathrm{Spin}(p,q)$. We indicate with $Spin_0(p,q)$ the component of the spin group connected to the identity.

Observe that, in this definition, we have implicitly excluded the degenerate case $p = q = 0$, while, if either $p = 0$ or $q = 0$, then $\mathrm{SO}_0(p,0)$ or $\mathrm{SO}_0(0,q)$ coincide respectively with $\mathrm{SO}(p)$ and $\mathrm{SO}(q)$. The attentive readers (are you included in this group? Does this group coincide with you?) should be rather unsatisfied by this definition for two reasons. The first concerns the exact sequence. Its existence is easily justified by standard results in the theory of Lie groups, but what about its uniqueness? This is by far not an obvious question. Secondly it is rather unclear why one should call spin group the double cover of $SO_0(p,q)$ and not its universal cover, which exists and it is uniquely defined, still on account of standard results in the theory of Lie groups. In the special and admittedly more interesting scenario $p = 1$ and $q = 3$ (or viceversa), one obtains that the spin group is the double cover of the component connected to the identity of the Lorentz group, namely

$$\mathrm{Spin}(1,3) \equiv \mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C}).$$

Since $\mathrm{SL}(2,\mathbb{C})$ is simply-connected, it coincides with the universal cover of $\mathrm{SO}_0(3,1)$ or equivalently of $\mathrm{SO}_0(1,3)$. An answer to all the questions asked requires to open a full, new chapter in differential geometry, which leads to a rather interesting and fashionable subject, called *spin geometry*. A good, probably by far the best reference in this field is [10].

The notion of spin group does not suffice by itself but it needs to be attached coherently to a smooth manifold M. In this context, we have seen in Definition 53 that, whenever M is oriented, $\mathrm{SO}_0(p,q)$ appears naturally in the bundle of oriented orthonormal frames. This prompts the following definition:

Definition 55. *Let (M,g) be a smooth, oriented manifold endowed with a metric of signature $(\underbrace{-,\dots,-}_{p},\underbrace{+,\dots,+}_{q})$. We call* **spin structure** *on M a pair (SM,π) where*

1. *$SM \equiv SM[M;\pi',\mathrm{Spin}_0(p,q)]$ is a principal bundle over M called the* **spin bundle***,*
2. *$\pi : SM \to \mathfrak{O}[M]$ is a morphism of principal bundles covering the identity:*

$$\begin{tikzcd} SM \arrow[rr, "\pi"] \arrow[dr, "\pi'"'] & & \mathfrak{O}[M] \arrow[dl, "\widetilde{\pi}"] \\ & M & \end{tikzcd}\ ,$$

such that, for all $S \in \mathrm{Spin}(p,q)$ and for all $p \in SM$

$$\pi(\varphi_{SM}(p,S)) = \varphi_{\mathfrak{O}[M]}(\pi(p), \Lambda(S)),$$

where φ_{SM} and $\varphi_{\mathfrak{O}[M]}$ are the group action respectively on SM and on $\mathfrak{O}[M]$, cf. Definition 46, while $\Lambda : \mathrm{Spin}(p,q) \to SO_0(p,q)$ is the covering group homomorphism as in Definition 54.

This definition prompts the question whether a manifold admits a spin structure. Certainly the problem does not lie in the spin bundle (one can always construct the trivial principal bundle $M \times \mathrm{Spin}_0(p,q)$), but rather in π whose existence is not at all obvious.

Remark 17. *The problem of the existence of a spin structure is a delicate one which has been thoroughly investigated. The outcome is the existence of a potential obstruction of topological nature which is ruled by the so-called* **second Stiefel-Whitney class** *of the underlying manifold M. In other words, if $H^2(M;\mathbb{Z}_2) = \{0\}$ then a spin structure exists. Unfortunately, this statement is meaningful only to those beloved readers familiar with singular homology and cohomology, which, alas, we shall probably not have time to discuss (as if time makes really sense in a written text). Nonetheless the brave one can consult [10, §2.2]. Although, in this reference, it is assumed that M is a Riemannian manifold, the assumption on the metric signature can be easily changed according to the scenario under investigation. Notice, in addition that, even if a spin structure exists, it might not be unique. Also this information is of topological nature and it is ruled by $H^1(M;\mathbb{Z}_2)$, the first Stiefel-Whitney class.*

For all practical purposes this discussion in Remark 17 is moot. As a matter of fact, it turns out that every globally hyperbolic spacetime as well as every three dimensional, metric complete, Riemannian manifold admits a spin structure. As a matter of fact, recall that their frame bundle is trivial, *cf.* Remark 16 and try to solve the following exercise.

Exercise 19. Let (M,g) be a smooth manifold endowed with a smooth metric of arbitrary signature. Prove that, if the bundle of oriented orthonormal frames is trivial, then there exists a spin structure (though it might not be unique).

Having established a good notion of spin structure, we can focus our attention to the case of physical interest where (M,g) is a globally hyperbolic spacetime while $\mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C})$. If one has in mind the Wigner-Bargmann construction of relativistic field in Minkowski spacetime in terms of unitary and irreducible representations of the Poincaré group, one should recall that the notion of spin of an elementary particle/field was tied to the choice of a unitary representation of the double cover of the Lorentz group, namely $\mathrm{SL}(2,\mathbb{C})$, induced either from $\mathrm{SU}(2)$ or from the two-dimensional Euclidean group $\mathrm{E}(2)$ depending whether the mass is positive or vanishing. In particular recall that for every two unitary and irreducible representations of $\mathrm{SU}(2)$, labeled as $D^j \in \mathrm{Hom}(\mathrm{SU}(2), \mathcal{BL}(\mathbb{C}^{2j+1}))$, it turns out that $D^{(j,j')} \doteq D^j \otimes D^{j'}$ identifies a unique representation of $\mathrm{SL}(2,\mathbb{C})$ on $\mathbb{C}^{2j+1} \otimes \mathbb{C}^{2j'+1}$. For Dirac fields (using CPT invariance) or spinors in general, the relevant representation is $D^{\left(\frac{1}{2},0\right)} \oplus D^{\left(0,\frac{1}{2}\right)}$ which acts on $\mathbb{C}^2 \oplus \mathbb{C}^2 \equiv \mathbb{C}^4$. This prompts the following definition.

Definition 56. *Let (M, g) be a four dimensional, smooth Lorentzian manifold, admitting a spin structure as per Definition 55. We call* **Dirac bundle** *the associated vector bundle*

$$DM \doteq SM \times_T \mathbb{C}^4,$$

where $T \doteq D^{(\frac{1}{2},0)} \oplus D^{(0,\frac{1}{2})}$. We call **spinor** *any $\psi \in \Gamma^\infty(DM)$. At the same time we define the dual Dirac bundle $D^*M \doteq [DM]^*$, cf. Exercise 6 and we call* **cospinor** *any $\phi \in \Gamma^\infty(D^*M)$.*

If you are interested in making a connection to the standard notion of spinors/Dirac fields used in theoretical physics, you should solve the following exercise:

Exercise 20. Let (M, g) be a four dimensional, smooth Lorentzian manifold with a trivial spin bundle. Prove that DM is a trivial vector bundle and that $\Gamma^\infty(DM) \simeq C^\infty(M; \mathbb{C}^4)$.

Remark 18. *Definition 56 is a first non trivial example how to construct a vector bundle whose sections can be interpreted as the kinematic configurations of a relativistic free field. A similar procedure works in all other scenarios and it reproduces all known and/or expected results. It might be tempting to proceed in the same direction also for constructing a Yang-Mills theory, but actually gauge fields are not best described with the language used in this subsection.*

3 Differential Forms on Manifolds

In Section 2 we have introduced a new geometric structure which makes precise the heuristic idea of assigning to each point of an underlying space (or spacetime) a vector space encoding information on internal degrees of freedom. Vector bundles include notable examples such as the tangent or the cotangent space of a smooth manifold. In addition the main lesson that we learned is that each natural operation on vector spaces has a counterpart at the level of vector bundles. This was the rational behind the Whitney sum, *cf.* Definition 27 or the tensor product of bundles as per Exercise 7.

Focusing instead on differential forms on $\mathbb{R}^n$, we have learned in Remark 1 and particularly in Equation (8) that we can realize differential k-forms as smooth maps from $\mathbb{R}^n$ to Ω^k_n. In the following we shall combine all structures that we have introduced to obtain a natural generalization of differential forms to manifolds.

First of all we need a slight generalization to arbitrary vector spaces of the structures outlined at the beginning of Section 1.

Definition 57. *Let V be a real vector space of dimension $\dim V = n \geq 1$ and let φ be any isomorphism between V and $\mathbb{R}^n$. We call* **alternating (or) exterior algebra of V***, $\mathcal{A}_{n,V}$ the unique algebra over the reals such that φ extends to an algebra isomorphism.*

Exercise 21. Prove that, under the hypotheses of Definition 57, $\mathcal{A}_{n,V}$ does not depend on the choice of φ, namely, the algebras obtained choosing two isomorphisms are isomorphic to each other.

As a consequence of Definition 57 $\mathcal{A}_{n,V}$ inherits all properties of the exterior algebra $\mathcal{A}_n$. In particular, in view of Proposition 3, it is positively graded and we can decompose it as

$$\mathcal{A}_{n,V} = \bigoplus_{k=0}^{n} \mathcal{A}^k_{n,V}.$$

We can now adapt this discussion to the specific scenario, we are interested in.

Definition 58. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$. for every $k \geq 1$, we call* **bundle of alternating k-tensors on M** *the vector bundle of rank $\binom{n}{k}$ $\Lambda^k(M) \doteq \bigsqcup_{p\in M} \Lambda^k_p$ where $\Lambda^k_p \equiv \mathcal{A}^k_{n,T^*_pM}$ is the vector space of positively graded elements of degree k of the alternating algebra associated to the vector space T^*_pM.*

In order to make sure that the reader is following carefully the discussion, we recommend a nice exercise which recapitulates more or less everything presented up to now.

Exercise 22. Prove that $\Lambda^k(M)$ is a real smooth vector bundle of rank $\binom{n}{k}$ using both Definition 20 and Proposition 26.

We can define the main object of investigation of this whole section:

Definition 59. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$. We call* **k-differential forms**, *$0 \leq k \leq n$, the elements of the vector space*

$$\Omega^k(M) \doteq \Gamma^\infty(\Lambda^k(M)),$$

where we set $\Lambda^0(M)$ as the trivial line bundle $M \times \mathbb{R}$.

In view of this definition, it is desirable to have a concrete description of a k-form in a local trivialization of $\Lambda^k(M)$. Let us start from $k = 1$ so that $\Lambda^1(M) = T^*M$. In this case, mutatis mutandis, we can use Proposition 24 and its proof to infer that, taken any local chart (U, φ), $U \subseteq M$ and an associated coordinate system $\{x_i\}_{i=1,\dots,n=\dim M}$, this induces both an associated trivialization of $T^*M \equiv T^*M[M, \pi, \mathbb{R}^n]$ and a coordinate system on $T^*M|_U$. In other words there exists a diffeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^n$ such that, for all $\omega \in \Omega^1(M)$,

$$\omega|_U = \sum_{i=1}^{n} \omega_i(x) dx_i,$$

where $\omega_i \in C^\infty(U)$ for all $i = 1, \dots, n$. Observe that, at each $p \in U$, $dx_i : T_pM \to T_{x_i(p)}\mathbb{R}$ is the differential of the local coordinates $x_i : M \to \mathbb{R}$, $i = 1, \dots, n$ and it is completely characterized by the relation

$$dx_i \left(\frac{\partial}{\partial x_j} \right) = \delta_{ij} \frac{d}{d\lambda},$$

where $\frac{d}{d\lambda}$ is the generator of $T_{x(p)}\mathbb{R}$ while $\left\{ \frac{\partial}{\partial x_j} \right\}_{j=1,\dots,n}$ is the natural basis of T_pM induced by the coordinate system.

Remark 19. *The geometric concept of differential (push-forward) here used is only apparently different from that of Section 1 and particularly of Definition 9, which is traded from calculus. As a matter of fact, if we restrict our attention to $M = \mathbb{R}^n$, given $f \in C^\infty(\mathbb{R}^n)$, the geometric differential, here indicated to avoid confusion with f_*, is a map*

$$f_* : T_q\mathbb{R}^n \to T_{f(q)}\mathbb{R}.$$

Given a coordinate system $\{x_i\}_{i=1,\ldots,n}$, we can write any $v \in T_q\mathbb{R}^n$ as $v = \sum\limits_{i=1}^{n} v_i \frac{\partial}{\partial x_i}$, where each $v_i \in \mathbb{R}$, $i = 1, \ldots, n$. Per construction

$$f_*(v) = \sum_{i=1}^{n} v_i \left.\frac{\partial f}{\partial x_i}\right|_q \frac{d}{d\lambda},$$

where $\frac{d}{d\lambda}$ is the generator of $T_{f(q)}\mathbb{R}$. One can now infer that, calling df the differential as per Proposition 9, it holds $f_*(v) = df(v)\frac{d}{d\lambda}$. In view of this simple connection, with a slight abuse of notation, henceforth we will equivalently speak of push-forward and of differential of a map $f : M \to N$ between two smooth manifolds, indicating it as df.

Having obtained a local description of 1-forms on a smooth manifold via Φ, we can exploit that at each $p \in U$, $\Phi : T_pU \to \mathbb{R}^n$ is a vector space isomorphism using it in the spirit of Definition 57 to induce on $\Lambda^\bullet_p \doteq \bigoplus_{k=0}^n \Lambda^k_p$ the structure of an algebra. This can be done inducing an isomorphism

$$\Phi^\bullet : \Lambda^\bullet_p \to \Omega^\bullet_n,$$

which is completely specified by the requirement that, for all $i, j = 1, \ldots, n$,

$$(\Phi^\bullet)^{-1}(dx_i \wedge dx_j) = \Phi^{-1}(dx_i) \wedge \Phi^{-1}(dx_j),$$

where, with a slight abuse of notation we indicate with $\wedge$ the algebra product both of $\Omega^\bullet$ and of $\Lambda^\bullet_p$. Notice that the restriction of $\Phi^\bullet$ to the elements of fixed degree k becomes a vector space isomorphism $\Phi^\bullet|_k : \Lambda^k_p \to \Omega^k_p$ which coincides with Φ if $k = 1$. In view of these considerations, we obtain a local description of a k-form for $k > 1$, namely for all $\omega_k \in \Omega^k(M)$,

$$\omega_k|_U = \sum_{i_1 < i_2 < \ldots < i_k} \omega_{i_1\ldots i_k} dx_{i_1} \wedge \cdots \wedge dx_{i_k}, \tag{18}$$

where each $\omega_{i_1\ldots i_k} \in C^\infty(U)$. As a consequence, we have proven the following

Proposition 60. *Let M be a smooth manifold. Then $\Omega^\bullet(M) \doteq \bigoplus_{k=0}^n \Omega^k(M)$ is an exterior algebra if endowed with the wedge product*

$$\wedge : \Omega^k(M) \times \Omega^{k'}(M) \to \Omega^{k+k'}(M), \quad (\omega, \omega') \mapsto \omega \wedge \omega',$$

such that for all local trivialization (U, Φ) of $\Lambda^k(M)$ and for all coordinate system $\{x_i\}_{i=1,\ldots,n}$ subordinated to Φ, it holds that, if

$$\omega|_U = \sum_{i_1<i_2<\ldots<i_k} \omega_{i_1\ldots i_k} dx_{i_1} \wedge \cdots \wedge dx_{i_k} \text{ and } \omega'|_U = \sum_{j_1<j_2<\ldots<j_{k'}} \omega_{j_1\ldots j_{k'}} dx_{j_1} \wedge \cdots \wedge dx_{j_{k'}},$$

$$(\omega \wedge \omega')|_U = \sum_{i_1<i_2<\ldots<i_k} \sum_{j_1<j_2<\ldots<j_{k'}} \omega_{i_1\ldots i_k}\omega_{j_1\ldots j_{k'}} dx_{i_1} \wedge \cdots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \cdots \wedge dx_{j_{k'}}.$$

The structure of k-differential forms is from a geometric viewpoint the natural generalization of the counterpart on $\mathbb{R}^n$, $n \geq 1$ discussed in Remark 1. At the same time the correspondence has its limit as one can infer noticing that $\Omega^k(M) \neq C^\infty(M) \otimes \Omega^k_n$ as one might naively expect at first glance. The reason lies in the global non trivial structure of the tangent and of the cotangent bundle. The following proposition makes this statement more precise.

Corollary 61. *Let M be a smooth manifold whose associated tangent bundle is trivial as per Definition 22. Then, for all $k \in \mathbb{N}$, it holds*

$$\Omega^k(M) \simeq C^\infty(M) \otimes \Omega^k_n,$$

where Ω^k_n is the algebra of homogeneous elements of degree k generated from Ω^1_n as in (7).

Proof. If TM is diffeomorphic to $M \times \mathbb{R}^n$, $n = \dim M$, per duality also T^*M is diffeomorphic to $M \times \mathbb{R}^n$. On account of Proposition 29, $\Omega^1(M) = \Gamma^\infty(T^*M)$ is isomorphic to $C^\infty(M; \mathbb{R}^n)$ which is, in turn isomorphic to $C^\infty(M) \otimes \Omega^1_n$ up to the identification $\Omega^1_n \equiv \mathbb{R}^n$. Consider now a global frame of T^*M, that is n-linearly independent sections $\omega_i \in \Gamma(T^*M)$ whose existence is guaranteed by Corollary 32. Consider now the collection $\{\omega_i \wedge \omega_j\}_{i<j}$. These form a global frame of $\Lambda^2(M)$. This can be seen thanks to Corollary 33, which guarantees that, for any point $p \in M$, there exists a neighbourhood $U \subset M$, $p \in U$, and a coordinate system $\{x_j\}_{j=1,\ldots,n}$ such that $\omega_i = f_i dx_i$ where $f_i \in C^\infty(U)$. As a consequence the 2-forms $\{f_i f_j dx_i \wedge dx_j\}_{i<j}$ are linearly independent since the coefficients never vanish and $\{dx_i \wedge dx_j\}$ is a basis of Ω^2_n. The reasoning can be now repeated slavishly for all integers k. $\square$

To conclude this section we wish to analyze the behaviour of differential forms with respect to maps between smooth manifolds. This is nothing but the renown pull-back. Yet, as a preliminary step, we need to motivate why in Definition 58 we are talking of tensors, even though the structure involved is that of an alternating algebra.

Definition 62. *Let V be any finite dimensional real vector space. We call* **alternating k-tensor** *any $\alpha \in (V^*)^{\otimes k}$, $k \in \mathbb{N}$ such that, for all $v_1, \ldots, v_k \in V$ and for every pair of indices i, j with $i \neq j$,*

$$\alpha(v_1, \ldots, v_i, \ldots v_j, \ldots, v_k) = \alpha(v_1, \ldots, v_j, \ldots v_i, \ldots, v_k).$$

The space of alternating k-tensors is indicated as $\Lambda^k(V^*)$. In addition we call

$$\mathrm{Alt} : (V^*)^{\otimes k} \to \Lambda^k(V^*), \tag{19}$$

the **alternation** *projection (or antisymmetrization map). This is such that, for any collection $v_1, \dots, v_k \in V$ and for any $\alpha \in (V^*)^{\otimes k}$,*

$$(\mathrm{Alt}(\alpha))\,(v_1, \dots, v_k) = \frac{1}{k!} \sum_{\sigma \in S_k} (\mathrm{sign}(\sigma))\, \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}),$$

where S_k is the permutation group of k-elements and $\mathrm{sign}(\sigma)$ is the sign of the permutation $\sigma \in S_k$.

The following is a rather direct consequence of this last definition:

Corollary 63. *Let M be a smooth manifold. For every $p \in M$ there exists an isomorphism of vector spaces between Λ^k_p and $\Lambda^k(T^*_p M)$ which is completely characterized by the requirement*

$$dx_{i_1} \wedge \cdots \wedge dx_{i_k} \mapsto \mathrm{Alt}\left(dx_{i_1} \otimes \cdots \otimes dx_{i_k}\right), \tag{20}$$

where Alt is the alternation projection as in Equation (19).

Proof. It suffices to observe that both Λ^k_p and $\Lambda^k(T^*_p M)$ are real vector spaces of dimension $\binom{n}{k}$ and the map defined via Equation (20) is manifestly injective. □

We can now give an alternative characterization of differential forms. This is a nice exercise which allows a reader to recollect all given definitions.

Exercise 23. Prove that, for every $k \geq 1$, $\bigsqcup\limits_{p \in M} \Lambda^l(T^*_p M))$ admits a natural structure of vector bundle of rank $\binom{n}{k}$ and that it is diffeomorphic to $\Lambda^k(M)$.

Remark 20. *A notable consequence of this last exercise is that a differential form can now be seen as an antisymmetric covariant k-tensor field. This perspective is probably less natural from an algebraic viewpoint, but it is certainly the common interpretation of differential forms in many applications, especially in physics.*

Henceforth we will use whenever necessary the interpretation of a differential k-form as a particular covariant k-tensor field without making explicit the diffeomorphism discussed in Exercise (23).

Proposition 64. *Let M and N be two smooth manifolds and let $F \in C^\infty(M; N)$. For all $\omega \in \Omega^k(N)$, $k = 0, \dots n = \dim N$, we call* **pullback** *of ω, the form $F^*\omega \in \Omega^k(M)$ such that, for each $p \in M$ and for every collection $v_1, \dots, v_k \in T_p M$*

$$F^*\omega(v_1, \dots, v_k) \doteq \omega(dF(v_1), \dots, dF(v_k)),$$

where $dF : T_p M \to T_{F(p)} N$ is the differential map. It holds that

1. *F^* is $\mathbb{R}$-linear,*

2. *F^* is an algebra homomorphism, that is, for all $\omega \in \Omega^k(N)$ and $\omega' \in \Omega^{k'}(N)$,*

$$F^*(\omega \wedge \omega') = F^*\omega \wedge F^*\omega'.$$

3. *Given any coordinate system $\{y_1, \dots y_n\}$ in $U \subseteq N$ and given any $\omega = \sum_{i_1<i_2<\dots<i_k} \omega_{i_1\dots i_k} dy_{i_1} \wedge \dots \wedge dy_{i_k}$, with each $\omega_{i_1\dots i_k} \in C^\infty(U)$ it holds that on $F^{-1}(U)$*

$$F^*\omega = \sum_{i_1<i_2<\dots<i_k} (\omega_{i_1\dots i_k} \circ F) d(y_{i_1} \circ F) \wedge \dots \wedge d(y_{i_k} \circ F).$$

Proof. The map F^* is linear per construction. The second property holds since, for every $p \in M$ and for every collection $v_1, \dots, v_{k+k'} \in T_pM$ it holds

$$F^*(\omega \wedge \omega')(v_1, \dots, v_{k+k'}) = (\omega \wedge \omega')(dF(v_1), \dots, dF(v_{k+k'})) =$$
$$\frac{1}{(k+k')!} \sum_{\sigma \in S_{k+k'}} \mathrm{sign}(\sigma) \omega(dF(v_{\sigma_1}), \dots, dF(v_{\sigma(k)})) \omega'(dF(v_{\sigma(k+1)}), \dots, dF(v_{\sigma(k')})) =$$
$$(F^*\omega \wedge F^*\omega')(v_1, \dots, v_{k+k'}).$$

The third property is instead an immediate consequence of the second once we prove it for 1-forms. To this end it suffices to consider a basis element of the form $\omega_i = \alpha dy_i$, with $\alpha \in C^\infty(U)$. The definition of pullback entails that, for every $v \in T_qU$, where $p = F(q)$ and calling $\{x_i\}_{i=1,\dots m=\dim M}$,

$$dF(v)|_{F(q)} = \sum_{i=1}^m \sum_{j=1}^n v_i \frac{\partial y_j \circ F}{\partial x_i} \frac{\partial}{\partial y_j}$$

and, consequently,

$$F^*\omega_i(v) = \omega_i(dF(v)) = \omega_i|_{F(p)}(dF(v)) = (\omega_i \circ F) d(y_i \circ F).$$

$\square$

An application of this proposition, which is useful in several practical scenarios, is discussed in the following exercise

Exercise 24. Let M and N be two smooth manifolds of the same dimension n and let $F \in C^\infty(M; N)$. Let $\{x_i\}_{i=1,\dots,n}$ be a coordinate system in $U \subseteq M$ and let $\{y_j\}_{j=1,\dots n}$ be a counterpart in $V \subseteq N$. Show that, if $U \cap F^{-1}(V) \neq \emptyset$, then, for every $u \in C^\infty(V)$ it holds

$$F^*(u\, dy_1 \wedge \dots \wedge dy_n) = (u \circ F) \det(DF) dx_1 \wedge \dots \wedge dx_n,$$

where $\det(DF)$ is the Jacobian determinant of F.

Another important structural property of the pull-back is the following:

Exercise 25. Let M, N be two smooth manifolds and let $F \in C^\infty(M; N)$ be a submersion, namely dF is everywhere surjective. Prove that, for all $k \in \mathbb{N} \cup \{0\}$, the pull-back $F^* : \Omega^k(N) \to \Omega^k(M)$ is injective.

3.1 The exterior differentiation on a manifold

As already outlined in Section 1, one of the advantages of giving a geometric realization of differential forms on $\mathbb{R}^n$, $n \geq 1$ and of their algebra structure via the wedge product is the existence of a distinguished operator which maps k- into $(k+1)$-forms, *cf.* Definition 10. A natural question is whether there exists a counterpart on the space of differential k-forms on an arbitrary manifold M. Intuitively the answer is affirmative and one could expect that, in every coordinate patch of M, which corresponds to a local trivialization of $\Lambda^k(M)$, it coincides with the one introduced in Definition 10.

In order to make precise this idea, the best course of action consists of generalizing to an arbitrary smooth manifold the universal characterization given in Theorem 12. We shall proceed in three separate steps: uniqueness, naturality and existence of the exterior differentiation.

Remark 21. *At this stage we have used for the first time the word* naturality. *This is a concept often invoked in algebra and geometry, which has a kind of heuristic and philosophic flavor, although it has a precise formulation in the theory of categories. To given an idea of the meaning of naturality, consider two sets A, B and a distinguished map $f : A \to B$. Let A and B be endowed with a map $d_A : A \to A$ and $d_B : B \to B$. These maps are called natural with respect to f if $f \circ d_A = d_B \circ f$.*

In many instances naturality plays an important role in understanding the formulation of models in physics.

We start from uniqueness:

Theorem 65. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$. If existent, it is unique the operator $\underline{d} = (d_0, ..., d_n) : \Omega^\bullet(M) \to \Omega^\bullet(M)$ acting component by component on any $\underline{\omega} \in \Omega^\bullet(M)$ in such a way that the following four properties are satisfied:*

1. *$\underline{d}$ is $\mathbb{R}$-linear,*
2. *$\underline{d}^2 = 0$,*
3. *$\underline{d}$ is an* **antiderivation**, *namely, for all $\omega \in \Omega^k(M)$ and for all $\omega' \in \Omega^{k'}(M)$*

$$\underline{d}(\omega \wedge \omega') = \underline{d}\omega \wedge \omega' + (-1)^{\deg \omega} \omega \wedge \underline{d}\omega',$$

where $\deg : \Omega^k(M) \to \mathbb{Z}$ is the natural generalization of Equation 1 to k-forms on M.

4. *for every $\underline{\omega} = (f, 0, ..., 0)$ where $f \in C^\infty(M) \equiv \Omega^0(M)$, it holds*

$$\underline{d\omega} = (0, d_0 f, 0, ..., 0),$$

where $d_0 f = df$ coincides with the differential of f, that is

$$df : \Gamma^\infty(TM) \to C^\infty(M), \quad X \mapsto df(X) \doteq X(f).$$

Proof. Suppose that the operator $\underline{d}$ exists. We can prove its uniqueness at the level of each component. We start by showing that, for any $k \in \mathbb{N} \cup \{0\}$, d_k acts locally. To this end, consider $\omega_1, \omega_2 \in \Omega^k(M)$ and suppose that there exists an open neighbourhood $U \subseteq M$ such that $\omega_1|_U - \omega_2|_U = 0$. Let us fix any $p \in U$ and, up to a shrinking of the open subset, we can assume that U admits a local chart. Let $\eta = \omega_1 - \omega_2$ and let $\psi \in C^\infty(M)$ be a bump function centered at p, *cf.* [3, Prop. 2.25]. Hence $\psi\eta = 0$ and, condition *1.* and *3.*, entail

$$0 = d_k(\psi\eta) = d_0\psi \wedge \eta + \psi d_k \eta.$$

Since the bump function can be chosen to be constantly equal to 1 in a neighbourhood of p, the above identity entails that $d_k\eta|_p = 0$. In other words $d_k\omega_1|_p = d_k\omega_2|_p$. This entails that the operator d_k is completely specified by its action on a cover of M via local charts. Let (U, φ) be any of such charts and let $\{x_i\}_{i=1,\dots,n}$ be a coordinate system in U. Using Equation (18), we know that

$$\omega|_U = \sum_{i_1 < i_2 < \dots < i_k} \omega_{i_1 \dots i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k},$$

where each $\omega_{i_1\dots i_k} \in C^\infty(U)$. By using once more a bump function, we can use this expansion to construct $\widetilde{\omega} \in \Omega^k(M)$ by replacing $\omega_{i_1\dots i_k}$ and $\{x_i\}_{i=1,\dots,n}$ with $\widetilde{\omega}_{i_1\dots i_k} \in C^\infty(M)$ and $\widetilde{x}_i \in C^\infty(M)$, for each $i = 1, \dots, n$. These are chosen to coincide respectively with $\omega_{i_1\dots i_k}$ and $\{x_i\}_{i=1,\dots,n}$ on U. As a consequence the action of d_k on $\omega|_U$ coincides with that on $\widetilde{\omega}$. Yet this is completely specified by using condition *1.-4.* since

$$d_k\widetilde{\omega} = \sum_{i_1 < i_2 < \dots < i_k} d_k(\widetilde{\omega}_{i_1\dots i_k} d\widetilde{x}_{i_1} \wedge \dots \wedge d\widetilde{x}_{i_k}) = \sum_{i_1 < i_2 < \dots < i_k} (d\widetilde{\omega}_{i_1\dots i_k}) \wedge d\widetilde{x}_{i_1} \wedge \dots \wedge d\widetilde{x}_{i_k}.$$

This entails that, on each coordinate neighbourhood of M, d_k is uniquely specified. $\square$

We can now establish the second part of the main result of this section, namely naturality on $\mathbb{R}^n$.

Theorem 66. *Let $F \in C^\infty(\mathbb{R}^m; \mathbb{R}^n)$ and let $\underline{d}$ and $\underline{\widetilde{d}}$ be the exterior differentiation respectively on $\Omega^\bullet(\mathbb{R}^m)$ and $\Omega^\bullet(\mathbb{R}^n)$. Then, for every $k \in \mathbb{N} \cup \{0\}$, the following diagram is commutative:*

$$\begin{tikzcd}
\Omega^k(\mathbb{R}^n) \arrow[r, "F^*"] \arrow[d, "\widetilde{d}_k"'] & \Omega^k(\mathbb{R}^m) \arrow[d, "d_k"] \\
\Omega^{k+1}(\mathbb{R}^n) \arrow[r, "F^*"'] & \Omega^{k+1}(\mathbb{R}^m)
\end{tikzcd},$$

where F^* is the pullback as in Proposition 64.

Proof. In view of Remark 1, since $\Omega^k(\mathbb{R}^n)$ is a vector space it is sufficient to prove the statement of a k-form of $\alpha dx_1 \wedge \dots \wedge dx_k$ where $\alpha \in C^\infty(\mathbb{R}^n)$, while $\{x_i\}_{i=1,\dots,n}$ is an arbitrary coordinate system of $\mathbb{R}^n$. Using item *3.* of Proposition 64, it holds

$$\begin{aligned} d_k\left(F^*(\alpha dx_1 \wedge \dots \wedge dx_k)\right) &= d_k\left((\alpha \circ F) d(x_1 \circ F) \wedge \dots d(x_k \circ F)\right) = \\ &= d(\alpha \circ F) \wedge d(x_1 \circ F) \wedge \dots d(x_k \circ F) \end{aligned}$$

At the same time, using that the pullback is an algebra homomorphism

$$F^*\left(\widetilde{d}_k(\alpha dx_1\wedge\cdots\wedge dx_k)\right)=F^*\left(\widetilde{d}\alpha\wedge dx_1\wedge\cdots\wedge dx_k)\right)=$$
$$=F^*(\widetilde{d}\alpha)\wedge d(x_1\circ F)\wedge\ldots d(x_k\circ F),$$

where we used implicitly that differentiation over functions satisfies the chain rule on account of Definition 9. To conclude that the two expressions are the same, it suffices to observe that, on the one hand, per definition of pullback $F^*(d\alpha)=\widetilde{d}\alpha\circ dF$, while, on the other hand that, the chain rule entails $d(\alpha\circ F)=\widetilde{d}\alpha\circ dF$. □

Theorem 67. *Let M be a smooth manifold of dimension $\dim M=n\geq 1$. There exists an operator $\underline{d}=(d_0,....,d_n):\Omega^\bullet(M)\to\Omega^\bullet(M)$, called* **exterior differentiation on M** *acting component by component on any $\underline{\omega}\in\Omega^\bullet(M)$ and satisfying the same defining conditions as in Theorem 65.*

Proof. Suppose $\omega\in\Omega^k(M)$. In view of the proof of Theorem 65, we know that each operator d_k acts only locally. Hence, consider a smooth chart (U,φ) where $\varphi:U\to V\subset\mathbb{R}^n$. We define

$$d_k\omega|_U\doteq\varphi^*d_k|_{\mathbb{R}^n}((\varphi^{-1})^*\omega),\tag{21}$$

where $d_k|_{\mathbb{R}^n}$ is the exterior differentiation on $\mathbb{R}^n$, *cf.* Definition 10, here indicated with a subscript to avoid possible confusion. The maps φ^* and $(\varphi^{-1})^*$ are pullbacks as per Proposition 64. On account of Theorem 12 d_k inherits the condition *1.-4.* from the counterpart on $\mathbb{R}^n$. Therefore we need to show that the given definition behaves well on overlapping coordinate patches. Hence, let (V,ψ) be another chart such that $U\cap V\neq\emptyset$. On account of Theorem 66 it hold that, being $\varphi\circ\psi^{-1}$ a diffeomorphism on $\psi(U\cap V)\subseteq\mathbb{R}^n$,

$$(\varphi\circ\psi^{-1})^*d_k|_{\mathbb{R}^n}((\varphi^{-1})^*\omega)=d_k|_{\mathbb{R}^n}\left((\varphi\circ\psi^{-1})^*\circ(\varphi^{-1})^*\omega\right)=d_k|_{\mathbb{R}^n}(\psi^*\omega),$$

where we used both that $(\varphi\circ\psi^{-1})^*=(\psi^{-1})^*\circ\varphi^*$ and that $\varphi\circ\varphi^{-1}=id|_U$ entails $(\varphi^*)^{-1}=(\varphi^{-1})^*$. Since, in addition,

$$(\varphi\circ\psi^{-1})^*d_k|_{\mathbb{R}^n}((\varphi^{-1})^*\omega)=(\psi^*)^{-1}\circ\varphi^*d_k|_{\mathbb{R}^n}((\varphi^{-1})^*\omega),$$

we have proven that

$$\varphi^*d_k|_{\mathbb{R}^n}((\varphi^{-1})^*\omega)=\psi^*d_k|_{\mathbb{R}^n}((\psi^{-1})^*\omega),$$

which implies that the given expression of $d_k\omega$ is well-behaved on overlapping patches. □

To conclude the section, we extend the statement on the naturality of exterior differentiation to arbitrary smooth manifolds:

Theorem 68. *Let M and N be two smooth manifolds and let $F \in C^\infty(M;N)$ and let $\underline{d}^M$ and $\underline{d}^N$ be an exterior differentiation respectively on $\Omega^\bullet(M)$ and $\Omega^\bullet(N)$. Then, for every $k \in \mathbb{N}\cup\{0\}$, the following diagram is commutative:*

$$\begin{tikzcd}
\Omega^k(N) \arrow[r, "F^*"] \arrow[d, "d_k^N"'] & \Omega^k(M) \arrow[d, "d_k^M"] \\
\Omega^{k+1}(N) \arrow[r, "F^*"'] & \Omega^{k+1}(M)
\end{tikzcd},$$

where F^* is the pullback as in Proposition 64.

Proof. Since the exterior differentiation acts locally, it suffices to verify the commutativity of the diagram in an arbitrary chart (U,φ), $U \subseteq M$ and (V,ψ), $V \subseteq N$. Using Equation (21) on $U \cap F^{-1}(V)$ we obtain for an arbitrary $\omega \in \Omega^k(N)$,

$$\begin{aligned}
F^*(d_k^N\omega)|_{U\cap F^{-1}(V)} &= F^*\psi^* d_k^N((\psi^{-1})^*\omega) = \varphi^*\circ(\varphi^{-1})^*\circ F^*\psi^* d_k^N((\psi^{-1})^*\omega) = \\
&= \varphi^*\circ(\psi\circ F\circ\varphi^{-1})^* d_K^N((\psi^{-1})^*\omega) =_{N\to M} \varphi^* d_k^M\left((\psi\circ F\circ\varphi^{-1})^*(\psi^{-1})^*\omega\right) = \\
&= \varphi^* d_k^M((\varphi^{-1})^*F^*\omega) = d_k^M(F^*\omega)|_{U\cap F^{-1}(V)},
\end{aligned}$$

where for simplicity of notation we omitted in the intermediate steps the restriction $|_{U\cap V}$ and where $=_{N\to M}$ indicates that, in that equality, we used Theorem 66. $\square$

3.2 Orientation on a manifold

Finally we have met our goal of making precise the notion of differential k-forms on a smooth manifold. It is due time to use them to obtain some useful, cute, funny, good-looking, inspiring, awesome, tantalizing, results. The first one, that we discuss, might be quite surprising, but it has far reaching consequences. For example, whenever we integrate over $\mathbb{R}^n$, we always write the volume forms as $d^nx = dx_1 \dots dx_n$. What does it mean really? It might be surprising but the symbol is not only related to Lesbegue integration (obvious, come on!) but mainly to the notion of orientation. If you have no idea what does it mean, you can think of orientation as the mathematical procedure to make precise the idea behind the right hand rule that you learned in classical physics... bizarre isn't it?

We shall divide our analysis in two steps: first we discuss the notion of orientation on a generic vector space, having $\mathbb{R}^n$ in mind as guiding example and subsequently we discuss what happens on an arbitrary smooth manifold. Here we follow [3, Chap. 15].

Remark 22. *Let us consider the vector space $\mathbb{R}^n$. In this case we have a distinguished canonical basis*

$$e_i = (0,\dots,0,\underbrace{1}_{i^{th}\text{-entry}},0,\dots,0)$$

and if we set $n = 3$ it is easy to observe that (e_1, e_2, e_3) is ordered according to the right hand rule. Hence, if we take any other orthonormal basis $\{e'_i\}_{i=1,\ldots,n}$, we consider it oriented as the canonical one if the matrix $A \in O(n)$ such that $e'_i = Ae_i$ for all $i = 1, \ldots, n$ has positive determinant.

It is important to stress that the peculiarity of this case rests in the existence of a distinguished basis which identifies a natural standard orientation. In many other concrete instances, this is not the case. Therefore one cannot expect to be able to identify a unique or a distinguished orientation.

In view of this remark, the following definition appears as natural

Definition 69. *Let V be a real vector space of dimension $\dim V = n$. We call* **orientation** *of V an equivalence class $[e_1, \ldots e_n]$ where $(e_1, \ldots, e_n)$ is a basis of V and $(e_1, \ldots, e_n) \sim (e'_1, \ldots, e'_n)$ if the associated change of basis matrix has positive determinant*

In view of this definition we say that, given a real vector space V of dimension $\dim V = n$,

- a pair $(V, [e_1, \ldots, e_n])$ is an **oriented vector space**.
- any representative $(e'_1, \ldots, e'_n)$ of $[e_1, \ldots, e_n]$ is called a *positively oriented basis*. Any other basis is referred to generically as being negatively oriented.

It is rather fascinating that there is close connection between the choice of an orientation on a vector space V and its associated exterior algebra and alternating k-tensors, *cf.* Definition 62.

Proposition 70. *Let V be a real vector space of dimension $\dim V = n \geq 1$. Then, for every alternating n-tensor $\alpha \in \Lambda^n(V^*)$, $\alpha \neq 0$, there exists an associated orientation, indicated as $\mathcal{O}_\alpha$ and defined as follows:*

- *$\mathcal{O}_\alpha$ is the equivalence class of bases $[e_1, \ldots, e_n]$ such that, for every representative*

$$(e_1, \ldots, e_n) \mapsto \alpha(e_1, \ldots, e_n) > 0.$$

Proof. Let $\alpha \in \Lambda^n(V^*)$ and consider any basis $(e_1, \ldots, e_n)$ such that $\alpha(e_1, \ldots, e_n) \neq 0$. It must exist since $\alpha \neq 0$ and, without loss of generality, we can also assume that $\alpha(e_1, \ldots, e_n) > 0$. If not, per linearity of α it suffices to change the sign to one of the elements of the basis. Consider the orientation $[e_1, \ldots, e_n]$ and let $(e'_1, \ldots, e'_n)$ be another representative. Let $B \in M(n, \mathbb{R})$ the matrix connecting the two basis. Per Definition 69 $\det(B) > 0$. Let $\{e^i\}_{i=1,\ldots,n}$ be the dual basis of V^* and let $\mathcal{A}^n_n(V^*) = \mathrm{span}_{\mathbb{R}}(e^1 \cdot \ldots \cdot e^n)$ be the vector space built out the degree n elements of the exterior algebra $\mathcal{A}_n(V^*)$ generated by $\{e^1, \ldots, e^n\}$ where we highlighted the explicit dependence on V^*. Adapting to the case in hand both Definition 62 and Corollary 63, we know that there must exist $\kappa > 0$ such that

$$\alpha = \kappa \mathrm{Alt}(e^1 \otimes \cdots \otimes e^n),$$

as well as an isomorphism $I : \Lambda^n(V^*) \to \mathcal{A}_n^n(V^*)$ such that

$$I(\alpha) = \kappa e^1 \cdot \ldots \cdot e^n.$$

Consider now the second basis and its dual $\{(e')^i\}_{i=1,\ldots,n}$. Proposition 7 entails the following chain of identities

$$I(\alpha') = \kappa (e')^1 \cdot \ldots \cdot (e')^n = \det(B)^{-1}\kappa e^1 \cdot \ldots \cdot e^n = \det(B^{-1}) I(\alpha),$$

where $\alpha' = \kappa \mathrm{Alt}((e')^1 \otimes \cdots \otimes (e')^n)$. Since I is an isomorphism it descends $\alpha = \det(B)\alpha'$ and consequently

$$\alpha(e'_1, \ldots, e'_n) = \det(B)\alpha'(e'_1, \ldots, e'_n) = \kappa \det B > 0.$$

$\square$

Remark 23. *There are instances where it is convenient to consider also the degenerate case $n = 0$. In this scenario $\Lambda^0(V^*) = \mathbb{R}$ and the direct translation of Proposition 70 entails the $\mathcal{O}_\omega = [1] \equiv 1$ if $\Lambda^0(V^*) \ni \omega > 0$ while $\mathcal{O}_\omega = [-1] \equiv -1$ if $\Lambda^0(V^*) \ni \omega < 0$.*

Remark 24. *Observe that, since $\dim \Lambda^n(V^*) = 1$ for every finite dimensional real space V, the choice of an orientation for V is tantamount to choosing one of the two connected components of $\Lambda^n(V^*) \setminus \{0\}$. Hence, calling $\mathcal{O}(V)$ the collection of all possible orientations associated to V, there is a bijection $I : \mathcal{O}(V) \to \mathbb{Z}_2$.*

We can now ask ourselves how the notion of orientation can be generalized to an arbitrary smooth manifold. We can make a few preliminary observations:

- an orientation of a smooth manifold M, must correspond to the assignment of an orientation to T_pM for all $p \in M$, indicated as $\mathcal{O}_p$,
- we expect this assignment $p \mapsto \mathcal{O}_p$ to be continuous in p. Yet, to make sense of this concept we need to induce a topological structure on the set of possible orientations. To this end observe, that, as outlined in Remark 24, for every finite dimensional vector space V, $\mathcal{O}(V)$ is in bijection with $\mathbb{Z}_2$ which is a topological Lie group with respect to the discrete topology. Yet, exploiting this information is far from being trivial.

In order to make precise these ideas, we start by generalizing the notion of orientation:

Definition 71. *Let E be a real vector bundle of rank m over a smooth manifold M. We call it* **orientable** *if there exists a collection of local trivializations $\{\Phi_i = \pi^{-1}(U_i) \to U_i \times \mathbb{R}^n\}_{i=1,\ldots,k}$ such that $\bigcup_{i=1}^k U_i = M$ and, for each $p \in U_i$, the linear isomorphism $\Phi|_{E_p} : E_p \to \{p\} \times \mathbb{R}^n$ is an orientation-preserving map with respect to the standard orientation of $\mathbb{R}^n$. We call a manifold orientable if such is its tangent bundle.*

From this definition we can infer that a manifold is orientable whether there exists at least one possible trivialization of the associated tangent bundle compatible with the standard orientation of $\mathbb{R}^n$. Yet nothing forces us to keep such orientation and we could make another choice. Hence it is convenient to speak of an **oriented manifold** $(M, \mathcal{O})$, to underline whenever a specific orientation has been selected.

It is rather obvious that Definition 71 does not offer a concrete way of checking whether a manifold is orientable or not. A more practical criterion is necessary and the quandary can be solved by generalizing Proposition 70 to an arbitrary smooth manifold.

Proposition 72. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$. Any $\omega \in \Omega^n(M)$ such that $\omega(p) \neq 0$ for all $p \in M$ individuates a unique orientation of M. We call ω a* **smooth orientation form***.*

Proof. Let (U, Φ) be any local trivialization of TM, chosen, without loss of generality so that (U, φ) is a chart of M. Let $\{e_i\}_{i=1,\ldots,n}$ be the standard basis of $\mathbb{R}^n$ and let $\{(\Phi^{-1})_*(e_i)\}_{i=1,\ldots,n}$ be the induced basis of T_pU for each $p \in U$. Since $\dim \Lambda^n_p = 1$ for every $p \in U$, this entails that, calling $\{dx_1, \ldots, dx_n\}$ the induced basis on T^*_pU, $\omega|_U = f dx_1 \wedge \cdots \wedge dx_n$ where $f \in C^\infty(U)$. Furthermore since ω is nowhere vanishing f must be of definite sign, chosen without loss of generality to be positive. Hence, for all $p \in U$,

$$((\Phi^{-1})^*\omega)(\Phi(p))(e_1, \ldots e_n) = \omega(p)\left((\Phi^{-1})_*(e_1), \ldots, (\Phi^{-1})_*(e_n)\right) = f(p) > 0.$$

In addition, a direct computation shows that, for every other basis of $\mathbb{R}^n$, say $(e'_1, \ldots, e'_n)$, falling in the same orientation class of the standard basis, it holds that $((\Phi^{-1})^*\omega)(\Phi(p))(e'_1, \ldots e'_n) > 0$ for every $p \in U$. By applying Proposition 70 we can conclude the proof. □

Exercise 26. Let M be a smooth manifold of dimension $\dim M = n \geq 1$. For any orientation of M, there exists an associated smooth orientation form.

Proposition 72 is noteworthy for two reasons. The first is the connection between the existence of an orientation with that of a global top-form. The second is for its application to the case where the underlying smooth manifold is endowed with a Riemannian structure.

Hence, let $(M, \mathcal{O}, g)$ be an oriented Riemannian manifold of dimension $n = \dim M \geq 1$ and let $\{E_i\}_{i=1,\ldots n}$ be a (smooth) local frame over $U \subset M$. Without loss of generality we can require it to be a *local oriented orthonormal frame* that, for all $p \in U$,

$$[E_1|_p, \ldots, E_n|_p] \in \mathcal{O} \quad \text{and} \quad g_p(E_i|_p, E_j|_p) = \delta_{ij}\ \forall i, j = 1, \ldots, n.$$

The following notable global result holds true:

Theorem 73. *Let $(M, \mathcal{O}, g)$ be an oriented Riemannian manifold of dimension $n = \dim M \geq 1$. Then, there exists a unique smooth orientation form $\omega_g \in \Omega^n(M)$ such that, for all local oriented orthonormal frame $\{E_i\}_{i=1,\ldots,n}$, it holds*

$$\omega_g(E_1, \ldots, E_n) = 1.$$

We call ω_g the **Riemannian volume form***.*

Proof. Let us prove uniqueness. Consider a ω_g abiding to the properties and conditions in the statement of the theorem. Let $\{E_i\}_{i=1,\dots,n}$ be any local oriented orthonormal frame in $U \subseteq M$ and let $\{E^i\}_{i=1,\dots,n}$ be 1-form on U, forming the dual basis of T_p^*M for every $p \in U$. There must exist $f \in C^\infty(U)$ such that

$$\omega_g|_U = fE^1 \wedge \cdots \wedge E^n.$$

Yet, since $\omega_g|_U(E_1, \dots, E_n) = 1$, then $f = 1$. This determines ω_g uniquely in a cover of M and hence globally.

Let us prove existence. As above, let $\{E_i\}_{i=1,\dots,n}$ be any local oriented orthonormal frame in $U \subseteq M$ and let $\{E^i\}_{i=1,\dots,n}$ be 1-form on U, forming the dual basis of T_p^*M for every $p \in U$ and let

$$\omega_g|_U \doteq E^1 \wedge \cdots \wedge E^n.$$

We need to prove that this definition does not depend on the choice of the frame. Let $\{\widetilde{E}_i\}_{i=1,\dots,n}$ be a second local oriented orthonormal frame in $U \subseteq M$ and let $\{\widetilde{E}^i\}_{i=1,\dots,n}$ be 1-form on U, forming the dual basis of T_p^*M for every $p \in U$. Let $\widetilde{\omega}_g|_U \doteq \widetilde{E}^1 \wedge \cdots \wedge \widetilde{E}^n$. In addition there must exist $A \in C^\infty(U; SO(n, \mathbb{R}))$ such that, for each $p \in U$, $\widetilde{E}_i|_p = \sum_{j=1}^{n} A_{ij}(p) E_j|_p$. The function A is $SO(n)$-valued since the two frames are orthonormal and consistently oriented. By Proposition 7 it holds that, at each $p \in U$

$$\widetilde{E}^1|_p \wedge \cdots \wedge \widetilde{E}^n|_p = \det(A(p)) E^1 \wedge \cdots \wedge E^n = E^1 \wedge \cdots \wedge E^n,$$

which entails that $\omega_g = \widetilde{\omega}_g$. Hence the definition of the top form does not depend on the frame chosen. □

A notable consequence of this theorem is its local version, which is probably well-known to the readers (if not, play dumb and nod... trust me... it is better... really... no kidding)

Corollary 74. *Let $(M, \mathcal{O}, g)$ be an oriented Riemannian manifold of dimension $n = \dim M \geq 1$. For any chart (U, φ) with an associated oriented coordinate system*[2] *$\{x_i\}_{i=1,\dots,n}$, it holds*

$$\omega_g|_U = \sqrt{\det(g)} dx_1 \wedge \cdots \wedge dx_n,$$

where $\det(g)$ stands for the determinant of the matrix whose entries are $g_{ij} = g\left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j}\right)$.

Proof. Let (U, φ) be any chart of M and let $\{x_i\}_{i=1,\dots,n}$ be any coordinate system. Since $\dim \Lambda_p^n = 1$ for any $p \in M$, there must exist $f \in C^\infty(U)$ such that

$$\omega_g|_U = f dx_1 \wedge \cdots \wedge dx_n.$$

[2] We call a coordinate system $\{x_i\}_{i=1,\dots,n}$ oriented if the associated basis of the tangent space $\left\{\frac{\partial}{\partial x_i}\right\}_{i=1,\dots,n}$ is oriented according to $\mathcal{O}$.

To compute f, we can choose from the beginning U so that there exists an oriented orthonormal frame $\{E_i\}_{i=1,\dots,n}$. We call $\{E^i\}_{i=1,\dots,n}$ the dual basis so that, according to Theorem 73, $\omega_g = E^1 \wedge \cdots \wedge E^n$. In addition there exists $A \in C^\infty(U; \mathrm{GL}(n,\mathbb{R}))$ such that $\left.\frac{\partial}{\partial x_i}\right|_p = \sum_{j=1}^{n} A_{ij}(p) E_j$. This entails that, using Proposition 7, at each $p \in U$,

$$\omega_g\left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}\right) = f = \det(A(p)).$$

At the same time, observe that

$$g_{ij} = g\left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j}\right) = g\left(\sum_{k=1}^{n} A_{ik} E_k, \sum_{l=1}^{n} A_{jl} E_l\right) = \sum_{k=1}^{n} A_{ik} A_{jk}.$$

At a matrix level, this translates to $g = AA^T$ which entails $\det g = \det(A)^2$. □

Remark 25. *Theorem 73 and Corollary 74 have several practical consequences. Yet here, we wish to highlight that, in many papers, there is a tendency, when integrating functions on a Riemannian manifold (see next section) to write the volume form either as $\sqrt{g}\, d^n x$ or as μ_g or even as $d\mu_g$. Besides the commonly accepted notation which avoids writing explicitly the wedge product, one should keep in mind that the first of the three writings is a real abuse of notation since it is valid only locally. Hence using μ_g or $d\mu_g$ is a better option, even though $\sqrt{g}\, d^n x$ is accepted almost universally even in the mathematics literature.*

Disclaimer: Henceforth we will adopt the same notation as in the literature, omitting to write explicitly the symbol $\mathcal{O}$ to indicate that a manifold is oriented. We will highlight it in words.

3.3 Integration on manifolds

All good things come to an end... but also the bad ones. This means that, regardless of what you think about the previous sections, everything is in place to solve one of the main open questions in differential geometry: how to integrate a function (or more generally a k-form) on a smooth manifold. We can give a complete answer, but a disclaimer is necessary. In the preceding discussions all manifold could be with empty or non empty boundary, since this played no role. It should not come as a surprise that, when discussing integration, this is instead a somehow important distinction. In the following we will only discuss the case where the underlying manifold M has empty boundary $\partial M = \emptyset$. The reason to lose in generality is to highlight the important structural aspects of integration on a manifold. The reader should keep in mind that, including a (smooth) boundary does not limit the range of applicability of the theory of integration... it only makes it more complicated since boundary terms do occur. A reader interested in understanding what happens in presence of boundaries, or even corners, can consult [11, Chap. 16].

In the following we shall divide our analysis in three key steps: integration on $\mathbb{R}^n$ with forms, integration on smooth manifolds and integration on Riemannian manifolds.

Consider $\mathbb{R}^n$, $n \geq 1$ and recall that a domain of integration is any bounded subset $D \subset \mathbb{R}^n$ such that $\mu(\partial D) = 0$ where μ is the Lebesgue measure. Consider $\Lambda^k(\mathbb{R}^n)$, *cf.* Definition 58 and, to be general, following Remark 1, we consider only continuous sections, that is

$$\Gamma(\Lambda^n(\mathbb{R}^n)) \equiv C^0(\mathbb{R}^n) \otimes \Omega_n,$$

Regardless of this extension, since $\Lambda^n(\mathbb{R}^n)$ is a rank 1 vector bundle, considering the standard Cartesian coordinates $\{x_i\}_{i=1,\dots,n}$, for any $\omega \in \Gamma(\Lambda^n(\mathbb{R}^n))$, there must exist $f_\omega \in C^0(\mathbb{R}^n)$ such that

$$\omega = f_\omega dx_1 \wedge \cdots \wedge dx_n.$$

Following the standard theory of Lebesgue integration, we can give a natural definition of what does it mean integrating ω.

Definition 75. *Let $D \subset \mathbb{R}^n$ be a domain of integration and let $\omega \in C^0(\overline{D}) \otimes \Omega_n$. Considering the standard Cartesian coordinates and writing $\omega = f_\omega dx_1 \wedge \cdots \wedge dx_n$, with $f_\omega \in C^0(\overline{D})$, we call* **integral of ω over D***,*

$$\int_D \omega = \int_D f_\omega \, dx_1 \wedge \cdots \wedge dx_n \doteq \int_D f_\omega \, d^n x.$$

Remark 26. *It is immediate to notice that the hypothesis of D being a domain od integration and of considering forms lying in $C^0(\overline{D}) \otimes \Omega_n$ could be easily relaxed allowing for a generic open subset $U \subseteq \mathbb{R}^n$ while considering as integrand $\omega \in \Gamma_0(\Lambda^n(U))$. Here the subscript 0 indicates that we admit only n-forms of compact support. In this case we set*

$$\int_U \omega = \int_D \omega$$

where D is any domain of integration such that $\operatorname{supp}(\omega) \subset D \subset U$. It is immediate to observe that the definition does not depend on the choice of D. One might also wonder whether D exists since $\operatorname{supp}(\omega)$ is in principle an arbitrary compact subset of $\mathbb{R}^n$. This is guaranteed by compactness which ensures that there exists a finite cover of $\operatorname{supp}(\omega)$ in open balls of $\mathbb{R}^n$ with respect to the standard Euclidean metric. Each of these individuates a domain of integration and, by standard measure theory, a finite union of these sets shares the same property.

Disclaimer: Henceforth, we (in the sense of I since you have no right or reason or actually... way to choose) shall consider only the case of integrals of compactly supported n-forms on open subsets $U \subseteq \mathbb{R}^n$. In addition, for simplicity we shall only work with smooth forms, although continuity suffices. We shall use therefore

$$\Omega^k_0(\mathrm{U}) \doteq \Gamma^\infty_0(\Lambda^k(\mathrm{U})). \quad \forall k \in \mathbb{N} \cup \{0\} \text{ and } \forall\, \mathrm{U} \subseteq \mathbb{R}^n$$

In the following proposition we show that the definition of integral of a top form is natural with respect to pull-back.

Proposition 76. *Let $U, V \subseteq \mathbb{R}^n$ be two open subsets such that there exists an orientation preserving diffeomorphism $G : U \to V$. If $\omega \in \Gamma_0(\Lambda^n(U))$, then*

$$\int\limits_V \omega = \int\limits_U G^*\omega,$$

where G^* stands for the pull-back, cf. Proposition 64.

Proof. Consider any domain of integration E such that $\mathrm{supp}(\omega) \subset E \subset \overline{E} \subset V$. Since G is a diffeomorphism it holds that $D \doteq G^{-1}(E)$ is also a domain of integration such that $D \subset \overline{D} \subset U$. To avoid confusion, let us denote with $\{x_i\}_{i=1,\dots,n}$ the standard Euclidean coordinates on D and with $\{y_i \doteq G \circ x_i\}_{i=1,\dots,n}$ those on E. By using the formula for the change of coordinates within an integral, it holds

$$\int\limits_E \omega = \int\limits_E f(y)\, d^n y = \int\limits_D (f \circ G)(x) |\det(DG)| d^n x =$$
$$= \int\limits_D (f \circ G)(x) \det(DG) dx_1 \wedge \cdots \wedge dx_n = \int\limits_D G^*\omega,$$

where $\det(DG)$ is the Jacobian determinant of G and where we used in the third equality that G is orientation preserving, while, in the last one, Exercise 24. □

It is easy to check that, whenever G changes the orientation, then an overall minus sign has to be accounted form, namely

$$\int\limits_V \omega = -\int\limits_U G^*\omega.$$

Step 2: Integration of forms on manifolds

Using the results of the previous step we can start from Definition 75 constructing a well-defined integration of a top-form on a manifold. As one might expect the idea is to use a partition of unity argument so to reduce any integral of a top form on a generic smooth manifold to a sum of integrals locally defined on open subsets of $\mathbb{R}^n$. We proceed step by step showing that this idea can be made precise and rigorous.

Proposition 77. *Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$ and let $\omega \in \Omega^n_0(M) \doteq \Gamma^\infty_0(\Lambda^n(M))$ be such that there exists a single smooth chart (U, φ) obeying $\mathrm{supp}(\omega) \subset U$. We call* **integral of ω over M**

$$\int\limits_M \omega \doteq \pm \int\limits_{\varphi(U)} (\varphi^{-1})^*\omega,$$

where $(\varphi^{-1})^*$ is the pullback as per Proposition 64 while the sign $\pm$ depends whether $\varphi(U)$ is a positively oriented chart or not. Then the definition does not the depend on the choice of the local chart provided that it contains the support of ω.

Proof. Suppose to consider two charts (U,φ) and (V,ψ) both containing $\mathrm{supp}(\omega)$. Without loss of generality we can prove the statement only for positively oriented charts, the other cases following suit. Hence $\psi\circ\varphi^{-1}$ is a positively oriented diffeomorphism from $\varphi(U\cap V)$ to $\psi(U\cap V)$. We can apply Proposition 76 obtaining

$$\int\limits_{\varphi(U)} (\varphi^{-1})^*\omega = \int\limits_{\varphi(U\cap V)} (\varphi^{-1})^*\omega = \int\limits_{\psi(U\cap V)} (\varphi\circ\psi^{-1})^*(\varphi^{-1})^*\omega =$$
$$\int\limits_{\psi(U\cap V)} (\psi^{-1})^*\circ\varphi^*\circ(\varphi^{-1})^*\omega = \int\limits_{\psi(U\cap V)} (\psi^{-1})^*\omega = \int\limits_{\psi(V)} (\psi^{-1})^*\omega,$$

where we used that $(\varphi^{-1})^* = (\varphi^*)^{-1}$. □

In order to extend this argument to the whole oriented manifold M, consider $\{(U_i,\varphi_i)\}_{i\in\mathbb{N}}$, a cover of M in open charts. Without loss of generality we can only analyze the case where all φ, $i\in\mathbb{N}$ are positively oriented and we observe that the support of every $\omega\in\Omega^n_0(M)$ intersects only a finite collection of elements of the cover. These preliminary observations lead to the following result:

Theorem 78. *Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$ and let $\{(U_i,\varphi_i)\}_{i\in\mathbb{N}}$, be a cover in open, orientation preserving, charts. Let $\{\psi_i\}_{i\in\mathbb{N}}$ be a partition of unity[3] subordinated to the cover. Then, for any $\omega\in\Omega^n_0(M)$, we define the* **integral of ω on M** *as*

$$\int\limits_M \omega \doteq \sum_{i=1}^{\infty}\int\limits_M \psi_i\omega = \sum_{i=1}^{\infty}\int\limits_{\varphi_i(U_i)} (\varphi_i^{-1})^*\psi_i\omega,$$

where (U_i,φ_i) is the local chart in which $\psi_i\omega$ has support. The definition of the integral does depend neither on the choice of open cover nor on that of partition of unity.

Proof. Consider $\{(V_i,\varphi'_i)\}_{i\in\mathbb{N}}$ a second cover of M in open, orientation preserving, charts and let $\{\phi_i\}_{i\in\mathbb{N}}$ be a partition of unity subordinated to it. Then, for any $\omega\in\Omega^n_0(M)$,

$$\int\limits_M \omega = \sum_{i=1}^{\infty}\int\limits_M \psi_i\omega = \sum_{i=1}^{\infty}\int\limits_M\sum_{j=1}^{\infty}\phi_j\psi_i\omega = \sum_{j=1}^{\infty}\sum_{i=1}^{\infty}\int\limits_M \phi_j\psi_i\omega = \sum_{j=1}^{\infty}\int\limits_M\sum_{i=1}^{\infty}\phi_j\psi_i\omega = \sum_{j=1}^{\infty}\int\limits_M \phi_j\omega,$$

where each exchange between sums and integral is admissible since only a finite number of non-vanishing terms exists due to the compactness hypothesis. □

[3]Existence of a partition of unity subordinated to a given cover is a standard result in differential geometry, *cf.* [11, Prop. 2.23].

The integral of forms over a manifold shares many properties of the standard integral of functions. The following exercise highlights them.

Exercise 27. Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$ and let $\omega, \omega' \in \Omega^n_0(M)$. The following properties hold true:

1. Linearity: for all $a, b \in \mathbb{R}$
$$\int_M (a\omega + b\omega') = a \int_M \omega + b \int_M \omega',$$
2. Orientation reversal: Denoting with $-M$ the same manifold with opposite orientation,
$$\int_{-M} \omega = - \int_M \omega,$$
3. Positivity: if ω is positively oriented
$$\int_M \omega > 0$$
4. Diffeomorphism Invariance (or Naturality): If N is a smooth, oriented manifold and $F : N \to M$ is an orientation preserving diffeomorphism, then
$$\int_M \omega = \int_N F^*\omega.$$

Assuming that the dedicated reader has been eager to extend all results obtained up to know to smooth manifold with non empty boundary, we can prove the main result of the theory of integration on manifolds... the true, the one, the only inimitable, omnipresent Stokes theorem:

Theorem 79 (**Stokes Theorem**). *Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$, with boundary ∂M endowed with the induced orientation. Then, for every $\omega \in \Omega^{n-1}_0(M)$,*
$$\int_M d_{n-1}\omega = \int_{\partial M} \omega,$$
where the right hand side is a short cut notation for indicating $\iota^*_{\partial M}\omega$, where $\iota_{\partial M} : \partial M \to M$ is the embedding map of the boundary ∂M into M. In addition, whenever $\partial M = \emptyset$ the right-hand is per definition vanishing.

Proof. The proof is divided in three steps which we discuss separately.

Step 1: We consider the special case where the underlying manifold is the upper half space $M = \mathbb{H}^n \doteq \mathbb{R}^{n-1} \times [0, \infty)$. For any but fixed $\omega \in \Omega_0^{n-1}(\mathbb{H}^n)$ we can find $R > 0$ and an associated hyper-rectangle $K \doteq [-R, R]^{n-1} \times [0, R] \supset \mathrm{supp}(\omega)$. By using the standard Euclidean coordinates $\{x_i\}_{i=1,\dots,n}$ with the convention that the domain of x_n is $[0, R]$, there must exist $\omega_i \in C_0^\infty(\mathbb{H}^n)$, $i = 1, \dots, n$ such that

$$\omega = \sum_{i=1}^{n} \omega_i \, dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \dots dx_n,$$

where the hat means that dx_i is omitted. By using Definition 10 it holds that

$$d_{n-1}\omega = \sum_{i,j=1}^{n} \frac{\partial \omega_i}{\partial x_j} \, dx_j \wedge dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \dots dx_n = \sum_{i=1}^{n} (-1)^{i-1} \frac{\partial \omega_i}{\partial x_i} \, dx_1 \wedge \cdots \wedge dx_i \wedge \dots dx_n,$$

where we used implicitly Corollary 11. Standard Lebesgue theory of integration yields

$$\int\limits_{\mathbb{H}^n} d_{n-1}\omega = \sum_{i=1}^{n} (-1)^{i-1} \int\limits_{K} \frac{\partial \omega_i}{\partial x_i} \, dx_1 \wedge \cdots \wedge dx_n = \sum_{i=1}^{n} (-1)^{i-1} \int\limits_{0}^{R} dx_n \int\limits_{-R}^{R} dx_{n-1} \cdots \int\limits_{-R}^{R} dx_1 \, \frac{\partial \omega_i}{\partial x_i} =$$

$$\sum_{i=1}^{n-1} (-1)^{i-1} \int\limits_{0}^{R} dx_n \int\limits_{-R}^{R} dx_{n-1} \dots \widehat{\int\limits_{-R}^{R} dx_i} \cdots \int\limits_{-R}^{R} dx_1 \, \omega_i|_{-R}^{R} + \int\limits_{-R}^{R} dx_{n-1} \cdots \int\limits_{-R}^{R} dx_1 \, \omega_n(x)|_0^R =$$

$$(-1)^n \int\limits_{-R}^{R} dx_{n-1} \cdots \int\limits_{-R}^{R} dx_1 \, \omega_n(x_1, \dots, x_{n-1}, 0),$$

where the hat symbol entails that the integral below is omitted and where ω_i vanishes at $x_i = R$ for all $i = 1, \dots, n$ since ω is compactly supported. Observing that the boundary $\partial\mathbb{H}^n \doteq \{(x_1, \dots, x_n) \mid x_n = 0\}$, it turns out that the pull-back of ω to the boundary reads

$$\iota_{\partial M}^* \omega = \omega_n(x_1, \dots, x_{n-1}, 0) dx_1 \wedge \dots \wedge dx_{n-1}.$$

By integrating this form along the boundary we obtain the statement of the theorem, recalling that, if n is even the coordinates $(x_1, \dots, x_n)$ are positively oriented along $\partial\mathbb{H}^n$ while, if n is odd they are not and thus a minus sign has to be accounted for, *cf.* [11, Example 15.26].

Step 2: We consider the special case where the underlying manifold is $M = \mathbb{R}^n$. In this case we can repeat the procedure of the preceding step, replacing the hyper-rectangle K with $K' = [-R, R]^n$, $R > 0$. Considering any $\omega \in \Omega_0^{n-1}(\mathbb{R}^n)$ it turns out that $\int\limits_{\mathbb{R}^n} d_{n-1}\omega = 0$. This proves the statement of the theorem in this case.

Step 3: We consider a generic smooth manifold M but we take an arbitrary $\omega \in \Omega_0^{n-1}(M)$ such that $\mathrm{supp}(\omega) \subset U$ where (U, φ) is a local chart, which we assume to be positively oriented. The other cases follow suit. Hence, by using Proposition 64 and Theorem 68, it descends

$$\int_M d_{n-1}\omega = \int_{\mathbb{H}^n} d_{n-1}(\varphi^{-1})^*\omega = \int_{\partial\mathbb{H}^n} (\varphi^{-1})^*\omega = \int_{\partial M} \omega,$$

where we used Step 1 in the last equality. Here we assumed that the chart intersects the boundary. If this were not the case, the same result could be obtained using Step 2 in place of Step 1.

Step 4: We consider both a generic smooth manifold M and an arbitrary $\omega \in \Omega_0^{n-1}(M)$. Consider now a cover of M via $\{(U_i, \varphi_i)\}_{i=1,\dots,n}$, local charts which are assumed to be positively oriented. The other case can be dealt with similarly. Consider $\{\psi_i\}_{i=1,\dots,n}$ a partition of unity subordinated to the cover. It holds

$$\int_{\partial M} \omega = \sum_{i=1}^{n} \int_{\partial M} \psi_i\omega = \sum_{i=1}^{\infty} \int_M d_{n-1}(\psi_i\omega) = \sum_{i=1}^{\infty} \int_M d_{n-1}\psi_i \wedge \omega + \psi_i\, d_{n-1}\omega =$$
$$\int_M d_{n-1}(\sum_{i=1}^{\infty} \psi_i) \wedge \omega + \left(\sum_{i=1}^{\infty} \psi_i\right) d_{n-1}\omega = \int_M d_{n-1}\omega,$$

where we used that, being ω compactly supported, only a finite number of the elements in all sums are non vanishing. In addition we used that the differential is a linear operator and that $\sum_{i=1}^{\infty} \psi_i = 1$. ☐

Corollary 80. *Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$ with empty boundary. Then, for every form $\omega \in \Omega_0^n(M)$ such that $\omega = d_{n-1}\eta$, $\eta \in \Omega_0^{n-1}(M)$, it holds*

$$\int_M \omega = 0.$$

Proof. This is a direct application of Stokes theorem. ☐

Corollary 81. *Let M be a smooth, oriented manifold of dimension $\dim M = n \geq 1$. Then, for every form $\omega \in \Omega_0^{n-1}(M)$ such that $d_{n-1}\omega = 0$, it holds*

$$\int_{\partial M} \omega = 0.$$

Proof. This is a direct application of Stokes theorem. ☐

Step 3: Integration of forms on a Riemannian manifold

At last, we specialize our analysis to the case when the underlying background is a Riemannian manifold (M, g) of dimension $\dim M = n \geq 1$. In this case Theorem 73 guarantees us the existence of a distinguished element in $\Omega^n(M)$, namely the Riemannian volume form ω_g. This information can be combined with the definition of the integral of a top form in Proposition 77 as follows:

Definition 82. *Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. For any $f \in C_0^\infty(M)$, we call* **integral of f over M**

$$\int_M f\omega_g,$$

where ω_g is the Riemannian volume form as in Theorem 73, while the integral is interpreted in the sense of Proposition 77 reading $f\omega_g \in \Omega_0^n(M)$.

Remark 27. *We should stress that in Definition 82 we require f to be smooth only to keep the same attitude as in the previous steps and sections. It is quite straightforward to observe that one can certainly extend the class of integrands to $C_0^0(M)$.*

Disclaimer: In these notes we will adopt different equivalent notations for the integral over a Riemannian manifold, as it often occurs in the literature. As a matter of facts we shall write

$$\int_M f\omega \equiv \int_M f d\mu_g \equiv \int_M f\sqrt{g}d^n x,$$

where in the last equivalence we are referring implicitly to Corollary 74.

We prove the counterpart in this setting of a well-known property of Lebesgue integrals over $\mathbb{R}^n$.

Proposition 83. *Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. For any $f \in C_0^\infty(M)$ such that $f \geq 0$, it holds*

$$\int_M f d\mu_g \geq 0,$$

where equality holds if and only if $f = 0$. In addition if we set $f = 1$ and

$$\mathrm{Vol}(M) \equiv \mathrm{Vol}(M, g) \doteq \int_M d\mu_g < \infty,$$

we call $\mathrm{Vol}(M)$ the **volume of M***.*

Proof. Let $f \in C_0^\infty(M)$ be supported in a single chart (U, φ). Then, choosing local coordinates $\{x_i\}_{i=1,\dots,n}$, we can use Corollary 74 to write

$$\int_M f d\mu_g = \int_{\varphi(U)} f(x)\sqrt{g} dx_1 \dots dx_n \geq 0,$$

whenever $f \geq 0$. At the same time the integral can vanish if and only if $f = 0$. To prove the same statement for a generic compactly supported function, it suffices to use the standard argument which calls for considering a cover of M in local charts $\{(U_i, \varphi_i)\}_{i=1,\dots,n}$ and a partition of unity $\{\psi_i\}_{i=1,\dots,n}$ subordinated to it. Since each $\psi_i \in C_0^\infty(U_i)$ and $0 \leq \psi_i(x) \leq 1$, the proof is straightforward. $\square$

The following exercise shows once more how the integral of functions on a Riemannian manifold shares most of the properties of the Lebesgue counterpart on $\mathbb{R}^n$.

Exercise 28. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. Prove that, for any $f \in C_0^\infty(M)$, it holds

$$\left| \int_M f d\mu_g \right| \leq \int_M |f| \, d\mu_g.$$

The careful reader should now dive in his past, with no fear of living again some of his or her own good dreams or nightmares... yes... I mean it... think about the course of electrodynamics and relativity. In that context, it emerged the possibility of associating on $\mathbb{R}^n$ to a k-form a distinguished $(n-k)$-form. We discuss succinctly how this idea extends to a generic Riemannian manifold.

Definition 84. *Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$ and let us identify $\Omega^0(M) \equiv C^\infty(M)$. We call* **Hodge operator** *the map*

$$\star : \Omega^0(M) \to \Omega^n(M), \quad f \mapsto \star f \doteq f\omega_g,$$

where ω_g is the Riemannian volume form.

Using Theorem 36 and Definition 34, one can realize that the Hodge operator identifies a vector bundle isomorphism between $\Lambda^n(M)$ and the trivial line bundle $M \times \mathbb{R}$. In addition the action of $\star$ can be extended as one should show via these exercises:

Exercise 29. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. Prove that g induces a pairing between k-forms

$$\langle , \rangle_g : \Omega^k(M) \times \Omega^k(M) \to C^\infty(M),$$

which is pointwisely determined by $\langle , \rangle_{g,p} : \Lambda^k(T_p^*M) \times \Lambda^k(T_p^*M) \to \mathbb{R}$ such that for all $\{\omega_i\}_{i=1,\dots,k}$, $\{\eta_j\}_{j=1,\dots,k}$, with $\omega_i, \eta_j \in T_p^*M$,

$$\langle \omega_1 \wedge \dots \wedge \omega_k, \eta_1 \wedge \dots \wedge \eta_k \rangle = \det(A),$$

where A is the $k \times k$ real valued matrix whose entries are $A_{ij} = g^{-1}(\omega_i, \eta_j)$

Exercise 30. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. For any integer k such that $0 \leq k \leq n$, there exists a unique vector bundle isomorphism $\star : \Lambda^k(M) \to \Lambda^{n-k}(M)$ such that, for all $\eta, \eta' \in \Omega^k(M)$,

$$\eta \wedge \star\eta' = \langle \eta, \eta' \rangle_g \omega_g,$$

where $\langle , \rangle_g$ is the metric-induced pairing on $\Omega^k(M)$. Prove that $\star$ coincides with that of Definition 84 if $k = 0$.

Exercise 31. Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. Prove that the inverse of the Hodge operator, $\star^{-1} : \Omega^{n-k}(M) \to \Omega^k(M)$ obeys to the relation $\star^{-1} \doteq (-1)^{k(n-k)}\star$ where the right hand side should be interpreted as the Hodge operator acting on $(n-k)$-forms.

Remark 28. *Observe that, to be precise, one should always indicate with a subscript the space on which the Hodge operator acts, e.g. $\star_k : \Omega^k(M) \to \Omega^{n-k}(M)$. In addition, it is important to notice that, on Lorentzian manifolds, Exercise 31 is different due to the occurrence of an additional minus sign to account for the signature of the underlying metric.*

With these ingredients we can discuss the counterpart of the divergence theorem on an arbitrary Riemannian manifold. As first we generalize the notion of **interior multiplication (by a vector field)** as follows: for every $\chi \in \Gamma(TM)$, we set

$$\iota_\chi : \Omega^k(M) \to \Omega^{k-1}(M), \quad \omega \to \iota_\chi \omega \equiv \chi \lrcorner \omega, \tag{22}$$

which is the $(k-1)$-form such that, for all $\{v_i\}_{i=1,\ldots,k-1}$, $v_i \in \Gamma(TM)$,

$$(\iota_\chi)\omega(v_1, \ldots, v_{k-1}) \doteq \omega(\chi, v_1, \ldots, v_{k-1}).$$

We can prove a technical lemma.

Lemma 85. *Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$ and let*

$$\beta : \Gamma(TM) \to \Omega^{k-1}(M), \quad \chi \mapsto \beta(\chi) = \chi \lrcorner \omega_g, \tag{23}$$

where $\lrcorner$ is defined in Equation (22). Let $S \subseteq M$ be an immersed hypersurface ($\iota_S : S \to M$) with the orientation subordinated to the choice of a unit normal vector field N. Let h be the induced metric on S. For any vector field $\chi \in \Gamma(TM)$, calling $\chi|_S$ the restriction to S, it holds

$$\iota_S^*(\beta(X)) = \langle X|_S, N \rangle_h \omega_h,$$

where ι_S^* is the pullback as per Proposition 64 while $\langle , \rangle_h$ is the metric induced pairing on S as per Exercise 29.

Proof. Consider any $X \in \Gamma(TM)$ and indicate with $X|_S$ its restriction to S. Decompose this vector field as

$$X|_S = X^\perp + X^\top, \quad X^\perp \doteq \langle X|_S, N\rangle_g N, \quad X^\top \doteq X - X^\perp.$$

Using this decomposition and Equation (23), we obtain

$$\beta(X|_S) = X^\perp \lrcorner\, \omega_g + X^\top \lrcorner\, \omega_g.$$

Pulling back to S this expression, *cf.* Proposition 64, we get per linearity

$$\iota_S^*\beta(X|_S) = \langle X|_S, N\rangle_g \iota_S^*(N \lrcorner\, \omega_g) + \iota_S^*(X^\top \lrcorner\, \omega_g).$$

If we focus on the first term, there must exist $f \in C^\infty(S)$ such that $\iota_S^*(N \lrcorner\, \omega_g) = f\omega_h$. Yet $f = 1$ since, if we take an oriented orthonormal frame $(E_1, \ldots, E_{n-1})$ of T_pS, $p \in S$, it descends that $(N, E_1, \ldots, E_{n-1})$ is an oriented orthonormal frame of T_pM. Hence

$$\iota_S^*(N \lrcorner\, \omega_g)(E_1, \ldots, E_{n-1}) = \omega_g(N, E_1, \ldots, E_{n-1}) = 1.$$

By Theorem 73, it descends that $\iota_S^*(N \lrcorner\, \omega_g)$ must coincide with the volume form on (S, h), $h = \iota_S^* g$.

If we focus on the second term, consider once more an oriented orthonormal frame $(E_1, \ldots, E_{n-1})$ of T_pS, $p \in S$. A direct evaluation shows that

$$\iota_S^*(X^\top \lrcorner\, \omega_g)(E_1, \ldots, E_{n-1}) = \omega_g(X^\top, E_1, \ldots, E_{n-1}) = 0,$$

since $\langle X^\top, N\rangle_g = 0$ and, consequently, at each $p \in S$, $X^\top$ must be a linear combination of $E_1, \ldots, E_{n-1}$. $\square$

Definition 86. *Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. We call* **divergence operator**

$$\mathrm{div} : \Gamma(TM) \to C^\infty(M), \quad \chi \mapsto \mathrm{div}(\chi) = \star^{-1} d(\beta(X)),$$

where $\star^{-1}$ is the inverse of the Hodge operator, while β is defined in Equation (23).

We can now prove the generalization of Gauss theorem to an arbitrary oriented smooth manifold.

Theorem 87 (**Divergence Theorem**)**.** *Let (M, g) be an oriented Riemannian manifold of dimension $\dim M = n \geq 1$. Then, for every $X \in \Gamma_0(TM)$,*

$$\int_M \mathrm{div}(X)\omega_g = \int_{\partial M} \langle X, N\rangle_g \omega_h,$$

where ω_h is the metric-induced volume form on ∂M, N is the outward pointing unit vector field on ∂M, while $\langle , \rangle_g$ is the standard metric induced pairing on $\Gamma(TM)$.

Proof. Observe that, on account of Definition 86, $\mathrm{div}(X) = \star^{-1}d(\beta(X)) = \star d\beta(X)$. The last equality descends from Exercise 31. As a consequence of Definition 84 it holds that, since $\star(d\beta) \in C^\infty(M)$,

$$\star(\star(d\beta(X))) = d\beta = (\star d\beta(X))\omega_g.$$

Hence by Stokes Theorem 79 and by Lemma 85, it descends

$$\int\limits_M (\mathrm{div}(X))\omega = \int\limits_M (\star d\beta(X))\omega_g = \int\limits_M d\beta = \int\limits_{\partial M} \iota_S^*\beta(X) = \int\limits_{\partial M} \langle X, N\rangle_g \omega_h.$$

□

3.4 Connections on vector bundles

At first glance, the title might suggest that this section is out of context, but actually we have just gathered all tools to discuss properly the notion of a connection either on a vector or on a principal bundle. My educated guess is that all my beloved readers have acquired by now a sufficient knowledge in differential geometry to know that, given a real vector bundle $E \equiv E[M; \pi, \mathbb{R}^n]$ and two points $p, q \in M$, although $\pi^{-1}(p)$ is isomorphic as vector space to $\pi^{-1}(q)$, we need to specify a rule to compare them subordinated to the smooth structure of the underlying manifold M. A similar problem occurs, if, in place of E, one considers a principal bundle $P \equiv P[M; \pi, G]$. This prompts the introduction of the notion of *connection* on a fiber bundle. Here we shall discuss this topic starting from the more intuitive case of a smooth, real vector bundle. Let us start from the main definition:

Definition 88. *Let $E \equiv E[M; \pi, \mathbb{R}^n]$, $n \geq 1$, be a real or complex, smooth vector bundle of finite rank n. We call* **linear (or Koszul) connection** *on E a linear map $\nabla_L : \Gamma^\infty(E) \to \Gamma^\infty(E \otimes T^*M)$ such that, for all $\sigma \in \Gamma^\infty(E)$ and for all $f \in C^\infty(M)$, it holds*

$$\nabla_L(f\sigma) = f\nabla_L\sigma + df \otimes \sigma,$$

where $E \otimes T^*M$ is the tensor product between vector bundles as per Exercise 7.

As it happens often for primary concepts, the above is just one among the many equivalent definitions of connections present in the literature. In between these there is one which needs to be highlighted.

Proposition 89. *Let $E \equiv E[M; \pi, \mathbb{R}^n]$, $n \geq 1$, be a real or complex, smooth vector bundle of finite rank n. It is equivalent either to assigning a Koszul connection as per Definition 88 or a bilinear map*

$$\nabla : \Gamma^\infty(TM) \times \Gamma^\infty(E) \to \Gamma^\infty(E), \tag{24}$$

such that, calling $\nabla_X : \Gamma^\infty(E) \to \Gamma^\infty(E)$ the map $\Gamma^\infty(TM) \ni X \mapsto \nabla_X \doteq \nabla(X, \cdot)$, it holds that, for all $f \in C^\infty(M)$ and for all $\sigma \in \Gamma^\infty(E)$

$$\nabla_{fX}(\sigma) = f\nabla_X(\sigma), \quad \textit{and} \quad \nabla_X(f\sigma) = f\nabla_X\sigma + X(f)\sigma. \tag{25}$$

The map ∇_X is called the **covariant derivative (relative to X)** *and a section $\sigma \in \Gamma^\infty(E)$ is said to be* **parallely transported** *along $X \in \Gamma^\infty(TM)$ if $\nabla_X \sigma = 0$.*

Proof. Suppose a connection as per Equation (24) has been assigned. For any $\sigma \in \Gamma^\infty(E)$, consider $\nabla(\cdot, \sigma) : \Gamma^\infty(TM) \to \Gamma^\infty(E)$. This is a linear map which identifies therefore an element of $\Gamma^\infty(E \otimes T^*M)$ which we denote as $\widetilde{\nabla}(\sigma)$. Consider now $f \in C^\infty(M)$ and $X \in \Gamma^\infty(TM)$. It holds that

$$\nabla(X, f\sigma) = f\nabla(X, \sigma) + X(f)\sigma = f\widetilde{\nabla}(\sigma)(X) + df(X)\sigma.$$

By the arbitrariness of X, it descends that

$$\widetilde{\nabla}(f\sigma) = f\widetilde{\nabla}\sigma + df \otimes \sigma,$$

namely we have identified a Koszul connection. Conversely suppose that a linear connection ∇_L has been assigned in agreement to Definition 88. Define

$$\nabla : \Gamma^\infty(TM) \times \Gamma^\infty(E) \to \Gamma^\infty(E), \quad (X, \sigma) \mapsto \nabla(X, \sigma) = \nabla_X(\sigma) \doteq \nabla_L(\sigma)(X).$$

For any $f \in C^\infty(M)$, $X \in \Gamma^\infty(TM)$ and $\sigma \in \Gamma^\infty(E)$, it holds

$$\nabla_X(f\sigma) = \nabla(X, f\sigma) = \nabla_L(f\sigma)(X) = f\nabla_L(\sigma)(X) + df(X)\sigma = f\nabla_X(\sigma) + X(f)\sigma,$$

and similarly

$$\nabla_{fX}(\sigma) = \nabla(fX, \sigma) = \nabla_L(\sigma)(fX) = f\nabla_L(\sigma)(X) = f\nabla_X(\sigma),$$

where we used the defining properties of a Koszul connection in the before last equality. □

Disclaimer: In view of Definition 88 and of Proposition 89 we shall use indifferently the notion of Koszul connection and Equation (24) and with a slight abuse of notation we shall always use the symbol ∇.

The first question that one should ask at this stage is whether connections do exist on a generic real or complex, smooth vector bundle. From now on we shall focus only on the real case since, mutatis mutandis, all results hold true also in the complex scenario. Luckily enough, though not surprisingly, the answer to our question is affirmative. As a matter of fact, given a smooth real vector bundle $E \equiv E[M; \pi, \mathbb{R}^n]$, $n \geq 1$, consider a cover $\{U_\alpha\}_{\alpha \in I}$, I being an index set, of M, such that $E|_{U_\alpha}$ is trivial for all $\alpha \in I$. Choosing $\{\psi_\alpha\}_{\alpha \in I}$ a partition of unity subordinated to this cover, it holds per linearity that, for any connection ∇, as per equation (24), and for any $X \in \Gamma^\infty(TM)$,

$$\nabla_X \sigma = \sum_{\alpha \in I} \psi_\alpha \nabla_X \sigma = \sum_{\alpha \in I} \nabla_X(\psi_\alpha \sigma).$$

This formula guarantees that, in constructing a connection ∇, we can focus our attention to each open neighbourhood U_α where $E|_{U_\alpha} \simeq U_\alpha \times \mathbb{R}^n$. Thereon, consider a local frame $e_j : U_\alpha \to \mathbb{R}^n$,

$j = 1, \ldots, n$, *cf.* Definition 30 and Remark 8. For any $Y \in \Gamma^\infty(TU_\alpha)$ and for all $j = 1, \ldots, n$ set $\nabla_{\alpha,Y}(e_j) = 0$. This identifies uniquely the map

$$\nabla_\alpha : \Gamma^\infty(TU_\alpha) \times \Gamma^\infty(E|_{U_\alpha}) \to \Gamma^\infty(E|_{U_\alpha}), \quad (Y, \eta) \mapsto \nabla_\alpha(Y, \eta) \doteq \sum_{j=1}^{n} Y(\eta_j)e_j,$$

where we have decomposed $\eta = \sum_{j=1}^{n} \eta_j e_j$, $\eta_j \in C^\infty(U_\alpha)$ for all $j = 1, \ldots, n$. It is immediate to verify that we have defined a connection on $E|_{U_\alpha}$ as per Proposition 89. By gluing together all these data, it turns out that, for all $X \in \Gamma^\infty(TM)$ and for all $\sigma \in \Gamma^\infty(E)$, we can write

$$\nabla_X \sigma = \sum_{\alpha \in I} \nabla_{\alpha, X|_{U_\alpha}}(\psi_\alpha \sigma). \tag{26}$$

This is known as the **trivial connection on E**.

Exercise 32. Let $E \equiv E[M; \pi, \mathbb{R}^n]$ be a smooth, real vector bundle of finite rank $n \geq 1$. Prove that the set of all linear connections on E is a vector space.

At this stage, one can also investigate whether the notion of connection is natural with respect to standard operations on a vector bundle, such as the pull-back. This is the case as we see in the following proposition.

Proposition 90. *Let $E \equiv E[M; \pi, \mathbb{R}^n]$ be a real, smooth vector bundle and let $\nabla : \Gamma^\infty(TM) \times \Gamma^\infty(E) \to \Gamma^\infty(E)$ be a connection. Let N be a smooth manifold and let $f \in C^\infty(N; M)$ be an embedding. On f^*E the pull-back bundle, cf. Definition 37, there exists an induced connection ∇^f whose associated covariant derivative is such that, for all $\sigma \in \Gamma^\infty(E)$,*

$$\Gamma^\infty(TN) \ni Y \mapsto \nabla^f_Y \quad \textit{such that} \quad \sigma \mapsto \nabla^f_Y(f^*\sigma) = \nabla_{df(Y)}(\sigma \circ f), \tag{27}$$

where $f^*\sigma \doteq \sigma \circ f$ is a pull-back section.

Proof. On account of Proposition 89 we need to verify that Equation (27) abides to the constraints specified in Equation (25). We start from pull-back sections. Let $h \in C^\infty(N)$, then, for any $Y \in \Gamma^\infty(TN)$ and for every $\sigma \in \Gamma^\infty(f^*E)$:

$$\nabla^f_Y(hf^*\sigma) = \nabla_{df(Y)}(h\sigma \circ f) = h\nabla_{df(Y)}(\sigma \circ f) + df(Y)(h)\sigma \circ f = h\nabla_{df(Y)}(f^*\sigma) + df(Y)(h)f^*\sigma,$$

where we used simply that $\nabla_{df(Y)}$ is per definition a covariant derivative. Similarly

$$\nabla^f_{hY}(f^*\sigma) = \nabla_{df(hY)}(f^*\sigma) = \nabla hdf(Y)(f^*\sigma) = h\nabla_{df(Y)}(f^*\sigma) = h\nabla^f(f^*\sigma).$$

To extend this formula to all smooth sections of f^*E, observe that, being E an embedding, we can use the tubular neighbourhood theorem, *cf.* [11, Th. 6.23] (barring a straightforward generalization to embedded submanifolds of a smooth manifold M), to infer that there must exist an open subset of M diffeomorphic to $f(N)\times(-\epsilon,\epsilon)$, $\epsilon>0$. Let us indicate with z the coordinate along the direction not lying on $f(N)$, so that $f(N)$ coincides with the locus $z=0$. Let $\chi\in C_0^\infty(-\epsilon,\epsilon)$ be such that $\chi(0)=1$. For any $\widetilde{\sigma}\in\Gamma^\infty(f^*E)$, we define $\sigma\in\Gamma^\infty(E)$ such that $\sigma(x,z)\doteq\chi(z)\widetilde{\sigma}(x)$, for any $(x,z)\in f(N)\times(-\epsilon,\epsilon)$. Per construction it holds that $f^*\sigma=\widetilde{\sigma}$. At this point it suffices to define

$$\nabla^f_Y\widetilde{\sigma}\doteq\nabla^f_Y f^*\sigma,\quad \forall Y\in\Gamma^\infty(TN).$$

This is a connection per construction, but it remains to be shown that the definition does not depend on the choice of χ. Consider a second cut-off function $\chi'\in C_0^\infty(\epsilon,\epsilon)$ and let $\sigma'\doteq\chi'\widetilde{\sigma}$ for every $\widetilde{\sigma}\in\Gamma^\infty(f^*E)$. Since, still per construction $f^*\sigma'=\widetilde{\sigma}$, it descends that

$$\nabla^f_Y(f^*\sigma-f^*\sigma')=\nabla^f_Y(0)=0,$$

which concludes the proof. □

Much in the same spirit of Riemannian and Lorentzian geometry we can associate a notion of curvature to that of connection. In this endeavor, it is convenient to start from the definition of connection given in Proposition 89.

Definition 91. *Let $E\equiv E[M;\pi,\mathbb{R}^n]$ be a real, smooth vector bundle and let $\nabla:\Gamma^\infty(TM)\times\Gamma^\infty(E)\to\Gamma^\infty(E)$ be a connection. We call* **curvature (of the connection ∇)** *the operator*

$$R:\Gamma^\infty(TM)\times\Gamma^\infty(TM)\times\Gamma^\infty(E)\to\Gamma^\infty(E),$$
$$(X,Y,\sigma)\mapsto R(X,Y,\sigma)=\left(\nabla_X\nabla_Y-\nabla_Y\nabla_X-\nabla_{[X,Y]}\right)\sigma,$$

where ∇_X, ∇_Y are the covariant derivatives relative to X and Y respectively while $[,]$ stands for the Lie bracket between vector fields.

The curvature associated to a connection enjoys some simple, notable properties.

Corollary 92. *Let $E\equiv E[M;\pi,\mathbb{R}^n]$ be a real, smooth vector bundle and let $\nabla:\Gamma^\infty(TM)\times\Gamma^\infty(E)\to\Gamma^\infty(E)$ be a connection, whose associated curvature is R. It holds that, for all $f,g,h\in C^\infty(M)$, for all $X,Y\in\Gamma^\infty(TM)$ and for all $\sigma\in\Gamma^\infty(E)$,*

$$R(X,Y)=-R(Y,X),\quad \textit{and}\quad R(fX,gY,h\sigma)=fghR(X,Y,\sigma).$$

Proof. The first identity is a direct consequence of Definition 91 and of the antisymmetry of the Lie bracket, *i.e.* $[X,Y]=-[Y,X]$ for all $X,Y\in\Gamma^\infty(TM)$. The second identity can be proven in two steps. Let us start by assuming that $f=g=1$. It holds that for all $h\in C^\infty(M)$ and for all $\sigma\in\Gamma^\infty(TE)$,

$$\nabla_X\nabla_Y(h\sigma)=\nabla_X(h\nabla_Y\sigma+Y(h)\sigma)=h\nabla_X\nabla_Y\sigma+X(h)\nabla_Y\sigma+Y(h)\nabla_X\sigma+X(Y(h))\sigma,$$

where we applied iteratively the second identity in Equation (25). Similarly exchanging X and Y

$$\nabla_Y\nabla_X(h\sigma)=\nabla_Y(h\nabla_X\sigma+X(h)\sigma)=h\nabla_Y\nabla_X\sigma+Y(h)\nabla_X\sigma+X(h)\nabla_Y\sigma+Y(X(h))\sigma,$$

while, recalling that $[X,Y] \in \Gamma^\infty(TM)$

$$\nabla_{[X,Y]}(h\sigma) = h\nabla_{[X,Y]}\sigma + [X,Y](h)\sigma.$$

Putting together these three last identities, we obtain

$$R(X,Y,h\sigma) = hR(X,Y,\sigma).$$

The second step of the proof consists of assuming that $h=1$. It holds that for all $f,g \in C^\infty(M)$ and for all $\sigma \in \Gamma^\infty(TE)$, the first identity in Equation (**??**) entails

$$\nabla_{fX}\nabla_{gY}\sigma = f\nabla_X\left(g\nabla_Y\sigma\right) = fg\nabla_X\nabla_Y\sigma + fX(g)\nabla_Y\sigma.$$

At the same time exchanging X and Y we obtain

$$\nabla_{gY}\nabla_{fX}\sigma = g\nabla_Y\left(f\nabla_X\sigma\right) = fg\nabla_Y\nabla_X\sigma + gY(f)\nabla_X\sigma.$$

Finally, if we recall the structural properties of the Lie brackets, it holds that $[fX,gY] = fg[X,Y] + fX(g)Y - gY(f)X$. This entails that

$$\nabla_{[fX,gY]}\sigma = \nabla_{fg[X,Y]+fX(g)Y-gY(f)X}\sigma = fg\nabla[X,Y]\sigma + fX(g)\nabla_Y\sigma - gY(f)\nabla_X\sigma,$$

where we used that ∇ is a bilinear map, *cf.* Equation (24). Putting together all these identities we obtain

$$R(fX,gY,\sigma) = fgR(X,Y,\sigma),$$

which concludes the proof. □

It is very instructive and of practical use in computations looking at the local form of a connection and of the associated curvature. In other words consider a cover $\{U_\alpha\}_{\alpha\in I}$, I being an index set, of M, such that $E|_{U_\alpha}$ is trivial for all $\alpha \in I$. Fixing α and suppressing the symbol for simplicity of the notation, we can find a local frame $\sigma_i \in \Gamma^\infty(E|_U)$, $i = 1,\dots,n$ where n is the rank of E. Given a connection ∇ as per Proposition 89, for any $X \in \Gamma^\infty(TU)$, per linearity there must exist $\omega_{ij}(X) \in C^\infty(U)$ for all $i,j = 1,\dots,n$ such that

$$\nabla_X\sigma_i = \sum_{j=1}^n \omega_{ij}(X)\sigma_j. \tag{28}$$

Observe that, at all $x \in U$, $\omega_{ij}(X)$ identifies a $n \times n$ matrix, which can be read as an element of $\mathfrak{gl}(n,\mathbb{R})$. In other words $\omega_{ij} : \Gamma^\infty(TU) \to \mathfrak{gl}(n,\mathbb{R})$ is a 1-form on U with values in the Lie algebra $\mathfrak{gl}(n,\mathbb{R})$. Patching together all these local 1-forms, we obtain $\omega \in \Omega^1(M;\mathfrak{gl}(n,\mathbb{C}))$, called the **connection 1-form of** ∇.

Remark 29. *It is important to observe that the connection form associated locally to ∇ is yet another way to relate Equation (24) to the Koszul connection ∇_L as per Definition 88. As a matter of fact the above assignment is nothing but the local construction of ∇_L starting from ∇.*

Working still locally, it is instructive to look at the expression of the curvature of ∇. More precisely, starting from Definition 91 and considering once more a local frame $\sigma_i \in \Gamma^\infty(E|_U)$, $i = 1, \ldots, n$ where n is the rank of E, it holds that, for any $X, Y \in \Gamma^\infty(TU)$, there must exist $\Lambda_{ij}(X, Y) \in C^\infty(U)$ for all $i, j = 1, \ldots, n$ such that

$$R(X, Y, \sigma_i) = \sum_{j=1}^{n} \Lambda_{ij}(X, Y)\sigma_j.$$

As above Λ_{ij} can be seen as identifying an element of $\Omega^2(U; \mathfrak{gl}(n, \mathbb{R}))$. All these data can be patched together to obtain $\Lambda \in \Omega^2(M; \mathfrak{gl}(n, \mathbb{R}))$ called the **curvature form of** ∇. The following theorem relates the connection 1-form to the curvature form.

Theorem 93. *Let $E \equiv E[M; \pi, \mathbb{R}^n]$ be a real, smooth vector bundle and let $\nabla : \Gamma^\infty(TM) \times \Gamma^\infty(E) \to \Gamma^\infty(E)$ be a connection. Then the associated connection 1-form $\omega \in \Omega^1(M; \mathfrak{gl}(n, \mathbb{R}))$ and curvature form $\Lambda \in \Omega^2(M; \mathfrak{gl}(n, \mathbb{R}))$ are related by the* **(Cartan second) structure equation**

$$d_1\omega + \omega \wedge \omega = \Lambda,$$

where d_1 is the differential acting on 1-forms on M.

Proof. It suffices to prove the statement locally, namely restricting the attention to $U \subseteq M$ such that there exists a local frame $\sigma_i \in \Gamma^\infty(E|_U)$, $i = 1, \ldots, n$ where n is the rank of E. Starting from the curvature associated to the connection ∇, it holds that, for every $X, Y \in \Gamma^\infty(TU)$,

$$R(X, Y, \sigma_i) = \sum_{j=1}^{n} \Lambda_{ij}\sigma_j.$$

At the same time we can express R using Definition 91 and Equation (28)

$$R(X, Y, \sigma_i) = \left(\nabla_X\nabla_Y - \nabla_Y\nabla_X - \nabla_{[X,Y]}\right)\sigma_i =$$
$$= \sum_{j=1}^{n} \left(\nabla_X\omega_{ij}(Y) - \nabla_Y\omega_{ij}(X) - \omega_{ij}([X, Y])\right)(\sigma_j) =$$
$$= \sum_{j=1}^{n} \left(X(\omega_{ij}(Y)) - Y(\omega_{ij}(X)) - \omega_{ij}([X, Y])\right)\sigma_j + \sum_{j,k=1}^{n} \left(\omega_{ij}(Y)\omega_{jk}(X) - \omega_{ij}(X)\omega_{jk}(Y)\right)(\sigma_k).$$

To conclude it suffices to observe that

$$d_1\omega(X, Y, \sigma_i) = \sum_{j=1}^{n} \left(X(\omega_{ij}(Y)) - Y(\omega_{ij}(X)) - \omega_{ij}([X, Y])\right)\sigma_j,$$

while

$$(\omega \wedge \omega)(X, Y, \sigma_i) = \sum_{j,k=1}^{n} \left(\omega_{ij}(Y)\omega_{jk}(X) - \omega_{ij}(X)\omega_{jk}(Y)\right)(\sigma_k).$$

$\square$

As the wise reader has certainly already noticed, if there exists a second Cartan's structure equation, there must exist a first one. Oh well... we shall not discuss it here, but you can definitely prove it.

Exercise 33. Let $E \equiv E[M;\pi,\mathbb{R}^n]$ be a real, smooth vector bundle of finite rank $n \geq 1$ and let $\nabla : \Gamma^\infty(TM) \times \Gamma^\infty(E) \to \Gamma^\infty(E)$ be a connection. Consider a local frame $\sigma_i \in \Gamma^\infty(E|_U)$, $i = 1, \ldots, n$ and let ω be the connection 1-form. Prove that

$$d\sigma_i = \sum_{k=1}^{n} \sigma_k \wedge \omega_{ik}.$$

On the contrary, to conclude we shall investigate the behaviour of a connection under change of coordinates. More precisely let $E \equiv E[M;\pi,\mathbb{R}^n]$ be a real, smooth vector bundle of finite rank $n \geq 1$. Consider in addition two local trivializations, that is $U_\alpha, U_\beta \subseteq M$ together with two local diffeomorphisms

$$\varphi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^n, \quad \text{and} \quad \varphi_\beta : \pi^{-1}(U_\beta) \to U_\beta \times \mathbb{R}^n.$$

Suppose in addition that a connection ∇ has been assigned and refer to ω_i, Λ_i, $i = \alpha, \beta$ as the connection and curvature forms on U_α and U_β respectively. If we consider the case where $U_\alpha \cap U_\beta \neq \emptyset$, Proposition 25 guarantees us that there exist transition functions

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to GL(n,\mathbb{R}).$$

With all these ingredients we can prove the following transformation rule whose form should ring a bell in the mind of those readers who are familiar with gauge theories.

Lemma 94. *Under the above assumptions it holds that*

$$\omega_\beta = g_{\alpha\beta}^{-1}\omega_\alpha g_{\alpha\beta} + g_{\alpha\beta}^{-1} dg_{\alpha\beta} \tag{29}$$

$$\Lambda_\beta = g_{\alpha\beta}^{-1}\Lambda_\alpha g_{\alpha\beta}, \tag{30}$$

where all products are between matrices, upon the identification of $\mathfrak{gl}(n,\mathbb{R}) \equiv M(n,\mathbb{R})$.

Proof. Consider local frames $\sigma_i \in \Gamma^\infty(E|_{U_\alpha})$ and $\tau_j \in \Gamma^\infty(E|_{U_\beta})$, $i, j = 1, \ldots, n$. Proposition 25 entails that on $U_\alpha \cap U_\beta$,

$$\tau_j = \sum_{i=1}^{n} g_{ij}\sigma_i,$$

where we suppressed for simplicity of notation the subscripts α and β. Let $X \in \Gamma^\infty(U_\alpha \cap U_\beta)$, then it holds that

$$\nabla_X \tau_j = \sum_{k=1}^{n} (\omega_\beta)_{jk}(X)\tau_k = \sum_{i=1}^{n} \left(X(g_{ij})\sigma_i + g_{ij} \sum_{k=1}^{n} (\omega_\alpha)_{ik}(X)\sigma_k \right),$$

which, getting rid of τ_j, becomes

$$\sum_{i,k=1}^{n} (\omega_\beta)_{jk}(X) g_{ik}\sigma_i = \sum_{i=1}^{n} \left(X(g_{ij})\sigma_i + g_{ij} \sum_{k=1}^{n} (\omega_\alpha)_{ik}(X)\sigma_k \right).$$

Since this expression holds true for any X and since $X(g_{ij}) = dg_{ij}(X)$, we obtain due to linear independence of the local frames that

$$g_{\alpha\beta}\omega_\beta = dg_{\alpha\beta} + \omega_\alpha g_{\alpha\beta},$$

where we have restored the subscripts α and β. A direct inspection of this formula, together with the property of $g_{\alpha\beta}$ of being invertible, entails that we have proven Equation (29). To prove Equation (30), we start from Equation (29) and we consider Cartan second structural equation, *cf.* Theorem 93. Hence

$$\Lambda_\beta = d_1\omega_\beta + \omega_\beta \wedge \omega_\beta = -g^{-1}(dg)g^{-1} \wedge \omega_\alpha g + g^{-1}d\omega_\alpha g - g^{-1}\omega_\alpha \wedge dg - g^{-1}(dg)g^{-1} \wedge dg+ \\ +(g^{-1}\omega_\alpha g + g^{-1}dg) \wedge (g^{-1}\omega_\alpha g + g^{-1}dg) = g^{-1}(d_1\omega_\alpha + \omega_\alpha \wedge \omega_\alpha)g = g^{-1}\Lambda_\alpha g,$$

which concludes the proof. □

Remark 30. *The attentive reader should notice that, from a physical and practical viewpoint the situation that we have depicted is not so different from that of vector bundles in the following sense. It was rather clear that vector bundles were the correct geometric structure to describe via their smooth sections the kinematic configurations of a field theory. Yet, once an underlying Riemannian or Lorentzian spacetime has been assigned, one needs a rationale to select a specific vector bundle of given rank in between the possible ones. In Section 2.4 we have shown that, for relativistic field theories, a good selection criterion comes from constructing an associated vector bundle to the bundle of oriented orthonormal frames.*

A similar problem occurs when discussing connections on a given vector bundle. There is no rationale to select one in between the many available possibilities. In Riemannian and Lorentzian geometry, for example, when considering the tangent (or the cotangent) bundle, the dilemma does not occur if we accept to work with the Levi-Civita connection which is the only torsion free and metric compatible connection on the tangent space.

A selection criterion needs to be found also in other scenarios since connections are the natural differential operators at the heart of the construction of a consistent dynamics for a field theory whose kinematic configurations are the smooth sections of a vector bundle.

As last step, it is worth reverting our attention once more to the Koszul connection ∇_L, *cf.* Definition 88. Proposition 89 guarantees us that ∇_L gives the same information of ∇ when acting on sections of the underlying vector bundle E. One might wonder how we should have proceeded if we had to define curvature starting from ∇_L. The following exercise has the goal to answer this question.

Exercise 34. Let $E \equiv E[M;\pi,\mathbb{R}^n]$ be a real, smooth vector bundle of finite rank $n \geq 1$ and let ∇_L be a Koszul connection as per Definition 88. Prove that there exists a unique extension of ∇_L to a linear map

$$\widetilde{\nabla}_{L,k} : \Gamma^\infty(E \otimes T^*M^{\otimes k}) \to \Gamma^\infty(E \otimes T^*M^{\otimes (k+1)}), \quad \forall k \in \mathbb{N} \cup \{0\}$$

as well as a unique $R_L \in \Gamma^\infty(T^*M^{\otimes 2} \otimes \mathrm{End}(E))$ such that

1. $\widetilde{\nabla}_{L,k+k'}(\alpha \otimes \sigma) = d_k\alpha \otimes \sigma + (-1)^k \alpha \wedge \widetilde{\nabla}_{L,k'}\sigma$, for all $\alpha \in \Omega^k(M)$ and for all $\sigma \in \Gamma^\infty(E \otimes T^*M^{\otimes k'})$,
2. $(\widetilde{\nabla}_{L,k+1} \circ \widetilde{\nabla}_{L,k})(\sigma) = R_L(\sigma)$ for all $\sigma \in \Gamma^\infty(E \otimes T^*M^{\otimes k})$.

Here $\mathrm{End}(E)$ is the bundle of endomorphism that is $\mathrm{End}(E) = \bigsqcup\limits_{p \in M} \mathrm{End}(E_p)$ where $\mathrm{End}(E_p)$ is the linear space of all endomorphism of E_p. The map $\widetilde{\nabla}_L$ is called **covariant exterior differential** and it is often indicated as D_L.

It is interesting to notice that, contrary to the standard exterior derivative on a manifold $D_{L,k+1} \circ D_{L,k} \neq 0$ for all $k \in \mathbb{N} \cup \{0\}$, but it holds that $R_L = D_{L,1} \circ \nabla_L$ where $\nabla_L = D_{L,0}$.

3.5 Connections on principal bundles

In Section 3.4 we have introduced the notion of connections on a vector bundle as a distinguished tool to compare different fibers. It is certainly desirable to look for similar structures on generic fiber bundles, but it is at the same time clear that we need to address the problem differently. As a matter of fact the definition of linear connection as per Definition 88 relies on $E \otimes T^*M$, where E is a vector bundle over M. This structure is well-defined since the fibers are finite dimensional linear spaces, hence endowed with a canonical tensor product. Certainly more appealing is Equation 24 which relies only on the existence of smooth sections. Trying to generalize this approach is certainly a reasonable course of action, but it is not the most instructive, especially from a geometric viewpoint. In the following we shall adopt a radically new approach. It is applicable to generic fiber bundles, but, in view of the relevance for the applications in physics, we shall focus instead only on principal bundles. The reader in his nobility of spirit shall forgive us for losing only apparently in generality. If not, well... he/she is free to adapt the construction below to an arbitrary fiber bundle. It is actually a rather instructive, albeit long and tedious exercise. The following discussion enjoys much from [1, 8, 14].

Let M be a smooth manifold of dimension $\dim M = n \geq 1$ and let G be a Lie group. Consider a principal G-bundle over M, namely $P \equiv P[M;\pi,G]$. Since P is a smooth manifold in its own right, we can consider the associated tangent bundle TP and the induced differential map

$$d\pi : TP \to TM, \quad (p,v) \mapsto (\pi(p), d\pi_p(v)).$$

It is intuitively clear that $d\pi$ is not injective since every vector which is tangent to a fiber of TP will have vanishing projection on the tangent bundle of the base space M. This can be made precise as follows:

Proposition 95. *Let $P \equiv P[M;\pi,G]$ be a principal bundle over a smooth manifold M and let $VP = \bigsqcup_{p\in P} V_pP$ where $V_pP = \{v \in T_pP \mid d\pi_p(v) = 0\}$ be the* **vertical bundle** *associated to P. Then, calling $\iota : VP \to TP$ the natural inclusion, the following is a short exact sequence of vector bundles*

$$0 \longrightarrow VP \xrightarrow{\iota} TP \xrightarrow{d\pi} TM \longrightarrow 0 \tag{31}$$

In other words each map is a vector bundle homomorphism as per Definition 34 such that $d\pi \circ \iota = 0$.

Proof. Both maps ι and $d\pi$ are bundle homomorphisms covering the identity on M being fiber-wise linear per construction. In addition $d\pi\circ\iota = 0$ per definition of vertical bundle. To conclude we need to investigate exactness at TM and at VP. In the latter this amounts to showing that ι is injective, which is a direct consequence of the map being nothing but an injection. In the former this translates in proving that $d\pi$ is surjective. To show this property consider $X \in TM$ and let $U \subseteq M$ be an open subset such that thereon there exists a trivialization both of TM and of TP. This entails that $P|_U \simeq U \times G$. Using Exercise 12 and keeping implicit all diffeomorphisms, $TP|_U \simeq \pi_1^*TU \oplus \pi_2^*TG$ where π_i, $i = 1, 2$ are the natural projections. Hence, since each point in π_1^*TU is nothing but an element $((z, g),(z, Y_z)) \in U \times G \times TU$, we can see that X identifies canonically the point in π_1^*TU of the form $((z, g),(z, X_z))$ for all $(z, g) \in U \times G$. By considering $0 \in \pi_2^*TG$ we have identified an element of $TP|_U$, say Y such that $d\pi(Y) = X$ per construction. Repeating the procedure on a cover of P we obtain the desired result. □

In this proposition we have made a statement which needs to be proven separately. Since it is not particularly complicated, we leave it as an exercise to the reader. In addition we suggest to verify also a rather interesting generalization of the splitting lemma discussed in Exercise 5.

Exercise 35. Let $E \equiv E[M;\pi,F]$ be a fiber bundle over a smooth manifold M as per Definition 40 and let $VP = \bigsqcup_{p\in P} V_pP$ where $V_pP = \{v \in T_pP \mid d\pi_p(v) = 0\}$. Prove that VP is a smooth real vector bundle.

Exercise 36 (Splitting Lemma for Vector Bundles). Assume that $E \equiv E[M;\pi,\mathbb{R}^n]$, $F \equiv F[M;\pi,\mathbb{R}^m]$ and $H \equiv H[M;\pi,\mathbb{R}^k]$ are three smooth real vector bundles over M of finite rank $n, m, k \geq 1$. Suppose that there exists a short exact sequence of vector bundles, namely

$$0 \longrightarrow E \xrightarrow{\alpha} F \xrightarrow{\beta} H \longrightarrow 0 \tag{32}$$

where both α and β are vector bundle homomorphisms as per Definition 34 such that $\beta\circ\alpha = 0$, α is injective and β is surjective. Prove that, if there exists a vector bundle homomorphism $\lambda : H \to F$ or $\eta : F \to E$, then the sequence splits, namely $F = E \oplus F$ where $\oplus$ stands for the Whitney sum as per Definition 27.

We have all necessary ingredients to address the issue of defining in this framework the notion of connection. In Section 3.4 we adopted the viewpoint of associating such notion to that of a covariant derivative acting on sections of the underlying vector bundle, *cf.* Proposition 89. In this way one has killed two birds with one stone since, together with a distinguished, geometric, differential operator, it has been established when a section is parallely transported along each direction. The key point, observed by Ehresmann in 1950 (yes, this is still very basic differential geometry, nothing fancy, nothing modern) is that, one can only focus the attention on establishing when a section is parallel in each direction and this suffices to construct a connection.

Definition 96. *Let $P \equiv P[M;\pi,G]$ be a principal bundle and let VP be the associated vertical bundle. We call* **(Ehresmann or principal) connection** *on P the assignment of HP a smooth vector sub-bundle of TP such that $TP = HP \oplus VP$, where $\oplus$ is the Whitney sum, cf. Definition 27, such that*

- *for every $p \in P$, the fiber H_pP of HP is invariant under the right action $r_g : P \to P$ of G, $g \in G$, namely $H_{r_g(p)}P = dr_g(H_pP)$.*

Remark 31. *In many approaches to the definition of principal connection, it is often preferred to focus on the local properties rather than on the global ones. In other words the splitting of the tangent bundle TP is replaced by these two joint requests:*

1. *for every $p \in P$, we assign a vector space $H_pP \subset T_pP$ such that $T_pP = V_pP \oplus H_pP$,*

2. *the assignment $P \ni p \mapsto H_pP$ is smooth.*

As always, given a definition, it is important to verify that it does not lead to an empty set. Given a principal bundle P, it is not particularly complicated to construct a splitting of TP, since P is per assumption paracompact. Hence it can be endowed with a fiducial Riemannian metric $\tilde{\eta}$ out of which we can define H_pP, $p \in P$ as the fiberwise orthogonal complement of V_pP via $\tilde{\eta}$. This selects an infinite set of potential connections out of which one needs to single out at least one compatible with the group action as per Definition 96. This is possible, though it is somehow a tedious procedure. The interested reader can find it in [8, Th. 2.1].

Having established the basic definition of principal connection, we shall discuss an alternative characterization which has several practical consequences both at the mathematical and at the physical level. To this end we need to recall a few tools from the theory of Lie groups. In particular, if G is an arbitrary Lie group, this comes endowed with a natural left action

$$G \ni g \mapsto l_g : G \to G,$$

such that its differential reads as $dl_g : T_hG \to T_{gh}G$ for all $h \in G$. Recalling that $T_eG \simeq \mathfrak{g}$, e being the identity element and $\mathfrak{g}$ being the Lie algebra, we can identify a distinguished 1-form $\omega \in \Omega^1(G;\mathfrak{g})$ such that, for every $v \in T_gG$,

$$\omega(v) \doteq dl_{g^{-1}}(v).$$

This is called the **Maurer-Cartan form**. Secondly G comes with a distinguished action on its Lie algebra, the *adjoint representation*

$$Ad : G \to \mathrm{GL}(\mathfrak{g}), \quad g \mapsto Ad(g), \text{ such that } Ad(g)(X) = gXg^{-1}, \quad \forall X \in \mathfrak{g}.$$

where all compositions are to be read as matrix multiplications. These two concepts are related as follows:

Exercise 37. Let G be a Lie group and let $\omega \in \Omega^1(G; \mathfrak{g})$ be the associated Maurer-Cartan form. Prove that, for all $v \in T_gG$ and for all $h \in G$, it holds that

$$(r_h^* \omega)(v) = \mathrm{Ad}(g)(\omega(v)),$$

where r_h^* is the pull-back via the right action $r_h : G \to G$ such that, for all $g' \in G$, $r_h(g') = g'h^{-1}$.

A rather straightforward result is the following:

Corollary 97. *Let $P \equiv P[M; \pi, G]$ be a principal bundle. For any $x \in M$, it holds that*

$$V_p \simeq \mathfrak{g},$$

where $\mathfrak{g}$ is the Lie algebra of G while p is any point lying in $\pi^{-1}(x)$.

Proof. Let $U \subseteq M$ be an open subset such that $x \in U$ and there exists a local trivialization of P. Hence $\pi^{-1}(U)$ is diffeomorphic to $U \times G$, which, in turn, entails that there exists a diffeomorphism $\Phi : TP|_{\pi^{-1}(U)} \to T(U \times G)$. On account of Exercise 12

$$T(U \times G) \simeq \pi_1^* TU \oplus \pi_2^* TG,$$

where π_i, $i = 1, 2$ is the projection on each component, while $\oplus$ stands for the Whitney sum, *cf.* Definition 27. Standard results in the theory of Lie groups guarantee in addition that $TG \simeq G \times \mathfrak{g}$. Furthermore, per construction $\pi = \Phi \circ \pi_1$ and since Φ is a diffeomorphism it entails that $d\pi \circ (\pi_2 \circ \Phi)^* = 0$. In other words, omitting the diffeomorphism, $\pi_2^* TG \subseteq \ker(d\pi)$. At the same time, since, due to the chain rule, $d\pi = d\Phi \circ d\pi_1$, it descends that $[d\pi\, (\pi_1 \circ \Phi)^*] = id|_{TU}$ which entails that the first component in the Whitney sum does not contribute to the kernel. In other words $\pi_2^* TG \simeq \ker(d\pi)$ and using Definition 37 together with the identification $TG \simeq G \times \mathfrak{g}$ it descends

$$\ker(d\pi)|_{TP|_U} = VP|_U \simeq U \times G \times \mathfrak{g}.$$

Fixing $p \in U$, we obtain the sought isomorphism. $\square$

Remark 32. *Starting from Corollary 97 together with the group action which exists per definition on any principal bundle $P \equiv P[M; \pi, G]$, it holds that for any $z \in \mathfrak{g}$, we can identify $Z \in \Gamma^\infty(TP)$ as follows: Let $\exp(tz) \in G$, $t \in \mathbb{R}$ be the 1-parameter family of group elements obtained via exponential map. This induces a corresponding 1-parameter family of actions $r_{\exp(tz)} : P \to P$, $t \in \mathbb{R}$. This identifies a vector field $Z \in \Gamma^\infty(TP)$ such that*

$$P \ni p \mapsto Z_p \doteq \left. \frac{dr_{\exp(tz)}(p)}{dt} \right|_{t=0}.$$

Per construction it turns out that $Z_p \in V_pP$ and thus we have identified a correspondence

$$\mathfrak{g} \ni z \mapsto Z \in VP.$$

These are called **fundamental vector fields** *at each $p \in P$.*

In view of this discussion and particularly of the identification of V_p with the Lie algebra $\mathfrak{g}$ for all $p \in P$, it appears rather natural that the Ehresmann connection can be read also as a coherent assignment to each vector field on TP of a Lie algebra element at each $p \in P$. This is nothing but a Lie-algebra valued 1-form on P. The next proposition makes this statement precise, although before we need an ancillary result in differential geometry.

Lemma 98. *Let M be a smooth manifold and let $\varphi_t \in \mathrm{Diff}(M)$ for all $t \in I$, where $I \subseteq \mathbb{R}$ such that $0 \in I$. If $X \in \Gamma^\infty(TM)$ is the generator of Φ_t, then, for any $\varphi \in \mathrm{Diff}(M)$, $d\varphi(X)$ is the generator of $\varphi \circ \Phi_t \circ \varphi^{-1}$.*

Proof. Let $p \in M$ and let $q = \varphi^{-1}(p)$. Since $X_q \in T_qM$ is the tangent vector to the curve $x(t) = \Phi_t(q)$ at $q = x(0)$, it follows that $d\varphi : T_qM \to T_{\varphi(q)}M = T_pM$ yields $d\varphi(X_q)$ is tangent to the curve $y(t) = \varphi \circ \Phi_t$ at $y(0) = q = \varphi^{-1}(p)$. Hence $d\varphi(X)$ is tangent to the curve $y(t) = \left(\varphi^{-1} \circ \Phi_t \circ \varphi\right)(p)$ at $p = y(0)$. $\square$

Theorem 99. *Let $P \equiv P[M; \pi, G]$ be a principal bundle over a smooth manifold M. There exists a 1:1 correspondence between principal connections on P as per Definition 96 and* **connection 1-forms**, *namely $\omega \in \Omega^1(P; \mathfrak{g})$ such that*

1. *for any $z \in \mathfrak{g}$, $\omega(Z) = z$ where Z is the fundamental vector field associated to z,*
2. *for any $g \in G$, $r_g^*\omega = \mathrm{Ad}(g)\omega$ where r_g is the Lie group right action[4] on P while Ad is the adjoint representation of G on the Lie algebra.*

Proof. Suppose a principal connection has been assigned. Hence, for each $p \in P$, we have identified a splitting $T_pP = H_pP \oplus V_pP$, which induces a natural projection $\pi_2 : T_pP \to V_pP \simeq \mathfrak{g}$. Hence we have constructed $\omega \in \Omega^1(P; \mathfrak{g})$ by setting that, for any $p \in P$ and for any $X \in \Gamma^\infty(TP)$,

$$\omega(X)(p) = \pi_2(X_p),$$

where we leave implicit the identification between the fiber of the vertical bundle and the Lie algebra, *cf.* Corollary 97. Per construction, it holds that, if Z is a fundamental vector field, then $\omega(Z)(p) = Z_p$ for all $p \in P$. Yet, following Remark 32, it holds per construction that $Z_p = z \in \mathfrak{g}$. Finally, we observe that on each horizontal vector field $Y \in \Gamma^\infty(HP)$, it holds that $\omega(Y) = 0$. Hence we can focus our attention to vertical vector fields and for any $Y \in \Gamma^\infty(VP)$ and for all $g \in G$, it holds that

$$\left(r_g^*\omega\right)(Y) = \omega((r_g)_*Y) = \mathrm{Ad}(g)\omega(Y),$$

[4] In comparison with Definition 46 we adopt here the notation r_g in place of $\varphi(g, \cdot)$ to indicate the right action of the Lie group on the principal bundle P.

where we used Lemma 98.

Conversely, assume that a connection 1-form $\omega \in \Omega^1(P;\mathfrak{g})$ has been assigned. For every $p \in P$, set

$$H_pP \doteq \{X \in T_pP \mid \omega(X) = 0\}.$$

Since the form is smooth so is the assignment $P \ni p \to H_pP$. At the same time, per definition, for any $Z \in V_pP$, it holds that $\omega(Z) = z \in \mathfrak{g}$. Since $z \neq 0$, it descends that $V_pP \cap H_pP = \{0\}$, which entails that $T_pP = H_pP \oplus V_pP$. Finally, for every, for any $g \in G$ and for any $X \in H_pP$, it descends that

$$\omega(dr_g(X)) = (r_g^*\omega)(X) = \mathrm{Ad}(g)(\omega(X)) = 0,$$

where we used the second defining property of the connection 1-form. As a consequence $dr_g(X) \in H_{r_g(p)}$. Since the right action is invertible, being a diffeomorphism on G, it descends $H_{r_g(p)}P = dr_g(H_pP)$ for all $p \in P$. □

With this characterization, it is way easier to prove existence of a connection on an arbitrary principal bundle.

Corollary 100. *Every principal bundle $P \equiv P[M;\pi,G]$ admits a connection.*

Proof. Let us start from a trivial principal bundle $P = M \times G$. Calling $\pi_2 : M \times G \to G$ the standard projection on the second factor and indicating with $\omega_0 \in \Omega^1(G;\mathfrak{g})$ the Maurer-Cartan form, consider $\omega \doteq \pi_2^*\omega_0 \in \Omega^1(P;\mathfrak{g})$. In view of Exercise 37, this abides to all requirements for being a connection 1-form. If P is a generic principal bundle, we can follow at this point the usual procedure of considering a cover $\{U_\alpha\}_{\alpha\in I}$, I being an index set, of M such that there exist local trivializations $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times G$. Let $\{\psi_\alpha\}_{\alpha\in I}$ be a partition of unity subordinated to the cover and let $\omega_\alpha = \Phi_\alpha^*\omega_{0,\alpha}$, where $\omega_{0,\alpha}$ is the Maurer-Cartan form on $U_\alpha \times G$. If we define $\omega \in \Omega^1(P;\mathfrak{g})$ such that

$$\omega = \sum_{\alpha\in I}(\psi_\alpha \circ \pi)\omega_\alpha,$$

this inherits the defining properties of a connection 1-form from the Maurer-Cartan building blocks. □

Remark 33. *The relevance of Theorem 99 cannot be emphasized enough. As a matter of fact, we have just introduced what in physics one calls* **gauge fields**. *As a matter of fact, if $P \equiv P[M;\pi,G]$ is a principal bundle over a smooth manifold M, the assignment of a connection is tantamount to constructing a connection 1-form $\omega \in \Omega^1(P;\mathfrak{g})$. Yet, our heuristic idea that a field is a map from the underlying Riemannian or Lorentzian spacetime to a suitable target vector space can be recovered, as follows. First consider a basis $\{t_i\}_{i=1,\dots,m=\dim\mathfrak{g}}$ for the Lie algebra $\mathfrak{g}$ and decompose*

$$\omega = \sum_{i=1}^{m}\omega_i t_i, \quad \omega_i \in \Omega^1(P).$$

Finally, recall that, locally each principal bundle is trivial. In other words consider a cover $\{U_\alpha\}_{\alpha\in I}$, I being an index set, of M such that there exist local trivializations $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times G$. As a consequence there exist local sections $\sigma_\alpha : U_\alpha \to U_\alpha \times G$ so that we can build

$$A_\alpha = \sigma_\alpha^* \omega|_{U_\alpha} \in \Omega^1(U_\alpha; \mathfrak{g}) \longrightarrow \sum_{i=1}^{m} (A_\alpha)_i t_i,$$

where $(A_\alpha)_i = \sigma_\alpha^*(\omega_i)|_{U_\alpha} \in \Omega^1(U_\alpha)$. This construction makes precise in a rather unexpected way the heuristic expectation that gauge fields are locally defined. In addition one can easily prove a counterpart of Lemma 94 when G is a matrix group, namely, that, whenever $U_\alpha \cap U_\beta \neq \emptyset$, calling $g_{\alpha\beta} : U_\alpha \cap U_\beta \to G$ the transition functions, it holds

$$A_\beta = g_{\alpha\beta}^{-1} A_\alpha g_{\alpha\beta} + g_{\alpha\beta}^{-1} d g_{\alpha\beta}, \tag{33}$$

where $g_{\alpha\beta}^{-1} d g_{\alpha\beta} = -\mathrm{Ad}(g_{\alpha\beta})(g_{\alpha\beta}^* \omega_0)$, ω_0 being the Maurer-Cartan form associated to G. This is rather long and tedious calculation left as an exercise for the brave reader. Suffice to say that we have now understood that the kinematic configurations of a gauge theory are nothing but the connection 1-forms. Hence the model building part resides in the choice of the structure group G while the gauge transformations are automorphism of the principal bundle acting via pull-back on the connection 1-form. As for vector bundles, we have no a priori rationale to choose a specific principal bundle and thus the questions how many possible non equivalent choices do exist and how we distinguish them are natural ones. In addition, we still need to assign dynamics on top of all this discussion, but this is definitely another story, since here we focused only on kinematics.

Having proven that connections do exist, it is natural to ask how many of them can we construct. Of course this is a heuristic question whose answer is plainly, infinite many, but we can translate it in asking ourselves whether the space of all connections has a distinguished structure. It is not easy to convince ourselves that it cannot be a vector space since the 0-section does not exist, as it would violate for example the first condition in Theorem 99. In answering to this question we need an auxiliary object, namely the **adjoint bundle** to a principal bundle $P \equiv P[M; \pi, G]$. This is an associated bundle, canonically constructed from the adjoint representation $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ as

$$\mathrm{Ad}(P) \doteq P \times_{\mathrm{Ad}} \mathfrak{g}.$$

Using Proposition 51, we observe that $\mathrm{Ad}(P)$ is vector bundle over M with the Lie algebra $\mathfrak{g}$ as the typical fiber.

Proposition 101. *Let $P \equiv P[M; \pi, G]$ be a principal bundle over a smooth manifold and let $\mathcal{C}(P)$ be the set of all principal connections over P. Then $\mathcal{C}(P)$ is an affine space modeled over $\Omega^1(M; \mathrm{Ad}(P)) \doteq \Gamma^\infty(\mathrm{Ad}(P) \otimes T^*M)$.*

Proof. We have already established existence of a connection in Corollary 100. Consider now two of them $\omega, \omega' \in \Omega^1(P; \mathfrak{g})$ and pick a cover $\{U_\alpha\}_{\alpha\in I}$, I being an index set, of M such that there exist local trivializations $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times G$. As a consequence there exist local sections $\sigma_\alpha : U_\alpha \to U_\alpha \times G$ so that we can build

$$A_\alpha = \sigma_\alpha^* \omega|_{U_\alpha} \in \Omega^1(U_\alpha; \mathfrak{g}) \ \longrightarrow A_\alpha = \sum_{i=1}^{m} (A_\alpha)_i t_i,$$

where $(A_\alpha)_i = \sigma_\alpha^*(\omega_i)|_{U_\alpha} \in \Omega^1(U_\alpha)$. Let U_β be another element of the cover such that $U_\alpha \cap U_\beta \neq \emptyset$. Using Equation (33)

$$A_\beta - A'_\beta = g_{\alpha\beta}^{-1}(A_\alpha - A'_\alpha) g_{\alpha\beta},$$

where $g_{\alpha\beta} : U_\alpha \cap U_\beta \to G$ are the transition functions. Considering a local section of $P|_{U_\alpha}$, say s_α and any $X \in \Gamma^\infty(U_\alpha)$, we observe that we can define a corresponding local section σ_α of $\mathrm{Ad}(P)|_{U_\alpha}$ as

$$U_\alpha \ni x \mapsto \sigma_\alpha(x) = [s_\alpha(x), (A_\alpha - A'_\alpha)(X)].$$

Still, on account of (33), one can see that on $U_\alpha \cap U_\beta$

$$[s_\alpha(x), (A_\alpha - A'_\alpha)(X)] = [s_\beta(x), (A_\beta - A'_\beta)(X)],$$

where we used that $s_\alpha = g_{\alpha\beta} s_\beta$ on the overlap. To conclude it suffices first to observe that we have just identified a global section of $\mathrm{Ad}(P)$. Secondly, if we set on each U_α

$$[s_\alpha(x), (A_\alpha - A'_\alpha)](X) \doteq [s_\alpha(x), (A_\alpha - A'_\alpha)(X)],$$

we have constructed a 1-form on M with values in $\mathrm{Ad}(P)$, namely an element of $\Gamma^\infty(\mathrm{Ad}(P) \otimes T^*M)$. This suffices to conclude the proof. $\square$

The following exercise allows the reader to become familiar working with $\mathcal{C}(P)$, highlighting in addition that, in some cases of physical relevance, such as for example Maxwell theory, its structure becomes simpler. This result complements Remark 33.

Exercise 38. Let $P \equiv P[M; \pi, G]$ be a principal bundle over a smooth manifold with an Abelian structure group and let $\mathcal{C}(P)$ be the set of all principal connections over P. Prove that

1. the adjoint bundle $\mathrm{Ad}(P)$ is trivial, that is

$$\mathrm{Ad}(P) \simeq M \times \mathfrak{g}.$$

2. $\mathcal{C}(P)$ is an affine space modeled over $\Omega^1(M; \mathfrak{g})$.

Before proceeding in our discussion of principal bundles, it is worth emphasizing that there exists a third and equivalent characterization of a connection which is more algebraic in spirit and it has the short exact sequence in Equation (31) as its starting point. This modern viewpoint (ah ah ah ... just kidding, it is from 1957!) starts from the observation that, given a principal bundle

$P \equiv P[M;\pi,G]$ over a smooth manifold M, this comes equipped with a right G-action which we indicate as $\varphi : P \times G \to P$. Calling $r_g : P \to P$ the map

$$r_g(p) = \varphi(p,g), \quad \forall g \in G,$$

we can construct the quotient

$$TP/G \ni [(p,v)] \quad \text{such that} \quad (p,v) \sim (p',v'), \text{ if } \exists g \in G \quad \text{such that} \quad p' = pg, \text{ and } v' = dr_g(v).$$

Observe that TP/G is still a vector bundle over TM if endowed with the quotient manifold structure and if we define the projection map $\widetilde{\pi} : TP/G \to TM$ such that $\widetilde{\pi}([p,v]) = \pi_T P(p,v)$ for all $[p,v] \in TP/G$, where $\pi_T P : TP \to TM$. With these data we can introduce the **Atyah sequence** of vector bundles

$$M \times \{0\} \xrightarrow{\alpha} \mathrm{Ad}(P) \xrightarrow{\iota} TP/G \xrightarrow{d\widetilde{\pi}} TM \xrightarrow{\beta} M \times \{0\} \tag{34}$$

where $M \times \{0\}$ plays the role of the trivial bundle with 0-dimensional fiber, while the various maps are defined as follows:

1. $\alpha(x,0) = [(p,0)]$ where $p \in \pi^{-1}(x)$,
2. $\iota([p,x]) = [(p,X_p)]$ where X_p is the fundamental vector field associated to $x \in \mathfrak{g}$,
3. $d\widetilde{\pi}$ is the differential of the projection map $\widetilde{\pi} : TP/G \to TM$,
4. $\beta(y,Y) = (\pi_{TM}(y),0)$, where $\pi_{TM} : TM \to M$ is the projection of the tangent bundle.

The Atiyah sequence is the modern ... uhm ... more advanced tool which encodes all the geometric properties of a principal bundle, including the notion of connection, as one can prove solving the following exercise, which is not really easy.

Exercise 39. Let $P \equiv P[M;\pi,G]$ be a principal bundle over a smooth manifold M. Prove that

1. the Atyah sequence is exact,
2. There is a 1:1 correspondence between connections and right splittings of the Atyah sequence, that is $\lambda : TM \to TP/G$ such that $d\widetilde{\pi} \circ \lambda = \mathrm{id}|_{TM}$.

As we have outlined in Remark 33 the only choice that one can make in this framework is that of the structure group and the gauge group acts on connections via pull-back. A physically motivated reader (are you falling in this class?) should be rather unhappy since the heuristic idea of gauge transformations which originates from Maxwell's theory is far from that of an automorphism of a principal bundle covering the identity. Luckily for you, for me and for everyone we are right in the position to quench your thirst of knowledge. We start by giving an equivalent and more informative characterization of the gauge group starting from Definition 48.

Proposition 102. *Let $P \equiv P[M;\pi,G]$ be a principal bundle over a smooth manifold M and let* $\mathrm{Gau}(P)$ *be the gauge group built out of the bundle automorphisms covering the identity, cf. Definition 48. There exists a group isomorphism*

$$\mathrm{Gau}(P) \simeq \Gamma^\infty(\mathrm{Conj}(G)),$$

where $\mathrm{Conj}(G) = P \times_\alpha G$ *is the associated bundle built out of the conjugate action*

$$\alpha : G \times G \to G, \quad (g,h) \mapsto \alpha(g,h) \doteq ghg^{-1}.$$

Proof. Observe that $\mathrm{Conj}(G)$ is a fiber bundle (not of principal type) with M as base space and G as fiber. A generic element is an equivalence class $[p,h]$, $p \in P$ and $h \in G$. In addition the fibers inherit the group structure since, given $h' \in G$, it holds that $[p,h][p,h'] = [p,hh']$ is well-defined since it does not depend on the choice of the representative on account of trivial relation $ghg^{-1}gh'g^{-1} = ghh'g^{-1}$.

Consider now $\sigma \in \Gamma^\infty(\mathrm{Conj}(G))$. This is a map

$$x \ni M \mapsto \sigma(x) = [p(x), h(x)],$$

to which it corresponds an element of $\mathrm{Gau}(P)$ as follows

$$\Phi : P \to P \quad (x,g) \mapsto (x, gh(x)).$$

To verify that we have identified an element of the gauge group it suffices to prove that Φ is an automorphism of P. It is a morphism of fiber bundles per construction and it is also equivariant since, for all $g' \in G$ it holds

$$\Phi(p(g')^{-1}) = (x, gh(g')^{-1}) = \Phi(p)(g')^{-1}.$$

Conversely, suppose that $\Phi \in \mathrm{Gau}(P)$. Hence, for any $p \in P$, there exists a unique map $f : P \to G$ such that $\Phi(p) = pf(p)^{-1}$. Let $p' = ph^{-1}$ for $h \in G$. It descends that

$$\Phi(p') = p'f(p')^{-1} = ph^{-1}f(p')^{-1} = p(f(p')h)^{-1}.$$

At the same time equivariance of principal bundle morphisms entails that

$$\Phi(p') = \Phi(ph^{-1}) = \Phi(p)h^{-1} = p(hf(p))^{-1},$$

which entails that $f(p')h = hf(p)$, namely $f(p') = hf(p)h^{-1}$. Hence we have individuated the map

$$\sigma : M \to P \times_\alpha G, \quad \widehat{\pi}(p) \mapsto [p, f(p)],$$

where $\widehat{\pi}$ is the projection map of the bundle $\mathrm{Conj}(G)$. Since the above constructions are compatible with the group operations, we have constructed the desired isomorphism. □

It is instructive to repeat this proof in a simple albeit physically relevant scenario:

Exercise 40. Let $P \equiv P[M;\pi,G]$ be a principal bundle over a smooth manifold M with an Abelian structure Lie group G. Prove that

$$\mathrm{Gau}(P) \simeq C^\infty(M;G).$$

Finally we leave to a reader the task to be convinced that the gauge group defined in these notes corresponds to the one used in theoretical physics. In particular we suggest to solve the following exercise which is a specialization of Exercise 38

Exercise 41. Let $P \equiv P[M;\pi,G]$ be a principal bundle over a smooth manifold M. Let $U \subseteq M$ be an open subset admitting a local trivialization $\Phi : \pi^{-1}(U) \to U \times G$. Let $\omega \in \Omega^1(P;\mathfrak{g})$ be a connection 1-form and let σ_U be a local section of $P|_U$. Let $A = \sigma^*\omega|_U \in \Omega^1(U;\mathfrak{g})$. Prove that for any $\Phi \in$ Gau(P),

$$(\Phi \circ \sigma^*)\omega|_U = \sigma^*((\Phi^*\omega))|_U = \mathrm{Ad}(g_U)A + g_U^{-1}dg_U,$$

where $g \in \Gamma^\infty(\mathrm{Conj}(G)|_U)$ is the section of the conjugated bundle associated to Φ via Proposition 102. Show in addition that, if $G = U(1)$ and $P = \mathbb{R}^n \times U(1)$, $n \geq 1$, this reduces to the following statements:

1. $\mathcal{C}(P) \simeq \Omega^1(\mathbb{R}^n)$,
2. for any $\Phi \in \mathrm{Gau}(P)$, there exists $\chi_\Phi \in C^\infty(\mathbb{R}^n)$ such that, for all $A \in \mathcal{C}(P)$,

$$\Phi^*A = A + d\chi_\Phi.$$

As a last topic of the section, we focus on discussing the notion of curvature associated to a connection on a principal bundle. In a physical language this is nothing but the field strength tensor and it plays a key role in constructing the Lagrangian densities of gauge theories such as those of Yang-Mills or of Chern-Simons type.

Definition 103. *Let $P \equiv P[M;\pi,G]$ be a principal bundle over a smooth manifold M and let $\omega \in \Omega^1(P;\mathfrak{g})$ be a connection 1-form as per Theorem 99. We call* **curvature (2-)form** *$\Lambda \in \Omega^2(P;\mathfrak{g})$ such that*

$$\Lambda = d_1\omega + \frac{1}{2}[\omega,\omega], \tag{35}$$

where d_1 in the exterior derivative acting on 1-forms over P while, given a basis $\{t_i\}_{i=1,\dots,m=\dim\mathfrak{g}}$ for the Lie algebra $\mathfrak{g}$,

$$[\omega,\omega] = \sum_{i,j,k=1}^m \omega_i \wedge \omega_j c_{ijk} t_k, \quad \textit{and} \quad \omega = \sum_{i=1}^m \omega_i t_i, \quad \omega_i \in \Omega^1(P),\ \forall i = 1,\dots,m.$$

Here c_{ijk} are the structure constants defined so that

$$[t_i,t_j]_{\mathfrak{g}} = \sum_{k=1}^m c_{ijk} t_k. \quad \forall i,j = 1,\dots,m$$

If $\Lambda = 0$, we call the associated connection, **flat**.

It is customary to refer to Equation (35) as the *structure equation* being the natural generalization of Theorem 93. The curvature form enjoys several notable properties:

Proposition 104. *Let $P \equiv P[M;\pi,G]$ be a principal bundle over a smooth manifold M and let $\omega \in \Omega^1(P;\mathfrak{g})$ be a connection 1-form as per Theorem 99. Let $\Lambda \in \Omega^2(P;\mathfrak{g})$ be the associated curvature form as per Definition 103. Then the following properties hold true:*

1. *For any $g \in G$ $r_g^*\Lambda = \mathrm{Ad}(g)\Omega$, where r_g is the right action and Ad the adjoint representation of G on $\mathfrak{g}$,*

2. *For any $p \in P$ and for any $X, Y \in T_pP$, calling X_h, Y_h their horizontal components with respect to the connection,*
$$\Lambda(X,Y) = d_1\omega(X_h, Y_h),$$
while for all $Z \in V_pP$, $\Omega(Z,\dot{)} = 0$,

3. *For any $S, T \in \Gamma^\infty(HP)$,*
$$\Lambda(S,T) + \frac{1}{2}\omega([S,T]) = 0.$$

4. *the* **Bianchi identity** *$d_2\Lambda = [\Lambda,\omega]$, where given a basis $\{t_i\}_{i=1,\dots,m=\dim\mathfrak{g}}$ for the Lie algebra $\mathfrak{g}$,*
$$[\Lambda,\omega] = \sum_{i,j,k=1}^{m} \Lambda_i \wedge \omega_j c_{ijk} t_k,$$
with $\omega = \sum_{i=1}^{m} \omega_i t_i$ and $\Lambda = \sum_{j=1}^{m} \Lambda_j t_j$, $\omega_i \in \Omega^1(P)$ and $\Lambda_j \in \Omega^2(P)$ forall $i, j = 1, \dots, m$.

Proof. 1. Consider $g \in G$ and act via pull-back on Equation (35). Since the right action identifies a diffeomorphism on P, Theorem 68 entails that
$$r_g^*(d_1\omega) = d_1 r_g^*\omega = d_1\mathrm{Ad}(g)\omega = \mathrm{Ad}(g)d_1\omega.$$
At the same time Equation (35) entails also
$$[r_g^*\omega, r_g^*\omega] = [\mathrm{Ad}(g)\omega, \mathrm{Ad}(g)\omega] = \mathrm{Ad}(g)[\omega,\omega] = r_g^*[\omega,\omega],$$
where we used that $\mathrm{Ad}(g)$ is a Lie algebra homomorphism. Putting together the last two equations one obtains the result sought.

2. Having fixed a connection, we know that $TP = HP \oplus VP$ and therefore every $X, Y \in \Gamma^\infty(TP)$ can be uniquely written as $X = X_h + X_v$ and $Y = Y_h + Y_v$, where $X_h, Y_h \in \Gamma^\infty(HP)$ while $X_v, Y_v \in \Gamma^\infty(VP)$. Since the curvature form is bilinear it holds
$$\Lambda(X,Y) = \Lambda(X_h,Y_h) + \Lambda(X_h,Y_v) + \Lambda(X_v,Y_h) + \Lambda(X_v,Y_v).$$
To start with we prove that, whenever one of the entries is a vertical vector field, then one obtains a vanishing contribution. If we consider $X_v, Y_v \in \Gamma^\infty(VP)$ then, for every $p \in P$, there

exists $x, y \in \mathfrak{g}$ such that X_v and Y_v are the associated fundamental vector fields. In addition it holds that $[X_v, Y_v]$ is the fundamental vector field associated to $[x, y]_{\mathfrak{g}}$. Using Equation (35) it descends

$$\Lambda(X_v, Y_v) = (d_1\omega)(X_v, Y_v) + \frac{1}{2}[\omega, \omega](X_v, Y_v).$$

Expanding in local coordinates and using property 1 in Theorem 99, one obtains that

$$(d_1\omega)(X_v, Y_v) = \partial_{X_v}\omega(Y_v) - \partial_{Y_v}\omega(X_v) - \omega([X_v, Y_v]) = \partial_{X_v}Y_v - \partial_{Y_v}X_v - \omega([X_v, Y_v]).$$

Since the fundamental vector fields are constant, their derivatives are vanishing. Hence, still working with local coordinates, it descends

$$(d_1\omega)(X_v, Y_v) = -\omega([X_v, Y_v]) = -\frac{1}{2}[\omega, \omega](X_v, Y_v),$$

which entails $\Lambda(X_v, Y_v) = 0$. Consider now that in place of Y_v we have Y_h. It descends

$$\begin{aligned}\Lambda(X_v, Y_h) &= (d_1\omega)(X_v, Y_h) + [\omega(X_v), \omega(Y_h)] = \\ &= \partial_{X_v}\omega(Y_h) - \partial_{Y_h}\omega(X_v) - \omega([X_v, Y_h]) = -\omega([X_v, Y_h]),\end{aligned}$$

where we used that a connection 1-form evaluates to 0 every horizontal vector field and that $\partial_{Y_h}\omega(X_v) = \partial_{Y_h}X_v = 0$ since the fundamental vector fields are constant. If we manage to prove that $[X_v, Y_h]$ is an horizontal vector field, we can draw the sought conclusion. To this end consider $x \in \mathfrak{g}$ associated to X_v and let $\exp(tx)$, $t \in \mathbb{R}$, be the one parameter group of elements in G built via exponential map. This induces a one-parameter flow on P via $r_{\exp(tx)}$. This entails that

$$[X_v, Y_h] = \lim_{t\to 0} \frac{dr_{exp(tx)}Y_h - Y_h}{t}.$$

Evaluating it on the connection 1-form yields

$$\omega([X_v, Y_h]) = \left.\frac{d}{dt}\right|_{t=0} dr_{exp(tx)}(\omega)(Y_h) = \left.\frac{d}{dt}\right|_{t=0} \mathrm{Ad}(exp(tx))(\omega(Y_h)) = 0,$$

since the connection 1-form maps to 0 every horizontal vector field. We are left with the last task, namely proving that

$$[\omega, \omega](X_h, Y_h) = 0.$$

Yet, as above, working in local coordinates, one can see that

$$[\omega, \omega](X_h, Y_h) = [\omega(X_h), \omega(Y_h)] = 0,$$

for the same reasons given above. This entails the sought identity.

3. Consider two horizontal vector fields $S, T \in \Gamma^\infty(HP)$. Using Equation (35), it descends

$$\Lambda(S, T) = (d_1\omega)(S, T) = \partial_S\omega(T) - \partial_T\omega(S) - \frac{1}{2}\omega([S, T]) = -\frac{1}{2}\omega[S, T],$$

since, once more we used that the connection 1-form evaluates to 0 every horizontal vector field.

4. Using once more Equation (35) it descends that

$$2d_2\Lambda = d_2([\omega,\omega]) = [d_1\omega,\omega] - [\omega, d_1\omega] = 2[d_1\omega,\omega] = 2[\Omega,\omega] - [[\omega,\omega],\omega].$$

The last term vanishes due to Jacobi identity on $\mathfrak{g}$ and this concludes the proof. □

As a not so obvious consequence of this last proposition, observe that, up to now, we have worked only with $\mathfrak{g}$-valued 1-forms on P. Yet this is not necessary as the following exercise suggests.

Exercise 42. Let $P \equiv P[M;\pi,G]$ be a principal bundle over a smooth manifold and let $\omega\mathcal{C}(P)$. We call **covariant exterior derivative** $d^\omega_k : \Omega^k(P;\mathfrak{g}) \to \Omega^{k+1}(P;\mathfrak{g})$, $k \in \mathbb{N}\cup\{0\}$ such that, for all $\alpha \in \Omega^k(P;\mathfrak{g})$ and for all $\{X_i\}_{i=1,\ldots,k+1}$, $X_i \in \Gamma^\infty(TP)$

$$d^\omega_k\alpha(X_1,\ldots,X_{k+1}) = d_{k+1}\alpha(X^h_1,\ldots,X^h_{k+1}),$$

where X^h_i, $i = 1,\ldots,k+1$ is the horizontal component induced by the connection ω while d_{k+1} is the differential on P. Prove that if $\Lambda \in \Omega^2(P;\mathfrak{g})$ is the curvature 2-form associated to ω, then

$$d^\omega_2\Lambda = 0.$$

In addition show that, if $G = U(1)$ and $P = \mathbb{R}^n \times U(1)$, $n \geq 1$, this expression reduces to the first Maxwell's equation $dF = 0$, where F is the curvature 2-form associated to a connection $A \in \Omega^1(\mathbb{R}^n)$.

Remark 34. *Observe that we have specified all ingredients necessary to discuss both relativistic free fields as well as pure Yang-Mills theories both at the kinematic and at the dynamical level. Yet, we are still missing one bit of information, namely how to couple these theories. In theoretical physics this translates in the minimal replacement formula which in QED reads that, to couple a spinor to a photon whose vector potential is A one needs to substitute*

$$d_1 \mapsto d_1 - eA\wedge,$$

where e is the coupling constant interpreted as the spinor charge. This has a geometric counterpart which is nothing but the statement that, if we consider P, a principal $U(1)$-bundle over a smooth manifold M and the Dirac bundle DM, we can induce a non trivial connection on the latter via one on P. This operation is not always possible, but it highlights that there exists a geometric rationale even in the coupling between matter and gauge fields.

4 The De Rham Cohomology on Smooth Manifolds

In the previous sections, we have constructed the spaces of all differential forms on a smooth manifold M, as sections of suitable vector bundles, *cf.* Section 3. Subsequently we have shown that, in full analogy to the case where $M = \mathbb{R}^n$, $n \geq 1$, we can define an operator which codifies the exterior differentiation, *cf.* Section 3.1.

The wise and attentive reader (yes... I am pretty much convinced that wisdom and attention are not necessary implied one by the other... but I am open to discuss this viewpoint) will recognize that we have not yet discussed any counterpart of the de Rham complex and of the associated cohomology groups, see Corollary 14 and Definition 15 respectively.

As starting point let us fix notations and conventions.

Definition 105. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$. Let $\Omega^k(M)$ indicate the space of k-differential forms as per Definition 59 and let $\Omega^k_0(M) \subseteq \Omega^k(M)$ be those of compact support. For all $k \in \mathbb{N} \cup \{0\}$, we call*

- $\Omega^k_d(M) \doteq \{\omega \in \Omega^k(M) \mid d_k\Omega = 0\}$ *the space of* **closed k-forms**,
- $d_{k-1}[\Omega^{k-1}(M)] \subseteq \Omega^k(M)$, *the space of* **exact k-forms**.

Similarly we call $\Omega^k_{0,d}(M)$ and $d_{k-1}[\Omega^{k-1}_0(M)]$ the space of **closed** *and* **exact** *compactly supported k-forms.*

Following Definition 13 we can prove a counterpart of Corollary 14 for an arbitrary smooth manifold.

Corollary 106. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$ and let $\Omega^\bullet(M) \doteq \bigoplus_{k=0}^{n} \Omega^k(M)$ be endowed with the operator $\underline{d} = (d_0, \dots, d_n)$ defined as in Theorem 65. This identifies a differential complex, called the* **de Rham complex***:*

$$\Omega^0(M) \xrightarrow{d_0} \dots \longrightarrow \Omega^{k-1}(M) \xrightarrow{d_{k-1}} \Omega^k(M) \xrightarrow{d_k} \dots \longrightarrow \Omega^n(M) \quad , \tag{36}$$

Proof. This is a direct consequence of Theorem 65 which guarantees, that, for all $k \in \mathbb{N} \cup \{0\}$, $d_{k+1} \circ d_k = 0$ with the convention that $d_k = 0$ if $k \geq n$. □

This corollary suggests to apply to an arbitrary smooth manifold Definition 15.

Definition 107. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$. For any $k \in \mathbb{N} \cup \{0\}$ we call* **k-th de Rham cohomology of M** *the vector space defined as the quotient*

$$H^k_{\mathrm{dR}}(M) \doteq \frac{\Omega^k_d(M)}{d_{k-1}[\Omega^{k-1}(M)]},$$

where the numerator and denominator are defined as per Definition 105. We call k-th **Betti number**[5]

$$\beta_k(M) = \dim H^k_{\mathrm{dR}}(M).$$

[5] The reader should be warned that, in the literature, the Betti number is usually referred to the dimension of the k-th singular cohomology group with real coefficients $H^k(M;\mathbb{R})$. Yet de Rham theorem guarantees that this is isomorphic as a vector space to $H^k_{\mathrm{dR}}(M)$.

Analogously we call **k-th compact de Rham cohomology of M**

$$H^k_{0,\mathrm{dR}}(M) \doteq \frac{\Omega^k_{0,d}(M)}{d_{k-1}[\Omega^{k-1}_0(M)]}.$$

One natural question, which might arise during the sleepless nights of our fearless readers (which sounds like leaders), is whether, at the level of de Rham cohomology, it survives part of the information encoded in the exterior algebra structure induced by the wedge product, *cf.* Proposition 60. The following exercise answers this question:

Exercise 43. Let M be a smooth manifold of dimension $\dim M = n \geq 1$. Prove that the graded vector space $H^\bullet_{\mathrm{dR}}(M) \doteq \bigoplus_{k=0}^{n} H^k_{\mathrm{dR}}(M)$ is an exterior algebra if endowed with the **cup product**

$$\smile : H^k_{\mathrm{dR}}(M) \times H^{k'}_{\mathrm{dR}}(M) \to H^{k+k'}_{\mathrm{dR}}(M), \quad ([\omega],[\omega']) \mapsto [\omega] \smile [\omega'] \doteq [\omega \wedge \omega'],$$

where $\wedge$ stands for the standard wedge product between forms on M, *cf.* Proposition 60. For this reason we refer to $H^\bullet_{\mathrm{dR}}(M)$ as the **cohomology ring of M**.

It goes with saying that the main difference between cohomologies of $\mathbb{R}^n$ and of M lies in the absolute absence in the latter case of a simple computational tool to characterize them. An equivalent of Exercise 3 does not exist except for the case $k = 0$. In this case, we can make precise the idea outlined in Remark 3 and in Exercise 4 that the 0-th de Rham cohomology yields information on the number of disconnected components of the underlying manifold.

Proposition 108. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$. Suppose $M = \bigcup_{k=1}^m M_k$, where each M_k is a smooth manifold. Then $H^0_{\mathrm{dR}}(M) \simeq \mathbb{R}^m$.*

Proof. Since $\Omega^{-1}(M) = \{0\}$ per definition, it holds that $H^0_{\mathrm{dR}}(M) = \Omega^0_d(M)$. In other words we look for $f \in C^\infty(M)$ such that $df = 0$. By considering a cover of each connected component in open subsets diffeomorphic to $\mathbb{R}^n$, one can deduce that f must be a constant function on each M_k. This entails the statement. $\square$

To bypass the computational hurdle outlined above, we shall make use of the defining property of smooth manifolds of admitting a cover in open subsets homeomorphic to suitable counterparts in $\mathbb{R}^n$. For these, recall Exercise 4. To make this idea precise we shall need a new kind of sequence.

As starting point consider a smooth, n-dimensional manifold $M = U \cup V$, where U, V are two open subsets. Using standard inclusions we can draw the following, commutative, diagram

$$\begin{tikzcd} & U \arrow[dr, "\iota_{U,M}"] & \\ U \cap V \arrow[ur, "\iota_{U\cap V,U}"] \arrow[dr, "\iota_{U\cap V,V}"'] & & M \\ & V \arrow[ur, "\iota_{V,M}"'] & \end{tikzcd},$$

which induces via pullback, *cf.* Proposition 64, the following commutative diagram for all $k \in \mathbb{N} \cup \{0\}$

$$\begin{tikzcd}
& \Omega^k(U) \arrow[dl, "\iota^{*,k}_{U\cap V,U}"'] & \\
\Omega^k(U\cap V) & & \Omega^k(M) \arrow[ul, "\iota^{*,k}_{U,M}"'] \arrow[dl, "\iota^{*,k}_{V,M}"] \\
& \Omega^k(V) \arrow[ul, "\iota^{*,k}_{U\cap V,V}"] &
\end{tikzcd} ,$$

All these data can be put together to prove the following important proposition.

Proposition 109. *For all $k \in \mathbb{N} \cup \{0\}$, the following* **Mayer-Vietoris sequence***, is exact in the sense of Definition 16:*

$$0 \longrightarrow \Omega^k(M) \xrightarrow{\iota^{*,k}_{\oplus}} \Omega^k(U) \oplus \Omega^k(V) \xrightarrow{\iota^{*,k}_{U\cap V,-}} \Omega^k(U \cap V) \longrightarrow 0\,,$$

where, $\iota^{*,k}_{\oplus}(\omega) \doteq (\iota^{*,k}_{U,M}\omega, \iota^{*,k}_{V,M}\omega)$, $\omega \in \Omega^k(M)$ while $\iota^{*,k}_{U\cap V,-}(\eta,\eta') = \iota^{*,k}_{U\cap V,U}\eta - \iota^{*,k}_{U\cap V,V}\eta'$.

Proof. To prove exactness of the sequence, we proceed step by step. Starting from the left, the first step consists of showing that $\iota^{*,k}_{\oplus}$ is injective. This is immediate if we observe that the action of $\iota^{*,k}_{\oplus}$ amounts to restricting a k-form on M to U and to V. Hence, for all $\omega \in \Omega^k(M)$, $\iota^{*,k}_{\oplus}\omega = 0$ entails that $\omega|_U = \omega|_V = 0$. Since $M = U \cup V$, $\omega(p) = 0$ for all $p \in M$.

Exactness at $\Omega^k(U) \oplus \Omega^k(V)$ amounts to proving that $\mathrm{Im}(\iota^{*,k}_{\oplus}) = \ker(\iota^{*,k}_{U\cap V,-})$. Yet, as we remarked, the image of $\iota^{*,k}_{\oplus}$ amounts to the restriction of $\omega \in \Omega^k(M)$ to U and V respectively. Since they coincide per construction on $U \cap V$, it entails that $\mathrm{Im}(\iota^{*,k}_{\oplus}) \subseteq \ker(\iota^{*,k}_{U\cap V})$. Conversely, if $(\eta,\eta') \in \Omega^k(U) \oplus \Omega^k(V)$ lies in the kernel of $\iota^{*,k}_{U\cap V,-}$, this entails that $\eta|_{U\cap V} = \eta'|_{U\cap V}$. Hence we can define

$$\widetilde{\eta} = \begin{cases} \eta & p \in U \setminus U \cap V \\ \eta' & p \in V \setminus U \cap V \\ \eta = \eta' & p \in U \cap V \end{cases} \in \Omega^k(M).$$

Observe that smoothness is derived from that of η and η'. In other words, we have shown per construction that $\ker(\iota^{*,k}_{U\cap V,-}) \subseteq \mathrm{Im}(\iota^{*,k}_{\oplus})$ since $\iota^{*,k}_{\oplus}(\tilde{\eta}) = (\eta,\eta')$.

To conclude, we need to discuss exactness at $\Omega^k(U \cap V)$ which amounts to proving that $\iota^{*,k}_{U\cap V,-}$ is surjective. In other words, for every $\sigma \in \Omega^k(U \cap V)$, we need to find $\eta \in \Omega^k(U)$ and $\eta' \in \Omega^k(V)$ such that $\sigma = \iota^{*,k}_{U\cap V,-}(\eta,\eta')$. To this end, fix a partition of unity (φ_U, φ_V) subordinated to the cover $\{U, V\}$. Define

$$\eta = \begin{cases} \varphi_V \sigma & \text{on } U \cap V \\ 0 & \text{on } U \setminus \mathrm{supp}(\varphi_V) \end{cases}, \quad \eta' = \begin{cases} \varphi_U \sigma & \text{on } U \cap V \\ 0 & \text{on } V \setminus \mathrm{supp}(\varphi_U) \end{cases}.$$

Per construction $(\eta,\eta') \in \Omega^k(U) \oplus \Omega^k(V)$ is the sought element. □

The Mayer-Vietoris sequence extends naturally to a counterpart between differential complexes, namely

$$0 \longrightarrow \Omega^\bullet(M) \xrightarrow{\iota^*_\oplus} \Omega^\bullet(U) \oplus \Omega^\bullet(V) \xrightarrow{\iota^*_{U\cap V,-}} \Omega^\bullet(U\cap V) \longrightarrow 0\,, \tag{37}$$

where $\iota^*_\oplus \doteq (\iota^{*,0}_\oplus, \dots \iota^{*,n}_\oplus)$ and similarly $\iota^*_{U\cap V,-} = (\iota^{*,0}_{U\cap V,-}, \dots, \iota^{*,n}_{U\cap V,-})$. Observe that the differential of both $\Omega^\bullet(M)$ and $\Omega^\bullet(U\cap V)$ will be indicated as $\underline{d} = (d_0, \dots, d_n)$ where each d_k is nothing but the differential acting on k-forms as per Theorem 67. On the contrary the differential of $\Omega^\bullet(U) \oplus \Omega^\bullet(V)$ is $\underline{d}_\oplus = (\underline{d}, \underline{d})$ where each entry acts on $\Omega^\bullet(U)$ and on $\Omega^\bullet(V)$ respectively. In addition Theorem 68 guarantees that both $\iota^*_\oplus$ and $\iota^*_{U\cap V,-}$ are chain maps, *cf.* Definition 17. By applying Proposition 18, for every $k \in \mathbb{N}\cup\{0\}$, there exists a corresponding sequence at the level of de Rham cohomologies, *cf.* Definition 107:

$$H^k_{\mathrm{dR}}(M) \xrightarrow{\tilde{\iota}^{*,k}_\oplus} H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V) \xrightarrow{\tilde{\iota}^{*,k}_{U\cap V,-}} H^k_{\mathrm{dR}}(U\cap V)\,,$$

where $\tilde{\iota}^{*,k}_\oplus$ and $\tilde{\iota}^{*,k}_{U\cap V,-}$ are the counterparts of $\iota^{*,k}_\oplus$ and $\iota^{*,k}_{U\cap V,-}$, defined as per Proposition 18. As last step, on account of the Zig-Zag Lemma 19 we obtain the long exact sequence in cohomology:

$$\begin{tikzcd}
\cdots \arrow[r] & H^k_{\mathrm{dR}}(M) \arrow[r, "\tilde{\iota}^{*,k}_\oplus"] & H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V) \arrow[r, "\tilde{\iota}^{*,k}_{U\cap V,-}"] & H^k_{\mathrm{dR}}(U\cap V) \arrow[dll, "\tilde{d}_k"'] \\
& H^{k+1}_{\mathrm{dR}}(M) \arrow[r, "\tilde{\iota}^{*,k+1}_\oplus"] & H^{k+1}_{\mathrm{dR}}(U) \oplus H^{k+1}_{\mathrm{dR}}(V) \arrow[r, "\tilde{\iota}^{*,k+1}_{U\cap V,-}"] & H^{k+1}_{\mathrm{dR}}(U\cap V) \arrow[dll, "\tilde{d}_{k+1}"'] \\
& H^{k+2}_{\mathrm{dR}}(M) \arrow[r] & \cdots &
\end{tikzcd} \tag{38}$$

where $\tilde{d}_k : H^k_{\mathrm{dR}}(U\cap V) \to H^{k+1}_{\mathrm{dR}}(M)$ is the connecting map built as in Lemma 19.

The author is feeling like a mind reader and he knows that all his beloved readers are wondering what is the practical use of the Mayer-Vietoris sequence and of the associated long counterpart in Equation (38). Contrary to what one might think at first glance, it is a very effective computational tool as we highlight with the following example:

Example 110. Let us consider the unit circle $\mathbb{S}^1 \doteq \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 = 1\}$. In order to follow the argument at the heart of Proposition 109, we consider the open subsets, considered in the two standard Mercatore charts, namely

$$U = \mathbb{S}^1 \setminus \{(0,1)\}, \quad V = \mathbb{S}^1 \setminus \{(0,-1)\}.$$

It is immediate to realize that $\mathbb{S}^1 = U \cup V$ and that $U\cap V$ is the disjoint union of two components, namely

$$U\cap V = L \cup R,$$
$$L \doteq \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 = 1,\ \text{and } x<0\}, \quad R \doteq \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 = 1,\ \text{and } x>0\}.$$

Since U, V, L, R are all diffeomorphic to open subsets of $\mathbb{R}$, we can apply Exercise 4 to infer that

$$H^0_{\mathrm{dR}}(U) = H^0_{\mathrm{dR}}(V) \simeq \mathbb{R}, \ \text{ while } \ H^0_{\mathrm{dR}}(U \cap V) \simeq \mathbb{R}^2.$$

At the same time Proposition 108 guarantees that $H^0_{\mathrm{dR}}(\mathbb{S}^1) \simeq \mathbb{R}$ while Exercise (4) yields $H^1_{\mathrm{dR}}(U) \oplus H^1_{\mathrm{dR}}(V) = \{0\}$, which entails that the long sequence (38) $(k = 0)$ reduces to

$$\begin{aligned}
\cdots &\longrightarrow \mathbb{R} \xrightarrow{\tilde{\iota}^{*,0}_{\oplus}} \mathbb{R} \oplus \mathbb{R} \xrightarrow{\tilde{\iota}^{*,0}_{U\cap V,-}} \mathbb{R}^2 \\
&\xrightarrow{\tilde{d}_0} H^1_{\mathrm{dR}}(\mathbb{S}^1) \xrightarrow{\tilde{\iota}^{*,1}_{\oplus}} 0 \xrightarrow{\tilde{\iota}^{*,1}_{U\cap V,-}} 0 \\
&\xrightarrow{\tilde{d}_1} 0 \longrightarrow \cdots
\end{aligned}$$

To evaluate the first cohomology group of the circle, we exploit exactness of the sequence. Since $\tilde{d}_0$ must be surjective it holds that, exactness at $H^1_{\mathrm{dR}}(\mathbb{S}^1)$ entails

$$H^1_{\mathrm{dR}}(\mathbb{S}^1) = \mathrm{Im}(\tilde{d}_0) \simeq \frac{\mathbb{R}^2}{\ker(\tilde{d}_0)}.$$

Exactness at $\mathbb{R}^2$ implies that $\ker(\tilde{d}_0) = \mathrm{Im}(\tilde{\iota}^{*,0}_{U\cap V,-})$. Proceeding by iteration $\mathrm{Im}(\tilde{\iota}^{*,0}_{U\cap V,-}) = \frac{\mathbb{R}\oplus\mathbb{R}}{\ker(\tilde{\iota}^{*,0}_{U\cap V,-})}$. It suffices to remember that $\tilde{\iota}^{*,0}_{U\cap V,-} : H^0_{\mathrm{dR}}(U) \oplus H^0_{\mathrm{dR}}(V) \to H^0_{\mathrm{dR}}(U \cap V)$ is such that $([\omega], [\omega']) \mapsto \tilde{\iota}^{*,0}_{U\cap V,-}([\omega], [\omega']) = [\iota^{*,0}_{U\cap V,U}\omega - \iota^{*,0}_{U\cap V,V}\omega']$. Since $H^0_{\mathrm{dR}}(U) \oplus H^0_{\mathrm{dR}}(V) = \Omega^0_d(U) \oplus \Omega^0_d(V)$ this entails that $\ker(\tilde{\iota}^{*,0}_{U\cap V,-})$ is constituted by pairs of constant functions on U and V coinciding on the intersection. In other words $\ker(\tilde{\iota}^{*,0}_{U\cap V,-}) = \mathbb{R}$ and thus

$$H^1_{\mathrm{dR}}(\mathbb{S}^1) \simeq \frac{\mathbb{R}^2}{\mathbb{R}} \simeq \mathbb{R}.$$

Since all higher cohomology groups of the circle must vanish for dimensional reasons, we have computed the whole cohomology ring of M.

Remark 35. *A physically oriented reader might wonder what is the purpose of computing the cohomology ring of a smooth manifold. The answer can be found in electromagnetism and in Maxwell equations in particular. As a matter of fact the key kinematic variable $F \in \Omega^2(M)$ which codifies the Faraday tensor is subject to the constraint $d_2F = 0$. This entails that $F \mapsto [F] \in H^2_{\mathrm{dR}}(M)$.*

A naïve conclusion might lead to the claim that, whenever we are working with a non trivial second de Rham cohomology space of the underlying manifold, we are not allowed to consider the vector potential $A \in \Omega^1(M)$ such that $F = dA$.... not so fast young man or woman... To understand the correct interpretation, let us consider the **codifferential** $\delta_k : \Omega^k(M) \to \Omega^{k-1}(M)$

where $\delta_k = *^{-1} \circ d_{n-k} \circ *$ and $*$ is the Hodge operator, cf. Exercise 30. The full set of Maxwell equations, for simplicity without an external current, is

$$d_2 F = 0 \quad \text{and} \quad \delta_2 F = 0.$$

If the underlying manifold M is such that $H^2(M) \neq \{0\}$, this means that each $[F] \in H^2(M)$ individuates an orbit such that $F' \sim F$ if and only if there exists $A \in \Omega^1(M)$ such that $F' - F = d_1 A$. Hence Maxwell equations translate to $\delta_2 d_1 A = -\delta_2 F$ where F is any representative of the equivalence class $[F]$. The choice of the representative is irrelevant once one constructs the space of all solutions to the dynamical equation ruled by $\delta_2 \circ d_1$. Observe that since $H^2_{\mathrm{dR}}(M)$ is a vector space, it suffices per linearity to control the procedure on basis elements. Yet this prompts two natural questions:

1. *How do I compute higher de Rham cohomology spaces? Can one use an argument similar to the Mayer-Vietoris sequence?*

2. *Are de Rham cohomology spaces finite dimensional?*

For Maxwell equations the questions are limited to the second de Rham cohomology space of the underlying smooth manifold, but we shall see that the answer is always positive, for all $H^k_{\mathrm{dR}}(M)$, $k \geq 0$. Incidentally there is another question which a more mathematically oriented reader might ask, namely what information on M is encoded in the higher cohomology groups. We have already shown in Proposition 108 that the 0-th cohomology encodes a topological information, namely the number of connected components of the underlying smooth manifold M. What is encoded in $H^k_{\mathrm{dR}}(M)$ for $k \geq 1$?

4.1 Homotopy invariance of cohomology

If we try to apply the same reasoning of the previous section to the computation of the cohomology ring of n-sphere $\mathbb{S}^n$, $n > 1$, we realize that the main problem lies in the evaluation of $H^k(I \times \mathbb{S}^{n-1})$ where $I \subset \mathbb{R}$ is an open interval of the real line. The answer to this question is a posteriori simple, namely it suffices to establish that de Rham cohomology spaces are *homotopy invariant*. This means that manifolds which are homotopy equivalent have the same cohomology spaces. The goal of this section is to make this statement precise.

To start with, we study the behaviour of cohomology groups with respect to maps between smooth manifolds. The key ingredient in this discussion is Theorem 68.

Proposition 111. *Let M and N be two smooth manifolds and let $F \in C^\infty(M; N)$. Then there exists a ring homomorphism $\underline{F}^*_{\mathrm{dR}} : H^\bullet(N) \to H^\bullet(M)$, which is completely specified as the linear extension of the maps:*

$$F^*_k : H^k_{\mathrm{dR}}(M) \to H^k_{\mathrm{dR}}(N), \quad [\omega] \mapsto F^*_k[\omega] = [F^*\omega], \quad \forall k \in \mathbb{N} \cup \{0\},$$

where F^* is the pullback between k-forms as per Proposition 64.

Proof. On account of Theorem 68 we know that, for all $k \in \mathbb{N} \cup \{0\}$, $F^* \circ d_k^N = d_k^M \circ F^*$. This entails in particular that $F^*[\Omega_d^k(N)] \subseteq \Omega_d^k(M)$ and $F^*[d_{k-1}^N[\Omega^{k-1}(N)]] \subseteq d_{k-1}^M[\Omega^{k-1}(M)]$. This implies, in turn, that, for all $k \in \mathbb{N} \cup \{0\}$, F_k^* is a well-defined map between cohomology spaces. Its linear extension specifies completely $\underline{F}^*_{\mathrm{dR}} : H^\bullet(N) \to H^\bullet(M)$. It remains to be seen that $\underline{F}^*_{\mathrm{dR}}$ is an homomorphism with respect to the cup product. Let $[\omega] \in H^k_{\mathrm{dR}}(N)$ and $[\omega'] \in H^{k'}_{\mathrm{dR}}(N)$. It holds that

$$\underline{F}^*_{\mathrm{dR}}([\omega] \smile [\omega']) = \underline{F}^*_{\mathrm{dR}}([\omega \wedge \omega']) = [F^*(\omega \wedge \omega')] =$$
$$= [F^*\omega \wedge F^*\omega'] = [F^*\omega] \smile [F^*\omega'] = \underline{F}^*_{\mathrm{dR}}([\omega]) \smile \underline{F}^*_{\mathrm{dR}}([\omega']),$$

where we used that F^* is an algebra-homomorphism, *cf.* item *2.* of Proposition 64. □

This proposition leads to the following corollary whose proof is immediate.

Corollary 112. *Let M and N be two smooth, diffeomorphic manifolds. Then there exists a ring isomorphism between $H^\bullet_{\mathrm{dR}}(M)$ and $H^\bullet_{\mathrm{dR}}(N)$.*

The above results prompt a natural question, namely whether, given two smooth manifolds M and N and given $F, G \in C^\infty(M; N)$, it is possible that $\underline{F}^*_{\mathrm{dR}} = \underline{G}^*_{\mathrm{dR}}$ without F and G being the same. In view also of Theorem 68, this amounts to finding, for all $k \in \mathbb{N} \cup \{0\}$ a map $\widetilde{h}_k : \Omega_d^k(N) \to \Omega^{k-1}(M)$ such that, for all $\widetilde{\omega} \in \Omega_d^k(N)$, it holds

$$G^*\omega - F^*\omega = d_{k-1}^M(\widetilde{h}_k(\omega)).$$

For all practical purposes constructing a map on a rather elusive and non explicit space such as $\Omega_d^k(N)$ is quite a daunting task. It is better to look directly for a map defined on the whole $\Omega^k(N)$ whose restriction to closed k-forms coincides with the one sought. This gives us some leeway which leads to the following definition

Definition 113. *Let M and N be two smooth manifolds and let $F, G \in C^\infty(M; N)$. We call* **homotopy operator/map (cochain homotopy) between $\mathbf{F}^*$ and $\mathbf{G}^*$** *a family of linear maps $\{h_k\}_{k \in \mathbb{N} \cup \{0\}}$ such that, for all $k \in \mathbb{N} \cup \{0\}$, $h_k : \Omega^k(N) \to \Omega^{k-1}(M)$ and, for all $\omega \in \Omega^k(N)$,*

$$G^*\omega - F^*\omega = d_{k-1}^M(h_k(\omega)) + h_{k+1}(d_k^N \omega). \tag{39}$$

The main application of this definition lies in the following proposition

Proposition 114. *Let M and N be two smooth manifolds and let $F, G \in C^\infty(M; N)$ be such that there exists an homotopy operator between F^* and G^*. Then $\underline{F}^*_{\mathrm{dR}} = \underline{G}^*_{\mathrm{dR}}$.*

Proof. This is a simple application of Equation (39), which guarantees that, for any $\omega \in \Omega_d^k(N)$, $k \in \mathbb{N} \cup \{0\}$, $F^*\omega - G^*\omega \in d_{k-1}^M[\Omega^{k-1}(M)]$. As a consequence $F_k^* = G_k^*$ which concludes the proof. □

In order to extract more juice from these results we construct two distinguished homotopy operators for a given fixed smooth manifold M. In the following we will consider implicitly the case $\partial M = \emptyset$, though this assumption is only for simplicity and it can be easily removed. First of all we invite the wise reader to solve the following non trivial exercise (Wisdom lies often in the ability of understanding where the line must be drawn separating what a reader should be thought or should learn by him/herself.

Exercise 44 (**Cartan magica formula).** Let M be a smooth manifold and let $\chi \in \Gamma^\infty(TM)$. Prove that, for every $k \in \mathbb{N}$ and for every $\omega \in \Omega^k(M)$, it holds

$$\mathcal{L}_V \omega = V \lrcorner (d_k \omega) + d_{k-1}(V \lrcorner \omega), \tag{40}$$

where $\lrcorner$ is the interior multiplication, while $\mathcal{L}_V$ is the Lie derivative along V, *cf.* [11, Sec. 9]. In addition show that

$$d_k(\mathcal{L}_V \omega) = \mathcal{L}_V(d_k \omega).$$

A proof of this last exercise can be found in [11, Prop. 14.33].

Lemma 115. *Let M be a smooth manifold and let $I \doteq [0,1] \subset \mathbb{R}$. For any $t \in \mathbb{R}$, let $\iota_t : M \to M \times \mathbb{R}$ be the map $p \mapsto \iota_t(p) = (p,t)$. Then, there exists a homotopy operator between ι_0^* and ι_1^*.*

Proof. Consider the smooth manifold $M \times \mathbb{R}$ and, recall that, according to Exercise 12, $T(M \times \mathbb{R})$ is isomorphic to the Whitney sum $\pi_1^* TM \oplus \pi_2^* T\mathbb{R}$, where π_1 and π_2 are the natural projections on each of the factors of $M \times \mathbb{R}$. Hence, calling with s the coordinate along $\mathbb{R}$, for any $p \in M$, we can identify $T_{(p,s)}(M \times \mathbb{R})$ with $T_pM \times T_s\mathbb{R}$. Let $\chi \in \Gamma^\infty(TM)$ be such that, for each $(p,s) \in M \times \mathbb{R}$, $\chi(p,s) = \left(0, \frac{\partial}{\partial s}\right)$.

With these data, we construct the sought maps $h_k : \Omega^k(M \times \mathbb{R}) \to \Omega^{k-1}(M)$, $k \in \mathbb{N}$ as follows:

$$h_k(\omega) = \int_0^1 \iota_t^*(\chi \lrcorner \omega)\, dt,$$

while $h_0 \equiv 0$. Here $\lrcorner$ stands for the interior multiplication. For each $q \in M$, this formula reads

$$h_k(\omega)(q) = \int_0^1 \iota_t^*((\chi \lrcorner \omega)(q,t))\, dt,$$

which tells us that the integrand $\iota_t^*((\chi \lrcorner \omega) : \mathbb{R} \to \Lambda_q^{k-1}$, where $\Lambda_q^{k-1} \equiv \mathcal{A}^k_{n,T_q^*M}$ is a vector space of positively graded elements of degree k of the alternating algebra associated to T_q^*M with $n = \dim M$, *cf.* Definition 58. Since all elements of the integrand are either smooth forms or smooth maps, it is straightforward to conclude that $h_k(\omega)$ is smooth. In order to show that

h_k is what we are seeking, we can use both Theorem 68 and Equation (40) to obtain

$$h_{k+1}(d_k\omega) + d_{k-1}(h_k(\omega)) = \int_0^1 (\iota_t^*(\chi \lrcorner d_k\omega) + d_{k-1}\iota_t^*(\chi \lrcorner \omega))\, dt =$$

$$\int_0^1 (\iota_t^*(\chi \lrcorner d_k\omega) + \iota_t^*(d_{k-1}(\chi \lrcorner \omega)))\, dt = \int_0^1 \iota_t^*(\mathcal{L}_\chi \omega)\, dt = \int_0^1 \frac{d(\iota_t^*\omega)}{dt} = \iota^* 1_\omega - \iota_0^*\omega,$$

where, in the last expression we used the properties of both ι_t and χ. $\square$

Exercise 45. Complete explicitly the last part of the proof of Lemma 115. In other words, writing $\Theta_t \in C^\infty(M \times \mathbb{R}; M \times \mathbb{R})$, $t \in \mathbb{R}$, as the flow map of χ, namely

$$M \times \mathbb{R} \ni (q, s) \mapsto \Theta_t(q, s) = (q, s + t),$$

and using that $\iota_t = \Theta \circ \iota_0$, prove that on $\Omega^k(M \times \mathbb{R})$, $k \in \mathbb{N}$

$$\iota_t^* \circ \mathcal{L}_\chi = \frac{d}{dt} \circ \iota_t^*.$$

As next step, recall that, given two topological spaces X and Y and $f_1, f_2 \in C^0(X; Y)$ a *homotopy* between these two functions is a map $H \in C^0(X \times I; Y)$ with $I = [0, 1]$ such that, for all $x \in X$, $H(x, 0) = f_1(x)$ while $H(x, 1) = f_2(x)$. If all maps involved are smooth we call H a **smooth homotopy**, while f_1 and f_2 are **smoothly homotopic maps**.

Proposition 116. *Let M and N be two smooth manifolds and let $F, G \in C^\infty(M; N)$ be smoothly homotopic maps. Then the cohomology rings induced by $\underline{F}^*_{\mathrm{dR}}$ and $\underline{G}^*_{\mathrm{dR}}$ coincide.*

Proof. In view of Lemma 115 we can conclude that, for every $k \in \mathbb{N} \cup \{0\}$, there exists an isomorphism between $H^k_{\mathrm{dR}}(M \times I)$ and $H^k_{\mathrm{dR}}(M)$, $I = [0, 1]$. In addition, since F and G are smoothly homotopic maps, there must exist a smooth homotopy $H : M \times I \to N$, such that $H(q, 0) = F(q)$ and $H(q, 1) = G(q)$, *cf.* [11, Th. 9.28]. To conclude the theorem observe that $F = H \circ \iota_0$ while $G = H \circ \iota_1$. Thus there exists an homotopy operator between F^* and G^* and, on account of Proposition 114 $\underline{F}^*_{\mathrm{dR}} = \underline{G}^*_{\mathrm{dR}}$, which concludes the proof. $\square$

A rather more useful practical consequence of Proposition 114 can be derived considering two *homotopy equivalent* smooth manifolds M and N, namely there exists $F \in C^0(M; N)$ and $G \in C^0(N; M)$ so that $F \circ G$ is homotopic to $\mathrm{id}|_N$ while $G \circ F$ to $\mathrm{id}|_M$. The map F is called an *homotopy equivalence* between M and N, while G is the *homotopy inverse* for F.

Theorem 117. *Let M and N be two homotopy equivalent, smooth manifolds. Then the pullback of any homotopy equivalence $F \in C^\infty(M; N)$ induces a ring isomorphism between $H^\bullet_{\mathrm{dR}}(N)$ and $H^\bullet_{\mathrm{dR}}(M)$.*

Proof. Per hypothesis there exist an homotopy equivalence $F \in C^0(M;N)$ with inverse $G \in C^0(N;M)$. Using Whitney approximation theorem, see [11, Th.6.26] both F and G are homotopic to a smooth map, say $F_\infty \in C^\infty(M;N)$ and $G_\infty \in C^\infty(N;M)$. As a consequence $F_\infty \circ G_\infty$ and $G_\infty \circ F_\infty$ are homotopic respectively to $\mathrm{id}|_N$ and to $\mathrm{id}|_M$. Using Proposition 114 it holds that

$$\underline{F}^*_{\mathrm{dR}} \circ \underline{G}^*_{\mathrm{dR}} = \mathrm{id}|_{H^\bullet_{\mathrm{dR}}(M)}, \quad \text{and} \quad \underline{G}^*_{\mathrm{dR}} \circ \underline{F}^*_{\mathrm{dR}} = \mathrm{id}|_{H^\bullet_{\mathrm{dR}}(N)},$$

which, together with Proposition 111 is tantamount to saying that $\underline{F}^*_{\mathrm{dR}} : H^\bullet_{\mathrm{dR}}(N) \to H^\bullet_{\mathrm{dR}}(M)$ is a isomorphism. □

Remark 36. *A direct consequence of this last theorem is that, given any pair of smooth manifolds, which are homeomorphic, their de Rham cohomology rings are isomorphic. The reason lies in the fact that each homeomorphism is a homotopy equivalence. The consequence of this fact is that de Rham cohomology is insensible to the differentiable structure of a manifold, while it depends on the underlying topology on which the notion of continuity, hence of homeomorphism, is based. For this reason we say that the de Rham cohomology spaces are* **topological invariants**.

Remark 37. *A less obvious and more physically relevant application of the above results lies in the following fact: for any smooth manifold Σ, there exists a ring isomorphism between $H^\bullet(\Sigma)$ and $H^\bullet(\Sigma \times \mathbb{R})$. This has far reaching consequences, since, relativistic field theories, including electromagnetism, are built on a distinguished class of spacetimes, called globally hyperbolic. Any of such backgrounds, say (M, g) is isometric to $\mathbb{R} \times \Sigma$, where, calling t the coordinate along $\mathbb{R}$, each $\{t\} \times \Sigma$ is a smooth Riemannian manifold, known as Cauchy surface. Yet, since the de Rham cohomology spaces are sensitive only to topology, we can conclude that, on a physical spacetime, all information coming from de Rham cohomology is encoded in the Cauchy surface. In a field theory language this could be translated in the statement that topological data are encoded in initial data. This is certainly true for Maxwell equations, where $dF = 0$, $F \in \Omega^2(M)$ entails that we are looking for $[F] \in H^2_{\mathrm{dR}}(M)$. Yet, now, we also know that if the initial datum $[\iota_t^* F] \in H^2_{\mathrm{dR}}(\Sigma)$, then $[F] \in H^2_{\mathrm{dR}}(M)$ where $\iota_t : \Sigma \to \mathbb{R} \times \Sigma$ is the map $\Sigma \ni p \mapsto \iota_t(p) = (t, p)$.*

There exists three scenarios where Theorem 117 becomes of immediate applications:

1. Let M be a **contractible** manifold, namely M is homotopy equivalent to a point. Then $H^k_{\mathrm{dR}}(M) = \{0\}$ for all $k \geq 1$,
2. Let $U \subset \mathbb{R}^n$ be a star-shaped open subset. Then $H^k(U) = \{0\}$ for all $k \geq 1$. This is nothing the *Poincaré Lemma* you used in Elettrodinamica e Relatività.
3. Let $M = N \times J$, where $J \subseteq \mathbb{R}$. Then $H^\bullet_{\mathrm{dR}}(N \times J) \simeq H^\bullet_{\mathrm{dR}}(N)$.

Example 118. Let us consider the n-sphere $\mathbb{S}^n = \{(x_1, \ldots, x_{n+1}) \mid \sum_{i=1}^{n+1} x_i^2 = 1\}$, $n > 1$. In order to compute its cohomology ring we follow one more Proposition 109 as for the case $n = 1$ studied in Example 110. We start from $n = 2$. Consider thus the open subsets

$$U = \mathbb{S}^2 \setminus \{(0, 0, 1)\}, \quad V = \mathbb{S}^2 \setminus \{(0, 0, -1)\}.$$

It is immediate to realize that $\mathbb{S}^2 = U \cup V$ and that $U \cap V \simeq \mathbb{S}^1 \times J$ where $J \subset \mathbb{R}$.

Since $U, V, U \cap V$ are all connected, we can apply Exercise 4 to infer that

$$H^0_{\mathrm{dR}}(U) \simeq H^0_{\mathrm{dR}}(V) \simeq \mathbb{R}.$$

At the same time Proposition 108 guarantees that $H^0_{\mathrm{dR}}(\mathbb{S}^2) \simeq H^0_{\mathrm{dR}}(U \cap V) \simeq \mathbb{R}$ while Exercise (4) yields $H^1_{\mathrm{dR}}(U) \oplus H^1_{\mathrm{dR}}(V) = \{0\}$. At the same time, by homotopy invariance, $H^1_{\mathrm{dR}}(U \cap V) \simeq H^1_{\mathrm{dR}}(\mathbb{S}^1) = \mathbb{R}$ on account of Example 110. The long sequence (38) $(k = 0)$ reduces to

$$\begin{aligned}
\cdots &\longrightarrow \mathbb{R} \xrightarrow{\tilde{\iota}^{*,0}_{\oplus}} \mathbb{R} \oplus \mathbb{R} \xrightarrow{\tilde{\iota}^{*,0}_{U\cap V,-}} \mathbb{R} \\
&\xrightarrow{\tilde{d}_0} H^1_{\mathrm{dR}}(\mathbb{S}^2) \xrightarrow{\tilde{\iota}^{*,1}_{\oplus}} 0 \xrightarrow{\tilde{\iota}^{*,1}_{U\cap V,-}} \mathbb{R} \\
&\xrightarrow{\tilde{d}_1} H^2_{\mathrm{dR}}(\mathbb{S}^2) \xrightarrow{\tilde{\iota}^{*,2}_{\oplus}} 0 \longrightarrow \cdots
\end{aligned}$$

It is immediate to extract the sequence

$$0 \xrightarrow{\tilde{\iota}^{*,1}_{U\cap V,-}} \mathbb{R} \xrightarrow{\tilde{d}_1} H^2_{\mathrm{dR}}(\mathbb{S}^2) \xrightarrow{\tilde{\iota}^{*,2}_{\oplus}} 0\ ,$$

which entails that $\tilde{d}_1$ is an isomorphism. Hence $H^2_{\mathrm{dR}}(\mathbb{S}^2) \simeq \mathbb{R}$. At the same time, following slavishly the same reasoning as in Example 110 we know that

$$H^1_{\mathrm{dR}}(\mathbb{S}^2) \simeq \mathrm{Im}(\tilde{d}_0) \simeq \frac{\mathbb{R}}{\ker(\tilde{d}_0)}.$$

Exactness at $\mathbb{R}$ implies that $\ker(\tilde{d}_0) = \mathrm{Im}(\tilde{\iota}^{*,0}_{U\cap V,-})$. Proceeding by iteration $\mathrm{Im}(\tilde{\iota}^{*,0}_{U\cap V,-}) = \frac{\mathbb{R}\oplus\mathbb{R}}{\ker(\tilde{\iota}^{*,0}_{U\cap V,-})}$. It suffices to remember that $\tilde{\iota}^{*,0}_{U\cap V,-} : H^0_{\mathrm{dR}}(U) \oplus H^0_{\mathrm{dR}}(V) \to H^0_{\mathrm{dR}}(U \cap V)$ is such that $([\omega], [\omega']) \mapsto \tilde{\iota}^{*,0}_{U\cap V,-}([\omega], [\omega']) = [\iota^{*,0}_{U\cap V,U}\omega - \iota^{*,0}_{U\cap V,V}\omega']$. Since $H^0_{\mathrm{dR}}(U) \oplus H^0_{\mathrm{dR}}(V) = \Omega^0_d(U) \oplus \Omega^0_d(V)$ this entails that $\ker(\tilde{\iota}^{*,0}_{U\cap V,-})$ is constituted by pairs of constant functions on U and V coinciding on the intersection. In other words $\ker(\tilde{\iota}^{*,0}_{U\cap V,-}) = \mathbb{R}$ and thus

$$H^1_{\mathrm{dR}}(\mathbb{S}^2) \simeq \frac{\mathbb{R}}{\mathbb{R}} \simeq \{0\}.$$

One can now show easily (seriously... it is easy) via an induction argument that

$$H^k_{\mathrm{dR}}(\mathbb{S}^n) = \begin{cases} \mathbb{R} & k = 0, n \\ \{0\} & \text{otherwise} \end{cases}$$

4.2 Compactly supported cohomology

Before concluding our brief survey of de Rham cohomology, we need to translate, whenever possible, the results of the previous section for the compact de Rham cohomology spaces, see Definition 107. For this specific topic, we recommend to consult [3]. From a structural viewpoint, the main difference when considering k-forms of compact support on a smooth manifold lies in their behaviour with respect to maps between smooth manifolds, say M and N. As a matter of fact, if we consider $F \in C^\infty(M;N)$ and $\omega \in \Omega^k_0(N)$, $F^*\omega$ is well-defined as per Proposition 64. Yet, since F is not in general a proper map there is no guarantee that $F^*\omega$ is compactly supported. In other words we can only say that $F^*\omega \in \Omega^k(M)$. This is certainly an undesirable feature.

Yet, a wise reader (usually wisdom entails the study of basic functional analysis...) might recall that since F is in particular a continuous map, for every compact set $K \subseteq M$, $F[K]$ is also compact. If we combine this information with the fact that F is invertible on its image $F[M]$, *i.e.* there exists $F^{-1} : F[M] \to M$, we can introduce the following natural map

Definition 119. *Let M and N be two smooth manifolds and let $F \in C^\infty(M;N)$. For every $k \in \mathbb{N} \cup \{0\}$ we call* **push-forward** $F_* : \Omega^k_0(M) \to \Omega^k_0(N)$ *such that*

$$\Omega^k_0(M) \ni \eta \mapsto F_*(\eta) \doteq (F^{-1})^*\eta,$$

where $(F^{-1})^*$ is the pull-back as per Proposition 64.

In order to understand the rationale behind this definition, it is also useful to consider the explicit action of the push-forward on $\omega \in \Omega^1_0(M)$ such that $\mathrm{supp}(\omega)$ is contained within a single coordinate patch (U, φ), with the further constraint that there exists a coordinate patch (V, ψ) with $V \subseteq N$ and $U = F^{-1}(V)$. Choosing local coordinates $\{x_i\}_{i=1,\dots,m=\dim M}$, there must exist $\omega_i \in C^\infty_0(U)$, $i = 1, \dots, m$ such that

$$\omega = \sum_{i=1}^{m} \omega_i(x) dx^i.$$

We can now compute explicitly the push-forward $F_*\omega \in \Omega^1_0(N)$. Starting from V we see that

$$F_*\omega|_V(y) = \sum_{i=1}^{m} \omega_i(F^{-1}(y)) d(x^i \circ F^{-1}). \quad y \in V$$

It is immediate to realize that, for all $i = 1, \dots, m$ $\mathrm{supp}(\omega_i \circ F^{-1}) = F[\mathrm{supp}(\omega_i)]$ which is compact. Hence we have constructed a compactly supported smooth 1-form on V which can be extended to N as 0.

Although this is a very special scenario in many respects, it tells us that the push-forward via $F \in C^\infty(M;N)$ is nothing but the realization of $\eta \in \Omega^k(M)$ via its image through F in $F[M]$, eventually extended by 0 to the whole N.

For all practical purposes, one might think that this is not a substantial difference, worth a detailed discussion. Actually this has far reaching consequences, especially at the level of cohomology spaces. The first hint of this claim lies in the change of direction of the arrows in the Mayer-Vietoris sequence in case of compactly supported forms:

Proposition 120. *Let M be a smooth manifold and let U, V be two open subsets such that $M = U \cup V$. For all $k \in \mathbb{N} \cup \{0\}$, the following* **Mayer-Vietoris sequence of forms with compact support***, is exact in the sense of Definition 16:*

$$0 \longleftarrow \Omega^k_0(M) \xleftarrow{+^k_{U,V}} \Omega^k_0(U) \oplus \Omega^k_0(V) \xleftarrow{\iota^k_{*,U,V}} \Omega^k_0(U \cap V) \longleftarrow 0\ ,$$

where $\iota^k_{*,U,V} = (\iota^k_{*,U}, -\iota^k_{*,V})$ is the pull-back on compactly supported k-forms built out of the standard inclusion maps $\iota_U : U \cap V \to U$ and $\iota_V : U \cap V \to V$, while $+^k_{U,V} : \Omega^k_0(U) \oplus \Omega^k_0(V) \to \Omega^k_0(M)$ is realized by the sum of the k-forms.

Proof. Fix any $k \in \mathbb{N} \cup \{0\}$. Exactness of the sequence at $\Omega^k_0(M)$ is tantamount to proving that $+^k_{U,V}$ is surjective. Consider (ψ_U, ψ_V), a partition of unity subordinated to the cover of M built out of U, V. For any $\eta \in \Omega^k_0(M)$, let $\eta = \eta_U + \eta_V$ with $\eta_U \doteq \psi_U \eta$ and $\eta_V \doteq \psi_V \eta$. Both η_U and η_V are smooth and $\mathrm{supp}(\eta_U) \subseteq \mathrm{supp}(\eta) \cap \mathrm{supp}(\psi_U)$. This is a compact subset since it is the intersection between a closed ($\mathrm{supp}(\psi_U)$) and a bounded subset ($\mathrm{supp}(\eta_U)$). The same conclusion can be drawn for η_V. Exactness of the sequence at $\Omega^k_0(U \cap V)$ is tantamount to proving that $\iota^k_{*,U,V}$ is injective. This is a direct consequence of the definition since, for any $\tilde{\eta} \in \Omega^k_0(U \cap V)$, $\iota^k_{*,U,V}\tilde{\eta} = (\tilde{\eta}, -\tilde{\eta})$ with the convention that the form is extended to 0 outside $U \cap V$. Therefore $\iota^k_{*,U,V}\tilde{\eta} = 0$ if and only if $\tilde{\eta} = 0$. Finally exactness at $\Omega^k_0(U) \oplus \Omega^k_0(V)$ is true if $\ker(+^k_{U,V}) = \mathrm{Im}(\iota^k_{*,U,V})$. Yet, if $(\eta_U, \eta_V) \in \ker(+^k_{U,V})$ this entails that $\eta_U = 0$ on $U \setminus (U \cap V)$ while $\eta_V = 0$ on $V \setminus (U \cap V)$. At the same time, on $U \cap V$ it must hold $\eta_U = -\eta_V$. This is exactly the image of the map $\iota^k_{*,U,V}$ which concludes the proof. $\square$

The Mayer-Vietoris sequence of forms with compact support extends naturally to a counterpart between differential complexes, namely

$$0 \longleftarrow \Omega^\bullet_0(M) \xleftarrow{+_{U,V}} \Omega^\bullet_0(U) \oplus \Omega^\bullet_0(V) \xleftarrow{\iota_{*,U,V}} \Omega^\bullet_0(U \cap V) \longleftarrow 0\ , \tag{41}$$

where $+_{U,V} \doteq (+^0_{U,V}, \cdots +^n_{U,V})$ and similarly $\iota_{*,U,V} = (\iota^0_{*,U,V}, \ldots, \iota^n_{*,U,V})$. Observe that the differential of both $\Omega^\bullet_0(M)$ and $\Omega^\bullet_0(U \cap V)$ will be indicated as $\underline{d} = (d_0, \ldots, d_n)$ where each d_k is nothing but the differential acting on k-forms as per Theorem 67. On the contrary the differential of $\Omega^\bullet_0(U) \oplus \Omega^\bullet_0(V)$ is $\underline{d}_\oplus = (\underline{d}, \underline{d})$ where each entry acts on $\Omega^\bullet_0(U)$ and on $\Omega^\bullet_0(V)$ respectively. In addition Theorem 68 and the linearity of the differential acting on k-forms guarantees that both $+_{U,V}$ and $\iota*, U, V$ are chain maps, *cf.* Definition 17. By applying Proposition 18, for every $k \in \mathbb{N} \cup \{0\}$, there exists a corresponding sequence at the level of compactly supported de Rham cohomologies, *cf.* Definition 107:

$$H^k_{0,\mathrm{dR}}(M) \xleftarrow{\tilde{+}^k_{U,V}} H^k_{0,\mathrm{dR}}(U) \oplus H^k_{0,\mathrm{dR}}(V) \xleftarrow{\tilde{\iota}^k_{*,U,V}} H^k_{0,\mathrm{dR}}(U \cap V)\ ,$$

where $\tilde{+}^k_{U,V}$ and $\tilde{\iota}_{*,U,V}$ are the counterparts of $+^k_{U,V}$ and $\iota^k_{*,U,V}$, defined as per Proposition 18. As last step, on account of the Zig-Zag Lemma 19 we obtain the long exact sequence in cohomology:

$$\begin{tikzcd}
\cdots \arrow[r] & H^k_{0,\mathrm{dR}}(U\cap V) \arrow[r, "\tilde{+}^k_{U,V}"] & H^k_{0,\mathrm{dR}}(U)\oplus H^k_{\mathrm{dR}}(V) \arrow[r, "\tilde{\iota}^{*,k}_{U\cap V,-}"] & H^k_{0,\mathrm{dR}}(M) \arrow[dll, "\tilde{\delta}_k"'] \\
& H^{k+1}_{0,\mathrm{dR}}(U\cap V) \arrow[r, "\tilde{\iota}^{k+1}_{*,U,V}"] & H^{k+1}_{0,\mathrm{dR}}(U)\oplus H^{k+1}_{0,\mathrm{dR}}(V) \arrow[r, "\tilde{+}^{k+1}_{U,V}"] & H^{k+1}_{0,\mathrm{dR}}(M) \arrow[dll, "\tilde{\delta}_{k+1}"'] \\
& H^{k+2}_{0,\mathrm{dR}}(U\cap V) \arrow[r] & \cdots &
\end{tikzcd} \tag{42}$$

where $\tilde{\delta}_k : H^k_{0,\mathrm{dR}}(M) \to H^{k+1}_{\mathrm{dR}}(U\cap V))$ is the connecting map built in full analogy to the one in Lemma 19.

Before looking into specific examples, it is important to stress that compactly supported cohomology shares the same algebraic properties of $H^\bullet_{\mathrm{dR}}(M)$, for any smooth manifold M. The goal of the next exercise is to make this statement precise, *cf.* Exercise 43.

Exercise 46. Let M be a smooth manifold of dimension $\dim M = n \geq 1$. Prove that the graded vector space $H^\bullet_{0,\mathrm{dR}}(M) \doteq \bigoplus_{k=0}^{n} H^k_{0,\mathrm{dR}}(M)$ is an exterior algebra if endowed with the **cup product**

$$\smile : H^k_{0,\mathrm{dR}}(M) \times H^{k'}_{0,\mathrm{dR}}(M) \to H^{k+k'}_{0,\mathrm{dR}}(M), \quad ([\eta],[\eta']) \mapsto [\eta] \smile [\eta'] \doteq [\eta \wedge \eta'],$$

where $\wedge$ stands for the standard wedge product between forms on M, *cf.* Proposition 60. For this reason we refer to $H^\bullet_{0,\mathrm{dR}}(M)$ as the **de Rham cohomology ring of M with compact support**.

Let us now consider the simple case $M = \mathbb{R}^n$ and let us compute the compactly supported cohomology spaces. We start from $k = 0$. In absence of support restrictions Proposition 108 guarantees us that $H^0_{\mathrm{dR}}(M)$ is a measure of the number of connected components of the underlying manifold. It is immediate to see that this fails when considering forms of compact support.

Lemma 121. *Let $M = \mathbb{R}^n$, $n \geq 1$. Then $H^0_{0,\mathrm{dR}}(\mathbb{R}^n) = \{0\}$.*

Proof. Since $H^0_{0,\mathrm{dR}}(\mathbb{R}^n) = \Omega^0_{0,d}(\mathbb{R}^n)$, we look for $f \in C^\infty_0(\mathbb{R}^n)$ such that $d_0 f = 0$. In other words f is constant, hence not of compact support on $\mathbb{R}^n$. $\square$

The generalization to an arbitrary manifold of this fact is simple and therefore we leave it as an exercise to the reader

Exercise 47. Let M be a smooth manifold and split it as $M = M_0 \times M_1$, where M_0 and M_1 are disconnected components of M, where M_0 includes all the compact ones. Then $H^0_{0,\mathrm{dR}}(M) \simeq \mathbb{R}^{m_0}$ where $m_0 = \dim M_0$.

Let us now focus on higher cohomology spaces on $\mathbb{R}^n$ and let us start from the key building block:

Proposition 122. *There exists an isomorphism between $H^1_{0,\mathrm{dR}}(\mathbb{R})$ and $\mathbb{R}$ which is realized by the (integration) map*

$$\int_{\mathbb{R}} : H^1_{0,\mathrm{dR}}(\mathbb{R}) \to \mathbb{R}, \quad [\omega] \mapsto \int_{\mathbb{R}}([\omega]) \doteq \int_{\mathbb{R}} \omega.$$

Proof. First of all, observe that the map is well-defined, since for every $f \in C_0^\infty(\mathbb{R})$, it holds that

$$\int_{\mathbb{R}} df = f|_{-\infty}^{\infty} = 0,$$

since f has compact support. The map is clearly surjective since it suffices to consider any $h \in C_0^\infty(\mathbb{R})$ whose integral along the real lines is 1. This function identifies $[\omega] = [h(x)dx]$, x being the standard Euclidean coordinate, such that

$$\int_{\mathbb{R}}([\omega]) = \int_{\mathbb{R}} h(x)dx = 1.$$

By rescaling ω with any $\lambda \in \mathbb{R}$, it holds

$$\int_{\mathbb{R}}([\lambda\omega]) = \lambda.$$

We need to prove that the map is injective. Hence suppose that $[\omega] \in H^1_{0,\mathrm{dR}}(\mathbb{R})$ is such that $\int_{\mathbb{R}} \omega = 0$. Since ω has compact support, there must exist $f_\omega \in C_0^\infty(\mathbb{R})$ such that $\omega = f_\omega(x)dx$. Define

$$F(x) \doteq \int_{-\infty}^{x} f_\omega(y)dy.$$

By the fundamental theorem of calculus $F \in C^\infty(\mathbb{R})$ and $dF = \omega$. It remains to be seen that the support is compact. Let $R > 0$ be such that $\mathrm{supp}(f_\omega) \subset [-R, R]$. Then, for all $x < -R$, the integrand vanishes on $(-\infty, x)$ and, therefore $F(x) = 0$. Similarly, for all $x > R$, since the integrand vanishes on (R, ∞), $F(x) = \int_{\mathbb{R}} f_\omega(y)dy = \int_{\mathbb{R}} \omega = 0$. Hence $\mathrm{supp}(F) \subset (-R, R)$, thus it is bounded. Being closed per definition, it is also compact. □

Having computed the compact cohomology spaces on $\mathbb{R}$, one can generalize the result, using an induction mechanism. We leave the proof to the wisdom of our readers (I am getting cocky... I think not even that someone might read these notes, but, actually that there will more than one reader!)

Exercise 48. Let $n \geq 1$. Prove that, for any connected open subset $U \subseteq \mathbb{R}?n$

$$H^k_{0,\mathrm{dR}}(U) = \left\{ \begin{array}{cc} \{0\} & 0 \leq k < n \\ \mathbb{R} & k = n \end{array} \right. .$$

Remark 38. *It is worth observing that the key difference between $H^\bullet_{\mathrm{dR}}(M)$ and $H^\bullet_{0,\mathrm{dR}}(M)$ for an arbitrary smooth manifold M lies in the absence of a counterpart of Theorem 117 for the latter. As a matter of fact, Lemma 121 and Proposition 122 already make clear that compact cohomology spaces are not topological invariants. As a matter of fact $\mathbb{R}^n$ is homotopy equivalent to a single point $\{p\}$. This is a compact set and thus, $H^k_{0,\mathrm{dR}}(M)(\{p\}) = H^k_{\mathrm{dR}}(M)(\{p\}) = \{0\}$ for all $k \geq 1$, while $H^0_{0,\mathrm{dR}}(M)(\{p\}) = H^0_{\mathrm{dR}}(M)(\{p\}) \simeq \mathbb{R}$.*

To conclude the section, let us apply the idea behind the Mayer-Vietoris sequence to an explicit case in compact cohomology, namely the circle $\mathbb{S}^1$. Being a smooth, compact manifold, we expect a priori that the cohomology space with or without compact support must coincide.

Example 123. Let us consider the unit circle $\mathbb{S}^1 \doteq \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$. In order to follow the argument at the heart of Proposition 109, we consider the open subsets, considered in the two standard Mercatore charts, namely

$$U = \mathbb{S}^1 \setminus \{(0,1)\}, \quad V = \mathbb{S}^1 \setminus \{(0,-1)\}.$$

It is immediate to realize that $\mathbb{S}^1 = U \cup V$ and that $U \cap V$ is the disjoint union of two components, namely

$$U \cap V = L \cup R,$$
$$L \doteq \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1, \text{ and } x < 0\}, \quad R \doteq \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1, \text{ and } x > 0\}.$$

Since U, V, L, R are all diffeomorphic to open subsets of $\mathbb{R}$, we can apply Exercise 48 to infer that

$$H^0_{0,\mathrm{dR}}(U) = H^0_{0,\mathrm{dR}}(V) = H^0_{0,\mathrm{dR}}(U \cap V) = \{0\}.$$

At the same time Exercise 47 guarantees that $H^0_{0,\mathrm{dR}}(\mathbb{S}^1) \simeq \mathbb{R}$ while Exercise 48 yields $H^1_{0,\mathrm{dR}}(U) \oplus H^1_{0,\mathrm{dR}}(V) = H^1_{0,\mathrm{dR}}(U \cap V) \simeq \mathbb{R} \oplus \mathbb{R}$. The long sequence (**??**) ($k = 0$) reduces to

$$\cdots \longrightarrow 0 \xrightarrow{\tilde{+}^0_{U,V}} 0 \xrightarrow{\tilde{\iota}^{*,0}_{U\cap V,-}} \mathbb{R} \xrightarrow{\tilde{\delta}_0} \mathbb{R} \xrightarrow{\tilde{\iota}^1_{*,U,V}} \mathbb{R} \oplus \mathbb{R} \xrightarrow{\tilde{+}^1_{U,V}} H^1_{0,\mathrm{dR}}(\mathbb{S}^1) \xrightarrow{\tilde{\delta}_1} 0 \longrightarrow \cdots$$

To evaluate the first cohomology group of the circle, we exploit exactness of the sequence at $\mathbb{R} \oplus \mathbb{R}$. This guarantees us that $\mathrm{Im}(\tilde{+}^1_{U,V})$ is surjective. Hence it suffices to compute its kernel, which by exactness obeys to $\ker(\tilde{+}^1_{U,V}) = \mathrm{Im}(\tilde{\iota}^1_{*,U,V})$. By direct inspection of its definition in Equation 41, one can see that $\mathrm{Im}(\tilde{\iota}^1_{*,U,V})) \simeq \mathbb{R}$. Hence

$$H^1_{0,\mathrm{dR}}(\mathbb{S}^1) \simeq \frac{\mathbb{R} \oplus \mathbb{R}}{\mathbb{R}} \simeq \mathbb{R}.$$

Since all higher compactly supported cohomology groups of the circle must vanish for dimensional reasons, we have compute the whole compactly supported cohomology ring of $\mathbb{S}^1$.

The following is warm-up exercise

Exercise 49. Compute via the Mayer-Vietoris sequence the compactly supported cohomology ring of the n-sphere $\mathbb{S}^n$, $n \geq 1$.

4.3 The Mayer-Vietoris argument and Poincaré duality

We are left with two open questions, one prompted by the analysis of the last subsection, namely

1. Given a smooth manifold M are the associated cohomology spaces finite dimensional?
2. What is the concrete role played by compactly supported cohomology?

In this last part of the section, we shall answer to both questions. The first can be addressed by using the so-called **Mayer-Vietoris argument**. This does not apply in general but only to a specific class of smooth manifolds, here defined

Definition 124. *Let M be a manifold. We call it of* **finite type** *if it admits a* finite good cover*, that is there exists a cover of M made of open subsets $\{U_\alpha\}_{\alpha\in I}$, where I is a collection of indices, such that every non empty finite intersection $U_{\alpha_1} \cap \cdots \cap U_{\alpha_n}$, $\{\alpha_i\}_{i=1,\ldots,n} \in I$, is diffeomorphic to an open subset on $\mathbb{R}^n$, $n = \dim M$.*

Although, at first glance this might be a restrictive request, it is actually quite hard not to fulfill it. As a matter of fact all smooth manifolds are of finite type since they admit a Riemannian structure, see [11, Prop. 13.3] and, thus, convex geodesic neighbourhoods can be used to construct a finite good cover. We leave to the reader dotting the i's and crossing the t's.

Exercise 50. Let (M, g) be a smooth Riemannian manifold. Prove that there exists a finite good cover built of geodesically convex neighbourhoods. Recall that $U \subseteq M$ is called *geodesically convex* if, for any $p, q \in M$, there exists a unique minimizing geodesic joining p to q.

Theorem 125 (**Mayer-Vietoris Argument**)**.** *Let M be a manifold of finite type as per Definition 124. Then for all $k \in \mathbb{N} \cup \{0\}$, $\dim H^k_{\mathrm{dR}}(M) < \infty$.*

Proof. The proof proceeds by induction with respect to the number of open subsets in the cover of M. In particular if M is covered by a single open subset, then it is diffeomorphic to an open subset of $\mathbb{R}^n$, $n = \dim M$. On account of Exercise 3, we know that all cohomology spaces are finite dimensional.

Suppose now that the statement is true for an open cover of k subsets U_i, $i = 1, \ldots, k$. Consider now an additional one U_{k+1}. By construction the manifold $X \doteq (U_1 \cup \cdots \cup U_k) \cap U_{k+1}$ admits a finite good cover made of k elements. Therefore, by the induction hypothesis, X, U_{k+1} and $U_1 \cup \cdots \cup U_k$ have finite dimensional cohomology spaces. For all $p \in \mathbb{N} \cup \{0\}$, the long exact sequence (38) entails that

$$H^p(U_1 \cup \cdots \cup U_{k+1}) = \ker(\tilde{\iota}_{\oplus}^{*,p}) \oplus \mathrm{Im}(\tilde{\iota}_{\oplus}^{*,p}) = \mathrm{Im}(\tilde{\delta}_{p-1}) \oplus \mathrm{Im}(\tilde{\iota}_{\oplus}^{*,p}),$$

where $\tilde{\delta}_{p-1} : H^{p-1}_{\mathrm{dR}}(X) \to H^p_{\mathrm{dR}}(U_1 \cup \cdots \cup U_{k+1})$ while $\tilde{\iota}_{\oplus}^{*,p} : H^p_{\mathrm{dR}}(U_1 \cup \cdots \cup U_{k+1}) \to H^p_{\mathrm{dR}}(U_1 \cup \cdots \cup U_k) \oplus H^p_{\mathrm{dR}}(U_{k+1})$. Yet $\tilde{\delta}_{p-1}[H^{p-1}_{\mathrm{dR}}(X)]$ is finite dimensional being so $H^{p-1}_{\mathrm{dR}}(X)$, while the same conclusion can be drawn for $\mathrm{Im}(\tilde{\iota}_{\oplus}^{*,p})$ since the induction hypothesis guaranteed that $H^p_{\mathrm{dR}}(U_1 \cup \cdots \cup U_k) \oplus H^p_{\mathrm{dR}}(U_{k+1})$ is finite dimensional. □

A similar reasoning can be applied for the compactly supported cohomology spaces:

Exercise 51. Let M be a manifold of finite type as per Definition 124. Then for all $k \in \mathbb{N} \cup \{0\}$, $\dim H^k_{0,\mathrm{dR}}(M) < \infty$.

We have addressed the first of the two questions, we asked ourselves at the beginning of the section. Actually Theorem 125 plays a key role in answering also the second question. First we need to recall a basic notion of linear algebra, which I fear that my beloved readers might not remember.

Definition 126. *Let V, W be two real vector spaces and let $\langle , \rangle : V \times W \to \mathbb{R}$ be a pairing. We call it (weakly non degenerate) if*

$$\langle v, w \rangle = 0 \quad \forall w \in W \Longrightarrow v = 0 \quad \text{and} \quad \langle v, w \rangle = 0 \quad \forall v \in V \Longrightarrow w = 0$$

We call it strongly non degenerate if the maps

$$V \ni v \mapsto \langle v, \cdot \rangle \in W^* \quad \text{and} \quad W \ni w \mapsto \langle \cdot, w \rangle \in V^*,$$

are isomorphisms of vector spaces.

Corollary 127. *If V and W are finite dimensional vector spaces, a pairing is weakly non degenerate if and only if it is strongly non degenerate.*

Proof. We need to show only the arrow $\Longrightarrow$ since the converse statement is a consequence of the maps $v \mapsto \langle v, \cdot \rangle$ and $w \mapsto \langle \cdot, w \rangle$ being injective. Suppose that $\langle , \rangle$ is weakly non degenerate. This entails that the map $V \ni v \mapsto \langle v, \cdot \rangle \in W^*$ is injective. Hence, up to an isomorphism, $V \subseteq W^*$, but similarly $W \subseteq V^*$. Yet since $\dim V = \dim V^*$ and $\dim W = \dim W^*$, the only possibility is $\dim V = \dim W^*$ which concludes the proof. □

The strategy to understand the usefulness of compact cohomology classes starts from the observation that cohomology spaces form a ring as per Exercise 43. This entails that, if we assume that the underlying manifold M is oriented and with empty boundary, we can introduce the following map:

$$\int : H^k_{\mathrm{dR}}(M) \otimes H^{n-k}_{0,\mathrm{dR}}(M) \to \mathbb{R}, \quad [\omega] \otimes [\eta] \mapsto \int([\omega] \otimes [\eta]) \doteq \int_M \omega \wedge \eta, \tag{43}$$

where $n = \dim M$ and where the integral is defined as per Theorem 78. Notice that Equation (43) is well-defined, namely there is no dependence on the chosen representative, since, for every $\alpha \in \Omega^{k-1}(M)$

$$\int_M d_{k-1}\alpha \wedge \eta = \int_M d_{n-1}(\alpha \wedge \eta) = 0,$$

where we used Stokes Theorem 79 and $d_{n-k}\eta = 0$. A similar result holds true considering in place of η, $d_{n-k-1}\rho$ with $\rho \in \Omega_0^{n-k-1}(M)$.

It is immediate to realize that $\int$ realizes a map from $H^k_{\mathrm{dR}}(M)$ to $\left(H^{n-k}_{0,\mathrm{dR}}(M)\right)^*$ and similarly from $H^{n-k}_{0,\mathrm{dR}}(M)$ to $\left(H^k_{\mathrm{dR}}(M)\right)^*$. The question is whether we can establish more than a simple inclusion between dual spaces. This is indeed the case, but to prove it we need two preparatory results, one left to our beloved readers.

Exercise 52. Prove the **Five lemma**, namely consider the following commutative diagram of vector spaces (or more generally of Abelian groups)

$$\begin{tikzcd}
\dots \arrow[r] & A_1 \arrow[r, "f_1"] \arrow[d, "\alpha"] & B_1 \arrow[r, "f_2"] \arrow[d, "\beta"] & C_1 \arrow[r, "f_3"] \arrow[d, "\gamma"] & D_1 \arrow[r, "f_4"] \arrow[d, "\delta"] & E_1 \arrow[r] \arrow[d, "\epsilon"] & \dots \\
\dots \arrow[r] & A_2 \arrow[r, "h_1"] & B_2 \arrow[r, "h_2"] & C_2 \arrow[r, "f_3"] & D_2 \arrow[r, "f_4"] & E_2 \arrow[r] & \dots
\end{tikzcd}$$

and assume that each row is exact. Then if $\alpha, \beta, \delta, \epsilon$ are isomorphisms, so is γ.

The second one states the following

Lemma 128. *For any $k \in \mathbb{N} \cup \{0\}$, the following two sequences can be paired to form a commutative diagram*

$$\begin{tikzcd}
\dots \arrow[r] & H^k_{\mathrm{dR}}(M) \arrow[r, "\tilde{\iota}^{*,k}_{\oplus}"] & H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V) \arrow[r, "\tilde{\iota}^{*,k}_{U\cap V,-}"] & H^k_{\mathrm{dR}}(U \cap V) \arrow[r, "\tilde{d}_k"] & H^{k+1}_{\mathrm{dR}}(M) \arrow[r] & \dots, \\
& \otimes & \otimes & \otimes & \otimes & \\
\dots & H^{n-k}_{0,\mathrm{dR}}(M) \arrow[d, "\int_M"] & H^{n-k}_{0,\mathrm{dR}}(U) \oplus H^{n-k}_{0,\mathrm{dR}}(V) \arrow[l, "\tilde{+}^{n-k}_{U,V}"'] \arrow[d, "\int_U + \int_V"] & H^{n-k}_{0,\mathrm{dR}}(U \cap V) \arrow[l, "\tilde{\iota}^{*,k}_{U\cap V,-}"'] \arrow[d, "\int_{U\cap V}"] & H^{n-k-1}_{0,\mathrm{dR}}(M) \arrow[l, "\tilde{\delta}_{n-k}"'] \arrow[d, "\int_M"] & \dots \arrow[l] \\
& \mathbb{R} & \mathbb{R} & \mathbb{R} & \mathbb{R} &
\end{tikzcd}$$

where M is a smooth manifold such that $M = U \cup V$ and where all the maps are defined as in the sequences (38) and (42).

Remark 39. *The statement of the lemma, should be interpreted as follows: For any $\omega \in H^k_{\mathrm{dR}}(U \cap V)$ and for any $\eta \in H^{n-k-1}_{0,\mathrm{dR}}(M)$, it holds*

$$\int_{U\cap V} \tilde{d}_k(\omega) \wedge \eta = (-)^{k+1} \int_M \omega \wedge \tilde{\delta}_{n-k}(\eta),$$

where the sign dependence in the equality is often emphasized by referring to it as sign commutativity*.*

Proof. This is a direct application of the explicit expression for the connecting map built in Lemma 19 for the sequences (38) and (42). In particular calling $\{\psi_U, \psi_V\}$ a partition of unity subordinated to U, V, it holds that for any $[\omega] \in H^k_{\mathrm{dR}}(U \cap V)$

$$[\tilde{d}_k(\omega)]|_U = [d_k(\psi_U \omega)], \quad [\tilde{d}_k(\omega)]|_V = [-d_k(\psi_V \omega)].$$

Similarly, for any $[\eta] \in H^{n-k-1}_{0,\mathrm{dR}}(M)$, it holds that $[\tilde{\delta}_{n-k}(\eta)]$ can be realized in $H^{n-k}_{0,\mathrm{dR}}(U) \oplus H^{n-k}_{0,\mathrm{dR}}(V)$ as $([d_{n-k-1}(\psi_U \eta)], [d_{n-k-1}(\psi_V \eta)])$. Observing that, since every representative η is closed, it holds $[d_{n-k-1}(\psi_U \eta)] = [d\psi_U \wedge \eta]$ and similarly $[d_{n-k-1}(\psi_V \eta)] = [d\psi_V \wedge \eta]$, $[d_k(\psi_U \omega)] = [d\psi_U \wedge \omega]$, while $[-d_k(\psi_V \omega)] = [-d\psi_V \wedge \omega]$. By using these identities, it holds that

$$\int_M \omega \wedge \tilde{\delta}_{n-k}(\eta) = \int_{U\cap V} \omega \wedge d\psi_V \wedge \eta = (-)^k \int_{U\cap V} d\psi_V \wedge \omega \wedge \eta.$$

Similarly since $\mathrm{supp}(\tilde{d}_k(\omega)) \subset U \cap V$,

$$\int_M \tilde{d}_k(\omega) \wedge \eta = - \int_{U\cap V} d\psi_V \wedge \omega \wedge \eta,$$

from which we can conclude

$$\int_{U\cap V} \tilde{d}_k(\omega) \wedge \eta = (-)^{k+1} \int_M \omega \wedge \tilde{\delta}_{n-k}(\eta),$$

that is sign-commutativity of the third diagram. Notice that sign-commutativity of the first two is a consequence of linearity of integrals and of the support properties of the integrands. ☐

Now we can prove the main theorem of this last part of the section

Theorem 129 (Poincaré Duality). *Let M be a smooth, oriented and connected manifold of dimension $\dim M = n \geq 1$. Then, for every $k \in \mathbb{N} \cup \{0\}$, there exists an isomorphism*

$$H^k_{\mathrm{dR}}(M) \simeq [H^{n-k}_{0,\mathrm{dR}}(M)]^*,$$

which is realized by the following map

$$I_k : H^k_{\mathrm{dR}}(M) \to [H^{n-k}_{0,\mathrm{dR}}(M)]^*, \quad [\omega] \mapsto I_k([\omega]) \text{ such that } I_k([\omega])([\eta]) = \int\limits_M \omega \wedge \eta.$$

Proof. As starting point, consider $U \subseteq \mathbb{R}^n$. In this case the existence of an isomorphism is a direct consequence of Exercise 4 and 48. In addition observe that $\{I_k\}_{k=0,\dots,n}$ realizes the isomorphism since it is a linear map and if $k = 0$, then $[\omega] \in H^0_{\mathrm{dR}}(U)$ entails that ω is the constant function. Hence I_0 is automatically injective and thus an isomorphism since $\dim H^0_{\mathrm{dR}}(U) = \dim H^n_{0,\mathrm{dR}}(U) = 1$.

Now, the proof can proceed by induction as in Theorem 125. As first step, we observe that the Five Lemma in Exercise 52, combined with Lemma 128, entails that, given two open subsets of a smooth manifold, say U and V, if Poincaré duality holds for U, V and $U \cap V$, then it holds also for $U \cup V$.

In addition, recall that, since M is smooth, it admits a finite good cover. At this point, suppose the statement is true for an open cover of k subsets U_i, $i = 1, \dots, k$. By construction the manifold $X \doteq (U_1 \cup \dots \cup U_k) \cap U_{k+1}$ admits a finite good cover made of k elements. Therefore, by the induction hypothesis, X, U_{k+1} and $U_1 \cup \dots \cup U_k$ obey to Poincaré duality. As noticed above this entails that $U_1 \cup \dots \cup U_{k+1}$ enjoys the sought duality and this concludes the proof. □

The final result of this whole section is a very useful formula to compute the cohomology of a manifold which is diffeomorphic to a Cartesian product of two lower dimensional manifolds.

Theorem 130 (Künneth Formula). *Let M and K be two smooth manifolds. Then $H^\bullet_{\mathrm{dR}}(M \times K) \simeq H^\bullet_{\mathrm{dR}}(M) \otimes H^\bullet_{\mathrm{dR}}(K)$, namely, for every $k \in \mathbb{N} \cup \{0\}$,*

$$H^k(M \times K) \simeq \bigoplus_{j=0}^{k} H^j(M) \otimes H^{k-j}(F).$$

Proof. Since both M and K are smooth, they have a finite good cover. Hence consider the smooth manifold $M \times F$ together with the natural projection $\pi_1 \in C^\infty(M \times F; M)$ and $\pi_2 \in C^\infty(M \times F; F)$. For all $k, k' \in \mathbb{N} \cup \{0\}$, these induce a map

$$\pi_1^* \otimes \pi_2^* : \Omega^k(M) \otimes \Omega^{k'}(F) \to \Omega^{k+k'}(M \times F), \quad \omega \otimes \rho \mapsto (\pi_1^* \otimes \pi_2^*)(\omega \otimes \rho) = \pi_1^*\omega \wedge \pi_2^*\rho.$$

Since the differential commutes with the pull-back, *cf.* Proposition 68, $\pi_1^* \otimes \pi_2^*$ descends to a well-defined map between cohomology spaces, namely

$$\Psi_{k,k'} : H^k_{\mathrm{dR}}(M) \otimes H^{k'}_{\mathrm{dR}}(F) \to H^{k+k'}_{\mathrm{dR}}(M \times F), \quad [\omega] \otimes [\rho] \mapsto \Psi_{k,k'}([\omega] \otimes [\rho]) = [\pi_1^*\omega \wedge \pi_2^*\rho],$$

which in turn identifies a well-defined application between the cohomology rings, namely

$$\underline{\Psi} : H^\bullet_{\mathrm{dR}}(M) \otimes H^\bullet_{\mathrm{dR}}(F) \to H^\bullet_{\mathrm{dR}}(M \times F).$$

We shall prove that $\underline{\Psi}$ is an isomorphism. To start with consider two open subsets $U, V \subset M$ such that $M = U \cup V$. Consider the associated Maier-Vietoris sequence and the induced counterpart in cohomology (38). Then for any $k \in \mathbb{N} \cup \{0\}$, considering any integer $n \geq 0$ and taking the tensor product with $H^{n-k}_{\mathrm{dR}}(F)$, we obtain the following long exact sequence:

$$\ldots \longrightarrow H^k_{\mathrm{dR}}(M) \otimes H^{n-k}_{\mathrm{dR}}(F) \xrightarrow{\tilde{\iota}^{*,k}_{\oplus} \otimes \mathrm{id}|_F} \left(H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V)\right) \otimes H^{n-k}_{\mathrm{dR}}(F) \xrightarrow{\tilde{\iota}^{*,k}_{U\cap V,-} \otimes \mathrm{id}|_F}$$

$$\longrightarrow H^k_{\mathrm{dR}}(U \cap V) \otimes H^{n-k}_{\mathrm{dR}}(F) \longrightarrow \ldots$$

Since the sequence is exact for all k, we can take direct sum over this integer obtaining another long exact sequence (we omit for notation simplicity the maps on the arrows)

$$\ldots \longrightarrow \bigoplus_{k=0}^{n} H^k_{\mathrm{dR}}(M) \otimes H^{n-k}_{\mathrm{dR}}(F) \longrightarrow \bigoplus_{k=0}^{n} \left(H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V)\right) \otimes H^{n-k}_{\mathrm{dR}}(F) \longrightarrow$$

$$\longrightarrow \bigoplus_{k=0}^{n} H^k_{\mathrm{dR}}(U \cap V) \otimes H^{n-k}_{\mathrm{dR}}(F) \longrightarrow \ldots$$

By inserting the map $\underline{\Psi}$ and adopting the notation $H^k_{\mathrm{dR}}(U, V) \equiv H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V)$, we obtain a commutative diagram:

$$\begin{tikzcd}
\bigoplus_{k=0}^{n} H^k_{\mathrm{dR}}(M) \otimes H^{n-k}_{\mathrm{dR}}(F) \arrow[r] \arrow[d, "\underline{\Psi}"] & \bigoplus_{k=0}^{n} H^k_{\mathrm{dR}}(U, V) \otimes H^{n-k}_{\mathrm{dR}}(F) \arrow[r] \arrow[d, "\underline{\Psi}"] & \bigoplus_{k=0}^{n} H^k_{\mathrm{dR}}(U \cap V) \otimes H^{n-k}_{\mathrm{dR}}(F) \arrow[d, "\underline{\Psi}"] \\
H^n_{\mathrm{dR}}(M \times F) \arrow[r] & H^n_{\mathrm{dR}}(U \times F) \oplus H^n_{\mathrm{dR}}(V \times F) \arrow[r] & H^n_{\mathrm{dR}}((U \cap V) \times F)
\end{tikzcd}$$

This translates to a commutative diagram between the long exact sequences. At this stage the proof proceeds by induction as for Poincaré duality. We omit the details since mutatis mutandis the reader can follow Theorem 129.

$\square$

5 Singular Homology and Characteristic Classes

In the previous section we have discussed de Rham cohomology, which plays a natural role in Maxwell's equations catching information on the topological aspects of the theory. Yet, an attentive (very attentive) reader might have noticed that, although the de Rham cohomology ring contains information on the topology of the underlying manifold, it is not a sufficient tool to distinguish between different backgrounds. As a simple example consider the Möbius strip $\mathcal{M}_2$, *cf.* Example 23 and the 2-torus $\mathbb{T}^2 \simeq \mathbb{S}^1 \times \mathbb{S}^1$. It is rather simple to prove that $H^\bullet_{\mathrm{dR}}(\mathcal{M}_2) \simeq H^\bullet_{\mathrm{dR}}(\mathbb{T}^2)$. In other words, de Rham cohomology is not able to catch all topological information and, therefore, one needs more refined tools for this purpose. In the following we will discuss some alternative constructions. Yet, with the idea in the back of our mind that we want to consider smooth manifolds, many of these new structures turn out to be equivalent, since they are particularly effective in describing more singular structures. Nonetheless they are still very informative in the cases of interest to us and, more importantly, they are the building block to discuss a new concept, namely characteristic classes. These play a leading role in classifying isomorphism classes of principal or of smooth vector bundles on a manifold. This allows us to answer one of the open questions we asked ourselves in the previous sections, namely, given a base space M and a typical fiber, say either a Lie group G or a finite dimensional real/complex vector space V, how many non equivalent principal/vector bundles can I construct starting from these data.

5.1 Simplicial Homology and Cohomology

In this section we discuss a structure which has been introduced as an alternative tool to homotopy groups in order to probe the topological structure of an underlying space. In the whole section our main reference will be a great classic, namely [6], which is probably the standard textbook to learn algebraic topology. For some selected topics which will be mentioned expressly, we recommend also [12] and [14].

The first approach, that we discuss, is certainly the oldest and the most intuitive. As far as the authors know, its first formulation dates back to Riemann and it is based on the intuitive notion that any sufficiently regular space, say a topological if not even a smooth manifold, can be approximated or rather decomposed by means of simpler objects, namely triangles, tetrahedra and so on and so forth. The complete formalization is due to Clifford in 1886 (yes... you are still studying 150 years old math... why? because you need to know it to understand the brand new results!)

Definition 131. *Let $\sigma = \{v_i\}_{i=0,\dots,k}$ be a collection of points in $\mathbb{R}^N$, $k, N \geq 1$ which are in* **general position** *that is all vectors $v_{i0} \doteq v_i - v_0$, $i = 1, \dots, k$ are linearly independent. We call* **k-simplex** *the set*

$$|\sigma| = \left\{ a_0 v_0 + \cdots + a_k v_k, \;\; a_i \geq 0 \text{ and } \sum_{i=1}^{k} a_i = 1 \right\}.$$

Each v_i, $i = 0, \dots, k$ is called **vertex** *of the simplex and k is its dimension. The collection of real numbers $\{a_i\}_{i=0,\dots,k}$ are called* **barycentric coordinates***. In addition any (not necessarily proper) subset $\tau \subseteq \sigma$ generates $|\tau|$, a subsimplex called* **face** *of $|\sigma$.*

Disclaimer: A k-simplex $|\sigma|$ is the smallest convex set containing the generating points $v_i \in \mathbb{R}^n$, $i = 0, \dots, n$. To emphasize their role it is customary also to adopt the notation

$$|\sigma| \equiv |v_0 v_1 \dots v_k| \equiv [v_0, \dots, v_k].$$

In the text we will try to stick to the notation of Definition 131, but, if needed, we might hop between the different possibilities.

Having introduced k-simplices as k-dimensional polytopes built as the convex hull of their vertices, we can ask ourselves how to glue them coherently. Note that in Definition 131 the vectors $\{v_i\}$, $i = 0, \dots, k$ are arbitrary provided that they are in general position. One canonical way to select ad admissible $(k+1)$-tuple consists of considering the standard basis elements $e_i \in \mathbb{R}^{k+1}$, $i = 0, \dots, k$. In this way we construct the so-called *standard simplex*.

Definition 132. *An* **(Euclidean) simplicial complex** *is a collection K of simplices in $\mathbb{R}^N$, $N \geq 1$ such that*

1. *given any $|\sigma| \in K$ also its faces belong to K,*
2. *if $|\sigma|, |\sigma'| \in K$, then either $|\sigma| \cap |\sigma'| = \emptyset$ or it is a face of both $|\sigma|$ and $|\sigma'|$,*
3. *for any $|\sigma| \in K$ and for any $x \in |\sigma|$, there exists $U \subseteq \mathbb{R}^N$ containing x and intersecting only a finite number of simplices in K (***local finiteness***).*

A simplicial complex K is called of dimension k (or k-complex) if it contains no simplex of dimension greater than k. In addition a simplicial k-complex is called **homogeneous (or pure)** *if every simplex of dimension smaller than k is the fact of a k-simplex. Finally we denote with $|K|$ the union of all simplices of K and its realization as a subset of $\mathbb{R}^N$ is called* **polyhedron***.*

Observe that the assumption of local finiteness in Definition 132 is not always present and it depends on the approach used. Here we followed [14] and [12], while in [6] it is not present. In all examples and applications that we have in mind, local finiteness is fulfilled and, since it simplifies some technical proofs, we prefer assuming it. The reader(s) who love generality (and I know some of you who do!) should keep in mind that one can get rid of local finiteness. At this stage we can ask ourselves two key questions:

1. what information is really encoded in a simplicial complex?
2. how can we use it to extract information from a topological/smooth manifold?

We start from the first one and the attentive reader should already have noticed that the word complex in Definition 132 is very much reminiscent of Definition 13. The analogy is not just at a semantic level as we shall discuss now. First we need a preliminary definition.

Definition 133. *Let σ be a k-simplex generated by $\{v_i\}_{i=0,\dots,k}$. We call* **orientation** *of σ an equivalence class of orderings of the elements of $\{v_i\}_{i=0,\dots,k}$ where two orderings $(v_{a_0}, \dots, v_{a_k})$ and $(v_{b_0}, \dots, v_{b_k})$ are said to be in correspondence if the $(k+1)$-tuples $(a_0, \dots, a_k)$ and $(b_0, \dots, b_k)$ are connected by an even permutation. An oriented simplex will be denoted as $\langle\sigma\rangle \equiv \langle v_0, \dots v_k\rangle$. If all simplices of a simplicial complex K are endowed with an orientation, we call K an* **oriented simplicial complex***.*

It is immediate to observe that there exists two possible orientations for a given simplex $|\sigma|$ and, chosen one, the opposite is usually denoted as $-\langle\sigma\rangle$. Having in mind Definition 13 the next obstruction that we face is the lack of a linear structure in a simplicial complex. This can be easily resolved but the price to pay is that we have to give up working with vector spaces. At this stage this might look like a folly, but, in the long run, it will pay off. In the following we introduce a standard concept in algebra, which some of the readers might not be familiar with.

Definition 134. *Let S be a set of possibly infinite element. We call* **free Abelian group** $\mathbb{Z}(S)$ *the formal $\mathbb{Z}$-linear combinations of elements of S.*

This definition allows for an abstract construction of an Abelian group starting from a set of elements. It falls short from being a vector space since only integer coefficients are allowed, but it is not difficult to observe that, given any ring K, *e.g.* $K = \mathbb{R}$ or $\mathbb{C}$,

$$\mathbb{V}_K(S) \doteq \mathbb{Z}(S) \otimes K,$$

is a K-vector space. Here $\otimes$ is a generalization of the standard notion of tensor product of vector spaces which applies to Abelian groups, see [6, pg. 216]. Inspired by Definition 134 we can introduce the following structures.

Definition 135. *Let K be an oriented simplicial complex. We call $C_l(K)$, $l \in \mathbb{N} \cup \{0\}$, the free Abelian group generated by all l-simplices $\langle\sigma\rangle \in K$. Each element of $C_l(K)$ is called an* **l-chain***.*

The importance of the space of l-chains is related to the existence of a natural map, called the **boundary operator (map)** defined as

$$\partial_l : C_l(K) \to C_{l-1}(K), \quad \langle v_0, \ldots, v_k\rangle \mapsto \partial_l\langle v_0, \ldots, v_k\rangle = \sum_{i=0}^{k}(-1)^i\langle v_0, \ldots, \widehat{v}_i, \ldots, v_k\rangle, \tag{44}$$

where the hat symbol entails that the vector below is omitted. The following simple property plays a key role.

Corollary 136. *Let K be an oriented simplicial complex. Then, for any $l \in \mathbb{N} \cup \{0\}$ it holds*

$$\partial_{l-1} \circ \partial_l = 0.$$

Proof. For any $\langle v_0, \ldots, v_k\rangle \in C_l(K)$ it holds applying twice Equation (44)

$$(\partial_{l-1} \circ \partial_l)(\langle v_0, \ldots, v_k\rangle) =$$
$$\sum_{j=0}^{i-1}\sum_{i=0}^{k}(-1)^{i+j}\langle v_0, \ldots, \widehat{v}_i, \ldots, \widehat{v}_j, \ldots v_k\rangle + \sum_{j=i+1}^{k}\sum_{i=0}^{k}(-1)^{i+j-1}\langle v_0, \ldots, \widehat{v}_i, \ldots, \widehat{v}_j, \ldots v_k\rangle = 0,$$

where the last equality is a consequence of the fact that each term in the sum appears twice with opposite signs. $\square$

If we compare the data gathered with Definition 13, we can readily observe that something is going amiss in the sense that the boundary operators are lowering the index from l to $l-1$ rather than raising it like the differential operator. This is the source of the key difference between homology and cohomology. Since the reader is most likely not familiar with some of the very basic algebraic structure that we need, we shall start from scratch although this shall force us into more definitions.

Definition 137. *Let G, H be two Abelian group. We call* **direct sum** *of G and H, the Abelian group $G \oplus H$ which is the set of all pairs $(g, h) \in G \times H$ endowed with the group structure completely specified by*

$$(g, h) \cdot (g', h') = (g \cdot_G g', h \cdot_H h'), \quad \forall (g, h), (g', h') \in G \times H$$

where $\cdot_G$ and $\cdot_H$ indicate respectively the group structures on G and on H.

In full analogy to Definition 2,

Definition 138. *Let $G^\bullet$ be an Abelian group. We call it* **($\mathbb{Z}$-)graded** *if there exists a family G_i, $i \in \mathbb{Z}$, of Abelian group such that*

$$G^\bullet \equiv G_\bullet = \bigoplus_{i\in\mathbb{Z}} G_i,$$

where the direct sum is defined as in Definition 137.

Disclaimer: Observe that we have conventionally defined the grading with respect to the integers as labels. One could limit the attention to $\mathbb{N}$ or to $\mathbb{N} \cup \{0\}$ without any substantial difference. Henceforth we will not specify explicitly the set of labels, although, we consider mainly $\mathbb{N} \cup \{0\}$.

Finally we can endow a graded Abelian group with an additional structure, reminiscent of Definition 13.

Definition 139. *Let $G_\bullet$ be a graded Abelian group as per Definition 138. We call it a* **chain complex** *if there exists a* **boundary operator (map)** *$\partial_\bullet : G_\bullet \to G_\bullet$ such that $\partial_\bullet = \{\partial_l\}_{l\in\mathbb{Z}}$ where $\partial_l \in \mathrm{Hom}(G_l, G_{l-1})$*

$$\ldots \xleftarrow{\partial_{l-1}} G_{l-1} \xleftarrow{\partial_l} G_l \xleftarrow{\partial_{l+1}} G_{l+1} \xleftarrow{\partial_{l+2}} \ldots \quad ,$$

is subject to the constraint $\partial_{l+1} \circ \partial_l = 0$ for all $l \in \mathbb{N} \cup \{0\}$.

A natural application of these concepts concerns simplicial complexes.

Proposition 140. *Let K be an oriented simplicial complex. Then the associated l-chains form a chain complex $(C_\bullet(K), \partial_\bullet)$ where $C_\bullet(K) \doteq \bigoplus_{l\in\mathbb{Z}} C_l(K)$ while $\partial_\bullet \doteq \{\partial_l\}_{l\in\mathbb{Z}}$ acts on $C_\bullet$ component by component with the boundary operators ∂_l as in Equation (44). Here we set $C_l(K) = \{0\}$ and $\partial_l = 0$ if $l < 0$.*

Proof. On account of Definition 135, each $C_l(K)$ is an Abelian group per construction and, thus $C^\bullet(K)$ is a graded Abelian group, *cf.* Definition 138. Since each boundary operator is linear per definition, *cf.* Equation (44), it identifies a group homomorphism. Corollary 136 concludes the proof. □

In full analogy to Section 1.2, it is natural to introduce two distinguished families of simplices associated to the l-chains of an oriented simplicial complex K, namely, for every $l \in \mathbb{N} \cup \{0\}$, we set

- $Z_l(K) \doteq \{\sigma \in C_l(K) \mid \partial_l \sigma = 0\}$ where ∂_l is the boundary operator as in Equation (44). The elements of $Z_l(K)$ are called **(l-dimensional) cycles** of K,
- $B_l(K) \doteq \partial_{l+1}[C_{l+1}(K)]$ whose elements are **(l-dimensional) boundary cycles** of K.

Definition 141. *Let K be a simplicial complex. For every $l \in \mathbb{N} \cup \{0\}$, we call* **l-th homology group**

$$H_l(K) = \frac{Z_l(K)}{B_l(K)} = \frac{\ker(\partial_l)}{\mathrm{Im}(\partial_{l+1})},$$

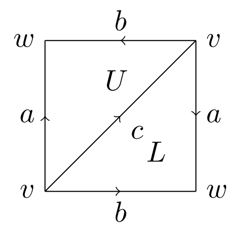
where ∂_l is the boundary operator map as in Equation (44).

Remark 40. *Definition 141 introduces a new structure, the homology groups, which have a clear resemblance with the de Rham cohomology group as in Definition 15 or in Definition 107 with the main difference that the relevant maps have an opposite direction since the boundary operator lowers the grading while the differential raises it in the relevant complex.*

From a mere mathematical viewpoint, homology groups are always coming "first" in the sense that they are naturally defined via basic geometric structures, while cohomology groups are derived from an homological counterpart. The example of de Rham cohomology groups is exceptional in two ways. On the one hand it has an immediate physical significance due to its connection with Maxwell's equations, making it easy to justify to an audience like you trained in theoretical physics. On the other hand it is based on the differential operator which is on its own a natural structure both on $\mathbb{R}^n$ and on an arbitrary smooth manifold M. Yet, notice that the space on which exterior differentiation acts is that of k-forms which in turn is defined out of smooth sections of T^*M and of its tensor powers. The cotangent bundle is naturally defined as a derived structure being the dual vector bundle built out of TM. In the following we will discuss how cohomology group arise as derived structures from an homological counterpart and how this philosophy applies also to the de Rham cohomology groups.

After this motivational remark, the author, who has absolute power in deciding what it is written here, thinks that it is important to make a few examples showing why we claimed that evaluating homology groups associated to a simplicial complex is computationally easy.

Example 142. Consider the following simplicial complex K



It is composed by the following constituents:

1. the oriented 2-simplices $U = \langle v, v, w\rangle$ and $L = \langle v, v, w\rangle$,
2. the oriented 1-simplices $a = \langle v, w\rangle$, $b = \langle v, w\rangle$ and $c = \langle v, v\rangle$,
3. the 0-simplices v and w.

In order to compute the homology group, it is immediate to infer for dimensional reasons that $H_l(K) = \{0\}$ if $l > 2$. At the same time by definition it holds that the boundary cycles are

$$B_2(K) = \{0\}, \quad B_1(K) = \mathbb{Z}2c \oplus \mathbb{Z}(a - b + c), \quad B_0(K) = \mathbb{Z}\{w - v\},$$

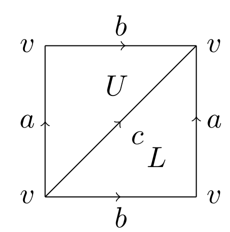
while the cycles are

$$Z_2(K) = \{0\}, \quad Z_1(K) = \mathbb{Z}(a - b + c) \oplus \mathbb{Z}c, \quad Z_0(K) = \mathbb{Z}v \oplus \mathbb{Z}w.$$

Applying Definition 141 we obtain

$$H_2(K) = \{0\}, \quad H_1(K) \simeq \frac{\mathbb{Z}2c}{\mathbb{Z}c} \simeq \mathbb{Z}_2, \quad H_0(K) \simeq \mathbb{Z}.$$

Example 143. Consider the following simplicial complex K



It is composed by the following constituents:

1. the oriented 2-simplices $U = \langle v, v, v\rangle$ and $L = \langle v, v, v\rangle$,
2. the oriented 1-simplices $a = \langle v, v\rangle$, $b = \langle v, v\rangle$ and $c = \langle v, v\rangle$,
3. the 0-simplex v.

In order to compute the homology group, it is immediate to infer for dimensional reasons that $H_l(K) = \{0\}$ if $l > 2$. At the same time by definition it holds that the boundary cycles are

$$B_2(K) = \{0\}, \quad B_1(K) = \mathbb{Z}(b + a - c), \quad B_0(K) = \{0\},$$

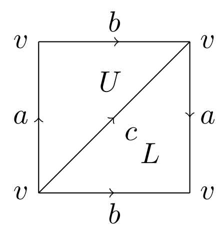
while the cycles are

$$Z_2(K) = \{0\}, \quad Z_1(K) = \mathbb{Z}a \oplus \mathbb{Z}b \oplus \mathbb{Z}c, \quad Z_0(K) = \mathbb{Z}v.$$

Applying Definition 141 we obtain

$$H_2(K) = \{0\}, \quad H_1(K) = \frac{\mathbb{Z}a \oplus \mathbb{Z}b \oplus \mathbb{Z}c}{\mathbb{Z}(b + a - c)} \simeq \mathbb{Z} \oplus \mathbb{Z}, \quad H_0(K) \simeq \mathbb{Z}.$$

Example 144. Consider the following simplicial complex K



It is composed by the following constituents (notice the opposite orientation of one of the edges in comparison to the complex in Example (143):

1. the oriented 2-simplices $U = \langle v, v, v \rangle$ and $L = \langle v, v, v \rangle$,
2. the oriented 1-simplices $a = \langle v, v \rangle$, $b = \langle v, v \rangle$ and $c = \langle v, v \rangle$,
3. the 0-simplex v.

In order to compute the homology group, it is immediate to infer for dimensional reasons that $H_l(K) = \{0\}$ if $l > 2$. At the same time by definition it holds that the boundary cycles are

$$B_2(K) = \{0\}, \quad B_1(K) = \mathbb{Z}(b + a - c) \oplus \mathbb{Z}(2a), \quad B_0(K) = \{0\},$$

while the cycles are

$$Z_2(K) = \{0\}, \quad Z_1(K) = \mathbb{Z}a \oplus \mathbb{Z}b \oplus \mathbb{Z}c, \quad Z_0(K) = \mathbb{Z}v.$$

Applying Definition 141 we obtain

$$H_2(K) = \{0\}, \quad H_1(K) = \frac{\mathbb{Z}a \oplus \mathbb{Z}b \oplus \mathbb{Z}c}{\mathbb{Z}(b + a - c) \oplus \mathbb{Z}(2a)} \simeq \mathbb{Z} \oplus \mathbb{Z}_2, \quad H_0(K) \simeq \mathbb{Z}.$$

These example should have convinced our beloved readers that, once a simplicial complex, has been assigned, establishing its associated homology groups is a possibly tedious, but algorithmic procedure. An interesting aspect is that, once an homology group has been constructed, there always exist a cohomological counterpart defined by duality. The key point is recalling that, on account of Proposition 140, the chain complex associated to a simplicial complex is built out of free Abelian groups. These do not have a vector space structure and thus assigning a dual space is not obvious. We solve this issue in a few steps, the first consisting of a generalization of Definition 13.

Definition 145. *Let $F^\bullet$ be a graded Abelian group as per Definition 138. We call it a* **cochain complex** *if there exists a* **coboundary operator (map)** $\delta_\bullet : F^\bullet \to F^\bullet$ *such that* $\delta^\bullet = \{\delta^l\}_{l\in\mathbb{Z}}$ *where* $\delta^l \in Hom(F^l, F^{l+1})$ *induces the sequence*

$$\dots \xrightarrow{\delta^{l-2}} F^{l-1} \xrightarrow{\delta^{l-1}} F^l \xrightarrow{\delta^l} F^{l+1} \xrightarrow{\delta^{l+1}} \dots \quad ,$$

subject to the constraint $\delta^{l+1} \circ \delta^l = 0$ *for every* $l \in \mathbb{Z}$.

With this definition in mind, we can derive a simple consequence:

Lemma 146. *Let* $(G_\bullet, \partial_\bullet)$ *be a chain complex as per Definition 139. For every* $l \in \mathbb{Z}$*, let* $F^l \doteq Hom(G_l, \mathbb{Z})$ *and let*

$$\delta^l : F^l \to F^{l+1}, \quad H \mapsto (\delta^l H)(\sigma) \doteq H(\partial_l \sigma),\ \forall \sigma \in G_l.$$

Then $(F^\bullet, \delta^\bullet)$ *is a* **cochain complex**.

Proof. Since the boundary operators ∂_l, $l \in \mathbb{Z}$ are group homomorphisms per construction, the coboundary maps δ^l inherit the same structural property. In addition using Corollary 136, it descends that, for all $\sigma \in G_l$

$$(\delta^{l+1}\delta^l H)(\sigma) = (\delta^l H)(\partial_{l+1}\sigma) = H(\partial_l \partial_{l+1}\sigma) = 0,$$

which concludes the proof. □

Lemma 146 applies naturally to Proposition 140 and it entails that we can construct the association

$$\text{simplicial complex L} \longrightarrow (C^\bullet(K), \delta^\bullet),$$

hence assigning a cochain complex to each simplicial complex. Each element of $C^l(K)$ is called an **l-cochain**.

At this stage, one more in full analogy to Section 1.2, it is natural to introduce two distinguished families of simplices associated to the l-cochains of an oriented simplicial complex K, namely, for every $l \in \mathbb{N} \cup \{0\}$, we set

- $Z^l(K) \doteq \{F \in C^l(K) \mid \delta^l F = 0\}$ where δ^l is the coboundary operator as per Lemma 146 built out of ∂_l defined in Equation (44). The elements of $Z^l(K)$ are called **(l-dimensional) cocycles** of K,

- $B^l(K) \doteq \delta^{l+1}[C^{l+1}(K)]$ whose elements are **(l-dimensional) boundary cocycles** of K.

Definition 147. *Let K be a simplicial complex. For every $l \in \mathbb{N}\cup\{0\}$, we call* **l-th cohomology group**

$$H^l(K) = \frac{\ker(\delta^l)}{\mathrm{Im}(\delta^{l+1})} = \frac{Z^l(K)}{B^l(K)},$$

where δ^l is the coboundary operator map as per Lemma 146 built out of ∂_l defined in Equation (44).

Observe that simplicial cohomology is more abstract and rather less explicit at a computational level in comparison to the homological counterpart. The problem is not really as severe as it might appear at this stage, but we postpone a detailed discussion to later stages. Our aim here was to show that cohomology groups are naturally defined once an homology theory has been discussed. We will show in particular that even de Rham cohomology is naturally related to the construction discussed above, although obtaining this result is not at all cheap. We conclude this section by suggesting to our readers to prove an exercise which shows the existence of a natural and canonical way to pair homology and cohomology groups.

Exercise 53. Let $(G_\bullet, \partial_\bullet)$ be a chain complex as per Definition 139 and let $(F^\bullet, \delta^\bullet)$ be the associated cochain complex as per Lemma 146. We call **Kronecker pairing** the map

$$\langle , \rangle : G_l \times F^l \to \mathbb{Z}, \quad (\sigma, H) \mapsto \langle \sigma, H \rangle = H(\sigma).$$

Show that

1. the coboundary operator δ^l is dual to the boundary map ∂_l with respect to Kronecker pairing, namely,

$$\langle \sigma, \delta^l H \rangle = \langle \partial_{l+1}\sigma, H \rangle, \quad \forall \sigma \in G_{l+1} \quad \text{and} \quad \forall H \in F^l.$$

2. the Kronecker pairing descends to a well-defined map

$$(,)' : H_l(G) \times H^l(F) \to \mathbb{Z}, \quad ([\sigma], [H]) \mapsto ([\sigma], [H])' = \langle \sigma, H \rangle,$$

where the homology and cohomology groups are defined in full analogy to Definition 147.

5.2 Singular Homology and Cohomology

In the previous section we have outlined a notion of homology based on simplices which has the net advantage of being computationally efficient. Yet it has also the strong deficiency of being completely detached from any underlying topological or smooth manifold of which we would like to probe the topology. For this reason we need to introduce another realization of the ideas at the heart of homology. A reader should be warned that the following analysis is an abridged version of what it is explained in standard algebraic topology courses. Here we follow the spirit of [14], but a truly complete exposition can be found in [6] and in [12, Ch. 13] though with a completely different goal in mind in comparison to the other references.

The starting point are still k-simplices but, rather than working with Definition 131, we consider a **standard k-simplex**

$$\Delta^k \doteq \{x = (x_1, \ldots, x_k) \in \mathbb{R}^k;\;\; x_i \geq 0, \text{ and } \sum_{i=1}^{k} x_i \leq 1\} \tag{45}$$

Observe that this realization of a simplex is equivalent to the following one

$$\Delta^k \equiv \{x = (x_0, x_1, \ldots, x_k) \in \mathbb{R}^{k+1};\;\; x_i \geq 0, \text{ and } \sum_{i=1}^{k+1} x_i = 1\}, \tag{46}$$

which is also commonly used in the literature and it becomes the only available one if we consider the case $k = 0$. As a matter of fact, starting from Equation (45), it suffices to set $x_0 = 1 - \sum_{i=1}^{k} x_i$ to obtain the desired correspondence. While simplicial homology was based on the idea of studying simplicial complexes obtained joining together different simplices, in this section we want to make precise the heuristic idea that a simplex can be used to probe a (topological) manifold by attaching it to it continuously.

Definition 148. *Let X be a topological space, we call* **singular k-simplex** *any $\sigma_k \in C^0(\Delta^k; X)$. We denote with $S_k(X)$ the free Abelian group generated by all singular simplices σ_k and its elements are called* **singular k-chains**.

The notion of singular simplices and chains tries to focus the attention that the embedding of Δ^k into X is only subordinated to the topology, but apart from the request of continuity, σ_k might be arbitrarily irregular. As for l-chains we can define a counterpart of Equation (44). To start with, for all $k \in \mathbb{N}$ we introduce the auxiliary maps between standard simplices

$$\epsilon_i : \Delta^{k-1} \to \Delta^k, \quad (x_1, \ldots, x_{k-1}) \mapsto \epsilon_i(x_1, \ldots, x_{k-1}) = (x_1, \ldots, x_{i-1}, 0, x_i, \ldots, x_{k-1}),\ \forall i = 1, \ldots, k$$

while

$$\epsilon_0 : \Delta^{k-1} \to \Delta^k, \quad (x_1, \ldots, x_{k-1}) \mapsto \epsilon_0(x_1, \ldots, x_{k-1}) = (1 - \sum_{i=1}^{k-1} x_i, x_1, \ldots, x_{k-1}).$$

As a consequence we define the **boundary operators**

$$\partial_k : S_k(X) \to S_{k-1}(X), \quad \sigma \mapsto \partial_k \sigma = \sum_{i=0}^{k} (-1)^i \sigma \circ \epsilon_i. \tag{47}$$

Exercise 54. Let X be a topological manifold. Then, for any $k \in \mathbb{N} \cup \{0\}$ it holds

$$\partial_{k-1} \circ \partial_k = 0,$$

where we set $\partial_{-1} \doteq 0$.

An attentive reader should have already observed that Equation (47) is formally identical to Equation (44). Of course this is not a coincidence and we will highlight further the existing correspondences in the next section. Following the same route as in the analysis of simplicial homology, we prove a counterpart of Proposition 140:

Proposition 149. *Let X be a topological space. Then the associated singular l-chains form a chain complex $(S_\bullet(X), \partial_\bullet)$ where $S_\bullet(X) \doteq \bigoplus_{k\in\mathbb{Z}} S_k(X)$ while $\partial_\bullet \doteq \{\partial_k\}_{k\in\mathbb{Z}}$ acts on $S_\bullet(X)$ component by component with the boundary operators ∂_k as in Equation (47). Here we set $S_k(X) = \{0\}$ and $\partial_k = 0$ if $k < 0$.*

Proof. On account of Definition 148, each $S_k(X)$ is an Abelian group per construction and, thus $S^\bullet(X)$ is a graded Abelian group, *cf.* Definition 138. Since each boundary operator is linear per definition, *cf.* Equation (47), it identifies a group homomorphism. Exercise 54 concludes the proof. $\square$

In full analogy to Section 1.2 and to the preceding one, it is natural to introduce two distinguished families of singular k-chains, namely, given any topological space X, for every $k \in \mathbb{N}\cup\{0\}$, we set

- $Z_k(X) \doteq \{\sigma \in S_k(X) \mid \partial_k\sigma = 0\}$ where ∂_k is the boundary operator as in Equation (47). The elements of $Z_k(X)$ are called **(k-dimensional) singular cycles** of X,
- $B_k(X) \doteq \partial_{k+1}[S_{k+1}(X)]$ whose elements are **(k-dimensional) singular boundary cycles** of K.

Definition 150. *Let X be a topological space. For every $k \in \mathbb{N}\cup\{0\}$, we call* **k-th singular homology group (with integer coefficients)**

$$H_k(X;\mathbb{Z}) \equiv H_k(X) = \frac{Z_k(X)}{B_k(X)} = \frac{\ker(\partial_k)}{\mathrm{Im}(\partial_{k+1})},$$

where ∂_k is the boundary operator map as in Equation (47). The collection of all singular homology groups forms a graded Abelian group $H_\bullet(X;\mathbb{Z}) \doteq \bigoplus_{k=0}^{\infty} H_k(X;\mathbb{Z})$.

Remark 41. *An attentive reader, like you (yes, ... you ... indeed!) might have noticed that, in the construction above and similarly in that of simplicial homology, the starting point is the identification of a free Abelian group with integer coefficients, cf. Definition 148. There is a certain freedom in this point of the construction since one might replace $\mathbb{Z}$ with any other Abelian group A, e.g $A = \mathbb{R}, \mathbb{C}$ or even $A = \mathbb{Z}_2$, $A = U(1)$. As a matter of fact there is no obstruction in this procedure thus singular k-chains with A-coefficients*

$$S_k(X;A), \quad \forall k \in \mathbb{N}\cup\{0\}$$

which can be endowed with the same boundary operator as in Equation (47). Neither the definition nor the properties depend on A provided it is an Abelian group. As a consequence, we have identified a chain complex

$$(S_\bullet(X;A), \partial_\bullet).$$

Similarly we can define $Z_k(X;A)$ and $B_k(X;A)$ as the k-dimensional singular cycles/boundary cycles with A-coefficients, $k \in \mathbb{N} \cup \{0\}$. This leads to constructing the **k-th singular homology groups with A-coefficients**

$$H_k(X;A) \doteq \frac{Z_k(X;A)}{B_k(X;A)}.$$

The natural questions at this point are two. The first concerns the reason of giving a distinguished role to the integers. This is due to the fact that the ensuing homology groups are $\mathbb{Z}$-module and thus they are sensible to torsion elements. These carry topological information on the underlying space X, which is lost whenever $A = \mathbb{R}$ or $A = \mathbb{C}$. In many physical applications in gauge theory these additional data play an important role. The second question instead is whether $H_k(X;A)$ is related to $H_k(X;\mathbb{Z})$. The answer is affirmative and it is one of the key results in basic algebraic topology going under the name of universal coefficient theorem*, see [6, Th. 3A.3].*

Having established a new type of homology theory, it is important to investigate its structural properties. If we take the analysis of de Rham cohomology as an inspiration, an important feature of the $H^\bullet_{\mathrm{dR}}(M)$, M being a smooth manifold, has been discussed in Section 4.1. Most notably homotopy invariance entails the independence of $H^\bullet_{\mathrm{dR}}(M)$ from the differentiable structure, making it a collection of topological invariants. In the case of singular homology the graded Abelian group $H_\bullet(X;\mathbb{Z})$, *cf.* Definition 141, depends per construction only on the underlying topology since X is not even endowed with a richer structure. Nonetheless, homotopy invariance is not a priori guaranteed and it is important to establish it since it represents an abstract property with far-reaching computational consequences. The following discussion will mimic closely that of Section 4.1 and a reader, who is not interested in the fine differences arising in the proofs of the statements compared to the same one for de Rham cohomology, can limit him/herself to browsing through the results. To start with we prove a naturality result for complexes of singular chains.

Proposition 151. *Let X and Y be two topological spaces and let $f \in C^0(X;Y)$. Then, for every $k \in \mathbb{N} \cup \{0\}$, there exists*

$$f_{\#,k} : S_k(X) \to S_k(Y), \quad \sigma \mapsto f_{\#,k}(\sigma) = f \circ \sigma, \tag{48}$$

which is a **chain map***, namely it induces the commutative diagram:*

$$\begin{array}{ccccccccc}
\dots & \longrightarrow & S_{k+1}(X) & \xrightarrow{\partial_{k+1}} & S_k(X) & \xrightarrow{\partial_k} & S_{k-1}(X) & \longrightarrow & \dots \\
 & & \downarrow{\scriptstyle f_{\#,k}} & & \downarrow{\scriptstyle f_{\#,k}} & & \downarrow{\scriptstyle f_{\#,k}} & & \\
\dots & \longrightarrow & S_{k+1}(Y) & \xrightarrow{\partial_{k+1}} & S_k(Y) & \xrightarrow{\partial_k} & S_{k-1}(Y) & \longrightarrow & \dots
\end{array}$$

Proof. First of all we observe that, for every $k \in \mathbb{N}\cup\{0\}$, $f_\# \in C^0(\Delta^k; Y)$ since it is a composition of continuous maps. In addition it extends uniquely as a linear map from $S_k(X)$ to $S_k(Y)$ as

$$f_{\#,k}\left(\sum_{i=1}^{N<\infty} n_i\sigma_i\right) = \sum_{i=1}^{N<\infty} n_i f_{\#,k}(\sigma_i),$$

where $n_i \in \mathbb{Z}$, while $\sigma_i \in C^0(\Delta^k; X)$ for all $i = 1, \dots, N$. To prove that, $f_{\#,k}$ is a chain map, it suffices to consider any but fixed $\sigma \in S_k(X)$ evaluating via Equation (47) and Equation (48)

$$f_{\#,k}(\partial_k\sigma)(x_1, \dots, x_{k-1}) = f_{\#,k}\left(\sum_{i=0}^{k}(-1)^i\sigma \circ \epsilon_i\right)(x_1, \dots, x_{k-1}) =$$

$$= \sum_{i=0}^{k}(-1)^i f_{\#,k}(\sigma) \circ (\epsilon_i(x_1, \dots, x_{k-1})) = \partial_k(f_{\#,k}(\sigma))(x_1, \dots, x_{k-1}),$$

which entails that $f_{\#,k} \circ \partial_k = \partial_k \circ f_{\#,k}$. This concludes the proof. □

Corollary 152. *Let X and Y be two topological spaces and let $f_{\#,k} : S_k(X) \to S_k(Y)$, $k \in \mathbb{N}\cup\{0\}$, be a chain map. For every $k \in \mathbb{N}\cup\{0\}$, it induces $\widetilde{f}_k \in \mathrm{Hom}(H_k(X;\mathbb{Z}); H_k(Y;\mathbb{Z}))$ which is completely defined by*

$$\widetilde{f}_k([\sigma]) \doteq [f_{\#,k}\sigma], \quad \forall[\sigma] \in H_k(X;\mathbb{Z}).$$

Proof. As a starting point we observe that, given a chain map $f_{\#,k} : S_k(X) \to S_k(Y)$, $k \in \mathbb{N}\cup\{0\}$, Proposition 151 entails that for every $\sigma \in Z_k(X)$

$$\partial_k(f_{\#,k}(\sigma)) = f_{\#,k}(\partial_k\sigma) = f_{\#,k}(0) = 0 \Longrightarrow f_{\#,k}(\sigma) \in Z_k(Y).$$

Similarly, for every $\sigma' \in B_k(X)$, there exists $\tau \in S_{k+1}(X)$ such that $\sigma = \partial_{k+1}\tau$. Hence

$$f_{\#,k}\sigma' = f_{\#,k}(\partial_{k+1}\tau) = \partial_{k+1}f_{\#,k}(\tau), \Longrightarrow f_{\#,k}\sigma' \in B_k(Y),$$

where we used once more that $f_{\#,k}$ is a chain map. These observations entail that $\widetilde{f}_k$ is well defined map from $H_k(X;\mathbb{Z})$ to $H_k(Y;\mathbb{Z})$ for every $k \in \mathbb{N}\cup\{0\}$. To conclude, it suffices to observe that for every $[\sigma], [\sigma'] \in H_k(X;\mathbb{Z})$

$$\widetilde{f}_k([\sigma] + [\sigma']) = \widetilde{f}_k([\sigma + \sigma']) = [f_{\#,k}(\sigma + \sigma')] = [f_{\#,k}(\sigma) + f_{\#,k}(\sigma')] =$$
$$= [f_{\#,k}(\sigma)] + [f_{\#,k}(\sigma')] = \widetilde{f}_k([\sigma]) + \widetilde{f}_k([\sigma']),$$

where we used the defining properties of the cohomology groups and of the maps $f_{\#,k}$ and $\widetilde{f}_k$ in the various identities. In other words we have shown that $\widetilde{f}_k$ is a group homomorphism. □

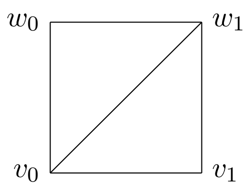
At this stage we can state the natural counterpart of Definition 113 at the level of chain complexes.

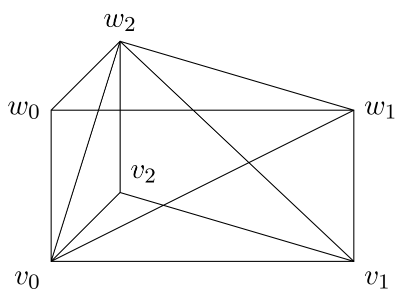
Definition 153. *Let X, Y be two topological spaces and let $f, g \in C^0(X;Y)$. We call* **chain homotopy** *(between $f_{\#,k}$ and $g_{\#,k}$) a family of maps $\{P_k\}_{k\in\mathbb{N}\cup\{0\}}$ such that, for all $k \in \mathbb{N}\cup\{0\}$, $P_k : S_k(X) \to S_{k+1}(Y)$ and*

$$\partial_{k+1}P_k + P_{k-1}\partial_k = g_{\#,k} - f_{\#,k}.$$

Theorem 154. *Let X and Y be two topological space and let $f, g \in C^0(X;Y)$ bet two homotopic maps. Then, for all $k \in \mathbb{N}\cup\{0\}$, $f_{\#,k}(H_k(X;\mathbb{Z})) = g_{\#,k}(H_k(X;\mathbb{Z}))$.*

Proof. The key ingredient is the subdivision of $\Delta^k \times I$, $I = [0,1]$ into the union of $(k+1)$-simplices. From a graphical viewpoint consider the following two examples ($k = 1, 2$):





From a technical viewpoint this amounts to considering two copies of the standard k-simplex as in Equation (46), namely $(v_0, v_1, \ldots, v_k)$ and $(w_0, w_1, \ldots, w_k)$ where the first is identified with $\Delta^k \times \{0\}$ and the second with $\Delta^k \times \{1\}$. Subsequently consider all k-simplices in $\Delta^k \times I$ obtained as $\Delta^k_i \doteq (v_0, \ldots, v_i, w_{i+1}, w_k)$, $i = 0, \ldots, k-1$. If we take the union of all these simplices together with $(v_0, v_1, \ldots, v_k)$ we have obtained the desired subdivision of $\Delta^k \times I$ (including $(w_0, w_1, \ldots, w_k)$). Observe that each Δ^k_i is not a standard k-simplex. Given a generic k-simplex as per Definition 131, say $|\sigma| = \langle z_0, z_1, \ldots, z_k\rangle$, we can introduce a linear homeomorphism

$$\underline{t} : \Delta^k \to |\sigma|, \quad \underline{t}(x_0, x_1, \ldots, x_k) = \sum_{j=1}^{k} x_j z_j.$$

The coefficients $\{x_j\}_{j=1,\ldots,n}$ are called the barycentric coordinates of the point $\sum\limits_{j=1}^{k} x_j z_j$. Consider these coordinates for each k-simplex Δ^k_i, $i = 0, \ldots, k-1$, these can be realized as the graph of the function

$$\varphi_i : \Delta^k_i \to I, \quad (x_0, \ldots, x_k) \mapsto \varphi_i(x_0, \ldots, x_k) = \sum_{j=i+1}^{n} x_j.$$

Observe that the graph of φ_i is included in that of φ_{i-1} ($\varphi_i \leq \varphi_{i-1}$) for all $i = 0, \dots, k-1$ and, from the string of inclusions $0 \doteq \varphi_k \leq \varphi_{k-1} \leq \cdots \leq \varphi_0 \leq \varphi_{-1} \doteq 1$, where 0 and 1 indicate the trivial and the identity maps, we have shown that $\Delta^k \times I = \bigcup_{i=0}^{k-1} \Delta_i^{k+1}$ where $\Delta_i^{k+1} = (v_0, \dots, v_i, w_i, \dots, w_k)$ (compare with the figures above for $k = 1, 2$). Having established the existence of this subdivision, consider an homotopy $F : X \times I \to Y$ such that $F(\cdot, 0) = f$ while $F(\cdot, 1) = g$. We can define the so-called *prism operators* for all $k \in \mathbb{N} \cup \{0\}$:

$$P_k : S_k(X) \to S_{k+1}(Y),$$
$$\sigma \mapsto P_k(\sigma)(v_0, \dots, v_i, w_{i+1}, \dots, w_k) = \sum_{i=0}^{k-1} (-1)^i F \circ (\sigma \times \mathbb{I})(v_0, \dots, v_i, w_i, \dots, w_k),$$

where $\sigma \times \mathbb{I} : \Delta^k \times I \to X \times I$ acts componentwise. We show that P_k is a chain homotopy as per Definition 153. Proceeding with a brute force method, it holds

$$\partial_{k+1} P_k = \sum_{j \leq i} (-1)^{i+j} F \circ (\sigma \times \mathbb{I})(v_0, \cdots, \widehat{v}_j, \cdots, v_i, w_i, \cdots, w_k) +$$
$$\sum_{j \geq i} (-1)^{i+j+1} F \circ (\sigma \times \mathbb{I})(v_0, \cdots, \cdots, v_i, w_i, \cdots, \widehat{w}_j, \cdots, w_k),$$

where the hat symbol means that the element below has been omitted. The sum can be divided in three contributions: if $i \neq j$ we obtain exactly $-P_{k-1}\partial_k$. If $i = j \neq 0$ there are two equal terms in the two sums with opposite sign, hence canceling each other. Finally if $i = 0$ and $j = n$ we obtain

$$F \circ (\sigma \times \mathbb{I})(\widehat{v}_0, w_0, \cdots, w_k) = g \circ \sigma = g_\#(\sigma),$$

and

$$-F \circ (\sigma \times \mathbb{I})(v_0, \cdots, \cdots, v_k, \widehat{w}_k) = -f \circ \sigma = -f_\#(\sigma).$$

Putting together these ingredients we have proven

$$\partial_{k+1} P_k + P_{k-1} \partial_k = g_{\#,k} - f_{\#,k}.$$

To conclude, suppose now that $\sigma \in Z_k(X)$. This last identity entails that

$$\partial_{k+1} P_k(\sigma) + P_{k-1} \partial_k(\sigma) = \partial_{k+1} P_k(\sigma) = g_{\#,k}(\sigma) - f_{\#,k}(\sigma),$$

which entails that the right hand side is a singular boundary k-cycle. In other words $g_{\#,k}(\sigma)$ and $f_{\#,k}(\sigma)$ determine the same homology class, which can be translated as

$$f_{\#,k}(H_k(X; \mathbb{Z})) = g_{\#,k}(H_k(X; \mathbb{Z})).$$

This concludes the proof. $\square$

The main application of this theorem occurs when the the map f is an homotopy equivalence, namely there exists $h \in C^0(Y;X)$ such that $fh \sim \mathrm{id}|_Y$ and $hf \sim \mathrm{id}|_X$, where $\sim$ indicates that there exist chain homotopies between $(fh)_{\#,k}$ and $(\mathrm{id}|_Y)_{\#,k}$ and between $(hf)_{\#,k}$ and $(\mathrm{id}|_X)_{\#,k}$.

Corollary 155. *Let X and Y be two topological space and let $f \in C^0(X;Y)$ be an homotopy equivalence. Then, for all $k \in \mathbb{N} \cup \{0\}$,*

$$H_k(X;\mathbb{Z}) \simeq H_k(Y;\mathbb{Z}),$$

where $\simeq$ indicates a group isomorphism.

Proof. This is a direct consequence of Theorem 154. $\square$

As we have explained at the beginning of this subsection, the key difference between simplicial and singular homology is that the first one is well-versed for computational purposes. Given a topological space, it is quite difficult to evaluate $H_k(X;\mathbb{Z})$, $k > 0$. The only exception is $k = 0$ as we establish with the following proposition.

Proposition 156. *Let X be a connected topological manifold. Then*

$$H_0(X;\mathbb{Z}) \simeq \mathbb{Z}.$$

Proof. Following Definition 150 it holds that $H_0(X;\mathbb{Z}) = \frac{\ker(\partial_0)}{\mathrm{Im}(\partial_1)}$. Yet, since $\partial_0 \equiv 0$ per construction, it descends that $H_0(X;\mathbb{Z}) = \frac{S_0(X)}{\mathrm{Im}(\partial_1)}$. Since each element of $S_0(X)$ is of the form $\sum\limits_{j=1}^{N} n_j\sigma_j$ where $N < \infty$, while, for each $j = 1, \ldots, N$, $n_j \in \mathbb{Z}$ and $\sigma_j \in C^0(\Delta^0;X)$, we can define the map

$$I : S_0(X) \to \mathbb{Z}, \quad \sum_{j=1}^{N} n_j\sigma_j \mapsto I\left(\sum_{j=1}^{N} n_j\sigma_j\right) = \sum_{j=1}^{N} n_j.$$

Per construction I is a surjective group homomorphism. If $\ker(I) = \mathrm{Im}(\partial_1)$, we reach the sought conclusion. Let $\sigma \in \mathrm{Im}(\partial_1) \subseteq S_0(X)$. There must exist $\widetilde{\sigma} \in S_1(X)$ such that

$$\sigma = \partial_1\widetilde{\sigma} = \widetilde{\sigma}(\epsilon_0 - \epsilon_1),$$

where we used Equation (47). In other words $\sigma = \sigma_0 - \sigma_1$ where $\sigma_i = (-1)^i\widetilde{\sigma}\epsilon_i$, $i = 0, 1$. This yields

$$I(\sigma) = 0 \Longrightarrow \sigma \in \ker I.$$

Suppose now $\sigma \in \ker I$. There must exist $n_j \in \mathbb{Z}$ and $\sigma_j \in C^0(\Delta^0;X)$ with $j = 1, \ldots, N < \infty$ such that

$$\sigma = \sum_{j=1}^{N} n_j\sigma_j, \quad \text{and} \quad \sum_{j=1}^{N} n_j = 0.$$

Since each $\sigma_j \in S_0(X)$, it identifies a point in X. The assumption that X is a topological manifold entails that it is path-connected. Hence, fixing a point $x_0 \in X$, we can find N-paths $\gamma_j : [0,1] \to X$ such that $\gamma_j(0) = x_0$ while $\gamma_j(1) = \sigma_j(\Delta^0)$. We observe that each γ_j can be read as a singular 1-simplex $\gamma_j : \Delta^1 \to X$ such that, using Equation (47), it holds

$$\partial_1 \gamma_j = \sigma_j - \sigma_0,$$

where $\sigma_0(\Delta^0) = x_0$. Consider now $\sum\limits_{j=1}^{N} n_j \tau_j$; it descends

$$\partial_1 \left(\sum_{j=1}^{N} n_j \tau_j \right) = \sum_{j=1}^{N} \sigma_j - \sum_{j=1}^{N} \sigma_0 = \sum_{j=1}^{N} n_j \sigma_j,$$

where we used the hypothesis that $\sum\limits_{j=1}^{N} n_j = 0$. We have shown that $\sigma \in \mathrm{Im}(\partial_1)$. $\square$

Having established the notion of singular homology, it is natural to look for a cohomological counterpart. Following the same spirit of the previous section, we observe that Lemma 146 applies naturally to Proposition 149 and it entails that we can construct the association

$$\text{topological space X} \longrightarrow (S^\bullet(X), \delta^\bullet),$$

hence assigning a cochain complex to each topological space. Each element of $S^k(X)$ is called a singular **l-cochain**.

Following the standard rationale, we introduce two distinguished families of cochains, namely, for every $k \in \mathbb{N} \cup \{0\}$, we set

- $Z^k(X) \doteq \{F \in S^k(X) \mid \delta^k F = 0\}$ where δ^k is the coboundary operator as per Lemma 146 built out of ∂_k defined in Equation (47). The elements of $Z^k(X)$ are called **(k-dimensional) singular cocycles** of X,
- $B^k(X) \doteq \delta^{k+1}[S^{k+1}(X)]$ whose elements are **(k-dimensional) singular boundary cocycles** of X.

Definition 157. *Let X be a topological space. For every $k \in \mathbb{N} \cup \{0\}$, we call* **k-th singular cohomology group**

$$H^k(X;\mathbb{Z}) \equiv H^k(X) = \frac{Z^k(X)}{B^k(X)} = \frac{\ker(\delta^k)}{\mathrm{Im}(\delta^{k+1})},$$

where δ^k is the coboundary operator map as per Lemma 146 built out of ∂_k defined in Equation (47).

Remark 42. *Similarly to Remark 41, we observe that the definition of the singular cohomology groups is tied to the construction of $S_k(X)$ as a free Abelian group with integer coefficients. If we* *consider a different Abelian group, e.g.* $A = \mathbb{R}, \mathbb{C}$ *or even* $A = \mathbb{Z}_2, U(1)$*, we can define singular* k*-cochains with* A*-coefficients*

$$S^k(X; A) \doteq \mathrm{Hom}(S_k(X; A); A), \quad \forall k \in \mathbb{N} \cup \{0\}$$

which can be endowed with the same coboundary operator as in Definition (145). Neither the definition nor the properties depend on A *provided it is an Abelian group. As a consequence, we have identified a cochain complex*

$$(S^\bullet(X; A), \delta^\bullet).$$

Similarly we can define $Z^k(X; A)$ *and* $B^k(X; A)$ *as the* k*-dimensional singular cocycles/boundary cocycles with* A*-coefficients,* $k \in \mathbb{N} \cup \{0\}$*. This leads to constructing the* **k-th singular cohomology groups with** A**-coefficients**

$$H^k(X; A) \doteq \frac{Z^k(X; A)}{B^k(X; A)}.$$

As in the case of homology groups it turns out that $H^k(X; A)$ *is related to* $H^k(X; \mathbb{Z})$ *via a version of the* universal coefficient theorem*, see [6, Chap. 3], especially Corollary 3A.4.*

We shall not discuss further the singular cohomology groups since a computational procedure to characterize them is at this stage quite hard. We limit ourselves (or actually should I say... write myself?) to proposing to the reader(s) an exercise highlighting a notable structural property, reminiscent of Exercise 43.

Exercise 55. Let X be a topological space. We call **cup product** the map

$$\smile : S^k(X) \times S^{k'}(X) \to S^{k+k'}(X), \quad \forall k, k' \in \mathbb{N} \cup \{0\}$$

such that, for all $\sigma \in C^0(\Delta^{k+k'}; X)$ and for all $\alpha \in S^k(X)$, $\beta \in S^{k'}(X)$, it holds

$$(\alpha \smile \beta)(\sigma) = \alpha(\sigma|_{(v_0,\ldots,v_k)})\beta(\sigma|_{(v_{k+1},\ldots,v_{k'})}).$$

Prove that this formula descends to a well-defined product of cohomology classes

$$\smile : H^k(X; \mathbb{Z}) \times H^{k'}(X; \mathbb{Z}), \quad ([\omega], [\omega']) \mapsto \smile ([\omega], [\omega']) = [\omega \smile \omega']$$

which is indicated with the same symbol with a slight abuse of notation. Prove in addition that the same construction works replacing $\mathbb{Z}$ with any ring A. The pair $(H^\bullet(X; \mathbb{Z}), \smile)$ is called the **singular cohomology ring** (with integer coefficients) of X.

A notable property both of singular homology and cohomology groups is their compatibility with respect to push-forward and pull-back. The proof of this statement is rather straightforward, to the point that I feel no shame to assign it to a reader as an exercise.

Exercise 56. Let X, Y be two topological spaces and let $F \in C^0(X;Y)$. Let $\sigma \in C^0(\Delta^k;X)$ be a singular k-simplex in X and let $F \circ \sigma$ be the counterpart in Y. Prove that the map F descends to the underlying homology and cohomology groups as follows. This means that for every $k \in \mathbb{N} \cup \{0\}$ and for every ring A,

$$F_* : H_k(M;A) \to H_k(N;A), \quad [\sigma] \mapsto F_*[\sigma] \doteq [F \circ \sigma],$$

while, for every $[\sigma] \in H_k(M;A)$

$$F^* : H^k(N;A) \to H^k(M;A), \quad [\alpha] \mapsto F^*[\alpha] \quad \text{such that} \quad (F^*[\alpha])([\sigma]) = [\alpha](F_*([\sigma])).$$

Remark 43. *The realization of singular cohomology groups as homomorphisms from the associated singular homology groups entails that they inherit some of the structural properties of $H_k(X;A)$ where $k \in \mathbb{N} \cup \{0\}$, X is the underlying topological space, while A is a ring, not necessarily coincident with $\mathbb{Z}$. Most notable is homotopy invariance, cf. Proposition 154, which has also some relevant physical consequences, similar to those outlined in Remark 36 for the de Rham cohomology ring. As a matter of fact Corollary 155 entails that, whenever $X = J \times Y$, where J is a connected open subset of $\mathbb{R}$, for every $k \in \mathbb{N} \cup \{0\}$ and for every ring A,*

$$H_k(X;A) \simeq H_k(Y;A), \quad H^k(X;A) \simeq H^k(Y;A).$$

The most notable instance where these isomorphisms play a role is in relativistic field theories, where the role of X is played by a globally hyperbolic spacetime M which is isometric to $\mathbb{R} \times \Sigma$. Here, each $\{t\} \times \Sigma$, $t \in \mathbb{R}$ plays the role of an initial value surface for hyperbolic PDEs such as for example the wave equation. The analysis in this section tells us in addition that all topological information encoded in the singular homology and cohomology groups originates from Σ and it cannot change under time evolution. The real point which is not clear at this stage is what kind of information is really encoded in singular homology and cohomology. As a matter of fact, in the case of De Rham cohomology, the answer was immediate due to the obvious connections to Maxwell's theory. At this level such interpretation is not obvious. At this stage a complete answer is still not possible, but we mention that there are two notable applications of singular cohomology. On the one hand we will show that, if $A = \mathbb{R}$ then we are just expressing de Rham cohomology in a different way. On the other hand, it turns out that, whenever we make this choice, we are neglecting some topological information which has physical consequences. The prime example is the Aharonov-Bohm effect, which cannot be detected or describe with $A = \mathbb{R}$, but it arises only if we allow for $A = \mathbb{Z}$ and $A = U(1)$. The whole story requires introducing a more advanced and sophisticated cohomology theory, which is known in the literature as Cheeger-Simons cohomology or equivalently as cohomology with integer periods. Understanding such theory requires studying the so-called **differential characters** *which we cannot discuss in these notes. An interested reader should consult [2]. Hence the question on the relevance (at least for these notes) of singular homology and cohomology stays. The answer lies in a question left open from the previous sections, namely how to classify and distinguish isomorphisms classes of vector and of principal bundles on a given manifold. The answer lies in the theory of characteristic classes which find their natural and most complete formulation within singular cohomology.*

At this stage we are left with two key open questions:

1. How is singular homology related to simplicial homology?

2. How is singular cohomology related to De Rham cohomology?

An exhaustive answer to the first question is far beyond the goal of these notes and it is to a certain extent an open question in algebraic topology. We will consider the restricted case of smooth manifolds for which an exhaustive answer exists. Since this includes all cases of physical interest, we can consider such answer satisfactory. The second question instead can be addressed in full generality via the renown *de Rham theorem*.

5.3 Relative Homology

In the previous two subsections, we have introduced two different notions of homology groups. The simplicial one is subordinated to the notion of a k-simplex and of a simplicial complex. It has the net advantage of being computationally very efficient, *cf.* Examples 142, 143 and 144. Yet it is completely detached from topological spaces and it does not seem to provide valuable information on these structures. On the contrary singular homology is tied to the notion of a standard k-simplex and of its continuous embeddings in a topological space. From a theoretical viewpoint it is a highly informative structure which is, alas, computationally a nightmare except for the 0-th homology group, *cf.* Proposition 156.

In this section we discuss how to relate these two theories taking the best of both worlds (which is incidentally the title of the finale of the third season as well as the premiere of the fourth one of Star Trek TNG... but probably, you have no idea what I am talking about... the Borg? Does it ring a bell? No... not the tennis player Björn Borg!). The idea is quite obvious at first glance. A direct inspection on simplicial and on singular homology suggests that a possible way to compare them is to consider a topological manifold X and to ask whether a simplicial complex K or rather its associated polyhedron $|K|$, *cf.* Definition 132, can be continuously embedded on it. Since each simplex can be put in correspondence to the standard one, *e.g.* via barycentric coordinates, *cf.* Definition 131, one can establish whether there exists a correspondence between the homology groups of $|K|$ and of the singular ones of X.

Of course this is easier said than done and, in the following, we try to formalize the idea. The staring point is a key concept in algebraic topology which makes precise the idea of attaching a simplicial complex to an underlying space

Definition 158. *Let X be a topological space and let K be a simplicial complex whose associated polyhedron is $|K|$. If there exists $T \in C^0(|K|; X)$ we call T a* **triangulation** *of X. Any topological manifold admitting a triangulation is called* **triangulable**.

The natural question which originates from this definition is whether a given topological space admits a triangulation. This is a very complicated problem which has a negative answer in general and, understanding the minimal additional requirements on X to make it triangulable is still an open problem. Yet, for all practical purposes, we do not need to work in such generality.

As a matter of fact we are interested in considering differentiable manifolds M, possibly endowed with a Lorentzian or with a Riemannian structure, which play no role here. In these case we can report a very important result whose proof is quite technical and we refer to the original papers in case of interest, [4, 16]:

Theorem 159 (Cairns-Whitehead). *Every smooth manifold M is triangulable.*

Having established the existence of an homeomorphism between a polyhedron subordinated to a simplicial complex K and a smooth manifold M, we can investigate how to relate $H_k(K)$ and $H_k(M;\mathbb{Z})$ for every $k \in \mathbb{N} \cup \{0\}$. First of all we invite our readers to solve the following exercises to understand how the triangulation of a manifold looks like concretely.

Exercise 57. Prove that the simplicial complexes in Examples 142, 143 and 144 identify a triangulation of $\mathbb{RP}^2$, $\mathbb{T}^2$ the 2-torus and of the Klein bottle respectively.

In the following we will still be working with arbitrary topological spaces, but the reader should keep in mind that we will be applying all results to spaces endowed with a smooth structured, so to be able to apply Theorem 159. Consider a topological space X and a subspace Y. The natural inclusion $Y \subset X$ induces a counterpart at the level of singular k-chains by restriction. In other words, for all $k \in \mathbb{N} \cup \{0\}$

$$S_k(Y) \subseteq S_k(X).$$

Since these are Abelian groups, it descends that $S_k(Y)$ is a normal subgroup of $S_k(X)$ and we can construct the quotient group $S_k(X,Y) \doteq \frac{S_k(X)}{S_k(Y)}$. The following lemma studies the properties of the boundary operator (44) in relation to this construction

Lemma 160. *Let X be a topological space and let $Y \subseteq X$ be a subspace. Then there exists an associated chain complex $(S_\bullet(X,Y), \partial_\bullet)$, cf. Definition 139, where $S_\bullet(X,Y) = \bigoplus_{k\in\mathbb{Z}} S_k(X,Y)$ while $\partial_\bullet \doteq \{\partial_k\}_{k\in\mathbb{Z}}$ acts on $S_\bullet(X,Y)$ component by component as the natural descent of the boundary operator (47) to $S_k(X,Y)$ where $\partial_k \equiv 0$ for all $k \leq 0$. Similarly we set $S_k(X,Y) = \{0\}$ if $k < 0$.*

Proof. First of all we observe that, since each $S_k(X,Y) = \frac{S_k(X)}{S_k(Y)}$, $k \in \mathbb{N} \cup \{0\}$ is a quotient of Abelian groups, it is inherits the structure of a commutative group, making the direct sum defining $S_\bullet(X,Y)$ meaningful. Let us consider the boundary operator $\partial_k : S_k(X) \to S_{k-1}(X)$. Per direct inspection of Equation (47), we can deduce that X can be replaced with Y in the sense that $\partial_k : S_k(Y) \to S_{k-1}(Y)$. Hence this entails that, for all $k \in \mathbb{N} \cup \{0\}$ we can define

$$\partial_k : S_k(X,Y) \to S_{k-1}(X,Y), \quad [\sigma] \mapsto \partial_k[\sigma] = [\partial_k \sigma].$$

Furthermore this definition entails that these boundary operator inherits all structural properties of the counterparts on $S_\bullet(X)$. In particular Exercise 54 entails that, also on $S_\bullet(X,Y)$, $\partial_{k-1} \circ \partial_k = 0$ for all $k \in \mathbb{N} \cup \{0\}$. This suffices to prove that we have identified a chain complex □

Disclaimer: Observe that we are using the same symbols ∂_k, $k \in \mathbb{N} \cup \{0\}$ for the boundary operators as in Equation (47) both when acting on $S_k(X)$, on $S_k(Y)$ or on $S_k(X,Y)$. This is a slight abuse of notation whose aim is to emphasize that the action of the boundary operator is substantially independent from the space on which the singular k-chains are defined, even if taking a quotient.

Following the by now standard procedure whenever we have identified chain or chain complexes, we introduce two notable sets:

- **relative cycles** $Z_k(X,Y) = \{[\sigma] \in S_k(X,Y) \mid \partial_k[\sigma] = 0\}$. A representative of $[\sigma] \in Z_k(X,Y)$ is $\sigma \in S_k(X)$ such that $\partial_k \sigma \in S_{k-1}(Y)$,
- **relative boundaries** $B_k(X,Y) = \{[\sigma] \in S_k(X,Y) \mid \exists [\tau] \in S_{k+1}(X,Y)$ for which $[\sigma] = \partial_{k+1}[\tau]\}$. A representative of $[\sigma] \in B_k(X,Y)$ is $\sigma \in S_k(X)$ such that $\sigma = \partial_{k+1}\tau + \sigma_Y$, where $\tau \in S_{k+1}(X)$ while $\sigma_Y \in S_k(Y)$.

Definition 161. *Let X be a topological space and let $Y \subseteq X$ be a subspace. For every $k \in \mathbb{N}\cup\{0\}$, we call* **k-th relative homology (of X modulo Y)**

$$H_k(X,Y) = \frac{Z_k(X,Y)}{B_k(X,Y)} = \frac{\ker(\partial_k)}{\mathrm{Im}(\partial_{k+1})},$$

where ∂_x is the boundary operator acting on $C_k(X,Y)$ as per Lemma 160. The collection of all relative homology groups forms a graded Abelian group $H_\bullet(X,Y) = \bigoplus_{k=0}^{\infty} H_k(X,Y)$.

Our next step consists of showing that all these spaces can be encoded in a long exact sequence, much in the spirit of Section 1.3 and particularly of Lemma 19. To this end it is convenient to consider a topological space X and a subspace Y indicating with $\iota : Y \to X$ the standard inclusion map. Lemma 160 guarantees us that we have three chain complexes $(S_\bullet(Y), \partial_\bullet)$, $(S_\bullet(X), \partial_\bullet)$ and $(S_\bullet(X,Y), \partial_\bullet)$ together with a short exact sequence:

$$0 \longrightarrow S_\bullet(Y) \xrightarrow{\underline{\iota}} S_\bullet(X) \xrightarrow{\underline{\pi}} S_\bullet(X,Y) \longrightarrow 0 \tag{49}$$

where $\underline{\iota} = \{\iota\}_{k\in\mathbb{Z}}$ is constructed out of the natural inclusion maps $\iota_k : S_k(Y) \to S_k(A)$, while $\underline{\pi} = \{\pi\}_{k\in\mathbb{Z}}$ is defined out of $\pi_k : S_k(X) \to S_k(X,A)$ which are the projection maps. Equation (49) gives rise to the following diagram:

$$\begin{tikzcd}
 & \cdots \arrow[d] & \cdots \arrow[d] & \cdots \arrow[d] & \\
0 \arrow[r] & S_{k+1}(Y) \arrow[r, "\iota_{k+1}"] \arrow[d, "\partial_{k+1}"] & S_{k+1}(X) \arrow[r, "\pi_{k+1}"] \arrow[d, "\partial_{k+1}"] & S_{k+1}(X,A) \arrow[r] \arrow[d, "\partial_{k+1}"] & 0 \\
0 \arrow[r] & S_k(Y) \arrow[r, "\iota_k"] \arrow[d] & S_k(X) \arrow[r, "\pi_k"] \arrow[d] & S_k(X,A) \arrow[r] \arrow[d] & 0 \\
 & \cdots & \cdots & \cdots &
\end{tikzcd}$$

where each of the square sub-diagrams is commuting. As next step we observe that each horizontal map in the above diagram gives rise to a suitable counterpart at a cohomology level. This statement can be proved mutatis mutandis as in Proposition 18

Exercise 58. Consider three chain complexes $(A_\bullet, \partial_\bullet^A)$, $(B_\bullet, \partial_\bullet^B)$ and $(C_\bullet, \partial_\bullet^C)$, together with two chain maps $\underline{f} : A_\bullet \to B_\bullet$ and $\underline{g} : B_\bullet \to C_\bullet$ where $\underline{f} = \{f_k\}_{k\in\mathbb{Z}}$, $\underline{g} = \{g_k\}_{k\in\mathbb{Z}}$ while

$$f_k \circ \partial_k^A = \partial_k^B \circ f_k \quad \text{and} \quad g_k \circ \partial_k^B = \partial_k^C \circ g_k. \quad \forall k \in \mathbb{N} \cup \{0\}$$

Furthermore suppose that there exists the short exact sequence

$$0 \longrightarrow A_\bullet \xrightarrow{\underline{f}} B_\bullet \xrightarrow{\underline{g}} C_\bullet \longrightarrow 0$$

Prove that, for every $k \in \mathbb{N} \cup \{0\}$, there exists an exact sequence

$$H_k(A_\bullet) \xrightarrow{\tilde{f}_k} H_k(B_\bullet) \xrightarrow{\tilde{g}_k} H_k(C_\bullet)$$

where $H_k(A_\bullet) \doteq \frac{\ker \partial_k^A}{\mathrm{Im}(\partial_{k+1})}$ (and similarly $H_k(B_\bullet)$ and $H^k(C_\bullet)$), whereas

$$\tilde{f}_k : H_k(A_\bullet) \to H_k(B_\bullet), \quad [a_k] \mapsto \tilde{f}_k[a_k] \doteq [f_k(a_k)],$$

while $\tilde{g}_k : H_k \to H_k(C_\bullet)$ is similarly defined.

Similarly we can rewrite at a homological level the zig-zag lemma 19. The proof is essentially identical and we do not feel guilty (well... not that I usually feel guilty... so it does not mean much) to leave also this step as an exercise to our beloved reader(s).

Exercise 59 (Zig-Zag for Homology)**.** Given a short exact sequence of differential complexes as in Exercise (58), for every $k \in \mathbb{N} \cup \{0\}$ there exists a linear map $\tilde{\partial}_k : H_k(C_\bullet) \to H_{k-1}(A_\bullet)$ so that the following long sequence is exact:

$$\begin{aligned}
\cdots \longrightarrow H_k(A_\bullet) \xrightarrow{\tilde{f}_k} H_k(B_\bullet) \xrightarrow{\tilde{g}_k} H_k(C_\bullet) \longrightarrow \\
\xrightarrow{\tilde{\partial}_k} H_{k-1}(A_\bullet) \xrightarrow{\tilde{f}_{k-1}} H_{k-1}(B_\bullet) \xrightarrow{\tilde{g}_{k-1}} H_{k-1}(C_\bullet) \longrightarrow \\
\xrightarrow{\tilde{\partial}_{k-1}} H_{k-2}(A_\bullet) \longrightarrow \cdots
\end{aligned} \tag{50}$$

If we apply this last exercise to the case of relative cohomology we obtain a long exact sequence similar to Equation (50) with the additional information that all homology groups for $k < 0$ are trivial per construction (here we use the notation $H_k(X) \equiv H_k(X; \mathbb{Z})$ and $H_k(Y) \equiv H_k(Y; \mathbb{Z})$):

$$
\begin{aligned}
&\cdots \longrightarrow H_k(Y) \xrightarrow{\iota_k} H_k(X) \xrightarrow{\pi_k} H_k(X,Y) \longrightarrow \\
&\xrightarrow{\tilde{\partial}_k} H_{k-1}(Y) \xrightarrow{\iota_{k-1}} H_{k-1}(Y) \xrightarrow{\pi_{k-1}} H_{k-1}(X,Y) \longrightarrow \\
&\xrightarrow{\tilde{\partial}_{k-1}} H_{k-2}(Y) \longrightarrow \cdots \longrightarrow H_0(X,Y) \longrightarrow 0
\end{aligned}
\tag{51}
$$

Important Remark: Relative homology has been constructed as subordinated to singular homology only for simplicity. Yet the whole procedure can be adapted verbatim to simplicial homology. The only difference is that we do not start from a topological space X together with a subspace Y. Rather, given an oriented simplicial complex K, we consider K_l as the union of all l'-simplices with $l' \leq l$. This is still an oriented simplicial complex together with a natural subcomplex K_{l-1}. Relative (simplicial) homology is built out of the pair (K_{l-1}, K_l). We sketch below the relevant maps since we will be needing them, but we do not enter into the details of the proofs since they are slavishly the same as in the case of relative singular homology. Hence our starting point is the inclusion of chain

$$
C_n(K_{l-1}) \subseteq C_n(K_l), \quad \forall l, n \in \mathbb{N} \cup \{0\}.
$$

Hence we can build the quotient Abelian group $C_n(K_l, K_{l-1}) \doteq \frac{C_n(K_l)}{C_n(K_{l-1})}$ and the associated chain complexes

$$
(C_\bullet(K_l), \partial_\bullet), \quad (C_\bullet(K_{l-1}), \partial_\bullet), \quad (C_\bullet(K_{l-1}, K_l), \partial_\bullet),
$$

where $\partial_\bullet \doteq \{\partial_k\}_{k\in\mathbb{Z}}$ is built out of the boundary operators in Equation (44) defined as the 0 operator if $k \leq 0$. These are related by a short exact sequences of chain maps:

$$
0 \longrightarrow C_\bullet(K_{l-1}) \xrightarrow{\iota_C} C_\bullet(K_l)) \xrightarrow{\pi_X} C_\bullet(K_{l-1}, K_l) \longrightarrow 0 \tag{52}
$$

where ι_C is induced by the inclusion map of chains, while π_C is induced by the natural projection $C_n(K_l) \to \frac{C_n(K_l)}{C_n(K_{l-1})}$. Applying Exercise 58 and Exercise 59 to the case in hand, we obtain the long exact sequence for simplicial homology:

$$
\begin{aligned}
&\cdots \longrightarrow H_k(K_{l-1}) \xrightarrow{\iota_k} H_k(K_l) \xrightarrow{\pi_k} H_k(K_l, K_{l-1}) \longrightarrow \\
&\xrightarrow{\tilde{\partial}_k} H_{k-1}(K_{l-1}) \xrightarrow{\iota_{k-1}} H_{k-1}(K_l) \xrightarrow{\pi_{k-1}} H_{k-1}(K_l, K_{l-1}) \longrightarrow \\
&\xrightarrow{\tilde{\partial}_{k-1}} H_{k-2}(K_{l-1}) \longrightarrow \cdots \longrightarrow H_0(K_l, K_{l-1}) \longrightarrow 0
\end{aligned}
\tag{53}
$$

We have all the ingredients to prove the equivalence between simplicial and singular homology under the assumption that the underlying space is triangulable, *cf.* Definition 158. We divide the analysis in two steps. The first can be summarized as follows:

Step 1: Consider a triangulable topological space X together with

- a simplicial complex K whose simplices are denoted as σ_i, $i = 1, \dots, N < \infty$ while their union is the polyhedron $|K|$,
- a triangulation $T \in C^0(|K|; X)$.

Since we work only with simplices which are in general position, *cf.* Definition 131, for every k_i-simplex $\sigma_i \in |K|$, there exists $L_i \in \mathrm{GL}(n, \mathbb{R})$, $i = 1, \dots, N$, such that $L_i(\sigma_i) = \Delta^{k_i}$ where Δ^{k_i} is the k_i-standard simplex, *cf.* Equation (46). In other words, combining all these data together, we obtain

$$T \circ L_i^{-1} : \Delta^{k_i} \to X, \quad \forall i = 1, \dots, N$$

which is a continuous map being such both T and L_i^{-1}, $i = 1, \dots, N$. Extending these maps per linearity we have established that given $(X, T : |K| \to X)$ there exists a family of canonical homomorphisms

$$\widetilde{T}_k \in \mathrm{Hom}(C_k(K); S_k(X)). \quad \forall k \in \mathbb{N} \cup \{0\}$$

To better appreciate the role of these homomorphisms, we suggest to the reader to show that these maps behave naturally with respect to the boundary operators as in Equation 44 and in Equation 47.

Exercise 60. Let $(X, T : |K| \to X)$ be a topological space admitting a triangulation and let $\mathcal{T}_k$ be the induced homomorphism between $C_k(X)$ and $S_k(X)$. Prove that it descends to $\mathcal{T}_k \in \mathrm{Hom}(H_k^{\Delta}(K); H_k(X; \mathbb{Z}))$, where

$$\mathcal{T}_k : H_k^{\Delta}(X) \to H_k(X; \mathbb{Z}), \quad [\sigma] \mapsto \mathcal{T}_k([\sigma]) = [\widetilde{T}_k(\sigma)]$$

and where we adopt the symbol H_k^{Δ} for the simplicial cohomology group, so to avoid potential notation clashes.

Step 2: Having established a relation between simplicial and singular k-chains and their associated homology groups, we need to prove that, actually, we have individuated an isomorphism. In this proof we limit our attention to the case where X is a topological manifold of finite dimension. This assumption can be dropped, see [6, Sec. 2.1], but the general case plays no role in our analysis and, thus, we do not consider it.

Theorem 162. *Let $(X, T : |K| \to X)$ be a finite dimensional topological manifold admitting a triangulation and, adopting the symbol H_k^{Δ} for the simplicial cohomology group, so to avoid potential notation clashes, let $\mathcal{T}_k \in \mathrm{Hom}(H_k^{\Delta}(K); H_k(X; \mathbb{Z}))$ be the induced homomorphism as per Exercise 60. This is a group isomorphism.*

Proof. Let X be as per assumption and let K_l, $l \in \mathbb{N} \cup \{0\}$, denote the union of all l'-simplices, $l' \leq l$, of K. Let us call $X_l \doteq \mathcal{T}(K_l)$ and let $H_n(X_l)$ and $H_n(X_l, X_{l-1})$ be the singular (*resp.* singular relative) cohomology groups generated by the cycles X_l. Putting together the long exact sequences in Equation (51) (bottom row) in Equation (53) (upper row) as well as the homomorphism in Exercise 60, we obtain for all $l, n \in \mathbb{N} \cup \{0\}$,

$$\begin{tikzcd}
\dots \arrow[r] & H^{\Delta}_{n+1}(K_l, K_{l-1}) \arrow[r, "\widetilde{\partial}_{n+1}"] \arrow[d, "\mathcal{T}_{n+1}"] & H^{\Delta}_n(K_{l-1}) \arrow[r, "(\iota_C)_n"] \arrow[d, "\mathcal{T}_n"] & H^{\Delta}_n(K_l) \arrow[r, "(\pi_C)_n"] \arrow[d, "\mathcal{T}_n"] & H^{\Delta}_n(K_l, K_{l-1}) \arrow[r, "\widetilde{\partial}_n"] \arrow[d, "\mathcal{T}_n"] & H^{\Delta}_{n-1}(K_{l-1}) \arrow[r] \arrow[d, "\mathcal{T}_{n-1}"] & \dots \\
\dots \arrow[r] & H_{n+1}(X_l, X_{l-1}) \arrow[r, "\widetilde{\partial}_{n+1}"] & H_n(X_{l-1}) \arrow[r, "\iota_n"] & H_n(X_l) \arrow[r, "\pi_n"] & H_n(X_l, X_{l-1}) \arrow[r, "\widetilde{\partial}_n"] & H_{n-1}(X_{l-1}) \arrow[r] & \dots
\end{tikzcd}$$

Observe that all vertical maps are still group homomorphisms due to Exercise 60. The next step consists on focusing on the first and on the fourth columns. Per construction $C_n(K_l, K_{l-1})$ consists only of l-simplices. Therefore if $n \neq l$ $C_n(K_l, K_{l-1})$ is the trivial Abelian group which entails that $H^{\Delta}_n(K_l, K_{l-1}) = \{0\}$. If $n = l$, $C_n(K_n, K_{n-1})$ is the free Abelian group generated by all n-simplices contained in the polyhedron $|K|$. Accordingly. $H^{\Delta}_n(K_n, K_{n-1})$ has the same description. Looking at the level of relative singular homology the pair (X_l, X_{l-1}) can be realized starting from considering $\bigsqcup_{\alpha \in I}(\Delta^k_\alpha, \partial\Delta^k_\alpha)$, where the index α labels the k-simplices in $|K|$ together with their boundaries. Per construction the triangulation T induces a map from $\bigsqcup_{\alpha \in I}(\Delta^k_\alpha, \partial\Delta^k_\alpha)$ to (X_k, X_{k-1}) which descends to an homeomorphism between the quotient spaces $\frac{\bigsqcup_{\alpha \in I}\Delta^k_\alpha}{\bigsqcup_{\alpha \in I}\partial\Delta^k_\alpha}$ and $\frac{X_k}{X_{k-1}}$. This descends to an isomorphism between the associated simplicial and singular homology groups, which entails that $H_n(X_l, X_{l-1})$ is trivial if $n \neq l$ while for $n = l$ is the free Abelian group generated by all l-cycles which are the image under T of an l-simplex. In other words the first and the fourth column in the diagram are isomorphisms. To conclude the proof, the trick consists of using the induction hypothesis with respect to the index l on the second and on the fifth column. More precisely let us start from $l = 0$. Since $K_{l-1} = X_{l-1} = \{0\}$ the second and the fifth column are isomorphisms. By invoking (yes... this is magic!) the Five Lemma, *cf.* Exercise 52, we conclude that $H^{\Delta}_n(K_0) \simeq H_n(X_0)$ for every $n \in \mathbb{N} \cup \{0\}$. We can now consider $l = 1$ and we can easily realize that the same argument entails that $H^{\Delta}_n(K_1) \simeq H_n(X_1)$. Proceeding by iteration we have shown that $H^{\Delta}(K_n) \simeq H_n(X_l)$ for every $l, n \in \mathbb{N} \cup \{0\}$. Since X is finite dimensional there exists $\bar{l} \in \mathbb{N}$ such that $H_n(X_{\bar{l}}) = H_n(X_{l'})$ for all $l' \geq \bar{l}$. In addition it holds per construction that $X_{\bar{l}} = T(|K|)$. Furthermore since a triangulation is per definition an homeomorphism it holds that $H_n(X_{\bar{l}}) = H_n(T(|K|)) = H_n(X, \mathbb{Z})$ for all $n \in \mathbb{N} \cup \{0\}$. □

5.4 The De Rham Theorem

Having established that singular and simplicial homology are related and having outlined the construction of the associated cohomology groups, we can investigate the relation with de Rham cohomology. To this end we will need to work mainly with the field of the reals, $\mathbb{R}$ rather than with $\mathbb{Z}$. This will have the net effect to get rid of all torsion elements in the underlying Abelian groups, computationally an advantage, but at the same time a loss of information at a topological level.

In order to characterize a relation between singular homology and de Rham cohomology, the first hurdle consists in an important structural difference. Singular and simplicial homology are subordinated to the topology of the underlying space and therefore all maps involved in the construction are continuous. In the case of de Rham cohomology, despite encoding topological information, the building blocks are differential forms, hence subordinated to the choice of an underlying smooth structure. In order to find a comparison criterion, the first step consists of eliminating this difference. Henceforth our ingredients will be the following:

- We will consider any but fixed smooth manifold M which always admits a triangulation, *cf.* Theorem 159,
- we consider a simplicial complex K and the associated polyhedron $|K|$. Observe that each k-simplex $|\sigma|$ in a convex open subset of $\mathbb{R}^N$, $N \geq 1$. Hence we can endow it and accordingly $|K|$ with the induced differentiable structure (and topology).

Having noticed that we can endow a polyhedron with a smooth structure, we can address the problem of establishing a relation between a smooth manifold and a simplicial complex compatibly to the differential structure. The first step consists of a specialization of Definition 158.

Definition 163. *Let M be a smooth n-dimensional manifold. Let K be a simplicial complex in $\mathbb{R}^N$. We call it a* **smooth (or $\mathbf{C^\infty}$-) triangulation** *of M if there exists $T : |K| \to M$ where $|K|$ is the polyhedron associated to K and if for every n-simplex $|\sigma| \in K$, $t|_{|\sigma|}$ is a smooth embedding. This means that that $t|_{|\sigma|}$ can be extended to smooth embedding from an open neighbourhood $U \subset \mathbb{R}^N$ to M where $|\sigma| \subset U$.*

As for the standard case, we do not prove existence of such kind of triangulations, but we limit ourselves to quoting a key result still due to Cairns and Whitehead [4, 16]

Theorem 164 (**Cairns-Whitehead (smooth version)**)**.** *Let M be a smooth manifold, possibly with a non empty boundary. Then there exists a smooth triangulation of M and, in addition, any triangulation of ∂M can be extended to one of the whole M.*

While this information allows to work with smooth structure at the level of simplicial homology, we still need to discuss what happens in the singular case. Here we start from the standard k-simplex, *cf.* Equation (45) or (46), which can be endowed with the induced smooth structure from $\mathbb{R}^k$. Accordingly we can introduce the natural generalization of Definition 148. At this stage and without any logical reason, I would like to say hello to Rubens asking him whether he has found this cameo dedicated to him. A reader not interested in this comment, might wish to consult [11, Chap. 18] for further details.

Definition 165. *Let M be a smooth manifold, we call* **smooth or ($\mathbf{C^\infty}$-) singular k-simplex** *any $\sigma_k \in C^\infty(\Delta^k; M)$. We denote with $S^\infty_k(M)$ the free Abelian group generated by all singular simplices σ_k and its elements are called* **smooth singular k-chains***.*

A direct inspection of Equation 47 unveils that the boundary operators can be safely restricted to smooth singular k-chains yielding for ever $k \in \mathbb{N} \cup \{0\}$

$$\partial_k : S_k^\infty(M) \to S_{k-1}^\infty(M).$$

Hence we can introduce both **k-dimensional smooth singular cycles** and **(k-dimensional) smooth singular boundary cycles** respectively

$$Z_k^\infty(M) = \mathrm{Im}(\partial_k|_{S_k^\infty(M)}), \quad B_k^\infty(M) = \ker(\partial_{k+1}|_{S_k^\infty(M)}). \quad \forall k \in \mathbb{N} \cup \{0\}$$

As a consequence there exists a natural counterpart of Definition 150 in the case in hand.

Definition 166. *Let M be a smooth manifold. For every $k \in \mathbb{N} \cup \{0\}$, we call* **k-th smooth singular homology group (with integer coefficients)**

$$H_k^\infty(M;\mathbb{Z}) \equiv H_k^\infty(M) = \frac{Z_k^\infty(M)}{B_k^\infty(M)},$$

where ∂_k is the boundary operator map as in Equation (47). The collection of all smooth singular homology groups forms a graded Abelian group $H_\bullet^\infty(M;\mathbb{Z}) \doteq \bigoplus_{k=0}^\infty H_k^\infty(M;\mathbb{Z})$.

It is natural to wonder whether there exists a relation between $H_k^\infty(M;\mathbb{Z})$ and $H_k(M;\mathbb{Z})$. One needs a rather lengthy analysis which is accounted for in [11, Th. 18.7]. Here we report only the final result.

Theorem 167. *Let M be a smooth manifold and let $\iota_k : S_k^\infty(M) \to S^\infty(M)$, $k \in \mathbb{N} \cup \{0\}$ be the natural inclusion map. This induces an isomorphism $\iota_{k,*} : H_k^\infty(M;\mathbb{Z}) \to H_k(M;\mathbb{Z})$.*

This theorem guarantees us that we can work at the level of smooth singular homology without loss of generality or of information on the underlying topology. At the level of singular cohomology this translates as follows:

Corollary 168. *Let M be a smooth manifold and let A be a ring. We call* **kth singular cohomology group with A-coefficients***, $k \in \mathbb{N} \cup \{0\}$,*

$$H_\infty^k(M;A) = \frac{\ker \delta^k|_{S_\infty^k(M;A)}}{\mathrm{Im}(\delta^{k+1}|_{S_\infty^{k+1}(M;A)})},$$

where $S_\infty^k(M;A) = \mathrm{Hom}(S_k^\infty(M;A);A)$ while δ^k is the coboundary operator built as per Lemma 146 out of $\partial_k|_{S_k^\infty(M;A)}$ the boundary operator as in Equation (47). It holds that, if $A = \mathbb{R}$

$$H_\infty^k(M;A) \simeq H^k(M;A).$$

Proof. This is a direct consequence of Theorem 167 together with the universal coefficient theorem, *cf.* [6, Th. 3.2], which guarantees that, for all $k \in \mathbb{N} \cup \{0\}$, $H^k(M;\mathbb{R}) \simeq \mathrm{Hom}(H_k(M;\mathbb{R}),\mathbb{R})$. □

The next step consists of analyzing how differential form can enter the game. This is done through the theory of integration. Let $\omega \in \Omega^k(M)$, *cf.* Definition 59 and let $\sigma \in C^\infty(\Delta^k; M)$ be a smooth singular k-simplex. We set

$$\int_{\sigma(\Delta^k)} \omega \doteq \int_{\Delta^k} \sigma^* \omega, \tag{54}$$

where Δ^k is a domain of integration on $\mathbb{R}^k$ since it is a bounded subset, σ^* indicates the pull-back as per Proposition 64 and the right hand side is built as per Definition 75. By linearity of the integral we can extend this formula to any smooth singular chain $c \in S_k^\infty(M)$, since, for all $i = 1, \ldots, N < \infty$, there exist $c_i \in \mathbb{Z}$ and $\sigma_i \in C^\infty(\Delta^k; M)$ such that $c = \sum_{i=1}^N c_i \sigma_i$. Hence

$$\int_c \omega \doteq \sum_{i=1}^N c_i \int_{\Delta^k} \sigma_i^* \omega.$$

Most notably we can formulate a counterpart of Theorem 79 for smooth singular chains.

Theorem 169 (**Stokes theorem for chains**)**.** *Let M a smooth manifold and let $c \in S_k^\infty(M)$, $k \in \mathbb{N}$. For every $\omega \in \Omega^{k-1}(M)$ it holds that*

$$\int_{\partial_k c} \omega = \int_c d_{k-1}\omega.$$

Proof. Per linearity of the integral, it suffices to prove the statement for $c = \sigma \in C^\infty(\Delta^k; M)$. It holds that

$$\int_{\sigma(\Delta^k)} d_{k-1}\omega = \int_{\Delta^k} \sigma^* d_{k-1}\omega = \int_{\Delta^k} d_{k-1}\sigma^*\omega = \int_{\partial_k \Delta^k} \sigma^*\omega,$$

where in the first equality we used Equation (54). The second identity descends from the naturality of the pull-back, *cf.* Theorem 68, while in the last one we used Theorem 79. To be precise, here we have used a version of Stokes theorem adapted to manifolds with corners. This is proven slavishly as the one outlined in these notes. Any doubting Thomas can consult [11, Th. 16.25]. Let $F_i : \Delta^{k-1} \to \partial\Delta^k$ $i = 0, \ldots, k$ be smooth parameterizations of the boundary faces of the standard k-simplex, chosen to be orientation preserving and abiding the requirements of [11, Prop. 16.8]. This last proposition entails that

$$\int_{\partial_k \Delta^k} \sigma^*\omega = \sum_{i=0}^k (-1)^i \int_{\Delta^{k-1}} F_i^*(\sigma^*\omega) = \sum_{i=0}^k (-1)^i \int_{\Delta^{k-1}} (\sigma \circ F_i)^*\omega =$$

$$= \sum_{i=0}^k (-1)^i \int_{(\sigma\circ F_i)(\Delta^{k-1})} \omega = \int_{\partial_k \sigma(\Delta^k)} \omega,$$

where we used the chain rule in the second identity. The other relations are special instances of Equation (54). ☐

We have all ingredients to start the main analysis of this section. The first step consists in introducing a relation between the de Rham cohomology group and the singular ones:

Proposition 170. *Let M be a smooth manifold and, for any $k \in \mathbb{N} \cup \{0\}$, let $H^k_{\mathrm{dR}}(M)$ and $H^k(M;\mathbb{R})$ be respectively the k-th de Rham cohomology group, cf. Definition 107 and the k-th singular cohomology group with real coefficients, cf. Definition 157 and Remark 42. Then the maps*

$$\mathcal{I}_k : H^k_{\mathrm{dR}}(M) \to H^k(M;\mathbb{R}), \quad [\omega] \mapsto \mathcal{I}_k([\omega]),$$

where

$$\mathcal{I}_k([\omega]) : H^\infty_k(M) \to \mathbb{R}, \quad [c] \mapsto \mathcal{I}_k([\omega])([c]) = \int\limits_c \omega, \tag{55}$$

are homomorphisms known as **de Rham homomorphisms**.

Proof. First of we observe that Theorem 167 entails that, for all $k \in \mathbb{N} \cup \{0\}$, $H_k(M;\mathbb{R}) \simeq H^\infty_k(M;\mathbb{R})$ and we recall that $H^k(M;\mathbb{R}) = \frac{Z^k(M;\mathbb{R})}{B^k(M;\mathbb{R})}$. Hence to see that the map $\mathcal{I}_k$ is well defined, we can start from

$$\widetilde{\mathcal{I}}_k : \Omega^k(M) \to S^k(M;\mathbb{R}) \doteq \mathrm{Hom}(S_k(M;\mathbb{R}),\mathbb{R}), \quad \omega \mapsto \widetilde{\mathcal{I}}_k(\omega)(c) \doteq \int\limits_c \omega.$$

Consider now $\omega = d_{k-1}\varphi$, $\varphi \in \Omega^{k-1}(M)$. It holds due to Theorem 169 and Lemma 146 that

$$\widetilde{\mathcal{I}}_k(d_{k-1}\varphi)(c) = \widetilde{\mathcal{I}}_{k-1}(\varphi)(\partial_k c) = [\delta^{k-1}\widetilde{\mathcal{I}}_{k-1}(\varphi)](c).$$

The last identity can be read saying that $\mathcal{I}_k : d\Omega^{k-1}(M) \to B^k(M;\mathbb{R})$. Similarly, if $\omega \in \Omega^k_d(M)$, it holds that, for any $c' \in S_{k+1}(M;\mathbb{R})$

$$[\delta^k\widetilde{\mathcal{I}}_k(\omega)](c') = \widetilde{\mathcal{I}}_k(\omega)(\partial_{k+1}c) = \widetilde{\mathcal{I}}_{k+1}k(d_k\omega)(c) = 0.$$

In other words we can read this identity as $\mathcal{I}_k : \Omega^k_d(M) \to Z^k(M;\mathbb{R})$. This entails that, for all $k \in \mathbb{N} \cup \{0\}$, $\widetilde{\mathcal{I}}_k$ descends to the quotient, making Equation (55) well-defined. Linearity of the integral entails that it is a group homomorphism. ☐

The next consists in showing that the de Rham homomorphism is natural with respect to pull-back.

Proposition 171. *Let M, N be smooth manifolds. Then, for every $k \in \mathbb{N} \cup \{0\}$ and for every $F \in C^\infty(M;N)$, the following diagram commutes*

$$\begin{tikzcd} H^k_{\mathrm{dR}}(N) \arrow[r, "F^*"] \arrow[d, "\mathcal{I}_k"'] & H^k_{\mathrm{dR}}(M) \arrow[d, "\mathcal{I}_k"] \\ H^k(N;\mathbb{R}) \arrow[r, "F^*"'] & H^k(M;\mathbb{R}) \end{tikzcd},$$

where $\mathcal{I}_k$ is the de Rham homomorphism as in Equation (55).

Proof. To start with observe that, for every singular smooth k-chain $\sigma \in C^\infty(\Delta^k; M)$, $F_*\sigma \doteq F \circ \sigma \in C^\infty(\Delta^k; N)$. Hence for every $[\alpha] \in H^k(N;\mathbb{R})$ and for every $[\sigma] \in H^k(M;\mathbb{R})$, it holds

$$(F^*[\alpha])([\sigma]) \doteq [\alpha]([F_*\sigma]),$$

which is well defined on account of Exercise 56. Let $[\omega] \in H^k_{\mathrm{dR}}(M)$, $k \in \mathbb{N} \cup \{0\}$. Yet Exercise 56 and equation (55) entail that, for every $[\sigma] \in H_k(M;\mathbb{R})$,

$$\mathcal{I}_k(F^*[\omega])([\sigma]) = \int_\sigma F^*\omega = \int_{F\circ\sigma} \omega = \mathcal{I}_k([\omega])([F \circ \sigma]) = \mathcal{I}_k([\omega])(F_*[\sigma]) = (F^*\mathcal{I}_k([\omega]))([\omega]),$$

which concludes the proof. □

Finally our last ingredient is the study of the interplay between the de Rham homomorphism and he coboundary operators. For this reason we need to consider once more a Mayer-Vietoris type of sequence. The starting point is a smooth manifold M together with $U, V \subseteq M$ such that $U \cup V = M$. Using standard inclusions we can draw the following, commutative, diagram

$$\begin{tikzcd} & U \arrow[dr, "\iota_{U,M}"] & \\ U \cap V \arrow[ur, "\iota_{U\cap V,U}"] \arrow[dr, "\iota_{U\cap V,V}"'] & & M \\ & V \arrow[ur, "\iota_{V,M}"'] & \end{tikzcd},$$

It is straightforward that these maps descend to homomorphisms between homology groups acting via push-forward. In other words, for every $k \in \mathbb{N} \cup \{0\}$ and for every ring A, it holds

$$\begin{tikzcd} & H_k(U;A) \arrow[dr, "\iota^U_{*,k}"] & \\ H_k(U \cap V;A) \arrow[ur, "\iota^{(U\cap V,U)}_{*,k}"] \arrow[dr, "\iota^{(U\cap V,V)}_{*,k}"'] & & H_k(M;A) \\ & H_k(V;A) \arrow[ur, "\iota^V_{*,k}"'] & \end{tikzcd},$$

where for simplicity we set $\iota^U_{*,k} \doteq (\iota_{U,M})_{*,k}$, $\iota^V_{*,k} \doteq (\iota_{V,M})_{*,k}$, $\iota^{(U\cap V,U)}_{*,k} = (\iota_{U\cap V,U})_{*,k}$ and $\iota^{(U\cap V,V)}_{*,k} = (\iota_{U\cap V,V})_{*,k}$. With this data, one can easily adapt Proposition 109 to the case in hand. The proof is mutatis mutandis the same and hence we leave it as an exercise:

Exercise 61. For all $k \in \mathbb{N} \cup \{0\}$, the following **Mayer-Vietoris sequence**, is exact in the sense of Definition 16:

$$0 \longrightarrow S_k(M) \xrightarrow{\iota^{\oplus}_{*,k}} S_k(U) \oplus S_k(V) \xrightarrow{\iota^{U\cap V,-}_{*,k}} S_k(U \cap V) \longrightarrow 0\,,$$

where, $\iota^{\oplus}_{*,k}(\sigma) \doteq (\iota^U_{*,k}\sigma, \iota^V_{*,k}\sigma)$, $\sigma \in S_k(M)$ while $\iota^{U\cap V,-}_{*,k}(\tau,\tau') = \iota^{U\cap V,U}_{*,k}\tau - \iota^{U\cap V,V}_{*,k}\tau'$, $(\tau,\tau') \in S_k(U) \oplus S_k(V)$. Prove that the result holds true also for the ring $A = \mathbb{R}$

The arbitrariness of k entails that we have individuated a short exact sequence of chain complexes ($A = \mathbb{Z}, \mathbb{R}$)

$$0 \longrightarrow S_\bullet(M;A) \xrightarrow{\iota^{\oplus}_*} S_\bullet(U;A) \oplus S_\bullet(V:A) \xrightarrow{\iota^{U\cap V,-}_*} S_\bullet(U \cap V;A) \longrightarrow 0\,, \tag{56}$$

which descends to a long exact sequence at the level of singular homology thanks to the zig zag lemma, *cf.* Exercise 59

$$\begin{aligned}
\cdots &\longrightarrow H_k(M;A) \xrightarrow{\tilde{\iota}^{\oplus}_k} H_k(U;A) \oplus H_k(V;A) \xrightarrow{\tilde{\iota}^{U\cap V,-}_k} H_k(U\cap V;A) \longrightarrow\,,\\
&\xrightarrow{\tilde{\partial}_k} H_{k-1}(M;A) \xrightarrow{\tilde{\iota}^{\oplus}_{k-1}} H_{k-1}(U;A) \oplus H_{k-1}(V;A) \xrightarrow{\tilde{\iota}^{U\cap V,-}_{k-1}} H_{k-1}(U\cap V;A) \longrightarrow\\
&\xrightarrow{\tilde{\partial}_{k-1}} H_{k-2}(M;A) \longrightarrow \cdots \longrightarrow H_0(U\cap V;A) \longrightarrow 0
\end{aligned} \tag{57}$$

where all maps are the natural counterparts at the level of singular homology groups of those in Equation (56), while $\widetilde{\partial}_k$ and $\widetilde{\partial}_{k-1}$ are the connecting homomorphisms constructed via the zig zag lemma, *cf.* Exercise 59. Most notably this long exact sequence admits a dual counterpart defined with respect to singular cohomology groups. We leave the proof of this fact to our readers suggesting to check the proof of Proposition 171 for the definition of a pull-back map acting on singular cohomology groups.

Exercise 62. Let M be a smooth manifold and let $A = \mathbb{Z}, \mathbb{R}$. Starting from Equation (57) prove the existence and well-posedness of the following long exact sequence

$$\begin{aligned}
\cdots &\longrightarrow H^k(M;A) \xrightarrow{\tilde{\iota}^{*,k}_{\oplus}} H^k(U;A) \oplus H^k(V;A) \xrightarrow{\tilde{\iota}^{*,k}_{U\cap V,-}} H^k(U\cap V;A) \longrightarrow\,,\\
&\xrightarrow{\tilde{\delta}_k} H^{k+1}(M;A) \xrightarrow{\tilde{\iota}^{*,k+1}_{\oplus}} H^{k+1}(U;A) \oplus H^{k+1}(V;A) \xrightarrow{\tilde{\iota}^{*,k+1}_{U\cap V,-}} H^{k+1}(U\cap V;A) \longrightarrow\\
&\xrightarrow{\tilde{\delta}_{k+1}} H^{k+2}(M;A) \longrightarrow \cdots
\end{aligned} \tag{58}$$

where all maps are the dual ones of these in the long exact sequence in Equation (57).

Proposition 172. *Let M be a smooth manifold and let $U, V \subseteq M$ be such that $U \cup V = M$. Then, for every $k \in \mathbb{N} \cup \{0\}$, the following diagram commutes*

$$\begin{tikzcd} H^{k-1}_{\mathrm{dR}}(U \cap V) \arrow[r, "\tilde{d}_{k-1}"] \arrow[d, "\mathcal{I}_{k-1}"'] & H^k_{\mathrm{dR}}(M) \arrow[d, "\mathcal{I}_{k-1}"] \\ H^{k-1}(U \cap V; \mathbb{R}) \arrow[r, "\widetilde{\delta}_{k-1}"'] & H^k(M; \mathbb{R}) \end{tikzcd} ,$$

where $\mathcal{I}_k$ is the de Rham homomorphism as in Equation (55), while $\tilde{d}_{k-1}$ is the connecting map in Equation (38) and $\tilde{\delta}_{k-1}$ that in Equation (58).

Proof. Let $[\omega] \in H^{k-1}_{\mathrm{dR}}(U \cap V)$ and let $[\sigma] \in H_k(M; \mathbb{R})$. From Equation (55) it descends

$$\mathcal{I}_k(\tilde{d}_{k-1}([\omega]))([\sigma]) = \mathcal{I}_k([\omega])((\partial_k)_*[\sigma]),$$

where we used Stokes theorem 169. Calling $c \in S^\infty_{k-1}(M; \mathbb{R})$ a representative of $(\partial_k)_*[\sigma]$ and $\theta \in \Omega^k(U \cap V)$ one of $\tilde{d}_{k-1}([\omega])$, the above identity reads

$$\int\limits_\sigma \theta = \int\limits_c \omega.$$

In view of the characterization of $(\partial_k)_*$ coming from the zig zag lemma, *cf.* Exercise 59, there must exist $e \in S^\infty_k(U)$ and $e' \in S^\infty_k(V)$ such that $c = \partial_k e = \partial_k e'$ and $e + e'$ is a representative of $[\sigma]$. At the same time the characterization of the connection homomorphism for de Rham cohomology, *cf.* Lemma 19 and Equation (38) entail that there exist $\eta \in \Omega^{k-1}(U)$ and $\eta' \in \Omega^{k-1}(V)$ such that $\omega = \eta|_{U\cap V} - \eta'|_{U\cap V}$. Hence we can set

$$\widetilde{\theta} = \begin{cases} d\eta & \text{on } U \\ d\eta' & \text{on } V \end{cases}$$

Hence $\partial_k e + \partial_k e' = \partial_k c = 0$ and $d_{k-1}\eta|_{U\cap V} - d_{k-1}\eta'|_{U\cap V} = d_{k-1}\omega = 0$. From this it descends

$$\int\limits_c \omega = \int\limits_{\partial_k e} \omega = \int\limits_{\partial_k e} \eta - \int\limits_{\partial_k e} \eta' = \int\limits_{\partial_k e} \eta + \int\limits_{\partial_k e'} \eta' = \int\limits_e d_k\eta + \int\limits_{e'} d\eta' = \int\limits_e \widetilde{\theta} + \int\limits_{e'} \widetilde{\theta} = \int\limits_\sigma \widetilde{\theta},$$

where the rightmost expression is nothing but $\widetilde{\delta}_{k-1}\left(\mathcal{I}_{k-1}([\omega])\right)([\sigma])$. This proves that the diagram is commutative. □

Before proceeding with the main statement, we need a rather direct technical result, which we leave as an exercise.

Exercise 63. Let M_j, $j = 1, \dots, N$ be a countable collection of smooth manifolds and let A be any ring. Prove that the natural inclusion maps $\iota_j : M_j \to \bigsqcup_{j=1}^{N} M_j \equiv M_1 \times \cdots \times M_N$, $j = 1, \dots, n$ induces the following isomorphisms for all $k \in \mathbb{N} \cup \{0\}$

$$H^k_{\mathrm{dR}}(M_1 \times \cdots \times M_N) = \bigoplus_{j=1}^{N} H^k_{\mathrm{dR}}(M_j), \quad H^k(M_1 \times \cdots \times M_N; \mathbb{R}) = \bigoplus_{j=1}^{N} H^k_{\mathrm{dR}}(M_j; \mathbb{R}).$$

Theorem 173 **(de Rham).** *Let M be a smooth manifold. For every $k \in \mathbb{N} \cup \{0\}$, the map $\mathcal{J}_k$ as in Equation (55) is an isomorphism.*

Proof. The proof is divided in different steps.

Step 1. We prove that, for any convex open subset U of $\mathbb{R}^n$, $n \geq 1$, $\mathcal{J}_k$ is an isomorphism for all $k \in \mathbb{N} \cup \{0\}$. Using Exercise 4 we know that $H^k_{\mathrm{dR}}(U) = \{0\}$ if $k \neq 0$, while $H^0_{\mathrm{dR}}(U) \simeq \mathbb{R}$. Working at the level of singular cohomology, we observe that U is homotopic to a single point, say $\{p\}$ and that singular homology and cohomology groups are homotopy invariant, *cf.* Corollary 155 adapted to the ring $A = \mathbb{R}$ and Remark 42. Hence Proposition 156 entails that $H_0(U; \mathbb{R}) \simeq \mathbb{R}$. If $k > 0$, then it is easy to realize that, since there exists only the constant k-simplex $\sigma_0 : \Delta^k \to \{p\}$, Equation (47) entails that $\partial_k \sigma_k = \{0\}$ if k is odd while it coincides with σ_{k-1} if k is even. Yet since $H_k(U; \mathbb{R}) = \frac{\ker \partial_k}{\mathrm{Im}(\partial_{k-1})} = \frac{\mathbb{R}}{\mathbb{R}} = \{0\}$. Accordingly since $H^k(U; \mathbb{R}) = \mathrm{Hom}(H_k(U; \mathbb{R}); \mathbb{R})$, then $H^k(U; \mathbb{R}) = \{0\}$. To conclude this step is suffices to observe that $\mathcal{J}_0$ is not the zero map. This is the case since, given $[f] \in H^0_{\mathrm{dR}}(M)$ and $[\sigma] \in H_0(M; \mathbb{R})$, then

$$\mathcal{J}_0([f])[\sigma] = f(\sigma(0)),$$

where we used that $\sigma : \Delta^0 \to U$ is nothing but $\sigma(0)$. Yet since f is a closed 0-form, it is the constant function. Fixing it equal to 1, entails that $\mathcal{J}_0([f])([\sigma]) = 1$.

Step 2. We prove that if M admits a finite cover $\{U_i\}_{i \in I}$ made of convex open subsets such that the map $\mathcal{J}_k$ acting on $H^k_{\mathrm{dR}}(U_i)$ is an isomorphism for all $k \in \mathbb{N} \cup \{0\}$ and for all $i \in I$, then the statement of the theorem holds true. We prove the result by induction on i. If $i = 1$, there is nothing to prove, since the statement is true per construction. Consider $i = 2$ and let $U, V \subset M$ be such that the hypotheses are met. Using the long exact sequences (38) and (58), together with Proposition 170 we can construct the following commuting diagram for all $k \in \mathbb{N}$:

$$\begin{tikzcd}
H^{k-1}_{\mathrm{dR}}(U) \oplus H^{k-1}_{\mathrm{dR}}(V) \arrow[r, "\tilde{\iota}^{*,k-1}_{U\cap V,-}"] \arrow[d, "\mathcal{J}_k"'] & H^{k-1}_{\mathrm{dR}}(U \cap V) \arrow[r, "\tilde{d}_{k-1}"] \arrow[d, "\mathcal{J}_k"'] & H^k_{\mathrm{dR}}(M) \arrow[r, "\tilde{\iota}^{*,k}_{\oplus}"] \arrow[d, "\mathcal{J}_k"'] & {} \\
H^{k-1}(U; \mathbb{R}) \oplus H^{k-1}(U; \mathbb{R}) \arrow[r, "\tilde{\iota}^{*,k-1}_{U\cap V,-}"'] & H^{k-1}(U \cap V; \mathbb{R}) \arrow[r, "\tilde{\delta}_{k-1}"'] & H^k(M; \mathbb{R}) \arrow[r, "\tilde{\iota}^{*,k}_{\oplus}"] & {}
\end{tikzcd}$$

$$\begin{tikzcd}
{} \arrow[r, "\tilde{\iota}^{*,k}_{\oplus}"] & H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V) \arrow[r, "\tilde{\iota}^{*,k}_{U\cap V,-}"] \arrow[d, "\mathfrak{I}_k"'] & H^k_{\mathrm{dR}}(U\cap V) \arrow[d, "\mathfrak{I}_k"] \\
{} \arrow[r, "\tilde{\iota}^{*,k}_{\oplus}"'] & H^k(U;\mathbb{R}) \oplus H^k(V;\mathbb{R}) \arrow[r, "\tilde{\iota}^{*,k}_{U\cap V,-}"'] & H^k(U\cap V;\mathbb{R})
\end{tikzcd} .$$

Since $U \cap V$ is a convex open subset of $\mathbb{R}^n$ being such U and V separately, we can use the induction hypothesis and Step 1. to conclude that the first, second, fourth and fifth vertical maps are all isomorphisms. Apply the five lemma, *cf.* Exercise 52, we conclude that also the middle one is an isomorphism. Proceeding with $i > 2$, the reasoning is similar letting the role of U being played by $\bigcup_{j=1}^{i-1} U_j$ while $V = U_j$.

Step 3. We prove that if M admits a basis of its topology $\{U_\alpha\}_{\alpha\in J}$, J being a countable index set, where $\mathfrak{I}_k$ is an isomorphism for every $k \in \mathbb{N}\cup\{0\}$ and for every $\alpha \in J$, then $\mathfrak{I}_k$ enjoys the same property on M. To this end we need a standard result in differential geometry, *cf.* [11, Prop. 2.28], namely that every smooth manifold M admits a positive *exhaustion function*. This is $f \in C^\infty(M;(0,\infty))$ such that for every $c > 0$, $f^{-1}((0,c])$ is a compact subset of M. Considering such function, we define two families of sets, $m \in \mathbb{N}\cup\{0\}$:

$$A_m \doteq \{x \in M \mid m \leq f(x) \leq m+1\}, \quad A'_m \doteq \{x \in M \mid m - \frac{1}{2} < f(x) < m + \frac{3}{2}\}.$$

For each point $p \in A_m$, there exists a basis subset containing p and contained in A'_m. Since A_m is compact being f an exhaustion function, we can cover it with a finite number of such basis elements. Their collection is indicated with B_m and, on account of Step 2. the map $\mathfrak{I}_k$ is an isomorphism on B_m for every $k \in \mathbb{N}\cup\{0\}$. Since $B_m \subset A'_m$, it holds that $B_m \cap B_{m'}$ if and only if $m' = m \pm 1$. Consider therefore

$$U = \bigcup_{m \text{ odd}} B_m, \quad V = \bigcup_{m \text{ even}} B_m.$$

To prove that $\mathfrak{I}_k$ is an isomorphism for all $k \in \mathbb{N}\cup\{0\}$ on both U and V knowing that it is such on each B_m, it suffices to invoke Exercise 63. Finally, consider $U \cap V$. This can be read as the disjoint union of the set $B_m \cap B_{m+1}$, $m \in \mathbb{Z}$. Each of these admits a finite cover such that $\mathfrak{I}_k$ is an isomorphism on all of the underlying sets for all $k \in \mathbb{N}\cup\{0\}$. These set are nothing but intersection of the basis sets of B_m and B_{m+1}. Hence we can now use Step 2. to conclude that $\mathfrak{I}_k$ is an isomorphism for all $k \in \mathbb{N}\cup\{0\}$ on $M = U \cup V$ since it is such on U, V and $U \cap V$ separately.

Step 4. Let $U \subseteq \mathbb{R}^n$, $n \geq 1$, be any open subset endowed with the induced topology. An associated basis is built out of Euclidean balls, which are convex. On account of Step 1. $\mathfrak{I}_k$ is an isomorphism on each of these for all $k \in \mathbb{N}\cup\{0\}$. The same applies to any finite intersection of basis elements since they are still convex open subsets of $\mathbb{R}^n$. Accordingly we can apply Step

3. to conclude that $\mathcal{I}_k$ is an isomorphism on U for all $k \in \mathbb{N} \cup \{0\}$. To conclude it suffices to consider a cover of M in open coordinate neighbourhoods. They and their finite intersections are diffeomorphic to open subsets of $\mathbb{R}^n$. Hence using the previous reasoning and once more Step 3., we can conclude that $\mathcal{I}_k$ is an isomorphism for all $k \in \mathbb{N} \cup \{0\}$ on each of these subsets and, thus, also on M. This concludes the proof □

5.5 Characteristic Classes

In outlining the theory of vector and of principal bundles, it has been emphasized more than once that what matters is not a single bundle but rather the equivalence classes built out of isomorphic bundles. Yet, given a smooth manifold M and a typical fiber, regardless whether it is a vector space or a Lie group, it is very hard to establish whether two bundle are actually isomorphic. Even more complicated is answering the question how many non trivial vector or principal bundles with a given typical fiber or structure group do exist on top of a smooth manifold M. One needs therefore a tool which allows to give at least a partial answer to such questions. In this endeavor we will make use of the theory of connections developed in Section 3.4 and in Section 3.5 combining it with that of de Rham and singular cohomology, *cf.* Section 4 and Section 5.2. Our analysis will be rather succinct and straight to the point avoiding many technical results and side remarks which will force us to write even longer notes. A reader who is really in pain for such lack of an in depth analysis can consult an old, but really nice book written by a true master [13].

As starting point we need to introduce a new, key ingredient, which will help us in developing the theory of characteristic classes.

Definition 174. *Let $n \geq 1$ and let $f : \mathfrak{gl}(n,\mathbb{R}) \to \mathbb{R}$ be a polynomial function. We call it* **invariant polynomial** *if for every $A \in GL(n,\mathbb{R})$*

$$f(A^{-1}XA) = f(X), \quad \forall X \in \mathfrak{gl}(n,\mathbb{R}).$$

Observe that invariant polynomial form a subalgebra of the polynomial one which is not empty since there exists two notable examples, the trace and the determinant functions:

$$\mathrm{Tr} : \mathfrak{gl}(n,\mathbb{R}) \to \mathbb{R}, \quad \text{and} \quad \det : \mathfrak{gl}(n,\mathbb{R}) \to \mathbb{R}.$$

We recall a notable result in linear algebra, but... wait... I forgot that you are like John Snow... you know nothing. So let us start from scratch proving the following structural property.

Proposition 175. *Let I_n be the algebra of invariant polynomials of $\mathfrak{gl}(n,\mathbb{R})$ as per Definition 174. Let*

$$\chi : \mathbb{R} \to I_n \quad t \mapsto \det_t(A) \doteq \det(t\mathbb{I} - A).\ \forall A \in \mathrm{GL}(n,\mathbb{R})$$

Expanding this formula as $\chi(t) = \sum_{i=0}^{n} (-1)^i t^i s_{n-i}(\cdot)$, with $s_0(A) = 1$ for all $A \in \mathfrak{gl}(n,\mathbb{R})$, it holds that $s_i(\cdot)_{i=1,\dots,n}$ are algebraically independent and they generate I_n, that is

$$I_n \simeq \mathbb{R}\left[s_1, \dots, s_n\right].$$

Proof. Consider a vector $\mathbb{R}^n \ni v = (v_1, \dots, v_n)$ and the so-called companion matrix

$$A_v \doteq \begin{pmatrix} 0 & \dots & \dots & 0 & -v_n \\ 1 & \ddots & & \vdots & -v_{n-1} \\ 0 & \ddots & \ddots & 0 & -v_2 \\ 0 & \dots & 1 & 0 & -v_1 \end{pmatrix},$$

to which it is associated $[\chi(t)](A_v) = t^n + \sum_{i=1}^{n} v_{n-i} t^{n-i}$. In other words $s_i(A_v) = (-1)^i v_i$. Since the values of each v_i are arbitrary we can generate $\mathbb{R}^n$ with the vectors $(s_1(A_v), \dots, s_n(A_v))$. To prove that the maps $s_i(\cdot)$ are algebraically independent, suppose there exist a non trivial polynomial function $p(s_1, \dots s_n)$ such that, for all $A \in \mathrm{GL}(n, \mathbb{R})$, $p(s_1, \dots, s_n)(A) = p(s_1(A), \dots, s_n(A)) = 0$. This holds true in particular for A_v and thus it descends that $p(v_1, \dots, v_n) = 0$ for all $v_1, \dots, v_n \in \mathbb{R}$. This can happen only if $p = 0$, which entails the statement. To show that we can generate the whole I_n, let f be an invariant polynomial and let $p : \mathbb{R}^n \to \mathbb{R}$ be the polynomial function such that

$$p(v_1, \dots, v_n) \doteq f(A_v).$$

Similarly define the polynomial $P : I_n \to \mathbb{R}$ such that

$$P(A) \doteq p(-s_1(A), s_2(A), \dots, (-1)^n s_n(A)).$$

Per construction P and f agree on companion matrices A_v and, due to to the invariance property also on all matrices of the form BA_vB^{-1} where $B \in \mathrm{GL}(n, \mathbb{R})$. Yet a truly standard result in linear algebra guarantees that every matrix with n distinct eigenvalues is conjugate to a companion one and this forms a dense subset of $\mathfrak{gl}(n, \mathbb{R})$. Since $f - P$ is the trivial operator on such dense subset, per continuity, it admits a unique extension to the whole $\mathfrak{gl}(n, \mathbb{R})$ as the zero operator. $\square$

An alternative convenient basis of I_n can be constructed as follows. For each $k \in \mathbb{N}$, let

$$I_n \ni \sigma_k \mid \sigma_k(A) = \mathrm{Tr}(A^k), \quad \forall A \in \mathfrak{gl}(n, \mathbb{R}). \tag{59}$$

It is easy to connect the collection $\{\sigma_k\}_{k \in \mathbb{N}}$ to the basis $\{s_i\}_{i \in \mathbb{N}}$, *i.e.*

$$\sigma_1 = s_1, \quad \sigma_2 = s_1^2 - 2s_2, \quad \sigma_3 = s_1^3 - 3s_1s_2 + 3s_3, \dots$$

In other words it holds that, for all $n \geq 1$,

$$I_n \simeq \mathbb{R}[\sigma_1, \dots, \sigma_n].$$

We have all ingredients to start studying characteristic classes. This is a very broad topic, which would need a separate course. Here we shall focus on three different scenarios which we will be analyzing separately: smooth real vector bundles, smooth complex vector bundles and principal bundles.

Let $E \equiv E[M;\pi,\mathbb{R}^k]$ be a real smooth vector bundle over a smooth manifold M of dimension $\dim M = n \geq 1$ as per Definition 20. Let ∇ be a curvature as per Equation 24 and let R be the associated curvature, *cf.* Definition 91. Recall that we can associate to R, $\Lambda \in \Omega^2(M;\mathfrak{gl}(n,\mathbb{R}))$, *cf.* Theorem 93. In addition consider a cover $U_{\alpha\alpha\in I}$, I being an index set, of M made of open subsets so that, whenever $U_\alpha \cap U_\beta \neq \emptyset$, Lemma 94 entails that thereon

$$\Lambda_\beta = g_{\alpha\beta}^{-1}\Lambda_\alpha g_{\alpha\beta},$$

where $\Lambda_\alpha, \Lambda_\beta$ are the restriction of the curvature 2-form to U_α and U_β respectively while $g_{\alpha\beta}: U_\alpha \cap U_\beta \to \mathrm{GL}(n,\mathbb{R})$ are the transition functions. Since the curvature is a form with values in the Lie algebra $\mathfrak{gl}(n,\mathbb{R})$, we can combine it with invariant polynomials introduced in Definition 174. The outcome is summarized in the following proposition

Proposition 176. *Let M be a smooth manifold of dimension $\dim M = n \geq 1$ and let $E \equiv E[M;\pi,\mathbb{R}^k]$ be a smooth real vector bundle on top of which it is assigned a connection. Let $\Lambda \in \Omega^2(M;\mathfrak{gl}(n,\mathbb{R}))$ be the associated curvature 2-form. The, for every invariant polynomial $f \in I_n$ of degree k, it holds that $f(\Lambda) \in \Omega^{2k}(M;\mathfrak{gl}(n;\mathbb{R}))$ is a closed form.*

Proof. Let Λ be as per hypothesis. On account of Cartan second structure equation, *cf.* Theorem 93,

$$d_2\Lambda = d_1\omega \wedge \omega - \omega \wedge d_1\omega = \Lambda \wedge \omega - \omega \wedge \Lambda,$$

which is known as **Bianchi's identity**. Here $\omega \in \Omega^1(M;\mathfrak{gl}(n,\mathbb{R}))$ is the connection 1-form, *cf.* Equation (28). Since the algebra of invariant polynomials is generated by $\{\sigma_k\}_{k\in\mathbb{N}}$ as in Equation (59), it suffices to prove that $d_2\sigma_k(\Lambda)$ vanishes. A direct computation shows that, calling $\Lambda^k = \underbrace{\Lambda \wedge \cdots \wedge \Lambda}_{k}$,

$$d_2\sigma_k(\Lambda) = d_2\mathrm{Tr}(\Lambda^k) = \mathrm{Tr}(d_2\Lambda^k) = \mathrm{Tr}(d_2\Lambda \wedge \Lambda^{k-1} + \Lambda \wedge d_2\Lambda \wedge \Lambda^{k-1} + \cdots + \Lambda^{k-1} \wedge d_2\Lambda) =$$
$$\mathrm{Tr}((\Lambda \wedge \omega - \omega \wedge \Lambda) \wedge \Lambda^{k-1} + \Lambda \wedge (\Lambda \wedge \omega - \omega \wedge \Lambda) \wedge \Lambda^{k-1} + \cdots + \Lambda^{k-1} \wedge (\Lambda \wedge \omega - \omega \wedge \Lambda)) =$$
$$\mathrm{Tr}(-\omega \wedge \Lambda^k + \Lambda^k \wedge \omega) = 0,$$

where we used that d_2 commutes with Tr per linearity, while, in the second equation we exploited that d_2 in an anti-derivation, *cf.* Theorem 65. In the third step we inserted Bianchi's identity, while, in the last one, we used the cyclicity of the trace. □

Since each pair (f,Λ) where f is an invariant polynomial and Λ is the curvature 2-form identifies a closed $2k$-form, we can identify its equivalence class in the de Rham cohomology, namely

$$[f(\Lambda)] \in H^{2k}_{\mathrm{dR}}(M).$$

The natural question is whether this class provides information on the topology of the underlying manifold. This is tantamount to asking what does it happens if a different connection is chosen. The following proposition answers the question

Proposition 177. *Let M be a smooth manifold and let $E \equiv E[M;\pi,\mathbb{R}^k]$ be a smooth real vector bundle. Let $[f(\Lambda)] \in H^{2k}_{\mathrm{dR}}(M)$ be the de Rham cohomology class individuated by an invariant polynomial and by the curvature 2-form. It does not depend on the choice of the connection on E.*

Proof. Suppose that we have chosen two connections ∇ and ∇' with associated curvature 2-forms, Λ and Λ'. Since both E and M are smooth manifolds, we can build a second smooth vector bundle $E\times\mathbb{R}$, whose base space is $M\times\mathbb{R}$ and whose typical fiber is the same as that of E. The projection map is

$$\widetilde{\pi} = \pi \times \mathrm{id} \in C^\infty(E\times\mathbb{R}; M\times\mathbb{R}).$$

In order to assign a connection to $E\times\mathbb{R}$, we employ Equation (24). To this end, calling t the coordinate along $\mathbb{R}$, we can use Exercise 12 to infer that $\Gamma^\infty(T(M\times\mathbb{R}))$ is spanned by $\frac{\partial}{\partial t}$ together with all the vector field $X \in \Gamma^\infty(TM)$ up to the identification of TM with $T(M\times\bar{t})$, $\bar{t}\in\mathbb{R}$. In other word for each $Y \in \Gamma^\infty(T(M\times\mathbb{R}))$,

$$Y = a\frac{\partial}{\partial t} + bX,$$

where $a, b \in C^\infty(M\times\mathbb{R})$, we can define a connection $\widetilde{\nabla}$ on $E\times\mathbb{R}$ via the covariant derivatives

$$\widetilde{\nabla}_Y : \Gamma^\infty(E\times\mathbb{R}) \to \Gamma^\infty(E\times\mathbb{R}), \quad \sigma \mapsto \widetilde{\nabla}_Y\sigma = a\frac{\partial\sigma}{\partial t} + b\left((1-t)\nabla_X\sigma + t\nabla'_X\sigma\right).$$

Calling $\widetilde{\Lambda}$ the curvature 2-form associated to $\widetilde{\nabla}$, we construct $[f(\widetilde{\Lambda})] \in H^{2k}_{\mathrm{dR}}(M\times\mathbb{R})$ for every invariant polynomial of degree k. Consider now the canonical embedding map

$$\mathbb{R} \ni \epsilon \mapsto \iota_\epsilon : M \to M\times\mathbb{R}, \quad p \mapsto \iota_\epsilon(p) = (p,\epsilon).$$

Let $\Lambda^\epsilon \doteq \iota_\epsilon^*\widetilde{\Lambda}$ and, since ι and f act on different spaces, it descends that

$$\iota_\epsilon^* f(\widetilde{\Lambda}) = f(\iota_\epsilon^*\widetilde{\Lambda}).$$

At the same time due to homotopy invariance of de Rham cohomology, *cf.* Theorem 117 and due to Proposition 114 and Lemma 115, we can conclude that

$$[f(\Lambda)] = \iota_0^*([f(\widetilde{\Lambda})]) = \iota_1^*([f(\widetilde{\Lambda})]) = [f(\Lambda')],$$

which is the sought statement. $\square$

Definition 178. *Let M be a smooth manifold and let $E \equiv E[M;\pi,\mathbb{R}^m]$ be a real, smooth vector bundle. We call* **characteristic class** *any $f(E) \doteq [f(\Lambda)] \in H^{2k}_{\mathrm{dR}}(M)$ where f is an invariant polynomial of degree k while Λ is the curvature 2-form built out of any connection of E.*

Observe that the notation $f(E)$ is justified by Proposition 177 which guarantees that there is no dependence from the choice of the connections and thus it is an intrinsic property of E. We can study a few structural properties of the characteristic classes of a real, smooth vector bundle. The first concerns its naturality with respect to bundle homomorphisms.

Proposition 179. *Let $E \equiv E[M;\pi,\mathbb{R}^m]$ and $E' \equiv E'[M';\pi',\mathbb{R}^{m'}]$ be two real, smooth vector bundle over two smooth manifolds M, M' and let $H \in C^\infty(E;E')$ be a bundle homomorphism covering $h \in C^\infty(M;M')$ as per Definition 34. It holds that, for every characteristic class $f(E)$,*

$$f(E) = h^*(f(E')).$$

Proof. Let ∇ be a connection on E' and let $h^*\nabla$ be the induced one on E, *cf.* Proposition 90. Accordingly if $\omega \in \Omega^1(M';\mathfrak{gl}(n,\mathbb{R}))$ is the connection 1-form associated to ∇, then $h^*\omega$ is that associated to $h^*\nabla$. Using Cartan second structure equation, *cf.* Theorem 93 together with the property that $h^*\omega \wedge h^*\omega = h^*(\omega\wedge\omega)$, *cf.* Point 2. in Proposition 64, it descends that the curvature form of E is $h^*\Lambda$, where Λ is that of E'. Since invariant polynomials do act on Lie algebra elements the statement of the proposition descends. □

It is a duty of a teacher to train and to test all students. I will do it proposing the following exercise which generalizes Proposition 179.

Exercise 64. Let $E \equiv E[M;\pi,\mathbb{R}^m]$ be a real smooth vector bundle over a smooth manifold M. Let N be a second smooth manifold and let $h \in C^\infty(N;M)$. Calling h^*E the pull-back bundle as per Definition 37, prove that, for every invariant polynomial f, it holds

$$f(h^*E) = h^*(f(E)).$$

Finally we show that several characteristic classes are trivial.

Proposition 180. *Let $E \equiv E[M;\pi,\mathbb{R}^m]$ be a real smooth vector bundle over a smooth manifold M. If f is an invariant polynomial of odd degree, then $f(E) = 0$.*

Proof. Recall that in Equation (59) we have individuated generators of the algebra of invariant polynomials, namely $I_n \simeq \mathbb{R}[\sigma_1,\dots\sigma_n]$. The statement of the proposition descends if we prove that $[s_k(\Lambda)] = 0$ for any odd $k \in \mathbb{N}$. Here Λ is the curvature 2-form associated to any connection on E. We can exploit this freedom by observing that, being E smooth, hence paracompact, it can be endowed with a metric $\langle,\rangle : \Gamma^\infty(E)\times\Gamma^\infty(E) \to C^\infty(M)$, namely a fiberwise positive inner product. Much in the same spirit of the construction of the Levi-Civita connection on the tangent bundle of a Riemannian metric, we consider a cover of M $\{U_\alpha\}_{\alpha\in I}$, I being an index set, where each U_α is chosen so that there exists a local frame: $\sigma_i : U_\alpha \to U_\alpha \times \mathbb{R}^m$, $i = 1,\dots,m$, *cf.* Definition 30 and Remark 8. Without loss of generality we can choose via a Gram-Schmidt procedure the frame elements as orthogonal, that is

$$\langle\sigma_i,\sigma_j\rangle = \delta_{ij}, \quad \forall i,j = 1,\dots,m$$

where δ is the Kronecker delta. In addition, consider on each U_α the trivial connection ∇^α defined in Equation (26). Consider two sections $s, s' \in \Gamma^\infty(E|_{U_\alpha})$. There must exist $a_i, b_i \in C^\infty(U_\alpha)$, $i = 1,\dots,m$ such that

$$s = \sum_{i=1}^m a_i\sigma_i, \quad s' = \sum_{i=1}^m b_i\sigma_i \Longrightarrow \langle s,s'\rangle = \sum_{i=1} a_ib_i.$$

For every $X \in \Gamma^\infty(TM)$, the Leibnitz rule entails that, being $\langle s, s' \rangle \in C^\infty(M)$,

$$X(\langle s, s' \rangle) = \sum_{i=1}^{m} (X(a_i)b_i + b_i X(a_i)). \tag{60}$$

At the same time

$$\nabla^\alpha_X s = \sum_{i=1}^{m} X(a_i)\sigma_i, \quad \nabla^\alpha_X s' = \sum_{i=1}^{m} X(b_i)\sigma_i,$$

which prove that the trivial connection is metric compatible, namely, for every $X \in \Gamma^\infty(TM|_{U_\alpha})$ and for every $\sigma, \sigma' \in \Gamma^\infty(E|_{U_\alpha})$

$$X(\langle \sigma, \sigma' \rangle) = \langle \nabla^\alpha_X \sigma, \sigma' \rangle + \langle \sigma, \nabla^\alpha_X \sigma' \rangle.$$

Recalling that, via a partition of unity argument, we can glue the trivial connections to a global one which we indicate with ∇, we consider the associated connection 1-form $\omega \in \Omega^1(M; \mathfrak{gl}(n, \mathbb{R}))$. Using Equation (28) together with Equation (60) for $s = \sigma_i$ and $s' = \sigma_j$

$$0 = \langle \sum_{k=1}^{m} \omega_{ik}(X)\sigma_k, \sigma_j \rangle + \langle \sigma_i, \sum_{k=1}^{m} \omega_{jk}(X)\sigma_k \rangle = \omega_{ij}(X) + \omega_{ji}(X).$$

In other words the connection 1-form is skew-symmetric and this property is inherited by Λ, the associated curvature 2-form on account of Cartan second structural equation, *cf.* Theorem 93. Every power of a skew symmetric matrix keeps this property and therefore if k is odd

$$\sigma_k(\Lambda) = \mathrm{Tr}(\Lambda^k) = 0,$$

which concludes the proof. $\square$

The following definition summarizes and concludes our analysis of characteristic classes of real vector bundles.

Definition 181. *Let $E \equiv E[M; \pi, \mathbb{R}^m]$ be a real, smooth vector bundle over a smooth manifold M. We call* **Pontrjagin class** *of degree k, $p_k(E) \in H^{4k}_{\mathrm{dR}}(M)$ corresponding to the invariant polynomial $\frac{1}{(2\pi)^{2k}} s_{2k}$ identified in Proposition 175. We call* **total Pontrjagin class**

$$p(E) = \sum_{i=0}^{\left[\frac{n}{2}\right]} p_i(E) \in H^\bullet_{\mathrm{dR}}(M),$$

where $\left[\frac{n}{2}\right]$ coincides with $\frac{n}{2}$ for n even and with $\frac{n-1}{2}$ for n odd, while $p_0(E) \doteq 1$.

In order to getting acquainted with this concept and to show a concrete way to compute in practical scenarios the Pontrjagin classes we recommend solving the following exercise

Exercise 65. Let $E \equiv E[M; \pi, \mathbb{R}^m]$ be a real, smooth vector bundle over a smooth manifold M. Prove that

$$p(E) = \left[\det\left(\mathbb{I} + \frac{\Lambda}{2\pi}\right)\right],$$

where $\Lambda \in \Omega^2(M; \mathfrak{gl}(n;\mathbb{R}))$ is the curvature 2-form associated to any connection ∇, while $\mathbb{I}$ indicates the 2-form such that for all $X, Y \in \Gamma^\infty(TM)$, $\mathbb{I}(X,Y)$ is the $n \times n$ identity matrix, while $\det : \Omega^2(M; \mathfrak{gl}(n,\mathbb{R})) \to \Omega^2(M)$ is the natural generalization of the determinant function.

Remark 44. *It is worth mentioning that we are taking a rather pragmatic and computationally oriented approach to introducing characteristic classes. In particular we have realized Pontrjagin classes as elements of de Rham cohomology. Yet, if one follows [13] and most of the recent literature, it is customary to follow an axiomatic approach to characteristic classes. In this way these are best seen as elements of a suitable singular cohomology group .For example Pontrjagin classes can also be read as elements of $H^{2k}(M;\mathbb{Z})$, $k \in \mathbb{N}$ and, in addition, a greater diversification is allowed. As we commented in Remark 17, existence and uniqueness of spin structures are subordinated to the Stiefel-Whitney class which are yet other characteristic classes though lying in the singular cohomology of the underlying manifold with $A = \mathbb{Z}_2$ as field. As last comment we want to emphasize that, although characteristic classes over relevant information on the topology of the underlying manifold, often they are not sufficient to disentangle different isomorphism classes of the underlying vector or principal bundle. Nonetheless they offer valuable information on the underlying spaces.*

Characteristic Classes for Complex Vector Bundles.

The analysis of the previous section is tied to the underlying smooth vector bundle being real. In the complex case, the procedure to introduce characteristic classes is slightly different. Here we analyze skipping all the steps where the differences with the preceding case are limited to complexification. These is nothing but taking the tensor product with the complex numbers. To quote some, though non exhaustive examples, letting M be a smooth real manifold, we call for every $k \in \mathbb{N} \cup \{0\}$.

$$\Omega^k(M;\mathbb{C}) \doteq \Omega^k(M) \otimes \mathbb{C}, \quad H^k_{\mathrm{dr}}(M;\mathbb{C}) \doteq H^k_{\mathrm{dR}}(M) \otimes \mathbb{C}.$$

Observe that we not requiring a priori that M is a complex manifold. If consider instead singular homology and cohomology, we can work from scratch with the field $A = \mathbb{C}$. Yet, as a rather simple application of the universal coefficient theorem [6], one can infer that, for every $k \in \mathbb{N} \cup \{0\}$,

$$H_k(M;\mathbb{C}) = H_k(M;\mathbb{R}) \otimes \mathbb{C}, \quad \text{and} \quad H^k(M;\mathbb{C}) = H^k(M;\mathbb{R}) \otimes \mathbb{C}.$$

Having established these conventions, the analysis is very similar to the preceding case and we sketch the main points.

- We consider $F \equiv F[M; \pi, \mathbb{C}^m]$ a complex vector bundle, *cf.* Definition 20 over a smooth manifold M, not necessarily complex. Hence F is a manifold but not necessarily a complex manifold. The space of smooth section $\Gamma^\infty(F)$ is a module over $C^\infty(M; \mathbb{C})$

- We consider the algebra of invariant polynomials defined as per Definition 174 though with $\mathfrak{gl}(m, \mathbb{R})$ and $\mathrm{GL}(m, \mathbb{R})$ replaces respectively by $\mathfrak{gl}(m, \mathbb{C})$ and by $\mathrm{GL}(m, \mathbb{C})$. Similarly to the real case it is generated either by $\{s_i\}_{i=1,\ldots,m}$, *cf.* Proposition 175, and by $\{\sigma_i\}_{i=1,\ldots,m}$, *cf.* Equation (59):
$$J_m \simeq \mathbb{C}[s_1, \ldots, s_m] \simeq \mathbb{C}[\sigma_1, \ldots \sigma_m].$$

- We consider a connection ∇ on top of E, *cf.* Proposition 89 and we built out of it the curvature R. These are uniquely associated respectively to $\omega \in \Omega^1(M; \mathfrak{gl}(n, \mathbb{C}))$ and to $\Lambda \in \Omega^2(M; \mathfrak{gl}(n, \mathbb{C}))$.

- The results of Proposition 176 and of Proposition 179 translate slavishly to the complex realm. In particular it holds that $[f(\Lambda)] \in H^{2k}_{\mathrm{dR}}(M; \mathbb{C})$ for every $f \in J_m$ of degree k. Here Λ is still any curvature 2-form.

- Repeating the proof of Proposition 180 for the case in hand, you fail. Odd characteristic classes are not a priori vanishing (*key difference*)

Putting together these data we obtain the following natural counterpart of Definition 181.

Definition 182. *Let $F \equiv F[M; \pi, \mathbb{C}^m]$ be a complex vector bundle over a smooth manifold M. We call* **Chern class** *of degree k, $c_k(F) \in H^{2k}_{\mathrm{dr}}(M; \mathbb{C})$ corresponding to the invariant polynomial $\left(\frac{-1}{2\pi i}\right)^k s_k$ identified in Proposition 175. We call* **total Chern class**

$$c(E) = \sum_{i=0}^{m} c_m(E) \in H^{\bullet}_{\mathrm{dR}}(M; \mathbb{C}),$$

where $c_0(E) \doteq 1$.

In order to getting acquainted with this concept and to show a concrete way to compute in practical scenarios the Pontrjagin classes we recommend solving the following exercise

Exercise 66. Let $F \equiv F[M; \pi, \mathbb{C}^m]$ be a complex, smooth vector bundle over a smooth manifold M. Prove that
$$c(E) = \left[\det\left(\mathbb{I} - \frac{\Lambda}{2\pi i}\right)\right],$$
where $\Lambda \in \Omega^2(M; \mathfrak{gl}(n; \mathbb{C}))$ is the curvature 2-form associated to any connection ∇, while $\mathbb{I}$ indicates the 2-form such that for all $X, Y \in \Gamma^\infty(TM)$, $\mathbb{I}(X, Y)$ is the $n \times n$ identity matrix, while $\det : \Omega^2(M; \mathfrak{gl}(n, \mathbb{C})) \to \Omega^2(M; \mathbb{C})$ is the natural generalization of the determinant function.

We can show that Chern classes are actually real cohomology classes.

Corollary 183. *Let $F \equiv F[M;\pi,\mathbb{C}^m]$ be a complex, smooth vector bundle over a smooth manifold M. Each Chern class $c_k(F) \in H^{2k}_{\mathrm{dR}}(M)$.*

Proof. Repeating slavishly the proof of Proposition 180, we can show that the trivial connection has associates connection 1-form ω and curvature 2-form Λ such that

$$\omega_{ij}(X) + \overline{\omega}_{ij}(X) = 0, \quad \Lambda_{ij}(X) + \overline{\Lambda}_{ij}(X) = 0, \quad \forall X \in \Gamma^\infty(TM).$$

The complex conjugate originates from the fact that the fiberwise positive inner product in Proposition 180 must be replace with a fiberwise Hermitean inner product. This entails that Λ is a skew-Hermitean matrix. Accordingly its determinant is real valued and, using Exercise 66, it descends that $c(F) \in H^\bullet_{\mathrm{dR}}(M)$. Hence the same must hold true for each Chern class. $\square$

This corollary is rather interesting since it entails that each Chern class $c_k(F)$ is real valued. Hence using the de Rham isomorphism, *cf.* Theorem 173, we can identify each $c_k(F)$ with a singular cohomology class with real coefficients, namely, keeping the said isomorphism implicit, $c_k(F) \in H^{2k}(M;\mathbb{R})$. At this point the connection with singular cohomology with integer coefficients can be via the universal coefficients theorem which guarantees that we can associate to $c_k(F)$ a corresponding element $c_k^{\mathbb{Z}}(F) \in H^{2k}(M;\mathbb{Z}) \otimes \mathbb{R}$. The following two exercises help clarifying some notable properties of Chern classes.

Exercise 67. Let $F \equiv F[M;\pi,\mathbb{C}^m]$ and $F' \equiv F'[M;\pi',\mathbb{C}^{m'}]$ be two smooth, complex vector bundles over a smooth manifold M. Prove the following **Whitney formula**

$$c_k(E \oplus F) = \sum_{i=0}^{k} c_i(E) c_{k-i}(F), \quad \forall k \in \mathbb{N} \cup \{0\} \tag{61}$$

where $E \oplus F$ is the Whitney sum, *cf.* Definition 27 and where $c_0(E \oplus F) = c_0(E) = c_0(F) = 1$. In addition show that Equation (61) entails that at the level of total Chern classes.

$$c(E \oplus F) = c(E)c(F).$$

Prove in addition that the same formulae hold true for real vector bundle with Chern classes replaced by the Pontrjagin ones.

Exercise 68. Let $E \equiv E[M;\pi,\mathbb{R}^m]$ be a smooth, real vector bundle over a smooth manifold M. Let $E_{\mathbb{C}} \doteq \bigsqcup_{p\in M} E_p \otimes \mathbb{C}$ be its complexification. Prove that, for all $k \in \mathbb{N} \cup \{0\}$,

$$p_k(E) = (-1)^k c_{2k}(E_{\mathbb{C}}),$$

where $p_0(E) = c_0(E) = 1$.

Characteristic Classes for Principal Bundles with $G = U(1), SU(2)$

The theory of characteristic classes for principal bundles is deeper and more complicated than that of vector bundles. It would require a whole course and we refer to [14] if a reader is interested. Our goal here is to sketch the analysis in two cases which are distinguished for their physical applications, namely when the structure group is either $U(1)$ or $SU(2)$. These principal bundles are particular instances of the so-called sphere bundles, namely fiber bundles whose typical fiber is an n-sphere. As a matter of fact recall that $U(1) \simeq \mathbb{S}^1$ while $SU(2) \simeq \mathbb{S}^3$.

In the following we will focus on the prototypical scenario ruled by the circle. The most refined analysis could be done using simplicial homology and cohomology, *cf.* Section 5.1, but, to keep this work short (recall... these are notes...) we work at the level of differential forms.

Hence to start with let $P \equiv P[M; \pi, U(1)]$ be a principal $U(1)$-bundle over a smooth manifold M of dimension $\dim M = n \geq 1$. From Theorem 50 we know that P is trivial, namely $P \simeq M \times U(1)$ if and only if it admits a global section. In addition Theorem 99 and Corollary 100 guarantee that assigning a connection ∇ to P is tantamount to constructing a connection 1-form $\omega \in \Omega^1(P; \mathfrak{g})$ subject to the constraints indicated in the said theorem. To each connection 1-form we can associate a curvature 2-form, $\Lambda \in \Omega^2(P; \mathfrak{g})$ via Equation (35). Since the structure group is Abelian and hence $\mathfrak{g} \simeq \mathbb{R}$ is commutative, this equations simplifies to

$$\Lambda = d_1^P \omega,$$

where the superscript P reminds us that we are acting on ω with the differential of P. Following Remark 33 and Proposition 101 it would be desirable to read all these data in terms of differential forms defined on the base space M. In the case of the connection 1-form this is nothing but the content of Exercise 38, item 2. in particular. In the case of the curvature 2-form, we need a separate proof.

Proposition 184. *Let $P \equiv P[M; \pi, U(1)]$ be a principal $U(1)$-bundle over a smooth manifold M and let $\omega \in \Omega^1(P)$ be a connection 1-form. There exists $\Theta \in \Omega^2(M)$ such that $\pi^*\Theta = d_1^P \omega$.*

Proof. Consider a cover of M, $\{U_\alpha\}_{\alpha \in I}$, I being an index set, such that $P|_{U_\alpha} = \pi^{-1}(U_\alpha) \simeq U_\alpha \times U(1)$. Without loss of generality we can assume that U_α is endowed with local coordinates x_i, $i = 1, \dots, n = \dim M$ and we call φ the coordinate on the fiber $U(1) \simeq \mathbb{S}^1$. Hence, given a connection 1-form ω it holds that its restriction to $\pi^{-1}(U_\alpha)$ can be written as

$$\omega|_{U_\alpha} = \sum_{i=1}^n f_i(x) dx_i + \theta_0,$$

where $f_i(x) \in C^\infty(U_\alpha \times \mathbb{S}^1)$ for all $i = 1, \dots, n$ while $\theta \in \Omega^1(\mathbb{S}^1)$ is such that $\int\limits_{\mathbb{S}^1} \theta = 2\pi$. Observe that locally $\theta_0 = d_\varphi$ and that it implements condition 1. in Theorem 99. In addition the function $f_i(x)$, $i = 1, \dots, n$ do not depend on the coordinate φ to implement condition 2. in Theorem 99. Taking the differential of ω as per Equation (35) we obtain that

$$d_1^P \omega|_{\pi^{-1}(U_\alpha)} = \sum_{i,j=1}^n \frac{\partial f_i}{\partial x_j} dx_i \wedge dx_j.$$

The right hand side identifies a 2-form $\Theta \in \Omega^2(U_\alpha)$. Repeating this procedure on each open subset and gluing together the results we obtain $\Theta \in \Omega^2(M)$ and the above identity reads globally as

$$d_1^P \omega = \pi^* \Theta,$$

which is meaningful and unambiguous since π^* is injective, *cf.* Exercise 25. □

Recalling Exercise 42 we know that the curvature 2-form $\Lambda \in \Omega^2(P;\mathfrak{F})$ is closed when the structure group is Abelian. Therein it is asked to prove that this is equivalent to the first Maxwell equation when the underlying space $M = \mathbb{R}^n$, $n \geq 1$. Here we show that this particular result holds true in general.

Corollary 185. *Let $P \equiv P[M;\pi,U(1)]$ be a principal $U(1)$-bundle over a smooth manifold M and let $\omega \in \Omega^1(P)$ be a connection 1-form. Then the associated curvature 2-form $\Theta \in \Omega^2(M)$ as per Proposition 184 is closed.*

Proof. Let ω be a connection 1-form and let $\Lambda = d_1^P\omega$. Let $\Theta \in \Omega^2(M)$ be such that $\Lambda = \pi^*\Theta$, *cf.* Proposition 184. Then, on account of Equation (35)

$$d_2^P \Lambda = 0 = d_2^P \pi^* \Theta = \pi^* d_2^M \Theta,$$

where we used the structural properties of the pull-back, *cf.* Theorem 68. Since the pull-back is injective, *cf.* Exercise 25, $\pi^* d_2^M \Theta = 0$ if and only if $d_2^M \Theta = 0$. □

In other words this corollary guarantees us that each curvature 2-form on a principal $U(1)$-bundle is a closed form and hence it identifies a de Rham cohomology class

$$P[M;\pi,U(1)] \longrightarrow \Theta \in \Omega^2_d(M) \longrightarrow [\Theta] \in H^2_{\mathrm{dR}}(M).$$

It is therefore reasonable to suspect that the curvature of a principal $U(1)$-bundle might be used as the starting point to define in this case characteristic classes. Yet, to this end we need to make sure that a counterpart of Proposition 177 holds true.

Lemma 186. *Let $P \equiv P[M;\pi,U(1)]$ be a principal $U(1)$-bundle over a smooth manifold M and let $\omega \in \Omega^1(P)$ be a connection 1-form. Let $[\Theta] \in H^2_{\mathrm{dR}}(M)$ be the cohomology class built out of the curvature 2-form individuated by ω as per Corollary 185. Such class in independent from the choice of the connection.*

Proof. On account of Exercise 38 we now that $\mathcal{C}(P)$, the set of all connections, is an affine space modeled over $\Omega^1(M)$. Hence, for every two connections 1-forms $\omega, \omega' \in \Omega^1(P)$, there must exist $\eta \in \Omega^1(M)$ such that $\omega' = \omega + \pi^*\eta$. At the level of curvature this entails

$$d_1^P \omega' = d_1^P \omega + d_1^P \pi^* \eta.$$

Yet, we can apply Theorem 68 to exchange the differential with the pull-back. Furthermore Proposition 184 together with Corollary 185 there exist $\Theta, \Theta' \in \Omega^2_d(M)$ such that the last identity becomes

$$\pi^*\Theta' = \pi^*\Theta + \pi^* d_1^M \eta,$$

which, being π^* injective, *cf.* Exercise 25, yields $\Theta' = \Theta + d_1^M \eta$. This entails that Θ and Θ' are representatives of the same de Rham cohomology class. $\square$

The following definition summarizes the results obtained.

Definition 187. *Let $P = P[M;\pi, U(1)]$ be a principal $U(1)$-bundle over a smooth manifold M. We call* **(real) Euler class** $e(P) \in H^2_{\mathrm{dR}}(M)$ *such that*

$$e(P) = -\frac{1}{2\pi}[\Theta],$$

where Θ is any 2-form over M corresponding to the curvature associated to a connection on P.

The Euler class can be considered as a full-fledged characteristic class as we establish in the following theorem.

Theorem 188. *The Euler class is a characteristic class for principal $U(1)$-bundles, namely it is natural with respect to pull-back. Hence, given two principal $U(1)$-bundles $P \equiv P[M;\pi, U(1)]$ and $P' \equiv P'[M';\pi', U(1)]$ and given a principal bundle morphism, cf. Definition 47, $\Phi \in C^\infty(P;P')$ covering $\phi \in C^\infty(M;M')$, then*

$$e(P) = \phi^* e(P').$$

Proof. Consider a connection 1-form $\omega \in \Omega^1(P')$. It descends that $\phi^*\omega \in \Omega^1(P)$ is a connection 1-form for P, whose associated curvature is

$$\Lambda = d_P^1 \phi^*\omega = \Phi^*(d_1^{P'}\omega') = \Phi^*\Lambda'.$$

This entails that, calling $\Theta \in \Omega^2_d(M)$ and $\Theta' \in \Omega^2_d(M')$ the 2-forms such that $\Lambda = \pi^*\Theta$ and $\Lambda' = \pi^*\Theta'$, it holds

$$\pi^*\Theta = (\Phi^* \circ (\pi')^*)\Theta' = ((\pi')^* \circ \phi^*)\Theta',$$

where we used the commutative diagram in Equation (16). Since π^* is injective, *cf.* Exercise 25 this entails that $\Theta = \phi^*\Theta'$ from which it descends the sought statement. $\square$

Similarly to the case of characteristic classes for smooth vector bundles also in this case the Euler class represents a distinguished invariant of a principal $U(1)$ bundle which has been constructed out of a differential form. Yet, still on account of de Rham isomorphism, *cf.* Theorem 173, up to an implicit isomorphism

$$e(P) \in H^2(M;\mathbb{R}).$$

Applying the universal coefficient theorem for cohomology [6], we can infer that to $e(P)$ we can associate a unique class in the singular cohomology with integer coefficients, called **(integer) Euler class**

$$e(P) \mapsto e_{\mathbb{Z}}(P) \in H^2(M;\mathbb{Z}).$$

It is interesting to remark that, were we have analyzed the same scenario using simplicial cohomology, we would have constructed directly $H^2(M;\mathbb{Z})$. We stress once more that this has notable advantages since singular cohomology with integer coefficients contains additional topological information related to torsion elements, which disappear once taking the tensor product with $\mathbb{R}$.

Remark 45. *Observe that the Euler class has a neat physical interpretation, being nothing but the first Maxwell equations. Yet its relevance goes far beyond this fact and it is encoded in a classifying theorem for principal $U(1)$-bundle. At this stage the author reckons that going through another dozen pages of proofs might be a little bit too much and I content (yes I am the author, so I am allowed to write "I content") myself in quoting this theorem without giving a proof which requires working at the level of simplicial and singular cohomology with integer coefficients. A proof of the following statements can be found in [14, Th. 6.22].*

Theorem 189. *There exists a 1:1 correspondence between isomorphism classes of principal $U(1)$-bundles, cf. Definition 47 and $H^2(M;\mathbb{Z})$ which is implemented by the integer Euler class, namely*

$$[P[M;\pi,U(1)]] \mapsto e_{\mathbb{Z}}(P).$$

Observe that this theorem tells us that, if the second singular cohomology group with integer coefficients is trivial, than, up to an isomorphism there exists only the trivial $U(1)$ bundle over M. This entails in addition that all connections are flat, cf. Definition 103. Yet, if all connections are flat, namely $H^2_{\mathrm{dR}}(M) = \{0\}$, this does not necessarily entail that the associated principal $U(1)$-bundle is flat up to isomorphisms. As a matter of fact, recalling that M admits a finite good cover being smooth, cf. Exercise 50, this entails that all singular cohomology groups are finitely generated. Hence we can apply the fundamental theorem of finitely generated Abelian group to conclude that ... wait... you do not know this Theorem? Oh... check then [9, §8] ... I was saying... that

$$H^2(M;\mathbb{Z}) = \mathbb{Z}^{\beta_2(M)} \oplus \mathbb{Z}_{q_1} \oplus \cdots \oplus \mathbb{Z}q_n,$$

where $\beta_2(M) \in \mathbb{N}$ is finite and it coincides with the second Betti number, cf. Definition 107, while each $\mathbb{Z}_{q_i}$, $i = 1, \ldots, n$ is a finite group and $q_1, \ldots q_n$ are powers of non necessarily distinct prime numbers. The first term is the direct sum is called the free part and the remaining ones the torsion part. It is now clear that taking the tensor product with $\mathbb{R}$ so to compare with de Rham cohomology via the de Rham isomorphism, cf. Theorem 173, will make us lose all information on torsion since

$$H^2(M;\mathbb{Z}) \otimes \mathbb{R} \simeq \mathbb{R}^{\beta_2(M)}.$$

This is the reason why de Rham cohomology is not sufficient to classify isomorphism classes of principal $U(1)$-bundles.

To conclude the section, we give a quick look to another case of relevance in theoretical and mathematical physics, namely principal $SU(2)$-bundle over a smooth manifold M. In this case a classification theory is possible and it is simplified by the fact that $SU(2)$ is diffeomorphic to the unit 3-sphere. In this case one can prove a generalization of Theorem 189 which we report here. The proof is rather complicated since it involves understanding the so-called *primary obstruction* and it requires advanced techniques in algebraic topology. The result below is the specialization of a more general result, see [14, Th. 6.30] though no proof nor references are given or see [15] for the general theory.

Theorem 190. *There exists a 1:1 correspondence between isomorphism classes of principal $SU(2)$-bundles, cf. Definition 47 and $H^4(M;\mathbb{Z})$.*

As last remark, I would like to emphasize the relevance of this theorem in physical applications. In many instances it is convenient to consider relativistic models on Riemannian manifolds, via Wick rotation. Without entering into the details of the admissibility of this operation, we stress that, if one considers four-dimensional Riemannian manifolds with no additional restrictions than, it is easy to conceive non trivial principal $SU(2)$-bundles. As an example if $M = \mathbb{S}^4$, it turns out that $H^4(\mathbb{S}^4,\mathbb{Z}) \simeq \mathbb{Z}$. This feature leads to interesting phenomena related to the so-called instanton configurations, which are often used in QCD, although in this case the structure group is $SU(3)$. Nonetheless, if one sticks to the idea of working in a Lorentzian framework, the underlying background (M, g) is always chosen to be globally hyperbolic, so to make sense of well-posed initial value problem for hyperbolic PDEs such as the wave equation. In this case M is isometric to $\mathbb{R} \times \Sigma$ with a distinguished form of the line element which plays no role here. Yet it is important that Σ is a three-dimensional Riemannian manifold. Since singular cohomology is homotopy invariant, *cf.* Remark 43, it descends that, for a globally hyperbolic spacetime M,

$$H^4(M;\mathbb{Z}) \simeq H^4(\Sigma,\mathbb{Z}) = \{0\},$$

since all k-th cohomology groups with k larger than the underlying space dimension are trivial. Hence, while non trivial principal $U(1)$-bundles appear in many relativistic scenarios, this is not the case when the structure group is $SU(2)$. If we consider a generic Lie structure group G, then the analysis above becomes more complicated. The construction of characteristic classes is very similar to the one leading to the Euler class for principal $U(1)$-bundles culminating in the *principal theorem of Chern-Weil theory*, see [14, Th. 6.47], but a classification result for isomorphism classes of principal G-bundles is very hard to obtain and it requires really advanced techniques in algebraic topology.

I would like to conclude these notes as I do often with my notes (no, I know Rubens... I have not concluded those on fluids in this way... oh wait... you have been mentioned for the second time), namely listing some topics which I would have loved to explain and I might do in the distant future. Yet they are certainly accessible to students, who are cordially invited to learn them on their own... if you got till this point without skipping to the end as I always do with books (it is a strange habit, isn't it?), you are certainly up to the task:

1. the *universal coefficient theorem*, which appears sometimes at key points, but I never stated it, let alone proved it,

2. the *Chern-Weil theory of characteristic classes* for principal G-bundle so to construct characteristic classes starting from simplicial and singular homology and cohomology. In

this way we could really make justice to the example of the *Hopf bundle (fibration)* as a typical example of a non trivial principal $U(1)$-bundle

3. the *Aharonov-Bohm effect and (more generally) Berry phases*, which are the prototypes of a topological effect in field theory,

4. the *dynamics of Yang-Mills theory*, namely constructing the Yang-Mills and the Chern-Simons top forms, focusing on the reason why they are basically the only natural choices available. These notes are focused on kinematics, but dynamics is really just round the corner,

5. the *well-posedness of the initial value problem* for Yang-Mills equations on four dimensional globally hyperbolic spacetimes as proved at the end of the last century by Chrusciel and Shatah in [5].

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